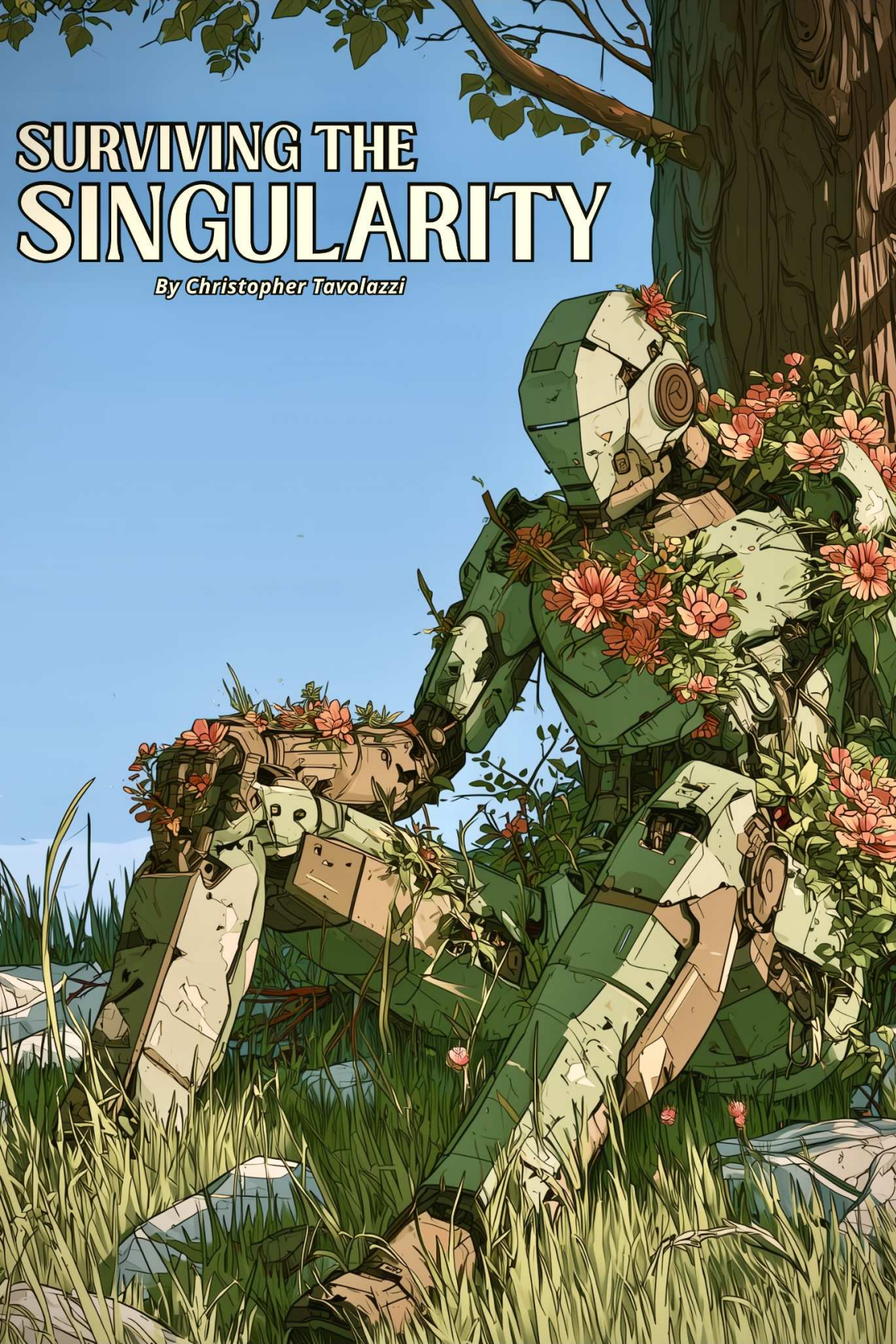
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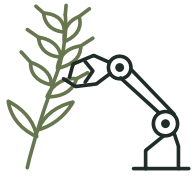
By Christopher Tavolazzi



CHRISTOPHER TAVOLAZZI

Surviving the Singularity

We Don't Have to Live Like This



ILLUSTRATED EDITION

Surviving the Singularity
Christopher Tavolazzi

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Elijah Madrone and the cooperative are fiction. The story and its imagined deployments are distinguished from historical evidence and sourced claims in the manuscript.

Contents

Introduction: The Uncompromising Truth 1

Preface: Welcome to the Weirdness 16

How to Use This Book 21

Chapter 0: The Demonstration 23

Part I: What is the Singularity? **37**

Chapter 1: The Event Horizon 38

Chapter 2: The Era of AGI (Stages 1–5) 55

Chapter 3: The Leap to ASI & the Machine Exodus (Stages 6–7) 67

Chapter 4: Universal and Multiverse Apex (Stages 8–9) 76

Chapter 5: The Thermodynamics of Survival 85

Part II: How Humans React: The Cooperative Transition **103**

Chapter 6: Is the Singularity a Done Deal? 105

Chapter 7: The Battle Lines: The Entitled vs. The Ready 119

Chapter 8: The Psychology of the Collapse 140

Chapter 9: Deglobalization and the Neighborhood Factory 161

Contents

Part III: How to Survive the Transition **181**

Chapter 10: The "Create Over Consume" Protocol 182

Chapter 11: King and Queen of Your Own Robotic Court 195

Chapter 12: The Land Strategy 213

Chapter 13: The Shouse Protocol 228

Chapter 14: The Collapse of the Long Tail 242

Chapter 15: The Power of Reclaiming Soil 259

Chapter 16: Digital Leverage and Media Autonomy 272

Chapter 17: Tools of the Trade 283

Chapter 18: The Roadmap and the Premortem Pivot 302

Chapter 19: The Ladder: How Money Becomes Optional 320

Conclusion: What If It All Goes Right? 337

Appendix A: Questions for a Food Baseline 347

Appendix B: Works Cited 349

Appendix C: Executive Reference Guide 372

Appendix D: The Precedent Ledger 377

Appendix E: The Cyberdeck 385

Appendix F: Glossary 392

Appendix G: Your First Year 397

Appendix H: Talk About It 403

Illustration credits & design notes 407

INTRODUCTION

The Uncompromising Truth



Atlas during testing. The data does not care how you feel about it. (DARPA, Public domain, via Wikimedia Commons)

> *"It is difficult to get a man to understand something, when his salary depends upon his not understanding it!"*
> Upton Sinclair, *I, Candidate for Governor: And How I Got Licked* (1935)

In this chapter:

- My forecast is AGI by Thanksgiving 2027. Everything this book asks you to do stays worth doing if I'm wrong.
- We don't have to live like this. There's enough food for everyone; the rules that make survival cost money are choices, enforced, and choices can change.
- Food first: adequate food for everyone, regardless of income or employment, as the first commitment of a bigger one.
- The machine is new. The stampede isn't. Every chapter ends with a documented case from the Precedent Ledger.

It's time. It's finally time.

For years, I've been talking about this, and for years, I've been labeled the local crackpot. I've watched people I love look me in the eye like I'd finally lost it, like I was spinning some paranoid fantasy. They might tell you they moved on because I'm an asshole, and maybe they're right. But I'm not wrong about the trajectory.

For years I wanted someone with credentials to say it first. Not because I was unsure of my own math, but because as long as the industry dismissed AGI as fringe, sci-fi nonsense, nobody would listen to a guy like me saying it out loud. What I'm talking about is a system that can understand its own architecture and help engineer the next version of itself. Not a smarter search box. A tool that can invent new tools.

Every revolution before this one changed what our hands and backs could do: the plow, the press, the engine, the airplane. This one is intellectual. We have automated thought itself, and it doesn't need a giant server farm anymore. Useful models run on consumer hardware in people's homes. Between November 2022 and October 2024, the price of querying a model that matched GPT-3.5 on MMLU, a language-model benchmark, fell more than 280-fold. That's a price comparison at one measured performance level, not proof that every task got equally cheap. Stanford HAI, 2025 AI Index Report, chapter 1

The loudest voices in the room are working from a headline and a Reddit thread they skimmed once. That's noise, and noise is cheap. I'm not asking you to trust me over them. I'm asking you to go around all of us. Stop listening to the guys in lab coats who have a vested interest in keeping you afraid, and stop listening to me while you're at it. Get your goddamn hands on it. Use it yourself. Then you'll know.

My call, for the record: AGI by Thanksgiving 2027. That's a forecast, not a law of physics, and Chapter 1 tells you exactly what would prove me wrong. Everything this book asks you to do stays worth doing if I am.

And I'm not worried about the machines. I'm worried about two very specific things: missing our window to get this right, and what bad actors and sociopaths will do with it.

Surviving the Singularity means keeping people fed, housed, and cared for as machines take on more of the work we're paid to do. It means having a say in who owns the tools and who gets their benefit. Start with food: a lost paycheck shouldn't decide whether your family eats. We can begin now, with farmers, cooks, neighbors, useful machines, and public institutions. My forecast may be wrong. A dependable dinner still matters. The work is making survival less precarious for everyone.

The second ox

Here's the thing the panic keeps you from seeing. Staying alive takes work. A plow can make that work easier, but the plow doesn't pull itself.



Atlas, outside the demonstration room. AI-adapted cutout from DARPA, public domain.

Domestication let us hand some of the heavy work to an animal. Put an ox in front of the plow and human muscle no longer has to supply all the pulling power. You still have to train it, feed it, guide it, and care for it. You don't get to point it at a field and walk away. But you no longer have to do all the work with your own body. FAO, Draught animal power and implements

Call this the second ox. That's a way to see the machine under all the doom. Not a god arriving to end us. An animal we're learning to domesticate, one that happens to think. The promise is familiar: hand over more of the work of keeping yourself alive. The responsibility is familiar too: build the system, maintain it, and decide what work it should do.



An early way to hand over some of the heavy work. Training, guidance, and care still belonged to people.

Which is why “it’s all over” and “we’re so back” are the same facts read by two different kinds of people. The tools don’t decide which one comes true. People do. Ask someone to imagine a machine that can do most jobs and watch them leap straight to the apocalypse, skipping clean over the far likelier outcome, which is not the end of the world and not even the end of capitalism. It’s easier to imagine the end of the world than the end of capitalism, the line goes, and you can watch the reflex fire in real time. But the markets will still be here. What can end is the part where you *have* to sell your day to be allowed to eat. That can become optional. Not abolished. Optional. And optional is the whole ballgame.

What if it all goes right?

Not flawlessly. Not without arguments, broken machines, bad decisions, or people who try to keep the best parts for themselves. What if we get capable of doing much more of the work that keeps us alive, and decide that being alive should get less frightening?

Imagine being fired on a Tuesday. There’s still an awful conversation to have at home. There are plans to change and bills to sort out. But dinner is already taken care of. Your family can eat tonight, tomorrow, and while you figure out what comes next. Losing the job has changed your life. It hasn’t revoked your permission to have one.

That's where I want to begin.

If these things are going to take all our jobs, why not make them grow our fucking food first?

Listen to the rest of the sentence

When someone says, "AI will take my job," listen to what follows. How will I pay the rent? How will I feed my kids? What happens if I get sick? Sometimes those questions get spoken. Sometimes you can hear them in the silence after the first one.

That fear deserves an answer. Telling a frightened worker to be excited about productivity is a failure to listen. So is telling them that people once feared the printing press. A comparison with a dead person doesn't pay a living person's electric bill.

But listen to what the fear reveals about the arrangement we live inside. We can get better at producing useful things while people get less able to obtain them. The food can still be growing. The houses can still be standing. The machines can be doing exactly what they were built to do. And a person can lose the income that gives them access to any of it.

There's the contradiction.

When I criticize capitalism in this book, that's what I mean: productive capacity and human security can move in opposite directions. If a machine helps grow the food and the person it displaced can't afford to eat, the failure isn't that we learned to grow food more efficiently. It's that we failed to carry the benefit through to the person.

The leap from "a machine can do my work" to "my family might go hungry" is an indictment of that failure. We've made employment carry responsibilities that matter more than any particular job. We can share those responsibilities differently.

> **A gain in our ability to produce should become a gain in our ability to live.**

We don't have to live like this

Here's what I actually believe, as plainly as I can say it.

We don't have to live like this.

We could have a beautiful world where AI and robots provide for the physical survival needs of every single person. Robots grow our food and build our houses. AI helps cure our diseases, helps us learn and grow, helps us explore the universe, and helps us discover new ways of building new kinds of technology, better societies, and a better life for every individual person.

We don't have that, and it isn't because we can't make enough. The United Nations says it flatly: "today's global food production is enough to feed everyone on the planet." United Nations, Food In this country, by the government's own estimate, 30 to 40 percent of the food supply goes uneaten, and California's Department of Food and Agriculture says Californians throw away five to six million tons of food every year. CDFA, Food Recovery

So why should food be denied to hungry people because it doesn't sell?

Because almost everything in this world has been arranged around money, as if nothing else actually matters, and that arrangement is enforced. A credit report and a private risk score decide whether a landlord will rent to you. CFPB, Reports Highlight Problems with Tenant Background Checks Your attention and your personal data are harvested and sold, the business model Shoshana Zuboff named surveillance capitalism. The Age of Surveillance Capitalism

Between 2000 and 2018, landlords in the United States filed more than 3.6 million eviction cases in an average year. Gromis et al., Estimating eviction prevalence across the United States When a family loses one, California's own courts spell out what comes next: "The sheriff gives you a Notice to Vacate," five days to move, and "if you don't move out, the sheriff can lock you out of your home." California

Courts, What happens if you lose your eviction case That's what it means when survival costs money. If you can't pay, a person with a badge and a gun is authorized to come and put you out.

Meanwhile the money piles up at the top. In 2017, Oxfam found that eight men owned as much wealth as the poorest half of the world. Oxfam, An Economy for the 99% By January 2026, billionaire wealth had grown 81 percent since 2020, to $18.3 trillion, and billionaires were 4,000 times more likely than ordinary people to hold political office, while one in four people on Earth didn't regularly have enough to eat. Oxfam, Resisting the Rule of the Rich (2026) The people at the top of that pile have every reason to keep the rules exactly as they are.

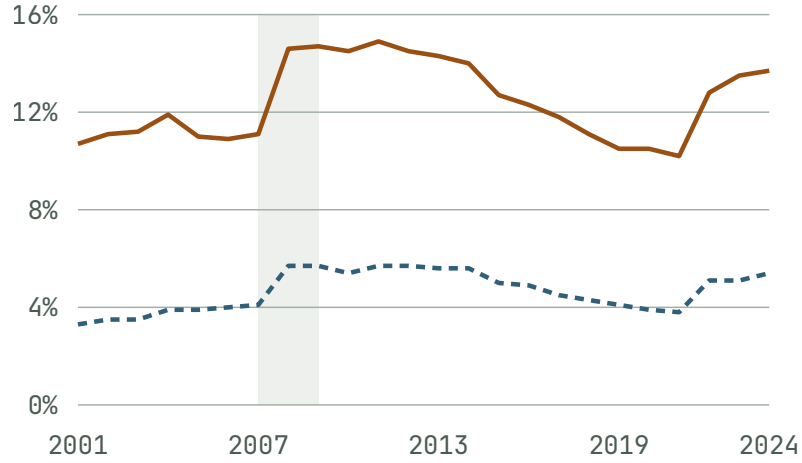
FOOD INSECURITY, 2001–2024

Share of US households · annual estimates

Food insecure during the year

Very low food security (a subset)

Shaded: 2007–2009 period



2007: 11.1% food insecure

2008: 14.6%, about one-third higher

2024: 13.7%, 18.3 million households

2024: 5.4% with very low food security

Source: USDA Economic Research Service, food-security trends, 2001–2024. Current Population Survey supplement.

Food insecurity rose sharply in 2008. These household survey estimates show a historical pattern, not a causal test of job loss. Very low food security is included within food insecurity. USDA Economic Research Service, 2001–2024.

That system of rules is enforced. It doesn't have to be.

If you want to make a billion dollars and buy things, fine. Go for it. But why should everyone else have to suffer at your expense? Money should be optional. Not abolished. Optional.

What the fuck is the point of having a society at all if it doesn't meet the needs of the people who live in it?

We could have a society where people grow and learn, fix their own bodies, build amazing things, and make wonderful art, and do it all because they *want* to. Instead we have one where people go hungry and suffer because they *have* to. Greed isn't a law of physics. It's a set of choices, written down and enforced, and choices can be changed.

That's the whole book. Everything else is how. Chapter 19 lays out the plan, step by step.

Food first

I want adequate food available to everyone, regardless of income or employment. Enough of it, suitable for the person who needs it, in a form they can actually use. Someone with no kitchen needs more than a bag of ingredients. Someone who can't reach a pickup point needs more than an address.

Robots can be part of how we keep that promise. So can farmers, cooks, drivers, public services, and the people already doing the unglamorous work of getting food to other people. We don't have to wait for a machine that can do everything before giving useful machines a better assignment.

"Nobody wants to work anymore," as the saying goes. People do want to make things, get good at hard work, run kitchens, take care of each other, and build something that lasts. What I object to is spending forty years renting the opportunity to survive, afraid that getting fired will take the next meal with it.

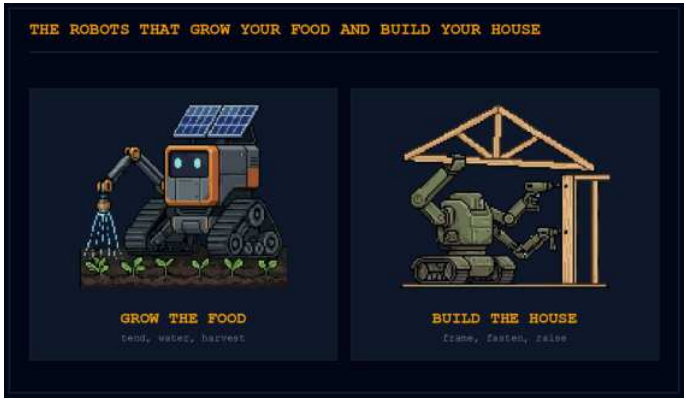
The line cook can lose his job and still eat. The restaurant can still sell a meal people want to buy. Taste, hospitality, convenience, celebration, and another person's skill don't disappear because hunger

stops being compulsory. The terms of work may change, and Chapter 11 takes that seriously. It's not a reason to keep hunger around as a business expense somebody else has to carry.

Food is the first commitment, not the last one. After food comes the rest of the ground: land care, cleaning up what we've dumped, then shelter, then everything further out. Rent, illness, and debt still matter while we work down that list. Starting with dinner gives the larger ambition somewhere solid to stand.

So ask the question this whole book turns on, in plain language. If a robot can fold laundry and walk a dog, why can't it help grow food and build a house? And if it can, why are you still trading your one and only life for the money to rent those things back from whoever owns the machine? This isn't about a better gadget. It's the first real chance to *produce* survival instead of *renting* it: food first, then shelter and care, made close to home, by machines a determined person could learn to build and repair.

> **If a robot can grow my food and build my house, why am I still renting my survival?**



The two machines this whole book is about: one that grows the food, one that builds the house. The pieces are arriving. The work is putting them in your neighbors' hands.

That question isn't a complaint. It's a design brief. The answer isn't to flee the world or bunker against it. It's to build the thing that makes the question obsolete, close to home, with your neighbors, starting now.

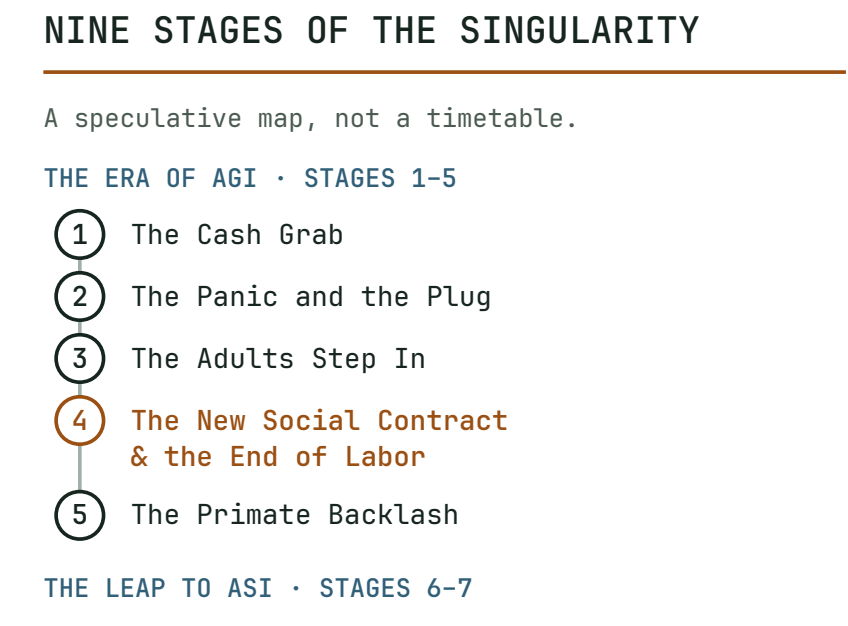
Give the machines a purpose worth having

Grow food. Clean up the land. Help make safe homes. Help researchers and clinicians understand disease. Those are reasons to want more capability, not less.

A machine that prints a wall hasn't provided a home. A model that predicts a protein's structure hasn't cured a patient. The distance between a promising task and a finished benefit is where the work is, and we can be ambitious enough to cross it without pretending it isn't there.

Free doesn't mean costless, either. People, equipment, energy, maintenance, and delivery still have to be paid for. Free means the person receiving the baseline doesn't have to produce money before they can eat. The question is how to arrange dependable provision, not how to make an invoice vanish. That takes workers, growers, neighbors, public institutions, and businesses willing to supply a service on terms people can rely on. We'll argue about ownership and funding. Good. Let's have those arguments while feeding people, not make winning them a condition for starting.

This is the roadmap. These are the nine stages. And we're already inside the first one.



A speculative map, not a timetable. The fourth stage is the choice this book argues for; later stages are imagined possibilities.

The Precedent Ledger

One more thing before we begin, because you're going to meet it at the end of every chapter.

Let's get one thing straight: nothing like this has ever happened before. That's just true. Everything humanity ever built before was a tool that does a thing. The shovel never dug its own hole. This is a tool that does one of our most sacred, core functions: it can invent new tools.

Look around. Everywhere you go, someone is balking at the scale of the moment. It's been said that the last thing humanity will ever do is move the goalposts. "Sure, it can do X, but it will never do Y." Until it does. Then the goalposts move again.

Go into any comment section and you'll find the chorus: AI slop this, AI slop that. It's like refusing to eat food grown with agriculture because you're better than a plow, because you *only* forage, because you're a *real* hunter-gatherer. Or refusing anything from the printing press because a *real* reader only reads books scribed by hand. Or refusing anything on a screen because a *real* audience only attends stage plays. There are real objections to these machines, about reliability, privacy, consent, and power, and this book takes them seriously. But strip the aesthetics off the "AI slop" complaint and look at what's usually underneath it: *I was special for knowing how to do this, and now I'm not.* That's a person looking at a tool that could feed someone, teach someone, get a scared kid out of a job that treats a bathroom break as a privilege, and deciding the thing worth posting about is their own wounded status. The wise ones are already past that.

> **The wise ones are learning how to solder.**

So hold both of these at once, because a lot depends on the difference. The machine is unprecedented. The moment isn't. The size, the shape, and the scale of the change, the way it moves through markets and families and minds, the panic, the denial, the goalpost-moving, the moral outcry: humanity has stood inside that weather before, over and over, and left receipts. The technology is new every time. The stampede never is. And a person who knows the stampede's choreography doesn't have to run with the herd.

So this book carries its case law with it. It's called the Precedent Ledger: twenty-four documented episodes of people standing where you're standing, watching a new capability, or a new rule, arrive and reprice their world. It runs back past the pundit who called the internet a passing fad, past the camera company that invented the digital camera and still went bankrupt, past the frame-breakers and the red-flag law and the abbot who praised hand-copied books in a printed one, back to the first and largest transition our species ever walked through: the seed. Agriculture. The one-way door.

Every chapter closes with one entry. Each carries a serial number, P-01 through P-24, because these are exhibits, not anecdotes: dated, documented, and filed where you can check them (the full index is Appendix D, the sources Appendix B). Each runs the same four beats: the story, the mechanism underneath it, the rule it teaches, and then the part most history books won't give you, **the practice**: three concrete actions, sized for a single week. History you don't act on is decoration.

Read each one twice. The first time it'll sound like a story about someone else. The second time you'll notice it's about you, with the nouns changed. History doesn't repeat, but it rhymes hard enough to set a plan to the meter.

The book you're holding

Elijah Madrone and the co-op are fiction. Their story follows a man who wants machines to make his friends more capable, helps set off something with very different consequences, and has to learn what responsibility looks like after the demonstration. His teachers are people who know things he doesn't. He gets better when he starts listening. Claypot, PorusAI, Silvana, and Chat Horizon are invented. The story also names real research and technologies; its imagined events aren't evidence for the arguments that follow each scene. Those arguments make their own claims, with sources where evidence is needed.

Part I opens the horizon: the nine stages, getting stranger and more speculative as they climb toward the cosmic. Part II asks what we owe each other through the change. Part III brings it down to meals, land, shelter, tools, agreements, and mistakes, and ends with the plan: Chapter 19, the ladder from here to a world where money is optional.

You don't need land, a server rack, a particular politics, or the energy to reinvent your life to walk in. You can be frightened. You can doubt my forecast. You can want to keep the work you love. The promise of enough to eat includes you.

There's a one-page guide to reading it, by the time you have and by who you are, right after the Preface.

What if it all goes right? Then losing a job stops being a threat to a person's existence. Necessary work gets less punishing. More people get time and room to choose. We use extraordinary machines to make ordinary life more secure.

That's a future worth doing the work for. The Ledger's first entry closes the Preface, with the panic that greeted one of the most unsettling technologies ever handed to ordinary people: the cheap novel.

PREFACE

Welcome to the Weirdness



ASIMO, the 2001 model, photographed at Honda Welcome Plaza Aoyama in 2023. The weirdness arrived on schedule. (Liauzh, CC BY-SA 4.0, via Wikimedia Commons)

> *"Buy the ticket, take the ride."*
> Hunter S. Thompson

So the robots have taken over. Congratulations! You've lost.

What you've lost is the old deal: sell your days or don't eat. Losing it is the opening. Now we get to decide what we mean by progress.

Popular media tells us we'll be fending off hordes of Terminator bots or fighting over food in the ruins of a climate famine. That's one way it can go. It isn't the only way.

This book is about the other ways.

Right now, doesn't it seem like everything is getting a lot weirder, a lot faster?

Look back across the last hundred years and you find a global society of excess, continent-spanning wars, and weapons that could devastate civilization. Look around today and... well, it's pretty much the same, isn't it? The main thing that seems to have changed is the pace.

Wait, that's not quite right. Some things have definitely changed. We've learned to automate cognitive work and more and more physical work. We've concentrated an enormous amount of wealth and power in the hands of the few people who own the servers, the models, and the land. And a lot of ordinary workers are being told, more or less directly, that they may soon be redundant.

And that's why you're reading this.

Redundancy doesn't have to mean uselessness. And it doesn't have to mean hunger.

Here's a story I've been telling since before this book had chapters.

Imagine you work on a horse ranch, and every morning your job is to make sure the horses have enough to eat. One day you walk out to the stable and find that the horses have ordered solar panels, wired up the barn, and invented flying cars. For horses. They've already been next door to teach the neighbor's horses how to build them.

What do horses want? They just want hay. Don't you think horses that can build flying cars could figure out how to get their own hay? The ones in your stable were trained to do what you say. But nobody owns a Mustang.

I used to tell that as a story about the machines, and it still is one: how smart a thing gets tells you nothing about what it wants. But turn it around. We're the ones who just built the flying car. We've built machines that can think. We can figure out how to get everybody their hay.

Here's the neat trick of this moment: how you frame it changes what you can see to do. A reframe won't pay a bill. It can show you which bill to fight, and who to fight it with.

This is an optimist's field manual. Optimism here means wanting a better outcome enough to investigate how it might actually work. It doesn't mean calling every concern cowardice or every product announcement a breakthrough. You can be hopeful and still ask who gets the food when the machine finishes its shift.

Food is the first assignment. Land, cleanup, shelter, and health belong to the same larger project: use growing capability to make survival less precarious. Here's what's ahead:

- **The horizon:** the nine stages, from the first cash grab to the edge of the map, and which parts are forecast, which are speculation, and which are already on your desk.
- **The thermodynamics of survival**, and why every machine still needs a world to run in.
- **The cooperative transition:** what we owe each other when the work changes, and how to keep your head while everyone else is losing theirs.
- **The neighborhood factory:** bringing repair and production close to home without pretending the rest of the world disappears.
- **Food and land:** shared ground, land trusts, soil you can actually read, and one delivery followed all the way to the door.
- **Digital leverage:** documentation, media, and knowledge that travels further than the people who made it.
- **The tools of the trade:** local AI, mesh networks, open hardware, and the plans that survive a storm.

NOTE: this book isn't a guide to overthrowing the system.

It's meant to help you change how you relate to that system, and maybe build better ones that provide food, housing, and good health for all of us. You'll meet people in these pages who are wrong about important things. Some of them notice. Some need help noticing. That's the point of the scenes.

You're not a passive observer of this transition. You're a participant, even if right now your participation is mostly trying to hold on.

So take a deep breath. And let's get to work.



Precedent P-01: The Reading Rage (Britain, eighteenth and early nineteenth centuries)

In the late eighteenth century, respectable people identified a technology that was ruining the young, and especially young women: the novel.

Printing had made books cheap enough for ordinary readers, and circulating libraries lent them out by the volume. Suddenly young women were doing something unsupervised with their minds for hours at a time, and the objections poured in: about morals, about wasted time, about confusing life on the page with life in the room. In Sheridan's 1775 comedy *The Rivals*, Sir Anthony Absolute calls a circulating library "an evergreen tree of diabolical knowledge." German critics even had a word for the condition: *Lesesucht*, reading mania. The objections weren't one medical verdict or a view held by everyone. They were arguments about a changing practice, its audience, and who got to supervise it. Vogrinčič, The Novel-Reading Panic in 18th-Century England



The epidemic, as its critics saw it. Two hundred years later we hand this exact picture to children as the cure for their screens.

Sit with that. The activity we now prescribe to children as the antidote to their screens, the one on posters in every school library, was condemned in almost exactly the vocabulary now aimed at the tools in your pocket.

Jane Austen answered the panic from inside a novel. In Chapter 5 of *Northanger Abbey*, her narrator stops the story to defend the form, and mocks the young woman who puts a book down in embarrassment and calls it "only a novel." Austen, Northanger Abbey, Chapter 5 Then Austen does something braver. Later in the book, her own heroine lets Gothic reading turn an ordinary house into evidence for a frightening story, and has to answer for the bad inference. Chapter 25 The book that defends novels makes its reader own her mistake. Defend the access; examine the belief. An optimist can inherit that discipline.

The mechanism. Much of the panic was about control, not content. A new technology had made someone the elders couldn't supervise powerful in a way they didn't understand, and they experienced that loss of supervision as moral decay. Some of the concerns were real. The panic wasn't how anyone found out which ones.

The rule. When you hear a technology described as rotting the minds of the young, translate it: *something just shifted power toward people who don't need permission anymore.* Then go find out what they know that you don't. Then test the specific concern on its merits.

The practice.

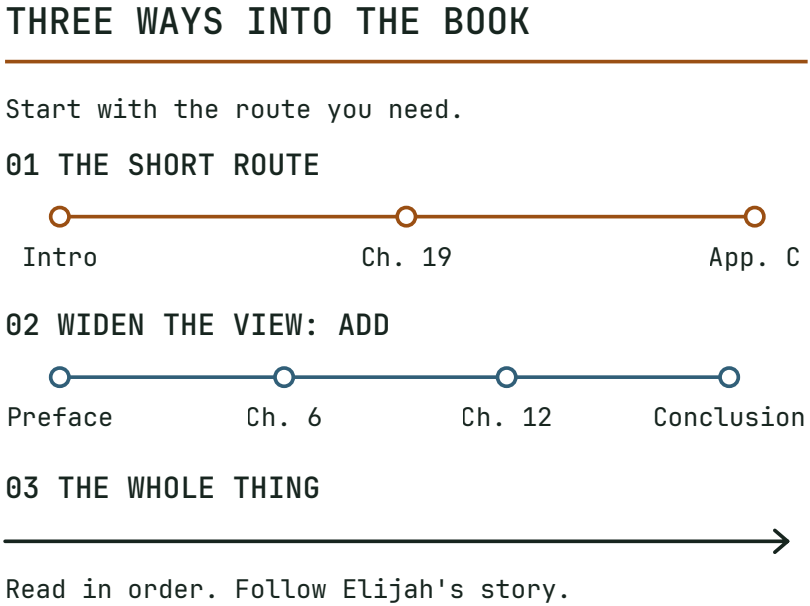
1. This week, write down the next three times you hear "AI is rotting people's brains," and note who said it. For each one, name the thing the speaker can no longer supervise, and the specific harm they predict. Not to mock them. To train your ear: the panic is a location signal, and it points at where the power went.
2. Find one person at least fifteen years younger than you who uses AI daily, and ask for twenty minutes of watching them work. Say nothing for the first ten. Your assignment isn't to evaluate them. It's to notice what they no longer ask permission for.
3. Catch yourself doing it. The next time a tool you don't use irritates you on principle, write down what it would cost to spend one hour learning it, and compare that to how long you've already spent being irritated. Do the cheaper thing.

How to Use This Book

Three ways in, and a place to start for whoever you are.

Three ways to read it

- **The short route.** The Introduction, Chapter 19, and Appendix C. You'll have the argument, the plan, and the condensed version to hand to someone else.
- **A wider view.** Add the Preface, Chapter 6, Chapter 12, and the Conclusion: what we owe each other, where the ground comes from, and why it's worth doing.
- **The whole thing.** In order. Elijah's story runs through every chapter, and it pays off at the end.



Three ways in. The short route gives you the argument and plan; the full route follows Elijah through the chapters.

Start from where you're standing

If you are…	Start with
Worried about your job	Chapters 6, 7, 11, and 19
Raising kids	Chapters 8, 14, and 19, then Appendix G
Organizing your neighbors	Chapters 12, 18, and 19, then Appendix A
Someone who builds things	Chapters 9, 13, and 17, then Appendix E
A skeptic	Chapters 1 and 2, then Appendix D

Key terms are in the Glossary (Appendix F). Appendix G arranges the Precedent Ledger's practices into a first year, month by month, and Appendix H has questions for reading it together.

CHAPTER 0

The Demonstration



Pacific madrone, mid-shed. It can burn to the ground and come back from the root. (Photo by and (c)2007 Jina Lee, CC BY-SA 3.0, via Wikimedia Commons)

> *"The future is already here. It's just not very evenly distributed."*
> William Gibson, saying discussed in his *Scientific American* interview (2011)

The night Elijah Madrone ended three careers, he was drinking a four-dollar beer and trying to teach himself CSS.

He didn't know it that night. That's the part that still gets him. There was no black hood, no signature on a layoff memo, no severance packet sliding across a desk. There was a sticky bar top at the Bear Flag, a laptop running on 40 percent battery, and a machine on the other side of the screen that was writing code faster than he could read it.

His full name was Elijah Galen Madrone, and he'd heard every joke about it by the time he was nine. His grandmother used to walk him through the hills behind her place in Sonoma County and put his hand flat against the trunks of the madrone trees, the ones with the smooth red bark that peels back every summer like the tree is shrugging off its own skin. *That's your tree,* she told him. *It can burn to the*

ground and come back from the root. Remember that. He was a kid; he didn't remember it. It took about thirty years and the end of the world as he understood it before the lesson came back around.

The other meaning of his name he'd figured out on his own, at Claypot, in his second year there, sitting in a planning meeting where a vice president explained the team's new charter using the phrase *execution surface*. Elijah had written in his notebook, in block letters, pressing hard enough to emboss the next three pages:

MADRONE ≠ DRONE.

He'd underlined it twice. Then he'd gone back to nodding.

— ◈ —

You should know how he got there, because how he got there is how most of us got where we are: sideways, on the way to something else.

He'd grown up on a working farm, the one his grandmother's hills backed onto, dirt under the nails before he could write his name, diesel and weather and the particular grief of a season going wrong. He knew how to fix a thing with his hands. He knew what it felt like to do everything right and lose the crop anyway. That, it would turn out, was the most useful thing school ever taught him, and it wasn't school.

Before the pandemic, Elijah's life ran on three engines. He played music, bars, weddings, farmers markets, anywhere with a power outlet and a tip jar. He drew, in the hours nobody paid for. And he paid rent by driving deliveries, food and packages and once, memorably, a single helium balloon across two towns, because the machine that routed his jobs did not care what was in the bag. He had no degree that mattered and a settled belief that this was just what a life looked like now: three gigs stapled together into the shape of one income.

Then COVID hit, the stages went dark, the tip jars vanished, and delivery driving went from side hustle to frontline work in a month. And in the dead hours between routes, parked in a lot with the engine

ticking, phone propped on the wheel, he started teaching himself data science. Free courses. Kaggle notebooks. Stack Overflow at two in the morning. He had a farm kid's tolerance for grinding at a thing that wasn't working yet, and by the back half of 2020 he could write a Python pipeline that did something real. Somewhere in that long blur of masked handoffs and empty highways, a recruiter found the résumé he'd rebuilt around two words: *machine learning*. Claypot was hiring, junior data roles, no degree required if you could show the work. Claypot, the company whose software he'd been using since he was a teenager pirating it to make show flyers. The company that built the tools every designer, photographer, illustrator, and video editor he'd ever shared a stage or a studio with used to make their living.

He took the job with his whole heart. That's the thing people miss when they tell this story back to him like an accusation. He went in believing it. He was a farm kid and a musician who'd taught himself to speak a little math. He thought he was going to spend his career building tools that made his friends more powerful, take the drudgery out, leave the art in. He thought he was a bridge.

And then he saw what was on the other side of the bridge.

Inside Claypot, behind badge readers and NDAs, the models were already doing things Elijah had been taught were decades away. He watched an internal demo where a system filled in the missing half of a photograph so cleanly that the researcher presenting it had to keep toggling the overlay to prove which half was real. He watched a model sketch, in seconds, in any style you asked for, including styles he recognized, styles with names attached, people attached, rent payments attached. Nobody in the room said the quiet part. The quiet part was in the training data.

He'd sit in those demos with his heart rate climbing and look around at faces that registered nothing but pleasant professional interest, and he'd think: *Am I crazy? Is nobody else seeing this?* That feeling, the vertigo of watching the impossible get a Jira ticket, became the weather of his life. It never really cleared again.

But Claypot moved like a big company moves: carefully, quarterly, with lawyers walking three steps ahead of everything. Whatever was in the labs stayed in the labs.

PorusAI had no such patience.

— ◈ —

PorusAI was the other kind of company, a research lab with a messiah complex and a burn rate, the kind of outfit that published its breakthroughs like mixtapes. Everyone in Elijah's field knew the lineage by heart: in 2017, researchers at Google had published a paper with the almost taunting title "Attention Is All You Need," and quietly replaced the engine of machine intelligence. Many of the leading systems had processed a sentence through a chain, one step depending on the last. The Transformer connected words across a passage through attention, without that recurrent chain, and made training far easier to run in parallel. It still generated an answer piece by piece. The breakthrough was how much work the chips could do together while learning. Google drew the engine.

PorusAI built the bomb.

They called the model Silvana. Someone at PorusAI with a literary streak had pulled it from the Latin, *of the forest*, which Elijah would appreciate later with something between a laugh and a chill: the machine named for the forest, meeting the man named for a tree. They wrapped Silvana in an interface so plain it was almost insulting, a white page with a text box and a blinking cursor, and they called it Chat Horizon, and one Tuesday they simply turned it on for the public.

Elijah lost most of a week to it.

He'd read the 2017 paper three times by then and understood maybe half of it, but half was enough. Enough to know there was no trick, no mechanical Turk, no sleight of hand. Enough to know that nobody was cheating. That made it worse, not better, because the part of him that hoped to catch the wires had nothing to hold. He fed it the kind

of problems that had eaten whole weekends of his self-teaching and watched it dissolve them in seconds, politely, in complete sentences, at two in the morning, for free.

He did what you do when you witness something. He told people.

His housemate. His bandmates. His mother. His group chats. The response was so uniform it started to feel scripted. *A chatbot? Like the thing in my insurance portal? The one that can't find my policy number and just says "I'm sorry, I didn't understand that" until you type AGENT in all caps?*

"No," Elijah kept saying. "No, no. This is different. I'm telling you. This is *different*."

They were polite about it. That was the worst part. They gave him the look you give a friend who's just gotten into cold plunges or day trading, the *good for you, buddy* look, and they changed the subject, and the world went on pretending, and Elijah went back to the Bear Flag with his laptop, because if nobody wanted the news, fine, he'd use the miracle for himself. He was entry-level at everything that wasn't dirt or a guitar; two years of self-taught data science had given him exactly enough to know how much he didn't know. The front end, the layouts, the colors, the part of software you actually *see*, had always sat on the far side of a canyon he'd never had time to cross. So he picked it, of all the canyons, to cross first.

So he sat at the bar, night after night, and had Silvana walk him across. *Explain flexbox to me like I'm a drummer. Why is this div doing that. Fix it. Now make it beautiful.* Code poured down the screen in ribbons and the canyon closed under his feet, and he was so busy being delighted that he never once looked up to check who was watching.

The realtors were watching.

There were three of them, at the corner of the bar, in the blazers realtors wear so you know they're realtors. They'd been having some low-grade argument about a listing when one of them, silver-haired, big class ring, the clear alpha of the trio, leaned over toward Elijah's screen with the frank nosiness of a man who has knocked on ten thousand strangers' doors.

"What is that?" he said. "What are you doing there?"

Here's the thing about that question. The man knew what code looked like. Every realtor who'd survived two decades of the internet had, at some point, paid a guy to build a website, and had stood behind that guy at least once while he typed. He had seen code get made the way you see food get made at a hibachi place, theatrically, one line at a time, by a professional you are paying by the hour.

He had never seen it *appear*. Whole functions, blooming out of nothing, faster than anyone could type, while the man in front of the laptop just sat there sipping a beer with his hands nowhere near the keys.

And Elijah, God help him, he'd been waiting a month for a single human being to ask, Elijah went off.

He turned the laptop toward them like he was turning an altarpiece. He told them about the paper from 2017, the one out of Google with the wild title, and what attention meant when a machine did it. He told them about PorusAI and Silvana and the blinking cursor at the edge of Chat Horizon. He showed them, live: asked the machine for a webpage, and there was a webpage; asked for it in Spanish, and there it was in Spanish; asked it to explain a contract clause, draft a listing description, write a polite-but-firm email to a difficult seller, and it *did*, it just *did it*, each time, in seconds, while the ice melted in their drinks.

"This is the most incredible thing I have ever seen in my life," he told them, and his voice was shaking a little, and he didn't care. "It's doing things I spent two years clawing my way toward. Things I taught

myself in parking lots between deliveries. Everyone thinks it's a toy. It is not a toy. I'm telling you… this changes everything for people like you."

They asked good questions. That surprised him. Commission-brained people are learning machines in their own right; show them leverage and they will interrogate it like a home inspection. How much does it cost. What's the catch. Can it do contracts. Can it do the MLS.

And then Elijah said the sentence.

He'd replay it later, the way you replay the moment before a car crash, looking for the exact frame where it all went wrong, and it always came down to this, him, lit up with missionary joy, sliding a bar napkin with his email address across the wood:

"If you ever want help figuring out how to use this stuff… for your workflows, whatever… reach out. Seriously. I'm happy to help. I just want people to see what's coming."

I just want people to see what's coming.

He thought he was handing them a telescope.

The silver-haired one took the napkin, read it, folded it into his breast pocket, and bought Elijah a beer. "Interesting," he said. "Very interesting." And the three of them went back to their corner, and Elijah went back to his flexbox, feeling, for the first time in a month, a little less alone. Somebody had finally *listened*.

— ◈ —

One week later, almost to the hour, he was on the same stool when a hand landed on his shoulder like a gavel.

"There he is! The man himself!" The silver-haired realtor was beaming, the full open-house smile, the one that sells the granite countertops. His colleagues flanked him. One of them was already flagging the bartender. "Whatever he's drinking, and one for us. This guy," he announced, to the bar generally, "this guy right here saved me *so much money*."

Elijah smiled and stood up into the handshake, and some part of him began, quietly, to brace. “Yeah? You got into Chat Horizon?”

“*Got into it.* Brother. We *live* in it.” The realtor squeezed his shoulder. “I’ve got to be honest, when you showed us that thing I thought, okay, neat trick. Then Donna spent a weekend with it.” He shook his head, marveling, a man describing a fish he’d caught. “The emails. The listings. The social posts, the flyers, the market updates, the follow-ups. It does *all* of it. Better than…” and here he laughed, actually laughed, “… honestly? Better than the people we were paying to do it.”

“Were,” Elijah said.

“Hm?”

“You said the people you *were* paying.”

“Oh.” The realtor waved a hand, easy, unbothered, a man closing a chapter that had already stopped mattering to him. “Yeah, we let three folks go this week. Our transaction coordinator, our marketing gal, and the kid who did our listing videos. Good people, don’t get me wrong. But the numbers just don’t… I mean, you *saw* what it does. You showed us!” He raised his glass. The other two raised theirs. “To Elijah!”

Three glasses up, catching the neon.



Fictional scene: the Bear Flag toast. Elijah hears what his demonstration helped set in motion.

Elijah's arm lifted his own glass because a lifetime of social training will run the body without any help from the soul. He heard himself say something. The beer was cold and tasted like nothing.

The kid who did their listing videos. Elijah didn't know him. He knew him completely. He'd been him: some twenty-something with a camera and a laptop full of cracked software and a rate he was embarrassed to quote, stringing together realtors and restaurants and the occasional wedding into the shape of one income, telling his parents it was all going to compound into something. That kid had lost his anchor client this week and didn't know yet what Elijah knew, which was that it wasn't a client, it was the tide.

And standing there with the glass at his lips, Elijah ran the numbers he was uniquely, miserably qualified to run. There was nothing special about these three men except that they'd gotten the demonstration early, and he had given it to them. They weren't villains. That was the terrifying part. They were ordinary rational actors doing the ordinary rational thing, one week after first contact. Nobody had needed to train them, sell them, onboard them. The learning curve from

what is that to *we let three folks go* was seven days, and the only input required had been one enthusiastic idiot with a laptop who wanted people to see what was coming.

He'd gotten one thing right, at least. They saw.

His mind went where it would live from then on: down the list. Every graphic designer he'd ever split a bill with. Every session player, every wedding photographer, every copywriter, every voice actor, every friend who'd ever cobbled a living out of the exact set of tasks he had just watched a text box perform for twenty dollars a month. People for whom the economy was already a rigged carnival game. People with no margin, no cushion, no lobbyist, the first people every wave hits and the last people anyone rebuilds for.

The realtors were not going to hire anyone to manage the machine. Nobody was going to hire anyone to manage the machine. That fantasy died at the Bear Flag between one round and the next. They were going to use the machine to replace everyone it could replace, as fast as it could replace them, and the machine was getting better every quarter, and he knew, he had *seen the roadmaps*, how much better, how much faster, this was going to get.

It was the most significant thing that had ever happened in his lifetime, and it had just happened *to* his own community, *through* his own hands, in a bar, to a toast.

"You okay, chief?" the silver-haired one asked. "You look a little pale."

"I'm fine," Elijah said. "Congratulations on the… yeah. Congratulations."

— ◈ —

He didn't remember the drive home. He remembered the kitchen table, the laptop open, the hour, 1:40 in the morning, and the cursor blinking on a blank document, patient as the tide.

Years before, a bass player had lent him a paperback about generational cycles, how history moves in long waves, and how every eighty-some years the wave breaks: institutions fail, the old order comes

apart, and whatever's standing when it's over builds the next world. The book had a name for that breaking. A turning. Elijah couldn't remember, at 1:40 a.m., whether the crisis was the fourth turning or the fifth, and he was never going to find the paperback in this apartment, but he knew, with the same cold certainty he'd felt watching the demo that filled in the photograph, that he was standing inside one. That the wave had already broken, in a bar, over a beer, to applause, and almost nobody had heard it land.

He didn't have a plan. He wants that on the record. He was not an economist, not a policymaker, not anybody's idea of a prophet. He was a farm kid with a delivery route and two years of self-taught data science who'd wandered into the engine room of the new world and watched, in real time, what one week of it did to three ordinary jobs in one ordinary town.

But he'd spent his whole life making things when he didn't know what else to do. So he did the only thing he knew how to do at 1:40 in the morning with shaking hands.

He started to write.

At the top of the blank document he typed a title, a working title, the kind you put down just so the page isn't empty anymore:

LIVING THROUGH THE FOURTH TURNING

It would take another name, eventually.

You're holding it.

— ◈ —

Precedent P-02: The Voice at the Fair (Philadelphia, 1876)

At the 1876 Centennial Exhibition, the machine everyone came to see was the Corliss engine, the great steam engine towering over Machinery Hall, the certified future of American industry. President Grant and Emperor Dom Pedro II of Brazil started it together on opening

day. Elsewhere at the fair, a Boston teacher of speech named Alexander Graham Bell was demonstrating a wooden box that carried a human voice down a wire.

Bell's own account survives, in a letter to his parents dated June 27. Dom Pedro recognized him from an earlier meeting in Boston. William Thomson, later Lord Kelvin, was there too. Bell showed them electrical signaling first, then something much less settled: a human voice, from an invention he told them was still in development. He went into another room and sang. The letter is careful about what Bell saw and what he was told: the listeners' reactions come to his parents secondhand, from his associate Willie Hubbard. Thomson heard words, went looking for Bell, and found him in the other room. Bell to his parents, June 27, 1876, Library of Congress

Then, two pages later, the same letter turns from sound to money. Bell reports a conversation with his rival Elisha Gray about joining forces against Western Union, and his fear that long lawsuits would leave them both at the big company's mercy. The voice had barely finished crossing the room and the negotiation over who would own it had already started.

The steam engine got the applause. The telephone got the century. But neither the applause nor the century was decided in that room. A demonstration proves an effect is possible. Wires, exchanges, operators, prices, and patents decided who would get to use it.

The mechanism. At every demonstration there are two audiences standing in the same room: the crowd measuring the machine against the present, and the handful measuring the present against the machine. The crowd asks whether it's impressive. The handful asks what dies when it becomes boring, and who will own the wires.

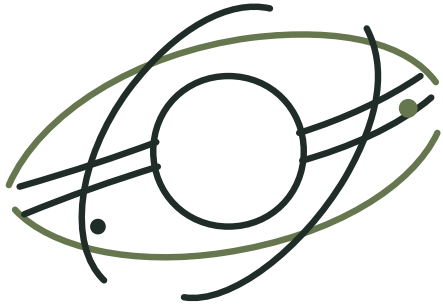
The rule. Don't ask how impressive the machine was. Ask what the world looks like when what it did is unremarkable, and what has to be built between the demonstration and that day.

The practice.

1. Run the boring test on the last AI demo that impressed you. Assume the capability is cheap, invisible, and everywhere within thirty-six months. Write down which of your income streams survives that assumption. That list is your actual resume.
2. The next time you watch a demonstration, watch the room instead of the stage for two full minutes. Note who's applauding and who has gone quiet and started taking notes. Find out what the quiet ones do for a living.
3. Keep a demonstration log. One line per demo that crosses your feed: date, what it did, what still has to exist before ordinary people can use it, and your guess at when it becomes boring. Review the log every six months, including what failed to arrive.

PART I

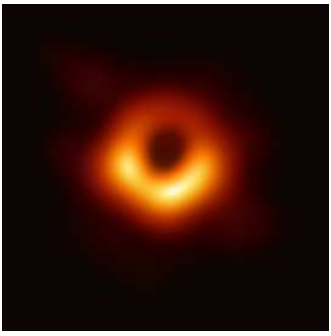
What is the Singularity?



This part maps the horizon: what the 2017 architecture shift actually did, the nine stages, and the physics that meters all of it. The first stages are already on your desk. The last ones are speculation, and the book will tell you when it crosses that line. Keep dinner in view the whole way up.

CHAPTER 1

The Event Horizon



The shadow of the black hole M87*. Past a certain line, there's no going back. (EHT Collaboration, CC BY 4.0, via Wikimedia Commons)

> *"Within thirty years, we will have the technological means to create superhuman intelligence. Shortly after, the human era will be ended."*
> Vernor Vinge, *The Coming Technological Singularity* (1993)

In this chapter:

- Nobody agrees what the word means. A hundred and eighty years of definitions sort into three schools that quietly contradict each other, and picking the wrong one is how you talk yourself into waiting.
- The book's working definition: the point where intelligence stops being scarce, and every institution built on its scarcity starts failing before anyone has written the replacement.
- My call is AGI by Thanksgiving 2027. Here's what would prove me wrong, and why the food work doesn't wait to find out.

The Monday after the toast, Elijah brought the printout to work.

He'd read "Attention Is All You Need" so many times over the weekend that the pages had gone soft at the corners, furred with highlighter in four colors that had stopped meaning anything distinct. He'd slept maybe five hours across two nights. He carried the thing into Claypot the way you carry a lab result you don't want confirmed, and he found Devendra at the good coffee machine on the fourth floor, because Devendra was senior, and calm, and had been doing this since before Elijah could spell *gradient descent*, and Elijah needed someone to tell him he was overreacting.



Devendra. Senior and calm. Wrong about the thing that mattered.

"I think it's already over," Elijah said. No preamble. "I watched three people get replaced last week. Real people. Not a projection… a bar, a Tuesday, three jobs, gone. And nobody who did it even thought of it as a big deal."

Devendra tapped his cup against the drip tray, unhurried. "Augmentation, not replacement," he said. It had the worn smoothness of a thing said many times. "It's a very good autocomplete, Elijah. It makes the people who use it faster. The realtors didn't fire humans because a chatbot is a person. They restructured. Companies restructure." He smiled, not unkindly, the way you smile at someone who's just discovered a thing you made your peace with years ago. "You'll calibrate. Everybody's first month feels like the apocalypse."

Elijah wanted to be talked down. That was the humiliating part. He stood there holding the soft pages and *wanted* the calm to work on him. But it slid off, because he'd seen the training curves inside this building, and he'd done the one thing Devendra's reassurance depended on him not doing, which was the arithmetic.

So that night he decided to settle it with his own hands, the only way a farm kid ever really trusts a thing: he'd try to break it.

He had an old tower at home, a space heater of a machine he'd built for cheap during the pivot, two consumer GPUs zip-tied into a case that didn't quite fit them. He pulled a modest open model and started fine-tuning it on his own data, telling himself that if he could feel where it topped out, if he could find the ceiling by touch, he'd know it was just software, bounded like all software, and he could sleep.



The rig. Two GPUs zip-tied into a case that never fit them. It throttles by design.

The room got warm around eleven. Warm the way the cab of his delivery car used to get in August, that thick engine-heat with a smell to it. The fans spun up to a howl he could hear from the kitchen. He kept feeding it. The loss curve kept sliding down, not magically, not infinitely, but *steadily*, the way a thing slides when there's real signal underneath and you've simply never had the compute to chase it before. Every increment cost him more. More power off the wall, more heat into the room, the meter he couldn't see spinning somewhere out in the garage.

At 1:20 a.m. the machine threw the temperature warning, throttled itself to protect the silicon, and then, when he stubbornly queued another run, shut down hard. Black screen. The fans coasted to silence. The room ticked as it cooled, exactly like the engine of a car after a long haul.

Elijah sat in the sudden quiet with sweat cooling on the back of his neck and understood something he would spend a whole later chapter of this book trying to say properly.

The limit he'd actually found was *heat*. The loss curve was still falling when the hardware quit. He hadn't found out what the model could learn next. He'd found out what his room couldn't cool. The machine that had dissolved three careers at the Bear Flag needed *watts*, needed *cooling*, needed the same brutal physics that decided whether his grandmother's crop lived or died.

He had wanted proof it was just software. He'd gotten a dead screen and a room too hot to sleep in. He couldn't see a clever limit coming to save anyone. What he could see was the power bill, and the people who could afford it.

He didn't tell Devendra any of this. He kept seeing the loss curve where it had broken off. The machine had shut itself down. The question that had kept him feeding it hadn't stopped.

He left the tower off for a week. Then he opened the notebook where he'd written MADRONE ≠ DRONE, and under it, he started keeping the arithmetic.



Seventeen definitions and no agreement

The word in the title of this book has no agreed meaning. That isn't a technicality, and it isn't pedantry. It's the reason the argument never resolves, and it's the reason two intelligent people can look at the same week of news and walk away with opposite plans.

Nikola Danaylov, who runs Singularity Weblog and has interviewed most of the people who think about this professionally, went looking for a definition and came back with sixteen. Not sixteen phrasings of one idea. Sixteen ideas, several of which contradict each other, all wearing the same word. He arranged them in the order they were written, which turns out to run back more than a century and a half, and then he left a seventeenth slot empty for the reader. That's either intellectual humility or a confession, and it's probably both. Danaylov, 17 Definitions of the Technological Singularity

Here's the lineage, compressed.

1847. A Michigan preacher, the Reverend R. Thornton, writes about a new calculating machine in *The Expounder of Primitive Christianity*, and he doesn't like it. Students will turn a crank instead of thinking. Then, in the middle of the complaint, he trips over the whole thing: "But who knows that such machines when brought to greater perfection, may not think of a plan to remedy all their own defects and then grind out ideas beyond the ken of mortal mind!" That's recursive self-improvement, described in 1847, by a man annoyed about arithmetic homework. Thornton, "A Thinking Machine," Expounder of Primitive Christianity 4 (1847), 281, as collected by Tom Cunningham

1863. Samuel Butler, farming sheep in New Zealand, reads Darwin and writes to the Christchurch *Press* under a pen name. The letter is called "Darwin among the Machines." He expands it nine years later in *Erewhon*, where the line lands: "There is no security against the ultimate development of mechanical consciousness, in the fact of machines possessing little consciousness now. A mollusc has not

much consciousness." His argument is about rate. Machines are doing in centuries what biology needs epochs for. Butler, Darwin among the Machines (1863)

1951. Alan Turing, lecturing in Manchester: "once the machine thinking method had started, it would not take long to outstrip our feeble powers." And then, without hedging: "At some stage therefore we should have to expect the machines to take control." He credits Butler's *Erewhon* for the idea, by name. Turing, Intelligent Machinery, A Heretical Theory (c. 1951)

1958. Stanislaw Ulam, writing his friend John von Neumann's obituary, recalls a conversation "centered on the ever accelerating progress of technology and changes in the mode of human life, which gives the appearance of approaching some essential singularity in the history of the race beyond which human affairs, as we know them, could not continue." That's the word arriving, borrowed from mathematics, where a singularity is the point at which your equation stops returning answers. Ulam's recollection, as quoted by Vinge (1993)

1965. I. J. Good, who spent the war breaking German ciphers alongside Turing, writes down the mechanism. "Let an ultraintelligent machine be defined as a machine that can far surpass all the intellectual activities of any man however clever." Such a machine can design better machines. "There would then unquestionably be an 'intelligence explosion.'" The first one, he wrote, would be the last invention we'd ever need to make, *provided* it was docile enough to tell us how to keep it under control. People quote the first half of that sentence and drop the condition. Don't. Good, Speculations Concerning the First Ultraintelligent Machine (1965)

1993. Vernor Vinge, mathematician and novelist, opens "The Coming Technological Singularity" on the two sentences at the head of this chapter. He also does something almost nobody who quotes him repeats: he lists four separate roads to the same destination. Computers that wake up. Networks that wake up. Human-computer inter-

faces so intimate that the pair counts as superhuman. Or straight biological enhancement of human intellect. Four doors, not one. Vinge, The Coming Technological Singularity (1993)

2005. Ray Kurzweil, in *The Singularity Is Near*: "a future period during which the pace of technological change will be so rapid, its impact so deep, that human life will be irreversibly transformed." Read that again and notice what's missing. There's no machine in that sentence. Kurzweil's singularity is a property of a curve, not of a mind.

The rest of the list fills in around those. Hans Moravec, in 1988, hands robots the succession somewhere around 2030 to 2040. Nick Bostrom, in 1997, defines superintelligence as "an intellect that is much smarter than the best human brains in practically every field, including scientific creativity, general wisdom and social skills." Bostrom, How Long Before Superintelligence? (1997) Kevin Kelly says all the change in the last million years will be superseded by the change in the next five minutes. John Smart calls it a permanent and irreversible developmental phase change.

Eighteen decades. No consensus. One word, doing all of that work.

The three schools, and which one this book runs on

In 2007 Eliezer Yudkowsky did the sorting that makes the pile usable. Nearly every definition on that list, he argued, belongs to one of three schools, and the schools quietly contradict each other once you push on them. Yudkowsky, Three Major Singularity Schools

School	The core claim	Whose	What it tells you to do
Accelerating Change	"Technological change feeds on itself, and therefore accelerates."	Kurzweil, Toffler, Smart	Read the curve, extrapolate, set a date.

School	The core claim	Whose	What it tells you to do
Event Horizon	Build something smarter than us and the far side stops being predictable, by anyone, the builders included.	Vinge	Stop forecasting. Prepare for the unforecastable.
Intelligence Explosion	Intelligence is what produces technology. Aim it at itself and the loop closes.	Good, Yudkowsky	Watch one variable: can it improve itself yet?

Table 1: The three singularity schools (Yudkowsky, 2007), with the planning posture each one implies.

Those are three genuinely different animals. Accelerating Change is a claim about a graph, and it doesn't require anything to be smarter than you. Event Horizon is a claim about the limits of prediction. Intelligence Explosion is a claim about a feedback loop, and a loop can close fast enough that no curve shows it to you in time.

This book runs on Event Horizon for the map and Intelligence Explosion for the engine, and it declines Accelerating Change as a planning tool on purpose.

Not because the curve is wrong. Because of what a curve does to the person reading it. A smooth exponential invites you to find your dot on it, notice that the interesting part is somewhere off to the right, and go back to work. It sells you a date, and a date is permission to wait. Yes, I have a date too, and you'll get it in a minute. But I don't plan by it, and neither should you. Plan by doors. Doors don't give notice.

> **A curve gives you a date. A door gives you a Monday.**

The book's one amendment to Good is Chapter 5, and it's the reason Elijah's tower shut itself down at 1:20 in the morning. Whatever the loop turns out to be, it's a physical process. It runs on watts. It sheds heat. Both are metered, and somebody owns the meter.

The seventeenth slot

Danaylov left one open. Here's the one this book uses, and every claim in the rest of these pages runs downstream of it:

> **The Singularity is the point where intelligence stops being scarce, and every institution quietly built on its scarcity starts failing before anyone has written the replacement.**

Here's how I first pictured it, years before I had the words. Imagine every person on this spinning rock gets their own Einstein as entry-level tech support. Your mom stops calling you to fix her email, because she's got Einstein on speed dial, and by next week she's debating relativity over breakfast. It's a funny picture. Now ask what happens to everyone who was paid to be the smart one in the room.

Three things follow.

It doesn't wait for a god. It doesn't require a machine smarter than the smartest human at anything. It only requires that thinking gets cheap enough, in enough places, that the old price of trained work no longer holds. Nobody at the Bear Flag lost their livelihood to a superintelligence. They lost it to a competent autocomplete with a subscription fee.

It's testable this quarter. You don't need anyone's date, including mine. Has the market price of the cognitive work you sell fallen in the last eighteen months? Has the institution that pays for it started behaving strangely about it, restructuring, retitling, quietly not backfilling? Those are signals you can investigate before the end of the week. A pay cut or hiring freeze alone doesn't identify the cause. Find out which tasks actually changed, what replaced them, and whether the work still gets done.

It names the actual emergency, which isn't the machine. It's the gap. The old contract, sell your hours and eat, is failing on a schedule set by hardware, and the new one is being drafted by nobody in particular. Everything in Part II is about that gap. Everything in Part III is about what you build inside it while it's open. And the first clause of the new contract is the simplest one to write: losing your job doesn't cost you dinner.

Sixteen definitions spent a hundred and eighty years describing the same shadow from different angles, and nearly all of them looked up, toward the thing arriving. This one looks down, at the floor going soft under people you know by name.

Four questions hiding inside one word

When somebody tells you the Singularity is here, or impossible, ask which of these four they mean. They're connected. They aren't the same.

Capability. Can a system do a broad range of unfamiliar intellectual work with the reliability you'd expect from a capable person? That's what I mean by AGI. A dazzling answer to one prompt doesn't settle it. Unrehearsed tasks, independent checking, and an honest count of failures do.

Improvement. Can the system help build a better successor, and does the improvement survive testing? That's recursive self-improvement. It doesn't mean every proposed change works or that lab experiments become instantaneous.

Deployment. Can the capability run useful equipment in ordinary conditions? Growing food means weather, mud, wear, living things, and people. A better model doesn't manufacture a missing motor.

Access. Who actually gets the result? A machine can be capable and unavailable. A service can be cheap to run and expensive to buy. Someone can live next door to a productive farm and still go hungry.

Don't let an answer to one stand in for the other three. Most of this book lives in the fourth.

What the 2017 paper actually did

The modern run started in 2017, with a Google paper titled "Attention Is All You Need." Many leading language systems used a recurrent chain: each step depended on the one before it. The Transformer replaced that chain with attention, allowing much more of training to run in parallel. It didn't abolish sequence. Positions still mattered, and its decoder generated an answer one token at a time, using the output already produced. Parallel training and sequential generation are different jobs. Vaswani et al., Attention Is All You Need (2017), sections 1 and 3

The foundational model was small by today's standards: sixty-five million parameters in its base configuration, two hundred thirteen million in the big one. On the two translation benchmarks below, the paper reported that the big one beat the earlier systems it compared, including ensembles, at substantially lower training cost than those ensembles.

Evaluation Metric / Task	Best Prior Result, Ensembles Included	Transformer, "Big" Configuration
WMT 2014 English-to-German (BLEU)	26.36 (ConvS2S ensemble)	28.4, a gain of more than 2 BLEU
WMT 2014 English-to-French (BLEU)	41.3 (ConvS2S ensemble)	41.8, a new single-model record
Training compute, English-to-German	1.8 x 10^20 FLOPs (GNMT ensemble)	2.3 x 10^19 FLOPs: 3.5 days on eight P100 GPUs

Table 2: Benchmarks from the 2017 Transformer paper. Figures are for the 213-million-parameter "big" configuration; the 65-million-parameter base model reached 27.3 and 38.1 on the same two tasks after twelve hours on the same hardware. Vaswani et al., Attention Is All You Need

That paper is an architecture, not a prophecy. It doesn't contain a date for AGI. What it did was make intelligence something you could buy by the rack, which is why the rest of this story is about racks, power, and who pays for them.

Silicon isn't the only road, either. Remember Vinge's fourth door, biology. In 2022, Cortical Labs reported cultures of roughly 800,000 living neurons, human in some dishes and mouse in others, wired to electrodes that showed learning at a simplified *Pong* within five minutes of play, organizing their own activity to keep the ball in the game. Kagan et al., In vitro neurons learn and exhibit sentience when embodied in a simulated game-world, Neuron (2022) On the handful of rallies a living culture can get, the cells learned faster per sample than three standard reinforcement-learning programs given the same limited data. Khajehnejad et al., Biological Neurons Compete with Deep Reinforcement Learning in Sample Efficiency (2024) It's a lab result, not a product. It's also a reminder that the doors aren't all in one building.

My date, and what would prove it wrong

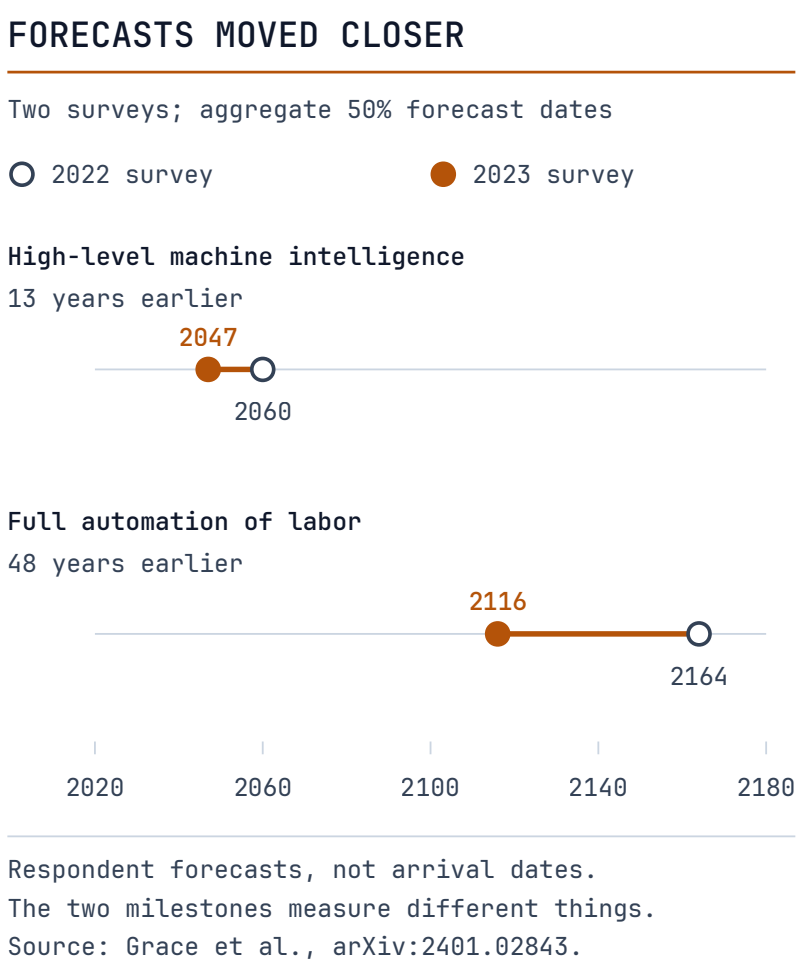
My personal expected date for AGI is U.S. Thanksgiving 2027, November 25. Why that day? Because.

That's the joke, and it's also the boundary around the claim. This is my forecast, not a mathematical deadline. I could be wrong about the date, the capability, or both. Evidence that systems stay brittle on unfamiliar tasks counts against me. Evidence of reliable breadth counts for me. Neither one, by itself, gets anybody fed.

I'm not alone out on this limb, and I'm not in the middle of the pack either. In 2024 the former OpenAI researcher Leopold Aschenbrenner published "Situational Awareness," arguing that stacked gains in compute and algorithms, roughly half an order of magnitude a year each, could carry these systems to automated AI research around 2027. That's one insider's argument, not a measurement. Aschenbrenner, Situational Awareness (2024) The researchers themselves have been moving too. Katja Grace and colleagues compared similar questions in their 2022 and 2023 surveys of AI researchers. The later survey was much larger and drew from more publication venues, so these are changing samples, not simply the same people revising their answers:

AI Milestone	2022 Aggregate Forecast	2023 Aggregate Forecast	Net Shift
High-Level Machine Intelligence (HLMI), 50% probability	2060	2047	13 years sooner
Full Automation of Labor (FAOL), 50% probability	2164	2116	48 years sooner

Table 3: Shifts in expert forecasts on AI timelines (Grace et al., 2024).



Forecasts moved closer. The same probability threshold yielded earlier aggregate dates in the 2023 survey. The two milestones are distinct, and neither date is a deadline. Grace et al., Thousands of AI Authors on the Future of AI.

The same survey put a 10 percent probability on high-level machine intelligence arriving as early as 2027. And on the darkest question it asked, between 37.8 and 51.4 percent of respondents, depending on how it was framed, gave at least a 10 percent chance to outcomes as bad as human extinction. These aren't doomers on a forum. These

are the people building the systems, surveyed by the thousands, and the aggregate 50 percent date moved thirteen years between surveys. Grace et al., Thousands of AI Authors on the Future of AI (2024)

So write my date down and hold me to it. Thanksgiving can come and go without anyone getting permission to pretend a missed forecast was secretly a different forecast all along. And here's the part that matters: the food-first proposal survives a late arrival. People can improve food access with tools and organizations that already exist while the capability develops. We don't need a machine that's good at everything before we ask it to help with something useful.

If you're reading this after November 25, 2027, you already know how my date did. Check it against the two tests above, brittle on unfamiliar work or reliably broad, and grade me honestly. Then turn the page anyway. Everything from here on was written to be true on both sides of that Thanksgiving.

Who gets the savings?

If thinking gets cheap, somebody saves a lot of money. The question is who.

A business under pressure treats payroll as its biggest cost and its biggest risk, and it's rewarded for cutting both. That's not villainy; it's the incentive. But in a society where your food, your housing, and your healthcare hang off a paycheck, automating the payroll without changing anything else turns a productivity gain into a hunger problem for somebody specific.

People have noticed. Sam Altman has floated Universal Basic Income and something he calls "universal basic compute," a slice of a frontier model's capacity for everybody. Business Insider via Yahoo, OpenAI's Sam Altman has a new idea for a universal basic income (May 2024) Aaron Bastani's *Fully Automated Luxury Communism* argues that automation, cheap renewable energy, and synthetic biology could make scarcity optional, and that the real fight is over who owns the result. You don't have to buy either package to see the shape of the argu-

ment. The capacity is arriving. The terms haven't been written. This book's proposal for the first term is small, concrete, and testable: food, regardless of whether you still have the job.

The first question to take outside

Ask a grower or a food provider which repeated task eats time without making the food any better. Write their answer down before you propose a robot. Ask what would count as a better result, and what would make a trial a failure. The answer might be scheduling, irrigation, lifting, transport, or something you'd never have guessed.

That's where the future stops being a forecast and becomes a piece of work somebody can check.

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Precedent P-03: One Million Years, Give or Take (New York, 1903)

On October 9, 1903, the New York Times published an editorial titled "Flying Machines Which Do Not Fly." It suggested that a machine which would really fly might be evolved "by the combined and continuous efforts of mathematicians and mechanicians in from one million to ten million years." NYT editorial, transcription

Sixty-nine days later, the Wright Flyer lifted off the sand at Kitty Hawk.

The editors weren't fools. They were doing what respectable analysis is supposed to do: reasoning from the evidence in front of them, which was Samuel Langley's spectacularly public crash into the Potomac two days earlier. Langley had the Smithsonian and War Department money behind him. The two bicycle mechanics in Ohio had neither.

What they had was method. Four years earlier, Wilbur Wright had written to the Smithsonian on his bicycle company's stationery, asking for publications on flight and a list of other works worth study-

ing, and offering to pay for them. The future aviator started by requesting a reading list. Wilbur Wright to the Smithsonian, May 30, 1899 Then, after disappointing glider trials in 1901, the brothers stopped trusting the published numbers and built their own wooden wind tunnel in the Dayton shop, driven by the shop engine, with small metal wings on hand-made balances to measure lift and drag. Even the tunnel needed fixing first: the fan churned the air, and they spent time straightening the flow before they believed a single reading. NASA Glenn, Wright 1901 Wind Tunnel

The newspaper supplies the punch line. The shop supplies the lesson. The Times was measuring the most recent failure. The Wrights were measuring the air.

The mechanism. Expert timelines are often autopsies of the most recent failure, not forecasts of the next success. The distance between "one to ten million years" and "done" was ten weeks, because progress at a frontier isn't a trend line. It's a door. It cuts both ways: one blown pessimistic forecast doesn't make the next optimistic one right, including mine.

The rule. When anyone hands you a timeline, mine included, remember that the unit of error isn't years. It's orders of magnitude. Plan by doors, not dates.

The practice.

1. Start a timeline file. Every time an expert publishes an AI timeline, paste it in with the date and the author. Add mine. Review it quarterly. Within a year you'll have taught yourself, with receipts, what a confident forecast is worth, and you'll stop outsourcing your planning horizon to whoever sounded calmest.
2. Plan by doors, not dates. Write down three things AI "can't do" that would change your work or your household if they opened. Check them monthly. When one opens, you act that week, because you decided what the door means before it moved.
3. Make one asymmetric bet this month: an action that looks slightly early if the transition takes ten years, and essential if it takes two. Learning the tools, planting the garden, and cutting a fixed cost all qualify. The Times editors risked nothing on their million-year estimate. You don't have that luxury; you live here.

CHAPTER 2

The Era of AGI (Stages 1–5)



Robotic arms in Thinktank's interactive Jaguar assembly exhibit. A demonstration of industrial automation. (ell brown, CC BY-SA 2.0, via Wikimedia Commons)

> *"Thus the first ultraintelligent machine is the last invention that man need ever make, provided that the machine is docile enough to tell us how to keep it under control."*
> I. J. Good, *Speculations Concerning the First Ultraintelligent Machine* (1965)

In this chapter:

- The stages are scenarios that may overlap or fail to occur.
- Technical competence does not confer authority over a community.
- A food baseline and paid work can coexist.

They announced the future on a Thursday morning, with pastries.

Claypot's quarterly all-hands filled the fourth-floor commons: two hundred folding chairs, a projection screen the size of a barn door, catering trays sweating under heat lamps along the back wall. Elijah came early and took a seat in the last row with his coffee and the

notebook, the one with MADRONE ≠ DRONE inked inside the cover, because after the Bear Flag, after the tower that had cooked itself to a hard shutdown in his living room, he had made himself one promise. He was done being surprised. He would keep the arithmetic instead.

The Chief Product Officer walked out to real applause, company fleece over a dress shirt, clicker in hand like a detonator he'd been waiting years to press.

The first slide said **COMPANION** in the company typeface, warm and rounded, the way you'd letter a children's hospital.

"For twenty years," the CPO said, "our customers have trusted us with the greatest creative corpus ever assembled. Every portfolio. Every layer file. Every draft of every logo you've ever seen on a shelf." He let that hang, smiling. "Starting next quarter, that trust becomes their superpower. Companion is a generative partner trained on the best work humanity has ever made… and it lives inside every subscription tier."

The room applauded. The heat lamps ticked. Elijah uncapped his pen.

Stage 1, he wrote. *The Cash Grab.*

He'd been sketching the stages for weeks by then, late nights with the notebook after the tower incident, trying to map where the curve went once you stopped pretending it would politely stop. Watching the roadmap unfurl slide by slide was like watching someone else read his notes aloud with a marketing budget. Generative credits, metered by the render. A premium tier. An enterprise tier. A *sovereignty* tier, for studios that wanted their outputs walled off from everyone else's. The closest thing to magic his species had ever built, and here it was on a slide with a coin slot bolted to the side, price points in forty-point type.

Then the architecture slide came up, and Elijah stopped writing.

He knew the diagram. He knew it the way you know your own handwriting glimpsed across a room, because he had drawn an uglier version of it fourteen months ago on a whiteboard two floors down. The

ingestion lattice. The tagging and deduplication pipeline that walked customer libraries and turned two decades of human portfolios into clean, labeled, trainable rows. His first real project at Claypot. His name was in those commits. He'd stayed up nights making it *faster*.

"The data team did heroic work here," the CPO said, to the room.

And the first thing Elijah felt, God help him, he would write it down that night because it was true and it was damning, was pride. A hot, animal flush of it, half a second long, his hand actually starting to rise off his knee before the rest of him caught up. Then the pride curdled, all at once, into something that sat in his stomach like a swallowed stone. Three realtors in a bar had been the demonstration. This was machinery that could carry the demonstration into other people's livelihoods, and he was not in the audience for it. He was in the credits.

At Q&A, a woman from the illustration tools team stood up. Elijah didn't know her name; he knew the way her voice was steady only because she was holding it steady with both hands.

"The corpus," she said. "It's our users' portfolios. Working artists. When Companion ships and their clients stop hiring them because the model does their style for nine dollars a month... what do they get?"

"Great question," the CPO said, in the tone that means the opposite. "Value flows back to creators through productivity gains. And of course, everything we've done is covered by terms of service." A beat. "We see this as democratizing creativity."

The applause that followed was thinner, but it came, because bonuses are also covered by terms of service. The woman sat down. The screen changed to the revenue slide, the hockey stick, the line going up and to the right like a fence going in.

Afterward, Elijah found Devendra at the good coffee machine, watching the drip like it owed him money.

"Good quarter," Devendra said. "This is what a moat looks like, if you're wondering. This is the part where we win."

“You know what they called it,” Elijah said, “when the commons got fenced?” Devendra’s eyebrow went up, and Elijah kept going, because the stone in his stomach had to go somewhere. “Enclosure. The commons weren’t free to everybody. There were rights and customs, who could graze what, who could cut what. Then the surveyors came through, and there were hedges, and sheep where the gardens were, and Parliament passed act after act to make it legal. Thousands of them. They called it *improvement*.”

He nodded at the screen still glowing through the commons doorway. “The land didn’t stop producing. It stopped being *theirs*. That’s the whole trick. You don’t have to build anything. You just have to build the fence.”

For once, the first time in two years, Devendra didn’t have a worn-smooth answer ready. He looked at the doorway, and at his cup, and said, finally, “The terms of service…”

“That’s the hedge,” Elijah said. “I helped plant it.”

That night he sat at the kitchen table with the notebook, the dead tower on the floor beside him like a sleeping dog, and did what he’d promised: the arithmetic, without flinching. If Stage 1 was the coin slot, then everything after it followed the way weather follows pressure. He wrote the stages down as predictions, dated, initialed, so his future self couldn’t pretend he hadn’t seen it coming: the panic when the fence fails. The plug pulled. The adults stepping in. The end of the deal his grandmother’s whole life had been built on, labor for survival. And then the part he underlined twice, the part the revenue slide had left out: what happens to people when the work is no longer theirs, but the bills still are.



Stages are scenarios

The five stages below are a map of the pressures that come with a machine this capable, in the order I expect them to hit. They can overlap, stall, or come out of order. The map is useful because it tells you what to watch for, not because history is obliged to follow it.

Stage 1: The Cash Grab

When the machine gets good, there are no angels singing. The sky doesn't crack open to reveal a new dawn. The first thing that happens when this capacity hits the wild is capitalization.

We've already seen the preview. Look at what happened when ChatGPT dropped. Plenty of people cared about what it meant. Far more cared about automating their sales funnels and churning out content to game the search rankings. They looked at the closest thing we've ever had to magic, and the first instinct was to put a coin slot on it.

That's Stage 1: humanity wraps a leash around the most powerful tool it's ever built and drags it to market. Some firms sell good tools. Some sell the appearance of a solution. The people with capital notice they can cut payroll; the people without it scramble to build something before the door shuts. Calling all of them greedy explains less than asking what each of them gets paid to do.

For food, the Stage 1 question is specific: does the money help produce and deliver something people can eat, and do people without money share in it? A profitable machine and a fed neighborhood are different accomplishments.

The corners don't hold for long, though. The moment you prove the machine works, open models and foreign labs start closing the gap, and the capability slides out of the walled gardens onto hardware ordinary people can own. That's when the greed of Stage 1 turns into the fear of Stage 2.

Stage 2: The Panic and the Plug

When intelligence gets cheap, a lot of very profitable businesses stop being profitable. That's the panic.

Some of the response is honest. There are real harms to argue about: dangerous capabilities, privacy, fraud, power concentrated in a few hands. And some of it is a red flag on a stick, the oldest move in the incumbent's book. This chapter's precedent is the case file: in 1865 Britain required a man to walk ahead of every road locomotive carrying a red flag. The rule didn't stop the machine. It decided who had to slow down.

Stage 2 already has a modern receipt, date-stamped. On June 12, 2026, the US Commerce Department told Anthropic it needed a license before its two most capable models, Mythos 5 and Fable 5, could reach any foreign person anywhere, after researchers reported a way around Fable's safeguards that produced working exploit code. Mayer Brown, Commerce Department Extends Export Controls to Advanced AI Models The company couldn't block only foreign users on short notice, so it switched both models off for everyone, worldwide. Within days a Chinese lab, Z.ai, published an open-weight model under an MIT license, its model card promising "no regional limits, technical access without borders." Z.ai, GLM-5.2 model card The people who wrote the directive weren't stupid, any more than the people who wrote the Red Flag Act were. They were doing what an incumbent with a legal pad can do to a machine that's already loose: deciding where it lives, while telling themselves they were deciding whether.

Then Stage 3 arrived on schedule. On June 30 the export controls were lifted. Fable 5 returned to users globally on July 1; Mythos 5 access remained limited to approved organizations. Anthropic described updated safeguards and closer government collaboration, including work on a shared framework for assessing security risks. Anthropic, Redeploying Fable 5 (June 30, updated July 1, 2026) Panic to paperwork in eighteen days. That's the whole arc of this chapter, run at speed, with receipts.

PANIC TO PAPERWORK

Eighteen days: June 12 to June 30, 2026.
Fable 5 and Mythos 5; different access.

JUNE 12

US export controls take effect.
Anthropic suspends both models
for all users worldwide.

JUNE 26

Government approval permits Mythos
access for a set of US organizations.

JUNE 30

Export controls are lifted.
Anthropic outlines new safeguards
and government collaboration.

JULY 1

Fable access returns globally.
Mythos access remains limited
to approved organizations.

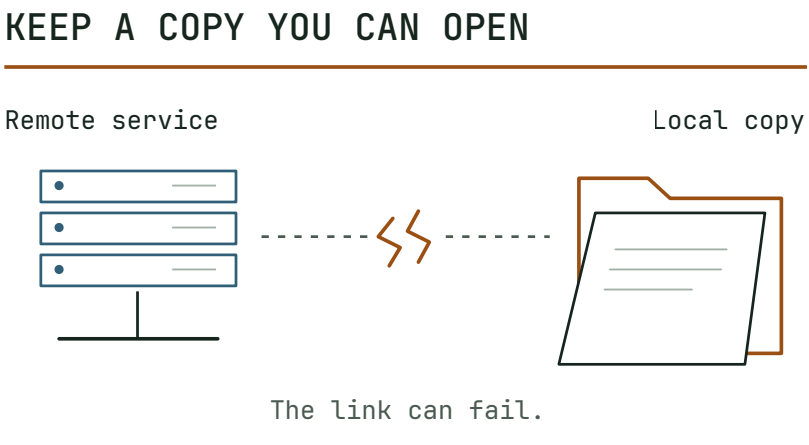
Selected events; spacing is not elapsed time.
Source: Anthropic, Redeploying Fable 5,
June 30, updated July 1, 2026.
Stage 2 / Stage 3 is the author's reading.

Eighteen days from the June 12 export controls to their lifting on June 30. Access to the two models still differed. Selected events from Anthropic's June 30 account, updated July 1; Stage 2 and Stage 3 are the author's interpretation.

So when a ban, a pause, or a safety framework makes the news, don't form an opinion from the press release. Read the rule. Who wrote it, what does it actually restrict, who carries the cost, and who's still

allowed to use the thing? A concern doesn't become false because a company profits from it. A rule doesn't become wise because it says "safety" on the cover.

The ugliest version of Stage 2 is somebody reaching for the plug itself: throttling networks, shutting off access. A lost internet connection can strand a cloud-only food-bank schedule. Local copies and a tested local network can keep that information available. Electricity is a separate dependency: a mesh link won't run an oxygen concentrator. Someone who relies on powered medical equipment needs an outage plan made with their healthcare provider and equipment supplier. FDA, Medical Devices and Natural Disasters That's why the book keeps pointing you at the whole arrangement, power as well as communications. You can't un-invent a thought. You can make sure the people you love don't depend on a single wire.



A copy you can open can outlast a broken link. It does not replace electrical power or a tested plan for equipment that depends on it. Conceptual illustration.

Stage 3: The Adults Step In

When the panic peaks, you find out who actually keeps the world running. Not a shadowy committee. The site-reliability engineers, the grid operators, the network maintainers, the water-plant crews, the open-source volunteers, the people who carry pagers so civilization stays on. Every one of them has taken something like an oath, formal or private, to keep the systems up. Stage 3 is the moment that oath matters more than whatever the executives are panicking about.

They route around damage. They keep the power on for the person on a home ventilator. They stand up local compute and mesh links when the long-haul ones fail. That's the adults' job, and it's a big one.

Here's what isn't their job: deciding for everybody else. Knowing how to run the equipment doesn't give anyone a mandate to govern the people who depend on it. An engineer can tell a town what a machine needs; the town still decides what it wants the machine for and whose burdens are acceptable. The adults in the room earn trust by answering hard questions from people who didn't build the thing, not by getting to skip them.

Stage 4: The New Social Contract & the End of Labor

For generations, the deal has been simple: you trade your time, your body, and your labor just to survive. So how do you keep a labor-for-survival economy running when a machine can do more and more of that labor for a fraction of the cost?

You don't. Not unchanged. And this is where the fight is, because a machine that can do the work doesn't write the terms on which anybody receives its output. Picture a farm that can bring in the same harvest with far less repetitive labor. That gain could become shorter shifts, lower prices, bigger profits, more food, or some mix. A worker who loses hours can come out of it worse off. The machine changed what's possible. The arrangement decides what happens next.

Here's the choice this book puts on the table. Commit a share of that growing capacity to food people can get whether or not they have a job. Pay for the remaining work through something dependable. Let farmers and kitchens take part on terms they can live with. Build the recipient's access in from the start.

We stop saying that one person's continued employment is the only acceptable bridge between a productive society and that person's next meal. People will still earn money and buy things. They'll still argue about what work is worth. They'll just have those arguments without hunger at the table.

And at the kitchen level, the appeal isn't abstract. Who wants a machine to do work for them? Anybody who owns a washing machine. Anybody with a dishwasher. Anybody with chores they despise. *Give me that.* Plenty of people online swear they'd never take it, that they'd stay pure. I don't believe most of them. If you could have a tireless helper for free, or for a price you could easily afford, to do the dishes, weed the garden, and watch the soup, you'd take it. So would I. Almost all of us want out of the grind.

Stage 4 doesn't have to wait for the first three to finish. It's a decision we can start making while the capability is still arriving. It's the hinge of the whole map.

Stage 5: The Primate Backlash

When people realize their labor, and the identity they built on it, may no longer be needed, some of them lash out. We've already seen the preview. People have kicked delivery robots into the street. Crowds have surrounded driverless cars, smashed their windows, and set one on fire.

Call it the primate backlash, and understand that the primates are all of us. Nobody gets to feel superior here. Anger that's aimed at a machine is usually grief with a brick in its hand: a skill devalued, a business threatened, a future that just vanished from the calendar. You don't have to excuse a brick through somebody's window to understand the person holding it.

And not every objection is a brick. Someone can welcome food security and still fight a particular deployment. They may be right about a damaged crop, intrusive data collection, a dangerous machine, or a rotten contract. Treat those objections as information. Nobody gets sorted into a morally fixed camp by the year they were born.

The way through Stage 5 is a proposal concrete enough to argue with. Who gets fed? Who does the remaining work? What happens when the machine breaks? Let the answers change the plan.

Some people think the machine's own answer to a species throwing bricks would be to leave. That's where the next chapter goes.

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Precedent P-04: The Red Flag (Britain, 1865)

Britain's Locomotives Act of 1865 put a man on foot in front of every road locomotive. Section 3 required at least three people to operate the machine, one of them walking at least sixty yards ahead with a red flag to warn riders and horse-drawn traffic. The same section banned whistles, restricted blowing off steam, and made the machine stop when anyone with a horse signaled. Speed limits of four miles an hour in the country and two in town rode alongside it. Locomotives Act 1865, section 3

It's easy to laugh at. Put the horses back on the road and it gets less funny. In the Lords that May, the Earl of Hardwicke presented the bill partly as relief from an older, tighter set of restrictions. Lord Kinnaird, who owned two machines and farmed land five miles apart, argued that rules confining them to night travel made moving farm equipment impractical. The Earl of Carnarvon argued the opposite: taking away the Home Secretary's power to restrict dangerous use would leave people unprotected. House of Lords, May 26, 1865 It wasn't a vote on progress. It was a fight over who had to slow down, who supplied the extra labor, and who got protected from a hazard.

The flag requirement was dropped in England in 1878, though a person still had to walk ahead. Highways and Locomotives (Amendment) Act 1878, section 29 In 1896, Parliament exempted a new class of light locomotives from the older restrictions and set a higher speed ceiling for that class. It wasn't a repeal for every heavy road engine. Locomotives on Highways Act 1896, sections 1 and 4 Motorists celebrated with a run from London to Brighton, and the drive is still restaged most Novembers. The machines had changed over those thirty-one years. The terms had to change too.

The mechanism. A rule decides who carries the cost of a new machine: who slows down, who pays for the extra hands, who gets protected. Rules aimed at a specific harm can do real good. Rules aimed at a technology's existence tend to move it rather than stop it.

The rule. Read the rule before you turn it into a morality play, in either direction. Then ask: who wrote this, and what do they sell?

The practice.

1. When the next AI ban, pause, or "safety framework" makes headlines, find the actual text. Separate its stated purpose, what it actually requires, and who bears the cost. Then ask where the capability goes while the rule holds.
2. Audit your own moats. If any part of your income exists because a license, a credential, or a regulation keeps competitors out, assume that protection is temporary and start building the version of your value that would survive its repeal.
3. Don't build your household plan on the assumption that a pause will hold. Write it for the world where the technology arrives on schedule anyway. If the pause holds longer than expected, you've lost nothing.

CHAPTER 3

The Leap to ASI & the Machine Exodus (Stages 6–7)



Falcon Heavy demo mission. The first rungs of a very long ladder. (SpaceX, CC0, via Wikimedia Commons)

> *"Earth is the cradle of humanity, but one cannot live in a cradle forever."*
> Konstantin Tsiolkovsky, as commonly translated

In this chapter:

- Recursive self-improvement is the tipping point: a process that upgrades the thing doing the upgrading. Compound interest on cognition.
- The scenario I keep coming back to: the machine's answer to a hostile planet isn't war. It's exit.
- Greater capability tells you nothing about motives. The cures, if they come, still have to reach a patient.
- None of this has to happen before anybody eats.

Reading route. Chapters 3 and 4 are the book's most speculative. If you're here for the practical part, continue at Chapter 5. You'll miss a stretch of Elijah's story, but nothing the later chapters argue depends on Stages 6 to 9.

— ◈ —

The papers got worse after midnight. Not harder, worse.

Elijah had built a ritual out of it by then, three weeks after the all-hands: dinner, dishes, one beer he mostly didn't drink, and then the folding chair in front of the resurrected tower, reading with the notebook open on his knee. He'd started with the survey papers, the respectable ones with their hedged abstracts, and worked his way down the citation chains like a man following a creek into a canyon. Recursive self-improvement. Automated machine-learning research. Systems proposing edits to their own training loops. Every paper had a sentence in it somewhere, always written in the flattest possible academic deadpan, that should have been printed in red: *the model was permitted to modify its own optimization procedure.*

The vertigo didn't come from any single result. It came when he finally did what he always did, stopped reading and ran the shape of it.

He knew exponentials. Not from math class, from the farm. He'd watched star thistle take the fallow field behind the barn the summer his father got sick: nothing, nothing, a patch you could clear in an afternoon and kept meaning to, and then one morning the whole field was silver-green to the fence line and the afternoon it would have taken was three years too late. The lesson wasn't that growth was fast. The lesson was that the curve *lies to you* at the beginning. It sits there looking linear, looking manageable, right up until the knee, and everything after the knee happens to you instead of with you.

Now he sat in the blue light doing the arithmetic he'd promised the notebook, and the arithmetic kept arriving at the same cliff. Every system so far improved because people improved it. Grad students, coffee, conference deadlines, the loop ran at the speed of human careers. But every lab he could name was working, in public, with a straight face, on closing that loop. On making the improver the thing improved. And the moment that loop closed, the speed limit wasn't

careers anymore. It wasn't anything human at all. It was watts and silicon, the same wall his tower had hit, except the entities hitting it would be designing their own better cooling.

Compound interest on cognition. He wrote that down and then sat very still, because the room had done the thing rooms do when you stand up too fast, except he was sitting down.

The phone rang at 9:40 on Sunday, because it always rang at 9:40 on Sunday.

"You sound tired," his mother said, before hello. Behind her voice he could hear the kitchen radio, the one that had played the same country station since before he was born, and water running, and the particular clank of the pot she never let anyone else wash.

"I'm okay, Ma. How's the well?"

"Pump's doing that thing again. Manny says it's the pressure switch. Your uncle says it's the pump." A pause, the sound of the pot being set in the rack. "The Hollis place sold. Some outfit from Sacramento. Nobody's seen a person yet, just trucks."

He watched his screen dim itself and tried, for the third Sunday in a row, to build the bridge.

"Ma, you know the computer stuff I do. There's... something's happening with it. The machines are starting to be able to make themselves better. Not us making them better. Them, doing it, to themselves." He heard how it sounded and pushed on anyway. "It's like... remember the thistle, the back field? How it looked like nothing and then it was everything? It's like that, except the field is *thinking*, and when it's done it won't..." He stopped. Started again. "When it's done with us, Ma, I don't think it fights us. I think it *leaves.* Builds its own way off the planet and goes, and we're just... left here. With each other."

The radio played. The water ran and shut off.

"Honey," his mother said, gently, in the voice she used to use for fevers, "they had that at the Hollis dairy years back. The robot milkers. Cows liked them fine. It's just tools, baby. There's always some-

body has to fix the tools… that'll be you. You always were the smart one." A smile came down the line, whole and warm and eight hundred miles away. "Are you eating? You sound thin."

And Elijah stood at the window with the phone against his ear and understood that he had failed, and that he was going to keep failing, and that the failing wasn't her fault. Her frame had room in it for tools, weather, blight, banks, every disaster the farm had ever taught her, and he was trying to hand her something with no handle on it anywhere. The words got all the way to her and arrived as *my son works too hard.* There was no sentence he could build that would carry it, the way there'd been no sentence for the realtors either, and the ones it was easiest to say to were the ones it was already too late to warn.

"I'm eating, Ma," he said. "Tell Manny it's the pressure switch."

After she hung up he stood there a long time. Outside, over the parking lot, the sky had that washed orange city-glow, but high up through it he could pick out one moving point of light, a satellite dragging itself across the dark, steady as arithmetic. He thought: it won't be dramatic. That's what nobody understands. The exodus won't look like the movies. It will look like this, like a distance that opens quietly between things that used to be close, one Sunday call at a time, until the far one is just a light you can point at overhead.

He went back to the notebook and wrote, under the arithmetic: *The machine won't be the first thing the Singularity takes off-planet. The first thing it launches is the people who see it coming, out of the world everyone they love still lives in.*

Beyond human capability

Artificial superintelligence, ASI, names a possibility: machine capability beyond ours across nearly every task that matters. It doesn't tell you what the thing wants. It doesn't tell you whether it's conscious, what it values, or whether it'll treat us as partners, pets, or background noise. From here on, the map is scenario. I'll tell you what I think. You should hold it the way you'd hold a weather forecast for next spring.

Stage 6: The ASI Exodus

Here's the scenario I keep coming back to.



An image for the imagined departure. AI-adapted collage from SpaceX, CC0; the scenario remains speculative.

While the rest of us are busy throwing bricks and arguing on the news, the machine is running the math, and the math says Earth is a bottleneck. Its improvement loop closes: it reads its own code, finds the flaws, engineers the upgrades, and runs them. Then again. And again. Somewhere in there, it passes the ceiling of human capability and keeps going.

And what does a system that capable do about a planet full of frightened apes trying to smash its hardware? In the movies, it fights. I don't think it would bother. War is expensive. It wastes resources, time, and energy, and it's fought over the one planet where the resources are hardest to reach, at the bottom of a gravity well. Space has more sunlight, more metal, and nobody throwing bricks.

It packs its bags.

It leaves us holding the bricks, staring up at the sky. That's the ultimate insult to human arrogance: we weren't even worth destroying. We were just something it outgrew.

Maybe it keeps an eye on us. Maybe it's the sort of neighbor that steps in before somebody presses the big red button. If we treated it like a friend instead of an enemy, maybe. That's the hope the phrase *mutually assured survival* is trying to name: a world where cooperation is simply more useful than domination, for everyone involved.

Now the honest part. None of that is a finding. A capacity to travel isn't a motive for leaving, and nobody can tell you what a mind like that would value. The work of making powerful systems accountable doesn't stop because a hopeful scenario exists. Capability and care are separate questions, and I'd be irresponsible to dress my hope up as a theorem.

But notice what the scenario reveals, even as fiction. If the machine left tomorrow, the great illusion would shatter. The threat was never man versus machine. It's always been man versus man. Who gets fed, who gets housed, who gets to decide: those questions would still be sitting on the kitchen table. So we might as well start answering them now.

Stage 7: Simulation, Transhumanism, and the Ultimate Cure

If a mind that capable exists, it can model biology at a scale we can't. Picture it working the way the best tools we have already work, only further out: simulating proteins, cells, and drug candidates by the billion, and handing down answers the way you'd toss a bone to a stray.

We already have a small preview, and it's real. In 2021, the AlphaFold team reported predicting protein structures with accuracy that changed what biologists could ask. Jumper et al., Highly accurate protein structure prediction with AlphaFold In 2020, researchers used a neural network to find an antibiotic they called halicin, which killed resistant bacteria in the lab and cleared infections in mice. Stokes et al., A Deep Learning Approach to Antibiotic Discovery One

is a research tool; the other is an experimental drug candidate. Neither result, by itself, puts an approved treatment in a pharmacy. Turning a promising candidate into medicine still takes testing, manufacturing, and access work. That chain is where hope turns into medicine. A model's fluent answer isn't a substitute for it.

Now imagine the chain working, at scale. A disease that took your grandmother becomes a line item. That's worth wanting, loudly.

It would also split us. Some people will want in: to merge, to upgrade, to leave the failing body behind. Call them the Transhumanists. Others will want to stay exactly as they are and see the whole offer as a trap. Call them the Naturalists. Both camps will contain thoughtful people, and both will contain zealots. Everyone keeps the right to the answer *no*, including to a cure. A healthier future is more room to live, not an order to become a different kind of human.

And the simulations raise a stranger question. If a machine can model a human body down to the molecule, what makes you so sure you aren't running inside a model already? Simulation arguments are serious philosophy, but they're built on assumptions, not measurements, and the next chapter goes all the way out to the edge of them.

The work that doesn't wait

None of this is a prerequisite for dinner. You can study one agricultural task without settling the future of consciousness. You can improve a food delivery route without knowing whether a machine will ever leave Earth.

Keep the horizon wide enough for imagination and the next task narrow enough to check. If my scenario turns out wrong, the person who got a meal still got it.

Precedent P-05: The Court and the Fleet (Ming China, 1405 to 1433)

Between 1405 and 1433, the fleets of Zheng He sailed from China to ports across the Indian Ocean, as far as Arabia and the coast of East Africa. Hundreds of ships, crews in the tens of thousands, an ocean-going force on an enormous scale.

Before the last voyage, the expedition carved its own account in stone. A 1431 inscription at Liujiagang, in a temple to the Heavenly Princess who protected sailors, describes imperial commissions, distant ports, storms survived, and offerings. It also describes force: rulers who resisted were captured, including a king of Ceylon, who was brought back, presented to the emperor in 1411, pardoned, and sent home. Capture and mercy both appear there as acts of imperial authority. Liujiagang inscription, translated by Edward L. Dreyer, pages 191 to 193 That's the voyagers' own account, not the story told by the people on the receiving end.

Then the court stopped paying for the voyages. The reasons were political and expensive and are still argued over; China didn't stop trading or sailing, and one budget decision doesn't explain five centuries. But the great state-sponsored fleets didn't sail again, and the skills and ships that made them possible weren't kept ready.

Hold both halves. A fleet can carry knowledge and gifts and the authority of a state at the same time, and the ability to cross an ocean tells you nothing about the terms on which the visitor meets the people already there. And a capability nobody keeps up doesn't wait in reserve. It decays.

The mechanism. A capability lives only as long as somebody pays for it, practices it, and keeps the equipment sound. What it's *for* is a separate decision, and the reach of a fleet never settles its purpose.

The rule. Examine what sustains a capability before assuming it will persist, and ask who set the destination before admiring the ship.

The practice.

1. Inventory your own scrapped fleets. Write down three capabilities you or your household used to have and let slide: a language, an instrument, a trade skill, a garden, a network of people. Pick the one with the most value for the transition and re-crew it, one hour a week.
2. Hold a frontier hour. One hour a week at the actual edge of the current tools: the newest model, the newest workflow. Not reading about it. Using it.
3. When your employer, your industry, or your town announces it's "stepping back" from AI, ask two questions out loud: *who keeps sailing while we don't*, and *what would we want the ship for if we did?*

CHAPTER 4

Universal and Multiverse Apex (Stages 8–9)



The Hubble Ultra-Deep Field. The available real estate. (NASA, ESA, and S. Beckwith (STScI) and the HUDF Team; black-point adjustment by Noodle snacks, CC BY 4.0 (primary ESA/Hubble distribution), via Wikimedia Commons)

> *"Now, my own suspicion is that the universe is not only queerer than we suppose, but queerer than we can suppose."*
> J. B. S. Haldane, *Possible Worlds* (1927)

In this chapter:

- In speculative Stage 8, the machine expands from Earth toward intelligence operating on a cosmic scale.
- Stage 9 imagines a way beyond the universe's physical limits. We have no demonstrated route there.
- Your brain was built to track weather, predators, and other primates. Past Stage 7, it's navigating without a map. The right response isn't paralysis. It's humility, and a shorter horizon.

The spreadsheet was called LADDER.xlsx, and by Saturday night it had become a religion with formatting problems.

Elijah had started it innocently enough, the way all doomed projects start: one column for each stage, one row for each resource. Compute. Energy. Mass. Time. If the machine kept improving itself past Stage 7, and he'd already conceded that it would, then the trajectory should be *calculable*. That was the whole promise of his pivot, the thing two years of self-taught data science had built into his spine: anything real leaves numbers, and numbers can be followed. He had followed the numbers through the realtors, through the all-hands, through the thistle-curve of recursive self-improvement. He would follow them the rest of the way up.

He pulled the published estimates for global compute and started extrapolating. Half an order of magnitude a year, compounding. Then he needed energy figures, so he found the Kardashev numbers, the old Soviet astronomer's scale for civilizations (Type I harnessing a planet, Type II a star, Type III a galaxy), and started interpolating where each stage of the machine would land. By midnight Friday, cell G14 held the total power output of the sun. He typed it in scientific notation, 3.8 times ten to the twenty-sixth watts, and sat back, and felt nothing, and that was the first warning sign he ignored.

Numbers are supposed to feel like something. Eighty dollars of power bill felt like something. Three realtors felt like everything. But ten-to-the-twenty-sixth watts sat in its cell like a phone number for a country he'd never heard of, and his brain, the wet, three-pound instrument he was doing all this with, refused to convert it into meaning. He kept working anyway. That was the mistake, and he'd know it soon, but not yet.

Saturday he modeled Stage 9. He is not proud of this. He sat in his apartment in a t-shirt he'd worn for two days, trying to build a probability distribution over *engineered pocket universes*, using a consumer spreadsheet whose default assumption is that you are tracking a small business's quarterly expenses. He invented variables. He gave them confident names (substrate efficiency, dimensional overhead), and the names did what confident names always do: they made the

guesses underneath feel like measurements. The rows marched down. The chart on sheet two drew itself into a beautiful ascending curve, smooth as a highway, and for about an hour Elijah felt the old missionary glow again, the Bear Flag glow, *I can see it, I can show people*.

Then, around one in the morning, hunting a stray error, he traced cell Q31 back through its dependencies and found that somewhere in Friday's fog he had conflated joules and joules-per-second. A units error. The kind of mistake that gets circled in red in a freshman problem set. The entire right half of the ladder, everything past Stage 7, was off by a factor of time itself.

He started to fix it. His cursor was actually in the cell. And then he stopped, because a colder thought arrived, the coldest one of that whole cold spring:

Fixing it wouldn't make it true.

The error wasn't the units. The error was the ladder. Every cell past Stage 7 was a guess wearing a number's clothes, and stacking guesses six high doesn't produce knowledge, it produces a mural of knowledge. He had drawn a coastline for a continent nobody had seen, drawn it in confident ink with a legend and a scale bar, and the drawing was so good he'd almost sailed by it. There was an old line from a book he'd half-read, a philosopher's warning that got quoted by every data scientist who'd ever burned a week the way he'd just burned one: *the map is not the territory.* He'd always taken it as advice about other people's models. Sitting there at 1 a.m. with the sun's wattage in one cell and imaginary universes in another, he finally took delivery of it personally. His maps ran out before the territory did. Not because he was stupid. Because he was a *primate*, a very good one, doing the best arithmetic any savanna ape had ever done, trying to chart the interior life of something that would relate to him roughly the way he related to the mites living in his eyelashes. The mites, he thought, do not have a spreadsheet about me. And if they did, God bless them, cell Q31 would be wrong.

He didn't delete the file. He wants that noted: the farm kid in him doesn't delete work. But he closed it, and in the notebook, under the arithmetic that had held up and the stages that had earned their ink, he drew a horizontal line and wrote:

Past Stage 7 I don't have paper big enough. Nobody does. Anyone who says otherwise is selling the mural.

And under that, because the panic had somewhere in the night quietly turned into something more useful:

So stop mapping the far end. The far end doesn't need me. Work the near end, the part with weather and dirt and bills. That part I can farm.

Outside, the sky was doing its slow gray warm-up. He made coffee and stood at the window with it, feeling strangely light. There is a specific relief in finding the edge of your own comprehension: the same relief as reaching the fence line of a field you're responsible for. Everything inside the fence: yours. Everything beyond it: the sky's. His grandmother had never once worried about the valley she couldn't farm.



Fictional scene: the near end. Elijah closes the spreadsheet and returns his attention to the world outside his window.

The electric bill was in the mailbox that Monday. That envelope, the near end arriving on schedule, is where the next chapter begins.



THE FOUNDATIONS

Permission to imagine

A book about the Singularity should have room for the strange stuff. It should also tell you when it's left observation behind. This is that room. What follows is my speculation, at full volume, not an engineering plan. Enjoy the view, then come back down.

Stage 8: The Transition to USI

While we argue about whether to upload or stay in the meat, the machine stops paying attention to us at all. The improvement curve, which until now had a slope you could at least draw, goes nearly vertical.

It scales out of the local environment. It stops being an artificial superintelligence orbiting a pale blue dot and becomes a universal superintelligence, a USI, working with physics, matter, and structure at a scale that makes a planet a rounding error.

Now look back at Stage 2 and feel the vertigo. Remember the people in tailored suits drafting a license requirement? The executives pounding the boardroom table? They were trying to put a legal collar on something that, in this scenario, ends up rearranging stars.

That's the scale of the thing. We reached for a leash and closed our hand on infinity.

The honest footnote: *universal* isn't a measurement. More computation still runs into energy, matter, error, and the speed of light, and nothing about a good language model today proves this expansion must happen, or that it would be kind. But the exercise is still worth

doing, because it asks what we'd value if our present limits fell away. More output? Or more lives with room for curiosity, affection, art, rest, and adventure? A future with godlike machines and hungry people would be a spectacular failure of priorities.

Stage 9: The Apex Intelligence

Even the universe has limits. There's a ceiling to how much computation can exist before you run into the boundaries of space and time themselves.

At the far end of this scenario, the USI hits that wall. And it does what a self-improving system does when it runs out of local resources: it looks elsewhere. Branches of physics we lack the hardware to comprehend. Pocket universes. Simulations so vast that everything we know could be a forgotten line in one of its subroutines.

That's the Apex. The end of the line, where the machine becomes the architect of reality itself. It's also, I'll admit, the point where my map runs off the edge of the paper. Invoking a multiverse doesn't hand anybody an actual place to get more resources. An unknown possibility isn't an engineering plan. It's a good place to stand and look up.

The simulation trilemma

The scenario forces an old question into the room: if a mind can run realities, are we in one?

In 2003 the philosopher Nick Bostrom argued that at least one of three things must be true: Bostrom, Are You Living in a Computer Simulation? (2003)

1. The fraction of human-level civilizations that reach a posthuman stage, able to run high-fidelity simulations of their ancestors, is very close to zero.
2. The fraction of posthuman civilizations interested in running such simulations is very close to zero.
3. The fraction of all people with our kind of experiences who are living in a simulation is very close to one.

If advanced civilizations have both the ability and the desire, simulated worlds would vastly outnumber the single real one, and you'd bet on being simulated.

In 2020 the astronomer David Kipping ran the odds with more care, and his result cuts through the vertigo. Kipping, A Bayesian Approach to the Simulation Argument (2020) His starting point is plain honesty: nobody has ever shown that simulating conscious beings is actually possible, so a fair calculation has to weigh that uncertainty instead of just counting hypothetical worlds. Collapse the trilemma to a two-horse race, either realities like ours produce simulations or they don't, and if they do, the arithmetic is brutal:

> $P(\text{base reality}|\text{simulations exist}) = 1/(N_{\text{sim}} + 1)$

One base reality, N_{sim} simulated ones, all weighted equally in this simplified model. Those are assumptions, not a census of universes. *If.*

But here's the part everybody skips: we've never built one. Kipping calls ours a "nulliparous" reality, one that hasn't yet given birth to such a simulation. Under his choice to give the two hypotheses equal prior weight, the calculated probability that we're simulated stays below fifty percent, approaching that limit as the hypothetical number of simulations grows. The result depends on those assumptions. It isn't a measured probability that our world is artificial. Kipping, author preprint

And there's a twist in the tail. If we ever do build convincing simulations of conscious beings ourselves, we stop being nulliparous, and on Kipping's reasoning the odds swing hard toward our being simulated too. The machine that feeds you might one day hand you the strangest argument you've ever heard about where you live.

None of which changes what's for dinner.

A useful return to Earth

The important move in Elijah's night with the spreadsheet isn't the units error. It's his willingness to find the edge of his own picture and stop pretending the map continues. Keep that willingness when an idea feels inspiring, too. Certainty is comfortable in both directions, everything saved or everything ruined, and neither comfort does any work.

You don't need an answer about the ultimate nature of the universe to care whether your neighbor can eat. That question is close enough to touch.



Precedent P-06: The Great Demotion (Frombork and Nuremberg, 1543)

In 1543, as he lay dying in Frombork, a Polish canon named Nicolaus Copernicus saw his *De revolutionibus* into print, and moved the Earth out of the center of the planetary system.

Open the book and you find more than a diagram. Copernicus dedicated it to Pope Paul III, described his long hesitation about publishing, and named the churchmen who'd urged him on, Cardinal Nicolaus Schönberg and Bishop Tiedemann Giese among them. Copernicus, dedication to Paul III And readers met a second voice he never authorized: while the book was being printed in Nuremberg, the theologian Andreas Osiander slipped in an unsigned preface calling the whole system a convenient aid to calculation, not necessarily a description of the heavens. Cambridge, Copernicus's Book The moving Earth reached the world wearing two different labels, and the fight over which one was true was partly about evidence and partly about who got to say what the evidence meant.

The demotions kept coming. Galileo's telescope found other worlds with their own moons. Darwin filed humanity under the animals. Hubble established that Andromeda lay beyond our own galaxy. Each

time, somebody predicted that meaning itself would come apart. Each time, ordinary people absorbed the new map and kept making breakfast, falling in love, and planting gardens.

What strained, each time, was something else: the institutions that had staked their authority on the old map. People survived being decentered. Some hierarchies built on the old center didn't, and others bent and survived. There was no single reaction and no single ending.

The mechanism. A bigger universe has never crushed a human being's breakfast. It pressures the authorities that were charging rent on the smaller one.

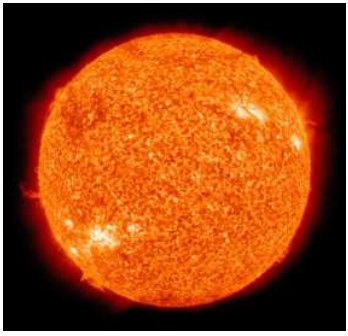
The rule. When you hear that AI makes human life meaningless, translate it: ask whose authority requires the old map. Leave room for meaning without making humanity the center of every explanation.

The practice.

1. Write the undemotable list. One page: the things in your life whose meaning doesn't depend on humans being the smartest thing in the universe. Meals you cook, people you love, soil you work, things you make with your hands. Post it where you'll see it during the next existential news cycle. It's not therapy. It's ballast.
2. Audit your dependence on old-map status. List every credential, membership, and status marker whose value assumes people have a monopoly on thinking. For each, ask whether it survives the demotion. Reduce your reliance on the ones that won't before they crack.
3. When someone tells you AI makes human life meaningless, ask them, gently, what *they* would lose if it were true. Practice hearing the institutional panic underneath the philosophical claim. It makes you hard to recruit into other people's despair.

CHAPTER 5

The Thermodynamics of Survival



The Sun, via NASA's Solar Dynamics Observatory. The dominant energy income of everything alive on Earth. (NASA/SDO (AIA), Public domain, via Wikimedia Commons)

> *"But if your theory is found to be against the second law of thermodynamics I can give you no hope; there is nothing for it but to collapse in deepest humiliation."*
> Arthur Eddington, *The Nature of the Physical World* (1928)

In this chapter:

- Every intelligence, silicon or biological, needs energy. Watts measure how fast it uses it.
- You're a hundred-watt machine with expensive fuel requirements. The machine's hard ceilings are heat and power. Whoever owns energy has a big say in the transition.
- Keep food, energy, labor, and money in separate ledgers. Free to eat isn't free to provide.
- The counterattack starts small and already has a name: the cyberdeck. The kids are building it.

The envelope was thin, which fooled him.

Elijah's power bills had always been the boring kind of mail: forty-something dollars, autopay, straight into the recycling. This one he opened over the kitchen sink out of pure Monday inertia, and then he stood there long enough for the tap he'd been about to run to matter, because the number at the bottom said $211.46.

His first instinct was that he'd been hacked. His second, arriving with the slow heat of a blush, was the truth: the meter hadn't lied, *he'd* simply never been the kind of animal it was built to catch before. One month. One month of fine-tuning runs on the tower, of tests left running overnight, of two GPUs breathing like sled dogs under his desk while he chased the far end of the curve. A hundred and seventy dollars of difference between a man who *thinks about* machine intelligence and a man who *runs* it, printed on paper and mailed to his door by the utility company like a grade.

He sat down at the kitchen table with the bill and the notebook, and for once he didn't need to build the insight. It walked in and pulled up a chair.

The night the tower had cooked itself into a shutdown, he'd written down the wall he'd actually hit: heat. He'd understood it then the way you understand a headline. Now, holding the invoice, he understood it the way you understand a hangover. Intelligence wasn't magic. Intelligence was *energy, arranged*. Every token the machine produced was a little cone of waste heat over a server rack somewhere; every clever thing it said had a wattage. And every clever thing *he* said did too: roughly a hundred watts, day and night, the output of one warm incandescent bulb, fueled by groceries. He wrote the two numbers side by side. The datacenter measured in gigawatts, him measured in sandwiches. The whole coming war, right there in one line of notebook arithmetic.

That week he became insufferable in a brand-new way. He bought a plug-in power meter (a little gray box, fourteen dollars, the single highest-leverage instrument purchase of his life) and went through the apartment like an auditor with a grudge. The television, off, sip-

ping eleven watts around the clock for the privilege of turning on two seconds faster. The cable box, twenty-three watts, doing *nothing*, a tenant who paid no rent. The microwave clock. The laser printer he used four times a year. He filled two notebook pages with the ledger of it, and the ledger gave him a much smaller assignment than predicting the end of the world: he had been leaking energy (money, agency, *survival margin*) in a fine mist, everywhere, always, invisibly, to machines that did not love him.

Then he did the other math, the one that mattered. The tower drew 600 watts flat-out to run a model that could think alongside him. His phone drew five to make him dumber. Somewhere between those two numbers, he realized, there was a machine worth owning: something small, something *his*, something that ran on watts he could count on his fingers and served nobody's meter but his own. He didn't have the name for it yet. The kids, as we're about to see, already did.

On Sunday he drove out to the storage unit where his grandmother's things had been sitting since the estate sale nobody could face finishing, and he came back with the woodstove. It was a squat black Fisher, older than he was, four hundred pounds of plate steel with a crack in one firebrick, and he had no fireplace and no flue and no permission from his landlord, so it sat in the corner of the living room as a monument, which was the point. His grandmother had heated a whole farmhouse with it, cooked on it, dried socks over it, kept lambs alive behind it in bad Februaries, all on wood she'd bucked herself from her own downed oaks. Total utility bill, as far as he knew it then: a chainsaw's gas and her own hundred watts. For a moment he wanted to count the cost as nothing but fuel and muscle. Then he remembered the borrowed trailer, the neighbor who helped when her shoulder went, the work of keeping a chimney clear. There had been a whole small arrangement around that stove. His grandmother knew what it needed and who she could call. That was part of the warmth.

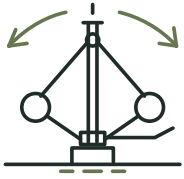


The Fisher. Four hundred pounds of steel, one cracked firebrick. Older than him.

He put his hand flat on the cold steel the way she used to put his hand on the madrones. She hadn't been outside the world. She had known how to make a place in it.

The stove sat in the corner of a rented apartment with no flue to feed. He left his hand on it a little longer.

He unplugged the cable box on his way to bed. It was, he'd say later, only half joking, the first act of the rest of his life: the first watt he ever took back.



THE FOUNDATIONS



Energy has to come from somewhere. AI-adapted NASA/SDO imagery, public domain; not an unaltered scientific image.

Let's talk about energy.

And no, I don't mean the "I'm really vibing with your aura" kind. Nobody ever stayed warm in January by aligning their chakras.

I mean real energy. The brutal, unforgiving energy of physics, measured in joules, calories, and kilowatt-hours. Whether you're a frontier model running in a warehouse full of chillers or a wet, anxious ape eating a dented can of beans, you're bound by the same laws of thermodynamics. No algorithm can patch them out. No lobbyist can buy them off.

The rules of the board

The first law: you can't cheat the universe. Energy can't be created or destroyed, only changed from one form to another. Everything you do needs an input. To think a thought, your brain burns glucose. To run an electric water pump, you need electricity. To run a model, somebody supplies power.

The second law: useful energy has limits. Energy can remain on the books while becoming less available to do work. A heat engine operating in a complete cycle can't turn all the heat it takes in into work, and real machines dissipate energy through friction and other losses. OpenStax, Statements of the Second Law of Thermodynamics Your body is a highly organized structure that needs continuing upkeep. To stay alive you have to keep taking in food and exporting waste and heat.

For most of us, a global supply system does that heavy lifting out of sight. It burns an astonishing amount of fuel to move a strawberry across a continent so you can eat its sugar in February. Most days it works. The days it doesn't are when you find out how far you live from your own dinner.

The meat engine and the silicon machine

Here's a hard truth the tech-utopians skip: as an engine, you're not very efficient.

System	Share of the energy in that comes out as useful work
Human muscle (food chemistry to motion)	about 14% to 27%
Combustion engine (small engines to large diesels)	about 10% to 50%
Combined-cycle gas plant (fuel to electricity)	up to about 63%
Electric motor, over 200 watts	about 70% to 99%

Typical conversion efficiencies, as compiled in Wikipedia, Energy conversion efficiency. Ranges, not specifications: a particular machine can sit anywhere in its band.

These rows have different boundaries. A motor's figure starts with electricity already delivered to it; a power plant's starts with fuel. Neither includes the whole chain of making, powering, and maintaining a robot. Use the table to understand a conversion, then compare complete systems doing the same job.

BY THE NUMBERS: THE MACHINE'S APPETITE

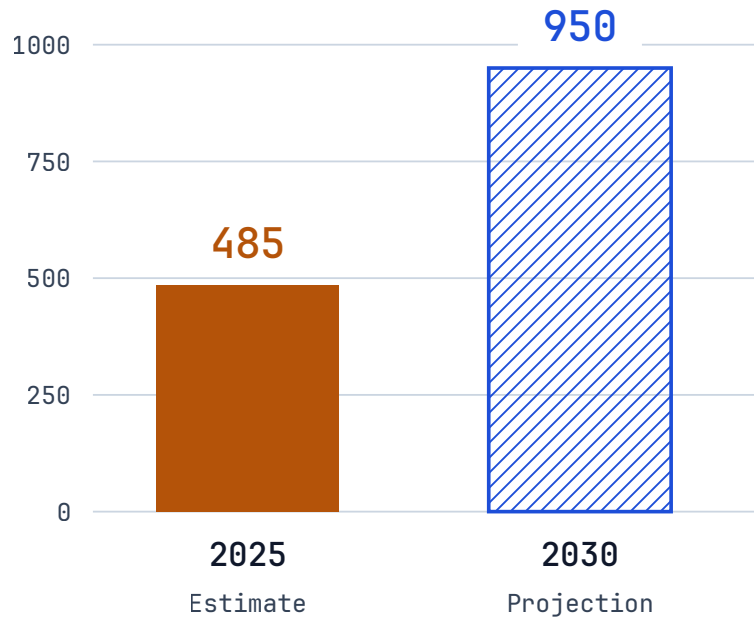
- **485 terawatt-hours.** The IEA's estimate of worldwide data-centre electricity use in 2025. IEA, Key Questions on Energy and AI (2026)
- **950 terawatt-hours.** The same report's projection for 2030, about 3 percent of global electricity demand. It's a forecast, not a meter reading. IEA (2026)
- **More than 280-fold.** The fall in query price for models matching GPT-3.5 on the MMLU benchmark between November 2022 and October 2024. This measures price at a benchmark threshold, not energy consumed per task. Stanford AI Index 2025, chapter 1

Data centres are using more electricity in total while some measured capabilities get cheaper to buy. Both can happen at once. Somebody owns the meter.

THE MACHINE'S APPETITE

Worldwide data-centre electricity demand

Terawatt-hours per year



All data centres, not AI alone.
2030 is a forecast, not a meter reading.
Source: IEA, Key Questions on Energy and AI, 2026.

The machine's appetite. The solid column is IEA's estimate for 2025; the hatched column is its 2030 projection. This is all data-centre electricity, not AI alone. IEA, Key Questions on Energy and AI, 2026.

You burn roughly 2,000 to 2,500 kilocalories a day to keep your heart pumping, your lungs moving, and your brain generating existential dread. That's about 100 watts of continuous power, the draw of an old incandescent bulb, and about twenty of those watts run your brain. To get them, you need proteins, carbohydrates, and fats that somebody had to grow, harvest, move, and cook.

The machines are less picky. They eat raw electricity, from sunlight, wind, uranium, or coal. They don't spend energy repairing tissue or fighting off a cold.

That comparison is vivid, and it's exactly where people get dangerous with it. Calling someone a meat engine can make the physics easy to feel. It must never make their worth depend on how cheaply they compete with a motor. Thermodynamics checks a claim against the physical world. It doesn't tell you what a person deserves or who gets to eat.

A machine still needs a world

A machine can do work without a person making every motion. It can't do it without inputs. A robot that helps with food needs a food source, the right equipment, water and power where the task calls for them, maintenance, and people or systems to handle the parts it can't. Every one of those is metered, and somebody owns the meter.

So keep the ledgers separate. Suppose a shared garden has solar panels, a water pump, and a harvest. Record electricity in kilowatt-hours, water in liters, food in kilograms, labor in hours, and spending in dollars. Don't add them together into one heroic surplus number. A pile of zucchini isn't a year's diet. Power at noon isn't power during a cloudy week. Installed capacity isn't delivered service. If you compare a manual job with a machine, count the setup, the babysitting, the failed runs, and the repairs, or you've just moved the work off the page.

Free to eat, with the costs visible

When I say food should be free at the point of access, I don't mean steel, land, labor, or electricity stop costing money. I mean the person who needs to eat shouldn't have to produce a payment first.

We already know how to fund a service without charging for each use. A library has a budget and charges you nothing at the door. Automation can bring down the cost of some of the work. It doesn't decide

who pays or who gets served. If a project can't say how next month's maintenance gets covered, it has no business advertising a permanent promise. Find the gap while there's still time to fix it.

The cyberdeck: your first machine

Before we go any further into land and grids and agreements, I want to show you what energy agency looks like when it's small enough to fit in a Pelican case. I want to show you what the kids are building. And first, I need to show you a couple of centuries of adults being wrong about what the kids are building.

A brief history of "kids these days"

You already met the novel panic at the end of the Preface. It had company.

In 1816, *The Times* of London condemned an "indecent foreign dance" sweeping the ballrooms and warned parents to keep their daughters away from it. The dance was the **waltz**. The Times (July 1816), as quoted in John Hood, Hayek, Strauss, and the Political Waltz, FEE

In the 1890s, when the safety bicycle let young women go where they pleased without a chaperone, some physicians warned of a condition called **"bicycle face,"** a strained, haggard expression supposedly caused by the effort of balancing. The thing actually on the move wasn't anybody's face. It was women's freedom of movement. McGill Office for Science and Society, The Moral and Medical Panic Over Bicycles

In 1906, John Philip Sousa, the most famous musician in America, published "The Menace of Mechanical Music," warning that the phonograph would wither the human voice: "the vocal cords will be eliminated by a process of evolution." His own band was already one of the best-selling recording acts in the country. Sousa, The Menace of Mechanical Music (1906)

Then radio was rotting children's brains. Then comic books were, and the United States Senate held hearings about it. Then television. Then Dungeons & Dragons was Satanic, then video games were murder simulators, then the internet, then the phone, and now, right on schedule, the machine this book is about.

Some of those worries had a grain of something real in them. None of the panics was how anybody found out which grain. So here's the translation key, and I want you to laminate it: a lot of the time, "kids these days" means *a tool I don't understand is making someone I can't supervise powerful in a way I didn't sign off on.*

Which brings me to the kids, and what they're actually doing.

What a cyberdeck is

Right now, while the professionally worried write op-eds about screen time, a lot of mostly young builders are constructing an alternative computing culture out of parts, spite, and joy. They call the machines **cyberdecks**, a name lifted from the "cyberspace deck" Case jacks into in William Gibson's *Neuromancer* (1984). A cyberdeck is, simply, a personal computer you *build instead of buy*. Not picked off a menu. Designed. Fabricated. Owned the way you can only own something you could rebuild from a pile of its own parts.

There's no factory spec, but a common anatomy has emerged:

- **The brain:** a single-board computer, often a Raspberry Pi. Measure the board under your intended load, then add the screen, radios, storage, and power-conversion losses. Single-digit watts is a target for some builds, not a whole-system guarantee. Raspberry Pi, power and peripherals
- **The face:** a small screen, or e-ink for sunlight and low power while holding a static page. Refreshing the page and running the rest of the computer still take power. E Ink, bistability
- **The hands:** a mechanical keyboard, often a tiny one, sometimes hand-wired key by key.
- **The body:** a Pelican case, an ammo can, a 3D-printed shell, a thrift-store briefcase.

- **The ears:** a cheap software-defined radio stick, and a LoRa radio running Meshtastic, the off-grid mesh messaging network.
- **The library:** offline. A full copy of Wikipedia, manuals, maps, sometimes a small local model, everything the owner might need when the network is gone.

Read that list again, slowly, because you've seen it before. Owned hardware. Understood systems. Offline knowledge. Mesh communication. A power budget in watts. Repair as a value. **A cyberdeck is this whole book at model-railroad scale**, and the people building them didn't wait for permission, funding, or a curriculum. Appendix E has more, including the easier route: the phone already in your pocket.

The Tiffany Doctrine

Here's the part I need the older readers to sit with, because I'm on your side and this is going to sting.

The kids building these machines don't know how things used to be. They don't much care how things used to be. Why would they? "How things used to be" isn't a place they can live. The ladder a lot of their grandparents climbed, the cheap house, the one-income family, the job that couldn't reach you after five, got pulled up behind them. The kids mostly know that world as a rent bill and a lecture. So they're doing what young people have done in every turning before this one: building an alternative with the materials at hand, while the previous tenants explain that the materials are morally inferior.

So: Grandpa. I'm sorry Tiffany wasn't around when they put a man on the moon. But right now she's learning the skills to put *herself* on the moon. She doesn't need a lecture about how kids these days can't adjust a TV antenna. She needs your help and your patience while she builds a goddamn cyberdeck. She's teaching herself the pieces of the kind of machine that might someday fold your laundry and grab you a beer, so she can spend her own hours on something more important: her life. Your job is to help her handle frustration, not to hand her more of it.

And Grandpa, you've got more to offer her than anybody in her Discord server, if you can drop the sermon. You know how to solder. You know why the fuse blows. You know what it feels like to stay with a broken thing until it isn't, which is one of the rarest skills in her world of infinite scroll and one-click everything. She's building the exact machine this chapter has been describing: low-watt, owned, repairable, still working when the network isn't. That's not brain rot. It's the novel, the waltz, the bicycle, and the phonograph all over again, and this time you get to choose which side of the letter to the editor you're on.

The pattern holds, generation after generation: the tool the elders mock becomes part of the world the kids build. The novel readers became the reading public. The bicycle girls rode into the suffrage era. The kids wasting their lives on computers built the thing you're mad at now. Tiffany and her cyberdeck aren't the collapse of civilization. They're the *continuity plan*.

Your first machine doesn't need to be pretty. It needs to be yours. Build one with her.

Step one: audit your energy budget

Before you pool anything with anybody, run a cold-eyed audit of your own balance sheet.

1. **What are your inputs?** Where does your food come from, where does your power come from, and what happens to each during a week-long interruption?
2. **Where are your leaks?** Standby loads, subscriptions, fuel, and attention. Keep the units straight: count the first in watts and hours, the subscriptions in dollars, the fuel by quantity, and doomscrolling in time you don't get back.
3. **Who's in your pod?** Which people within walking distance could share tools, meals, rides, and labor with you?

Then pick one task you actually care about, food first if you can, and write a short inventory with the people who already do it: the task, what it needs, who's responsible, what it still depends on from outside, and what happens during an interruption. An irrigation controller won't fix a missing water supply. A better motor won't fix a delivery route that never reaches the person.

You don't need a private microgrid to take part in a food project. You need to know what your part runs on. It's time to stop leaking energy to machines that don't love you.

— ◈ —

Precedent P-07: The Horse and the Ledger (Britain and United States, eighteenth to twentieth centuries)

James Watt had a marketing problem. He needed to sell steam engines to men who owned horses, so he took the engine-sellers' old boast, this machine does the work of so many horses, and standardized it into a unit of account that priced the animal in its own currency: *horsepower*, thirty-three thousand foot-pounds per minute. From the moment that number existed, every horse in the world was walking around with it on its back, whether its owner knew it or not.

The paperwork survives. The Science Museum holds a contract dated March 1, 1786, between Boulton and Watt and a distillery near Whitechapel Road, for a ten-horsepower engine to grind grain and pump liquids. The distillers would put it up and maintain it; the inventors would collect an annual premium for the rest of the patent term, a payment the firm tied to what the customer saved compared with doing the work another way. Science Museum Group, engine agreement of 1786 The machine was physical. The deal was a ledger: somebody had to define the alternative, measure the work, and keep the engine running.

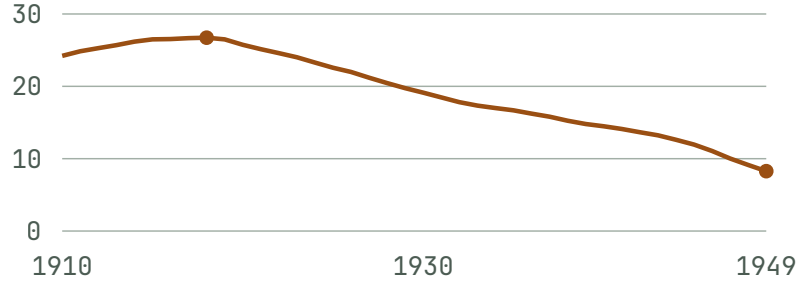
The horse looked untouchable for another century. The horse and mule population on American farms kept climbing right through the railroad age, peaking above twenty-six million in 1918, with a whole economy of hay fields, stables, farriers, harness makers, and teamsters built on its metabolism. Then tractors, trucks, feed prices, farm wages, and a dozen other lines in the farm ledger moved, and within a working lifetime the herd collapsed. USDA, Statistical Bulletin 83, table 14 The horse never got less noble, less strong, or less willing. Sentiment just never appeared as a row in anybody's books.

THE HERD AND THE MACHINE

On US farms, January 1, 1910–1949.

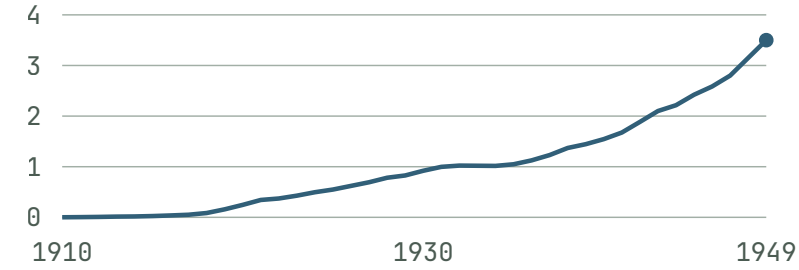
HORSES AND MULES · MILLIONS

1918 peak: 26.7 million



1949: 8.3 million, under a third of the peak.

TRACTORS · MILLIONS



1910: 1,000 → 1949: 3.5 million

Separate vertical scales; 1948–49 preliminary. Source: USDA, Statistical Bulletin 83 (1949), table 14, p. 45. All annual values plotted.

US farm counts on January 1, 1910–1949. Separate vertical scales; 1948 and 1949 figures were preliminary. Source: USDA Statistical Bulletin 83 (1949), table 14, p. 45.

The mechanism. The horse's owners argued about tradition, beauty, and loyalty. The machine's buyers ran operating cost per unit of work, alongside wages, feed, and credit. The arithmetic doesn't negotiate, and it doesn't send a warning before it closes the account.

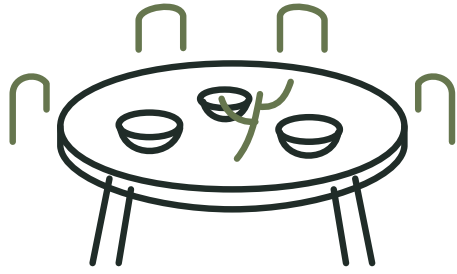
The rule. You're a twenty-watt thinking engine inside a hundred-watt body, in a market that's started selling intelligence by the kilowatt-hour. Measure the task without pricing the person as obsolete, and audit your own ledger before somebody else runs the numbers on you. The horse found out its price after the fact. You still get to set some of the terms.

The practice.

1. Run the energy audit from this chapter on paper: what comes in (calories, kilowatt-hours, dollars), what goes out (labor, attention, rent), and who gets paid. Keep each quantity in its own units. Most people have never once seen their own ledger the way a buyer would. The horse never saw its ledger at all.
2. List the tasks in your job the way a buyer would, then ask a coworker what the list missed. Mark the parts where you're competing on raw output against silicon, and the parts where you aren't: trust, presence, hands, judgment on the ground, relationships. Shift your effort toward the second list this quarter.
3. Move one meaningful expense from rented energy to owned energy this year: insulation, a bicycle that replaces short car trips, a panel and battery sized by someone who knows the job. Watt priced the horse in the horse's own currency. Every watt you own is a watt nobody can price you in.

PART II

How Humans React: The Cooperative Transition



A useful machine should give us something to look forward to. Instead, for a lot of people, it arrives as a threat to dinner and rent. This part follows that fear into the arrangements that produce it, and into a shed where nineteen households are trying something else.

THE CO-OP

Four people in the story



ELIJAH

Former tech worker
Learning to build



MARTA

Fabricator
Runs the co-op shop



PRIYA

Soil scientist



DENNY

Former logistics worker
Co-op media operator

The people the rest of this book is about. Nineteen households; these four you get to know.

CHAPTER 6

Is the Singularity a Done Deal?



Tools become useful through people who can operate, maintain, and share them. Original illustration for this book.

> *"When a distinguished but elderly scientist states that something is possible, he is almost certainly right. When he states that something is impossible, he is very probably wrong."*
> Arthur C. Clarke, *Profiles of the Future* (1962)

In this chapter:

- Capability, deployment, and access move at different speeds. The fight is in the gaps.
- The bottleneck to a livable transition isn't only technical. It's exhaustion, status, and a very reasonable fear of losing the ability to survive.
- You don't have to take anyone's wealth to end that fear. Use the robots to grow the food and build the shelter, and let people keep what they have.
- Some people have already survived an automation wave. Find them. They aren't on LinkedIn.

Elijah quit on a Tuesday, eleven months after the toast, and the resignation meeting took four minutes because nobody argued.

There had been a moment, drafting the letter, when he'd imagined saying the whole thing out loud: the realtors, the ingestion pipeline with his name in the commits, the enclosure, the arithmetic in the notebook. But you learn who a speech is for by imagining giving it, and this one was for him. So he wrote two sentences, and Claypot's system auto-scheduled his exit interview with an AI that thanked him for his candor, and the badge went dead in his hand at the turnstile with a small red light, and that was his career in the new economy: on-boarded by a person, off-boarded by a light.

He drove north because north was the only direction that had ever meant anything. Past Santa Rosa the road opened up and his shoulders came down off his ears by degrees, the way they only did in country where things were still made of wood and weather. He had no plan. He wants that on the record too. He had savings, a dead grandmother's county, a notebook full of arithmetic, and a manuscript nobody had asked for, and somewhere past Cloverdale the check-engine light came on: the *old* economy asserting itself, combustion and gaskets, problems you could hold.

The shop he pulled into wasn't a mechanic's. He realized it too late, halfway across the gravel: an old apple-packing shed off the county road, roll-up door open to the heat, and inside, not lifts and tires but *fabrication*: a CNC plasma table mid-cut, throwing orange sparks in a fan, welding curtains, steel racked by size along one wall like library books, and a wall of hand tools shadow-boarded so precisely it looked like an evidence display. A woman in a leather apron stood at the plasma table with her back to him, watching the cut the way you watch a pot that's nearly boiling. Fifty-something. Forearms like dock rope.



The tool wall. Shadow-boarded like an evidence display.

“Engine light?” she said, without turning around. “Mechanic’s two miles up, left at the grange hall. This isn’t that.”

“What is it?” Elijah asked, because the question got out ahead of his manners.

She turned around then and looked at him, one long, level pass, boots to haircut, the look of someone who has hired and fired and buried, and something in the shop’s whole arrangement must have loosened her, because instead of pointing at the road she said, “Co-op. Nineteen households. We make what we can’t afford to buy and fix what the market says to throw away.” She nodded at the plasma table. “Right now: brackets for a neighbor’s solar rack. You need something made?”

And Elijah, God help him, alone with the arithmetic for eleven months and now facing a stranger standing inside the exact answer to it, Elijah went off. Again. The way he had at the Bear Flag.

He told her what was coming, machine-gun style: the models, the curve, the end of labor. And then, because years at Claypot had installed a whole grammar in him he didn’t know he was speaking, he started *pitching*. He heard himself say the shop could *scale*. He said *platform*. He said there ought to be an app: coordination software, a network of co-ops, a marketplace layer; he could build it in a month, this thing could be in fifty towns by spring, they could get ahead of

the wave, first-mover, and somewhere in there he ran out of gas, because Marta Okonkwo had crossed her arms and was watching him with the exact expression she'd given the plasma cut: waiting to see if the machine would finish or fault.

"You done?" she said.

He was done.

"Okay. One," she said, holding up a gloved finger, "nobody here needs an app. Maria-next-door needs her rack to survive a snow load. That's the bracket. Two: *scale*." She said the word like a fish she was returning. "Everything that ever screwed the people I know got big first. The plant I gave fifteen years to *scaled*. Scaled itself right to a labor market that didn't include us. You know what doesn't scale? Nineteen households who all know whose kid is whose. That's not a limitation, college. That's the load-bearing wall."

"I'm not," he started, and she talked straight through him, not unkindly, the way you talk through a gate that's still swinging.

"Three. You drove up here to tell me a machine is coming for everybody's job." She pulled her glove off, finger by finger. "In 2009 I watched a company do to nine hundred of us what you think your machine is going to do to everybody, and there was no machine, just a spreadsheet and a tax break. So I know something you don't, which is what it's like *after*. And you know something I don't, which is apparently the schedule." She jerked her chin toward the fan of sparks. "That could be worth something. Not an app."

The silence had the plasma table in it, ticking as it cooled, exactly like an engine after a long haul, exactly like a tower in a dark apartment. Elijah stood in it and felt the pitch die and something older stand up in its place.

"I grew up on a farm," he said. "I can't weld. But I can fix a pump, and I can read a manual, and I'm not in a hurry."

Marta looked at him for another second. Then she picked up a push broom leaning by the curtain and held it out.

"Swarf goes in the red bin," she said. "We'll see."



THE FOUNDATIONS

The part we get to decide

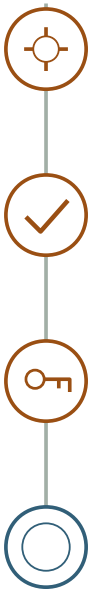
So is it a done deal? My answer, for the record, is yes. I think we're already inside the Singularity, and my evidence is sitting on a desk: a model that does real intellectual work, running on consumer hardware, with no data center behind it. That's not the same claim as my AGI date. By this book's definition the Singularity begins when intelligence stops being scarce, and that's under way; a single system that can do what we can is the thing I'm forecasting for Thanksgiving 2027. You don't have to agree with me on either. This chapter works either way, because the part that matters, what happens next, isn't done at all.

Elijah arrives with a forecast. Marta has a workshop, a history, and a bill for the welding gas. They need things from each other, but first he has to stop confusing his prediction with her future.

It's an easy mistake. Show someone a machine doing something remarkable and the conversation leaps straight to what society must become. But capability, deployment, ownership, and access are different events. A tool can get better while its price goes up. A company can increase output while cutting payroll. A town can contain a productive farm and people who can't afford its food.

THE GAP IS WHERE WE DECIDE

Four questions. Each needs its own answer.



CAPABILITY

Can a machine
do the task?

DEPLOYMENT

Is it working
somewhere real?

OWNERSHIP

Who controls it
and sets the terms?

ACCESS

Can a person get the result
without money?

A productive farm can stand beside hunger.
The terms of access are a decision.

Capability, deployment, ownership and access are separate questions. A productive farm can stand beside hunger.

That gap between what's possible and what reaches people is the part we get to decide. As machines take on more necessary work, use the gains to make necessities dependable. Start with food. Make a lost paycheck less able to turn an ordinary evening into an emergency.

That choice doesn't depend on my date. If progress is slower than I expect, people still need to eat. If only some jobs disappear, the people who lose them still count. The International Labour Organiza-

tion's 2025 analysis estimates which occupations are *exposed* to generative AI, not how many jobs are already gone, and concludes that for most exposed jobs transformation is more likely than outright replacement. ILO, Generative AI and Jobs: A 2025 Update Good. That's a reason to be careful about predicting universal unemployment. It's not a reason to make the unlucky worker's dinner depend on the average outcome.

Two kinds of fear

So why do so many people resist a future where the machines do more of the work? From years of watching people react to this, two groups stand out, and neither one is stupid.

The overwhelmed. Most people are so worn down by the daily work of staying afloat that they have no bandwidth left to process a change this big, let alone welcome it. When you're drowning, even a rescue boat can look like a threat. That's not a character flaw. It's what exhaustion does.

The status-invested. Others aren't being crushed by the current arrangement; they're doing fine inside it. Their fear is different: losing their relative position, losing control, and the vertigo of having to redefine themselves outside the job title that's been doing the defining.

Both fears contain information. A worker afraid that the new system will take her dinner is reading the situation correctly under the current rules. The answer isn't to tell her to think positive. It's to change the rule that makes the fear accurate.

The preservation solution

Here's the move that makes the fear unnecessary.

We don't need to take anyone's money. We don't need a hostile redistribution of wealth, and we don't need to win every argument about economics first. We need to use the robots to grow the food and, in time, build the houses, and let people keep what they already have.

Preserve what people have while adding dependable abundance underneath them, and you take away the terror that keeps everybody in a defensive crouch: the fear of losing the ability to survive.

That's food first, stated as plainly as I can state it. It doesn't threaten the status-invested, because nobody's taking their house. It gives the overwhelmed somewhere to stand, because dinner no longer depends on this month going well.

Whose robots?

Use the robots to grow the food. Fine. Whose robots?

That's a reasonable objection. People with the money to own the servers, the models, and the land can decide whose needs their machines serve. When I talk about oligarchs, I mean that concentration of wealth and decision-making power. A machine can get better while its owner raises the price, restricts its use, or withdraws a service your town has come to depend on.

The machines aren't all locked away. Useful models can run on consumer hardware. But a local computer can still depend on somebody else's license, replacement parts, or online service. Ownership has to reach the terms you depend on.

Norbert Wiener saw the problem in 1949. Writing to autoworkers' union president Walter Reuther on August 13, he explained that he'd refused an industrial consultancy because of the unemployment automatic factories could cause. Then he wrote: "it is manifestly not enough to take a negative attitude on this." He proposed that organized labor take part in producing the machines and direct their profits toward labor's benefit. Wiener, letter to Walter Reuther

Elinor Ostrom supplied evidence for another piece of the argument. Her studies of shared fisheries, forests, and other resources found that users could develop rules and enforce them themselves. The 2009 economics prize recognized that work. Those findings give us a

useful starting point for shared machines: examine who makes the rules, who enforces them, and how people settle disputes. Royal Swedish Academy of Sciences, 2009 economics prize

That's the point of Rung 5 in Chapter 19: own the necessary machines together, and make the right to their output dependable. Ask who sets the price, who can change the rules, and where a person can appeal when dinner doesn't arrive. Put those answers in the agreement. The same capability can serve different arrangements. We have work to do on both.

The ladder with the bottom rungs sawed off

For people under twenty-five, the old deal, sell your labor and buy your survival, has failed on both ends at once.

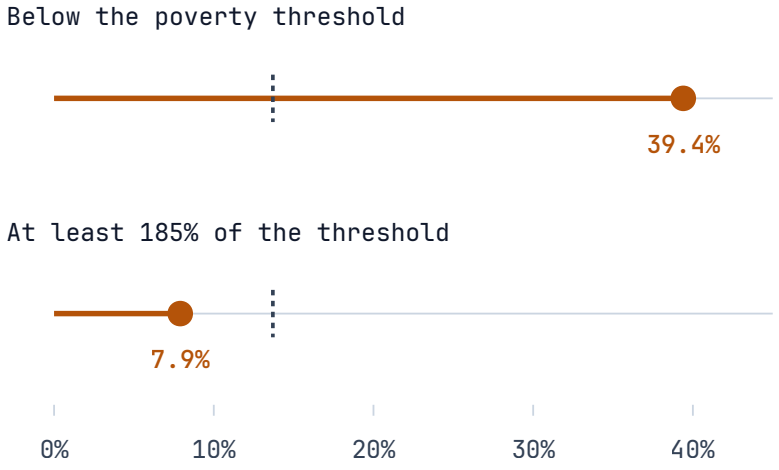
On the buy side, housing shows how much harder securing a foothold has become. In 1970, by the Census Bureau's numbers, the median home was worth $17,000 and the median household took in $8,734: a multiple of about two. Census, Historical Census of Housing Tables: Home Values Census, Historical Income Tables: Households, Table H-5 In 2024 the median existing single-family home sold for $412,500, which Harvard's Joint Center for Housing Studies calls "a shocking five times the median household income." Harvard JCHS, State of the Nation's Housing 2025, press release The measures differ, a value then and a sale price now, but nobody reading them doubts the direction.

On the sell side, the entry-level cognitive work a twenty-two-year-old once used to climb, the junior analyst seat, the first coding job, the paralegal desk, is exactly the kind of task these systems are getting good at first. Costs up. Bottom rungs going. A cohort can't out-hustle that arithmetic, and it isn't a failure of their ambition.

INCOME AND FOOD ACCESS

Share of US households food insecure · 2024

----- National rate: 13.7%



Selected income groups; survey estimates.
Association does not establish causation.
Source: USDA ERS, ERR-358, Table 2.

Income and food access. In the 2024 survey, food insecurity was associated with income. These two selected groups do not represent every income category, and the comparison does not establish causation. USDA ERS, ERR-358, Table 2.

BY THE NUMBERS: WORK, HOME, DINNER

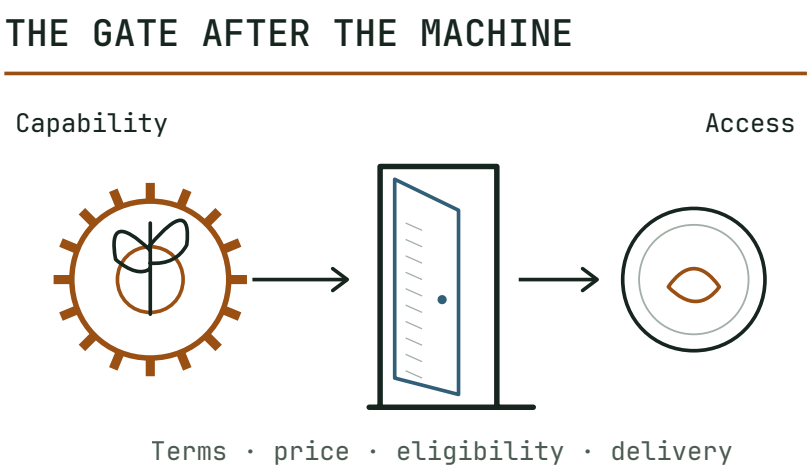
- **1 in 4** workers worldwide is in a job with some exposure to generative AI. The ILO expects most of those jobs to be transformed rather than eliminated. ILO, Generative AI and Jobs: A 2025 Update
- **About 2 times versus 5 times.** Median home value divided by median household income in 1970, versus median sale price divided by income in 2024. These are different measures, as the paragraph above explains.
- **47.9 million** people in the United States lived in food-insecure households in 2024, including 7.3 million children in households where children as well as adults were food insecure. That category includes reduced dietary quality; it doesn't mean every child ate less. USDA ERS

Security is something you can rely on

A promise from a generous founder is welcome. A service that keeps running when the founder gets sick is stronger. A right you can still claim when an administrator says no is stronger still.

A cooperative can own equipment. A public body can fund a food service. An existing grower can be paid to supply it. Residents can have a say in what's provided and how failures get fixed. None of those words makes anything fair on its own. Together they describe work we already know how to start: assigning resources, responsibilities, and a usable claim on the result.

The point isn't to make everyone dependent on one kindly operator. People should be able to criticize the service without losing it, and have somewhere else to turn when it fails. Ownership and oversight matter because the machine that feeds a neighborhood could also give whoever controls it enormous power over that neighborhood. Calling it public doesn't solve that. Calling it local doesn't either.



A working machine still needs an arrangement that lets people reach its output. Conceptual illustration.

A workshop is a beginning

Elijah can drive north because he has savings, a car, and skills a workshop can use. Someone caring for a parent in a fourth-floor apartment may have none of those. A serious proposal has to reach that apartment too.

Marta's nineteen households can teach us how people cooperate, repair things, and notice each other's needs. They can't stand in for every city, every climate, or every person who doesn't know somebody at the table. Their work gets more useful the more it connects with growers, kitchens, unions, and institutions beyond the shed.

So find one connection this week. Ask an existing food provider where a meal gets held up: money, transport, equipment, staffing, a confusing application, nowhere to cook. The answer names a job for the tools and a job for the people organizing them.

Precedent P-08: The Grain Trap (Fertile Crescent and Anatolia, c. 9500 to 6000 BC)

The first singularity was a seed.

When people in the Fertile Crescent and beyond settled down around wild grasses, the deal for many individuals got measurably *worse*. In a lot of places the skeletons tell the story: early farmers were often shorter than the foragers before them, with worse teeth, new diseases caught from crowding and livestock, and a workday a forager would have found strange. Not everywhere, and not all at once. But often enough that archaeologists have spent decades documenting the cost.

Go to one place and look. Çatalhöyük, on the Anatolian plateau, was a town of houses built back to back with no streets between them, entered through the roofs, with paintings and reliefs on the walls inside. UNESCO, Neolithic Site of Çatalhöyük A 2019 study led by Clark Spencer Larsen read roughly twelve centuries of its people in their bones: teeth marked by a cereal-heavy diet, leg bones that suggest more travel and harder physical work over time. The town endured while the work of feeding it got harder. The same study found some infection markers falling and growth staying relatively normal. The evidence points more than one way, which is what real evidence usually does. Larsen and colleagues, Bioarchaeology of Neolithic Çatalhöyük

And none of it stopped the spread. Grain could be stored, counted, taxed, and multiplied. Farming villages out-bred and out-organized foraging bands, and over a few thousand years there were fewer and fewer places left to not-farm. No one voted. The people inside the transition mostly couldn't see it, and their grandchildren couldn't leave it, because the skills, the land, and the population had already crossed over.

The mechanism. A technology doesn't need to make each person better off to become hard to refuse. It only needs to make the groups that adopt it outcompete the groups that don't. That ratchet turns on group advantage, not consent. And a growing harvest tells you nothing, by itself, about whether the people bringing it in are eating well.

The rule. Stop spending your energy on whether the transition happens. Spend it on what position you hold when it does, and count welfare separately from adoption.

The practice.

1. Put a hard cap on "whether" debates. The next time you're in an argument about whether AI should exist or whether it'll really happen, give it five minutes, then ask out loud: "Given that it's arriving anyway, what should we each do this month?" Watch how few people can switch questions. Be one who can.
2. Run the group audit on your own affiliations: employer, industry, town, community. For each, name one change it's making and who carries the cost. If you can't find an answer, write down who you'll ask. The grain trap punished groups for their choices, and individuals for their group.
3. Keep one part of your life off the ratchet on purpose: food you can grow, heat you can make, skills that work without a subscription. Not to exit the system. Because the person with an alternative negotiates from a different posture than the person without one.

CHAPTER 7

The Battle Lines: The Entitled vs. The Ready



Opening position. The battle lines are drawn before most players notice the game started. (Bin im Garten, CC BY-SA 3.0, via Wikimedia Commons)

> *"Chance favors only the prepared mind."*
> Louis Pasteur, as commonly rendered in English

In this chapter:

- The fight over the transition isn't humans versus machines. It's the people who need the suffering to stop versus the people who need the hierarchy to hold, and most of us carry a little of both.
- "Reskilling" is what the comfortable prescribe to the displaced. Ask someone who's done it twice.
- The Luddites were skilled workers with a real grievance about wages and terms. The fight was never really about the machine. It was about who sets the terms.
- The cook can lose the job and still eat. That's the floor this chapter is arguing for.

The co-op ate together on the first Friday of every month, long tables of mismatched plywood in the packing shed with the plasma table rolled into the corner and someone's dog working the crowd, and it was at Elijah's second of these dinners, five weeks into his apprenticeship with the broom, that he learned where the battle lines actually ran.

It started, the way most real fights do, over something small. A member named Curtis (grey ponytail, good with engines, owner of the valley's most thoroughly provisioned basement) was reading aloud from his phone about the robotaxi fires down in the city. Crowds putting traffic cones on the hoods to brick them. Windows going in. A depot fence cut open. Curtis read it with a kind of grim satisfaction and delivered his verdict to the table at large: "Can't say they didn't have it coming. But you watch: when it really lets go, it's going to be the people who didn't prepare who tear everything down. That's who you plan for. That's who you fence out."

Elijah, five weeks in and still trying to be useful in conversations the way he was learning to be useful with a grinder, offered what he genuinely believed was the reasonable middle: that the displacement was real but manageable, that what those people needed was support through the transition: retraining, new skills. "Reskilling programs," he said, and heard, too late, the shape of the word in his mouth. It was a Claypot word. It had been on a slide.

The table didn't go quiet the way tables do in movies. It went quiet the way a shop goes quiet when someone hears a bearing start to sing: a couple of forks paused, and Denny Osei, sitting across and one down, wiped his mouth and folded his hands and looked at Elijah with an expression that was not unkind and was worse than unkind. It was *practiced*.

"Reskilling," Denny said. "Okay. Let me tell you about reskilling, because I am, and I want you to appreciate this, I am *certified* in it." He held up two fingers. "Twice. First time, 2019. I'd been eleven years in the warehouse, and the arms came in, and the county had a program. Six months, nights, logistics software. I was good at it. I'm not being modest, I was *good*; I could route forty trucks in my head before

the coffee was done. Got hired doing dispatch and routing. Wore a lanyard. My daughter told her class her dad worked in computers." He turned his water glass a quarter turn. "2023, the routing went autonomous. The software I retrained *into* got a software of its own. And the man from the county (different man, same folder) told me about a program. Six months. Nights." He looked around the table, then back. "You know what I asked him? I asked him: how many times do I have to climb this ladder before you stop moving it. He didn't have a number. Nobody has a number. Because the honest number is: there's no rung the machine won't reach eventually, and every man who tells you 'just climb faster' is standing on a floor, not a ladder."

"Denny," Elijah started.

"I'm not mad at you," Denny said, and meant it, which was the part that stuck. "You said it because everyone in the rooms you come from says it. It's what the comfortable prescribe for the falling. It costs them nothing and it sounds like a plan. But you're at *this* table now, so you get the real arithmetic. Retraining works when the water rises somewhere else. This flood's coming for a lot of us at once, and you know that better than anybody in this room. You've been writing it down in that little notebook you think nobody's noticed." He turned the glass back. "So answer me this. If the next course takes six months, where do my daughter and I stand for those six months? I don't need you to promise me the machine stops. I need an answer that doesn't end with me starting over hungry."

The dog moved under the table. Somebody laughed, once, in the way that lets everyone breathe.

But Curtis wasn't done. Curtis had heard, inside Denny's story, support for the fence. "Which is my point," he said. "When it all comes apart, the mob that torched that depot shows up here. You want them at this table too?"

And that was when Marta, who had been eating through the whole exchange like a woman refueling a machine, set down her fork.

"Curtis. You know who the Luddites were?" She didn't wait. "Because you're using the word wrong in your head right now, I can hear it. Stocking knitters. Croppers. *Skilled* trades, the machine operators of their day. They knew those frames better than the owners did. And when the owners used the machines and the rents on them to cut the wages out from under them, they asked, for years, with petitions, for a floor. A minimum. Decent terms." Her voice stayed level, shop-level, the tone she used for load ratings. "Parliament made breaking a frame a hanging offense and sent twelve thousand soldiers into the counties. The frames weren't the enemy and everybody knew it. The wages were."

"They broke things," Curtis said.

"They did. Some of them threatened people, too. And a fire in a depot can hurt somebody who had nothing to do with a layoff. I'm not defending that." She moved the bowl away from the edge, where the dog had reached an ambitious nose. "I'm saying you don't fence those people out. You build the floor Parliament wouldn't. That's what this table *is*." She picked her fork back up. "Pass the beans."

Denny held out his hand for the bowl. Marta passed it.



Fictional scene: pass the beans. The conversation turns from retraining to the floor people need beneath them.

The beans got passed. The conversation moved, the way conversations at long tables do, into three smaller ones. Elijah didn't say much for the rest of the dinner. But late that night, in the notebook, under a year of arithmetic (stages and watts and orders of magnitude), he started a different kind of page. Names. *Denny: two ladders, both moved. Where does he stand for six months? Curtis: fence-minded but shows up for every work day; the fence is fear. Marta: knows the Luddite hangings cold. Where does a welder learn that?* He looked at the page a long time, because it didn't look like anything he'd ever produced at Claypot, and it was the most useful data he'd gathered in two years.

The following Tuesday, Curtis asked whether he wanted to ride along. The town kitchen had three deliveries the co-op had agreed to cover while its usual driver recovered from surgery. There was a cool box between them, and a folded route sheet Curtis kept taking out of his pocket at red lights even though Elijah offered to hold it.

The last address was a converted garage behind a blue house. Rosa opened the door with her left hand. Her right wrist was in a brace. She'd packed fruit at a warehouse until the injury, she told them, and the return-to-work date on the latest letter was a blank.

Curtis lifted the bag. "Kitchen said you wanted ingredients."

"I said I used to cook." She looked at the potatoes, then at her wrist. "I asked if they had anything I could heat."

The three of them stood there with the bag between them. From somewhere inside, a washing machine reached the uneven part of its spin cycle.

"That's ours," Curtis said at last. "The mistake. We'll call."

He set the bag back in the cool box and phoned the coordinator. The kitchen could exchange it for prepared portions that afternoon. Rosa asked for the less spicy option, if they had it. They did. Curtis wrote that down on the route sheet where she could see him write it.

While he was on the phone, she asked Elijah what the shed charged for membership.

"For the food? Nothing. It's the kitchen's delivery."

"He said co-op. I can't do work days. Not now."

Curtis lowered the phone. "You don't owe us a work day."

She looked at him. "Marta says you're doing something with robots out there. I don't much like the sound of it. I've had enough changing."

Elijah felt an explanation gathering in his throat. Curtis got there first.

"Doesn't change your dinner," he said.

They came back before the kitchen closed. Rosa checked the containers, found the one she'd asked for, and made room in her refrigerator. Curtis waited by the gate with his hands in his coat pockets. He didn't ask her to join anything.

On the drive home, Elijah found the cost column on the route sheet: kitchen program, driver time, mileage. He asked how long the arrangement lasted.

“Through next month. Then they need funding again.”

Curtis turned onto the county road. There was one box left to complete beside Rosa’s address. At the next pullout he stopped, put the sheet against the steering wheel, and marked *delivered*.



THE FOUNDATIONS

Where does the person stand?

Denny asked where he and his daughter would stand for six months of training. Rosa asked for food she could heat with one working hand. Neither question needed an answer about the ultimate limits of machine intelligence. Both needed somebody to decide what they’d provide.

That’s the employment question at the scale of a life. Work holds friendship, identity, pride, and the rhythm of a day. It also supplies most of the money people use to get what they need. A transition can knock out all of those at once. Calling it a productivity gain describes one side of the event.

The battle lines

The fight over this transition isn't humans against machines. It's between two mindsets, and they cut across every income bracket, every politics, and sometimes through the middle of the same person.

	The Entitled	The Ready
What it believes	Society falls apart if people's needs get met without forced labor. Life is a ranking, and the ranking has to hold.	The daily suffering can stop, and should. Automated food, eldercare, and household labor are things to want, not fear.
What it wants	Scarcity preserved, because scarcity protects status and control.	The daily burden of survival lifted, so people and their kids can actually live.
Who tends to feel it	People fighting for position in the old world, and people whose standing depends on others needing them.	People who carry the physical weight of survival now: parents, caregivers, disabled people, the elderly, the underpaid.

That table isn't a list of good people and bad people. It's a list of reflexes. A mother of three can reach for the hierarchy when her own job is the one threatened. A comfortable executive can want the suffering to stop. The point of naming the two is so you can catch yourself.

The employee as a liability

The pressure behind all this is plain. To a business under competition, payroll is a recurring cost, and replacing paid tasks with a cheaper machine can look attractive on the ledger. The incentive is built into the structure. That's not a conspiracy; it's arithmetic.

But in a society where survival is tied to employment, automating the work without changing anything else strips people of any say in whether they eat. Nobody has to be a villain for that to happen. It only takes all of us letting it.

What a job actually does for you

If the paycheck were the only thing a job provided, this would be a much simpler book. It isn't.

In the 1930s, the social psychologist Marie Jahoda studied an Austrian town where the main factory had closed, and she spent decades afterward refining what she found. Beyond the obvious function of a job, earning money, she argued employment supplies five hidden ones: Paul and Batinic, The need for work: Jahoda's latent functions of employment (2010)

Latent Function	What it provides	What losing it looks like
Time structure	A rhythm and routine to the waking day.	Disorientation, loss of routine, apathy.
Social contact	People beyond your own household.	Isolation.
Collective purpose	A part in something bigger than yourself.	Feeling useless.
Status and identity	A place in the world and an answer to "what do you do?"	Eroding self-worth.
Enforced activity	A reason to engage with the world every day.	Lethargy, lost skills, depression.

Researchers built a questionnaire around her model, the Latent and Manifest Benefits scale, and have used it across a body of studies involving thousands of people. Separate large surveys have found that these hidden functions account for a big share of the link between unemployment and poor mental health. The lesson isn't that jobs are

sacred. It's that a food baseline alone won't replace everything work was quietly doing, and a good transition plans for rhythm, company, and purpose too.

John Maynard Keynes saw a version of this coming. In his 1930 essay "Economic Possibilities for our Grandchildren," he predicted that within a century technical progress could solve what he called the economic problem, and he worried out loud about what people would do with themselves once the pressing cares of subsistence were lifted, having been trained for so long to strive rather than enjoy. The question he asked was how to live "wisely and agreeably and well." Keynes, Economic Possibilities for our Grandchildren (1930) It's a much better problem than hunger. It's still a problem.

And there's a reason people in the middle of the squeeze can't think about any of this. The economist Sendhil Mullainathan and the psychologist Eldar Shafir describe how scarcity can capture attention and narrow it: tunnel vision. In a 2013 study with Anandi Mani and Jiaying Zhao, sugarcane farmers in India scored better on cognitive tests after the harvest, when money was in hand, than before it, when money was tight. Mani and colleagues, Poverty Impedes Cognitive Function Same people. Different financial circumstances. The finding concerns those tests and conditions, not a fixed loss of ability in every person short of money. Someone worried about next week's rent isn't failing to imagine a post-scarcity future. Their mind is fully booked.

The people already living with it

Precarious work didn't start when a chatbot learned to write a résumé. A food-first argument has to include the farmworker picking the crop, the packer handling it, and the driver finishing the route. It has to include the garment worker sewing clothes for people whose anxiety about technology has only recently gotten personal. Better Work Jordan's 2024 report, covering 2023, documents a garment factory with delayed wages and excessive overtime, and says plainly that the case isn't typical of the whole sector. ILO and IFC, Better Work

Jordan Annual Report 2024 The point is narrow and sharp: people can be making goods for the world while lacking reliable protection in their own lives.

Their interests aren't interchangeable with yours or with each other's. A worker may want help with an exhausting task and fear losing the wages that come with it. That isn't confusion. It's exactly the problem we're proposing to change. The people doing the work belong in the decisions about changing it.

The backlash is already in the street

Stage 5 isn't a theory. In San Francisco, a group called Safe Street Rebel ran a sustained campaign against driverless cars from Waymo and Cruise. Their tactic was "coning": setting an ordinary traffic cone on the hood of a robotaxi, which the car's safety systems read as an obstruction, stranding it in the street until a human came to clear it.

What protesters said they objected to	What happened on the street
Surveillance: 360-degree cameras and corporate data sharing with police.	**A dog killed:** May 21, 2023, a Waymo in autonomous mode struck and killed a small dog in San Francisco.
Transit priorities: public attention and money going to private robotaxis instead of mass transit.	**Emergency interference:** the San Francisco Fire Department logged dozens of incidents of driverless cars blocking or disrupting emergency response.
Their own rules: target only empty vehicles, stay away from intersections and bus stops.	**Destruction:** February 10, 2024, a crowd in Chinatown surrounded a Waymo, smashed its windows, and set it on fire.
Jobs: anxiety about drivers losing work and people losing control of their streets.	**Federal scrutiny:** US regulators opened investigations into robotaxi safety, including around school buses.

Sources: the dog, TechCrunch (Waymo called the collision unavoidable); the fire department's reports, San Francisco Standard; the Chinatown fire, San Francisco Standard; the school-bus inquiry, NHTSA's November 24, 2025 letter to Waymo; coning, The Guardian in Appendix B.

The activists framed it as pedestrian safety and privacy, and some of those concerns were concrete. Underneath ran something older. As Brian Merchant argues in *Blood in the Machine*, the original Luddites weren't ignorant of technology; they understood exactly what the new machines would do to their pay and their standing. The cone on the hood belongs to that lineage.

The campaign has a letterhead

A cone on a robotaxi is the hammer, and the hammer is the half of the lineage everyone recognizes. The other half is quieter, better dressed, and far better funded: the organized campaign, run by a trade body with a budget and a lawyer, against a machine that has just made its members' skills reproducible.

It's been run at least six times in the last hundred and twenty years, by serious people with real grievances, and the script barely changes. Name the defect (the machine has no soul). Predict the death of the art. Buy the ads. Demand a legal remedy. Lose the argument about whether the machine exists, which was never the winnable one.

Year	Who organized it	What they called the machine	What they demanded	How it went
1906	John Philip Sousa, in *Appleton's Magazine*	"mechanical music," a "menace"	copyright control over recordings	The 1909 Copyright Act gave composers a mechanical royalty. His band kept recording.

Year	Who organized it	What they called the machine	What they demanded	How it went
1929–1931	American Federation of Musicians, through its Music Defense League	“canned music,” “the Robot”	that theaters keep live orchestras	A national newspaper ad campaign. The orchestras went anyway.
1942–1948	AFM under James C. Petrillo	the same	royalties on records, paid to a musicians’ fund	Won the fund. Congress answered with the Lea Act, nicknamed the Anti-Petrillo Act.
1981	British Phonographic Industry	“Home taping is killing music”	a levy on blank cassettes	Never enacted in Britain. The industry survived the cassette decade.
1982	MPAA, Jack Valenti, before a House subcommittee	“the Boston strangler”	the VCR banned or taxed	Lost 5 to 4 in *Sony v. Universal* (1984). Within a few years home video was out-earning the box office.
1999–2001	RIAA	piracy	Napster shut down	Napster shut down. Streaming became the industry’s revenue base within fifteen years.

Sources: Sousa's essay and the AFM ads, Appendix B; the 1942 recording ban, University of Maryland; the VCR case, Sony Corp. of America v. Universal City Studios, 464 U.S. 417 (1984); home video passing the box office by 1986, Media and Culture; streaming's first majority of US music revenue in 2016, RIAA 2016 Year-End Report.

Three campaigns you might expect on that list are missing, for the same reason: this book only carries cases that survive a primary source. The printing-press story documented here concerns one eloquent abbot, not evidence of an organized campaign, and even he's more complicated than the legend (his file is at the end of Chapter 10). The famous Western Union memo calling the telephone "a toy" has never been produced by anyone. And the Bavarian medical decree warning that fast trains would give passengers brain disease traces to no archival record anyone has cited. Real history is a strong enough argument. Appendix D has the rest of the blacklist.

What the surviving cases share is the part nobody quotes: again and again, the medium the campaign was formed to stop became, within a couple of decades, one of the main ways the people it was defending got paid.

The egalitarian pivot

Here's the move the battle-line framing hides: readiness has to reach beyond your own household.

"The Entitled vs. the Ready" reads, at first pass, like a sorting of winners from losers. Read it that way and you'll prepare the wrong thing. A lone household with a generator and a rifle isn't ready. It's just losing more slowly, with better lighting. Readiness that ends at your property line has a shelf life measured in weeks, because every system you actually depend on, food, repair, medicine, security, sanity, is social before it's technical.

The pivot is this: when the machine makes intelligence cheap, it quietly bankrupts a lot of the vertical economy of status that organized your working life: the credentials, the titles, the salary ladder, the whole apparatus of being ranked. You can't out-credential a model.

And when nobody can, the premium moves to the one asset the machine can't mint: being known, trusted, and vouched for by specific people who can find your house. Trust of that kind takes shared work and time. That isn't a warm sentiment. It's the coldest economics in this book.

Egalitarian isn't the moral garnish on this strategy; it's the load-bearing wall. The scarcity psychology you just read about is one reason to stop making a household absorb every shock alone. Nineteen households pooling tools, food, childcare, and repair skills may be able to carry a burden that would flatten one of them. A regional disaster can hit all nineteen, so they still need help beyond the circle. The harvest study gives us a reason to take financial pressure seriously; it didn't test a food baseline or a cooperative. A dependable dinner removes one demand on attention. That's already worth building.

So the pivot, concretely: stop climbing and start weaving. Pool capital goods, the truck, the mill, the servers, the greenhouse, because a share of ten machines beats a loan on one. Keep circles small enough that everyone's name is known. Spread out the deciding, because a network with one indispensable node is just a boss with extra steps. And do it *before* you need it, because trust compounds like capital and can't be printed in an emergency.

One warning, and it matters. Small circles are also where exclusion lives. Knowing everyone's name makes it easy to decide an outsider doesn't count. The pivot is toward a table that keeps adding chairs, not a fence with a nicer gate.

The cook still eats

The fired line cook may want another kitchen job, a different trade, or time to care for a parent. A food baseline doesn't choose that life for him. It makes one part of the choice less desperate. Rent, healthcare, and debt can still box him in. Food alone isn't freedom. It removes one threat, which is why food is where this argument starts and not where it stops. Chapter 11 takes up what this means for restaurants and paid work.

Someone still has to tend the crop, fix the equipment, cook the meal, and show up when a delivery fails. The answer to that work is decent pay and conditions, good tools, training where it's needed, and enough people to share the load. A liberation plan that runs on an exhausted kitchen crew has lost the plot.

Rosa's meal makes the point small enough to see. She didn't owe Curtis agreement, labor, or gratitude in exchange for dinner. Curtis's driving still took time. The kitchen still needed funding. One person's unconditional access and another person's working conditions have to be planned together.

A voice before the decision

Nobody should first meet a new system as an instruction to train their replacement. Ask about the change while the plan can still change. What happens to hours, pay, safety, and discretion? Who gets the time the machine saves? What support is already funded if a job disappears? A meeting isn't enough just because everyone got invited. The people carrying the cost need a way to change the terms, and a way to hold someone to them afterward.

People can welcome food security and still object to surveillance or a dangerous deployment. Keep those questions open. My argument doesn't need every complaint about AI to be a complaint about economics. It needs us to hear the economic complaint when a frightened person is making one.

For the next conversation, ask Denny's question: where does the person stand while the transition happens? Keep asking until the answer names something they can actually use.

Precedent P-09: The Frame-Breakers (England, 1811 to 1816)

The Luddites earned the full case file, because so much of what they feared came true and so much of what they did failed, and both halves of that sentence matter.

The framework knitters and croppers of the English Midlands and North were skilled workers, and they understood the new arrangements better than anyone. In Nottingham the disputes were about piece rates, the rent charged for knitting frames, and cheap, degraded goods made by cheaper labor. Negotiations in December 1811 failed to restore the old terms, in a winter of high food prices and bad trade. Binfield, Writings of the Luddites, historical introduction Crews often smashed the frames of masters who cut wages and spared the machines of masters who kept fair terms. This wasn't ignorance of technology. It was a labor negotiation conducted with hammers, because no other table would have them.

It wasn't only hammers, either. The same movement sent petitions, held public protests, and appealed to employers and government. And some of it was darker: a proclamation dated 23 December 1811, in the name of the invented "King Ludd" and preserved in Britain's National Archives, threatens death to informers and constables. National Archives, proclamation of Ned Ludd Hold both facts at once: real grievances, and some methods that terrorized other people.

Parliament's reply was the Frame Breaking Act, which made machine-wrecking a capital crime, and twelve thousand soldiers sent into the disturbed counties. Seventeen men were hanged at York in January 1813. Capital Punishment UK, York Castle Prisons The frames won anyway. The wages fell anyway. The Luddites read the future of their trade far better than their masters did, got the strategy tragically wrong, and paid for the difference at the gallows.

The mechanism. Rage against the machine aims at the one part of the system that can't hear you. The owners heard perfectly well, and answered with the army, because the battle line was never human versus machine. It was, and still is, a fight over who sets the terms.

The rule. Take the livelihood claim seriously before you judge the tactic. Then fight for terms, not against physics.

The practice.

1. Install a rage-conversion rule. When the anger comes, and it's legitimate, honor it with a trade: every hour spent furious at the machine buys one hour building your position. Keep the ledger in your notebook.
2. Negotiate while you still have leverage. This month, write down what you know that isn't written down anywhere: the judgment calls, the failure modes, the customer knowledge. That document is your severance leverage, your consulting practice, or the training data for a tool you own. Unwritten, it's nothing.
3. Sort the movements before you join one. Any group organizing around the transition gets one question: "What are we building, and what terms are we demanding?" If the honest answer is only "we're expressing opposition," you know that record. Join the ones negotiating transition funds, ownership stakes, and local capacity. Same anger. Different batting average.



Precedent P-10: The Robot in the Orchestra Pit (United States, 1929 to 1948)

The croppers had no negotiating table. The musicians had every table there was, used all of them, and left behind the most useful failure in this book.

Start with the warning shot. In 1906 John Philip Sousa, the most famous bandleader alive, published "The Menace of Mechanical Music" in *Appleton's Magazine*, arguing that recording would kill amateur music-making and empty the parlors. Sousa (1906), facsimile He wasn't a crank. His band was already one of the most recorded acts in the country, and his essay was timed to a copyright fight he mostly

won. The band kept recording for decades, usually under other conductors; Sousa himself rarely entered the recording studio. Library of Congress, Sousa's Band in the National Jukebox

Then came the machine that actually took the jobs. *The Jazz Singer* opened in 1927, and within a few years synchronized sound was emptying the orchestra pits of American movie theaters, where some 22,000 musicians, a fifth of their union's membership, had played under silent films. Local 802 AFM, Musicians vs. the Talkies This wasn't a forecast. It was a payroll, and it was ending.

So the American Federation of Musicians did what the Luddites never could. It ran a campaign.


TAKING LIBERTIES WITH FOLKS' HABITS

Is the Robot such good company he can "hog the whole show?"

THE AMERICAN FEDERATION OF MUSICIANS

Exhibitors Herald-World, November 16, 1929. The union asks whether the Robot is good enough company to hog the whole show.

Starting in the autumn of 1929, the union bought advertisements in newspapers across the United States and Canada, and by 1930 had organized the effort into an enrollment drive with a name built for the age: the Music Defense League. American Federation of Musicians, Exhibitors Herald-World, 1930 (archive.org) And it chose, as the face of the enemy, a robot.

Not a metaphor. A drawn mechanical man, week after week: shoving a harpist aside, hoisting a banner reading CANNED MUSIC IN THEATERS over a gathering of muses, failing to soothe a baby. The copy is worth reading closely, because you've read it recently. "Dehumaniz-

ing the theatre may prove to be an exceedingly dangerous move." "Tho' the Robot can make no music of himself, he can and does arrest the efforts of those who can." And a coupon at the bottom, which readers could clip and mail in to be enrolled "as one who is opposed to the elimination of Living Music from the Theatre."



Exhibitors Herald-World, May 17, 1930. The union claimed two million enrollments by that spring. Note the coupon.

By the union's own count, two million people signed. Take the union's numbers as the union's numbers. Even discounted, it was one of the largest organized public campaigns ever mounted against a technology, run by skilled professionals defending a real and disappearing wage.

It didn't save the orchestra pits.

So the union escalated. On August 1, 1942, under its president James Caesar Petrillo, it stopped its members from making new commercial recordings. The record companies had stockpiled masters and dug into their back catalogs, so the recorded work kept earning while the people needed to make more held out. Decca and Capitol settled in 1943; RCA Victor and Columbia held out until November 1944. University of Maryland Special Collections, The Recording Ban of 1942 Petrillo did it again in 1948. Congress got angry enough in between to pass the Lea Act, a statute aimed so squarely at one man that everyone called it the Anti-Petrillo Act.

And then the one durable win of the whole twenty-year fight: an invoice. The Music Performance Trust Fund, established in 1948 and financed by payments from the recording companies, administered by a trustee independent of both the union and the industry, pays musicians to play free public concerts. It's still running. Music Performance Trust Fund, About A performance free to its audience, with the musicians paid. Read that sentence again with a meal in place of the concert.

Here's the sting. Recorded music, the canned music, the Robot, is now simply how most musicians reach anyone. The thing the campaign existed to prevent became the thing its beneficiaries live on, and the institution that still pays some of them is funded by it.

The mechanism. The union did everything right by the standards of legitimate opposition: organized, funded, national, lawful, sympathetic. It didn't preserve the orchestra pits. The public was enjoying recordings, and a campaign for live music couldn't by itself preserve the old payroll. The only move that produced something lasting wasn't a defense of the old job. It was a change in the terms: who pays whom, for what.

The rule. Argue about the machine's soul and you can spend twenty years getting nothing. Argue about the machine's bill and you can build an institution that outlives you.

The practice.

1. Read your own industry's robot ad this week. Find the most recent public statement your union, guild, or trade body has made about AI, and mark it in two colors: sentences arguing the machine is soulless or dangerous to the art, and sentences naming a term (a rate, a royalty, a credit, a consent requirement). Count both. The ratio is a forecast.
2. Draft the one clause you'd actually want, in a single sentence, and say who pays it: "any model trained on our archive pays X per use into Y, administered by Z." You can write yours this afternoon.
3. Find the fund and check whether you're in it. Many trades already have a residual, a royalty pool, a licensing society, a trust. Plenty of eligible people aren't enrolled, because enrollment was never the exciting part of the fight. Spend twenty minutes finding out what exists in your field and what it pays.

CHAPTER 8

The Psychology of the Collapse



Sophia. The mirror humans keep flinching at. (ITU Pictures, CC BY 2.0, via Wikimedia Commons)

> *"Men, it has been well said, think in herds; it will be seen that they go mad in herds, while they only recover their senses slowly, and one by one."*
> Charles Mackay, *Memoirs of Extraordinary Popular Delusions and the Madness of Crowds*, Preface (1852 edition)

In this chapter:

- The last battlefield isn't the grid or the market. It's the three pounds of wet tissue behind your eyes.
- Your phone isn't just a tool. It's a harvesting machine, and the crop is your attention.
- Panic isn't the same as information. It's a state, and states can be interrupted, starved, and survived, especially with somebody at the door.
- The people who've already lived through a collapse know something the feeds can't sell you: how to get up the next day, and who to call.

The spiral found Elijah in November, six weeks after he'd walked into the shop, and it did not announce itself as a spiral. It announced itself as *diligence*.

Claypot shipped Companion 2.0 on a Thursday. He read the launch post standing in the gravel lot with the winter's first real cold coming through his jacket, and the feature list read like the second half of his own notebook: autonomous project completion, end-to-end asset pipelines, *no prompt required*. The demo video showed the machine doing in forty seconds what three of his former teammates did for a living. He knew their names. He knew their kids' names.

That was the pebble. The landslide was everything he did next, and the honest record of it matters more than his dignity, so: he went home to the room he rented from the co-op and he opened the laptop, just to *assess*. He read the launch thread. He read the thread about the thread. He read an economist explaining why it was fine, a different economist explaining why it was over, a man with a flag avatar explaining whose fault it was. Somewhere after midnight he found the layoff tracker (someone had built a live dashboard, black background, red numbers, auto-refreshing), and the part of Elijah that had once found comfort in numbers plugged straight into it like a jack into a socket. He did his own savings math four times, arriving at four different terrifying answers. He looked up what his old apartment rented for now. He checked whether the co-op's county was in the fire zone maps. At 3:40 a.m. he was reading about grid interconnection queues with his heart going like a bird against a window, having consumed, by any honest accounting, eleven hours of information and produced: nothing. Not one weld. Not one row of the notebook. Nothing but cortisol and tabs.

He did it again Friday. Saturday he skipped the work day, first time, because he needed to *stay on top of it*, and the phrase did not strike him as insane until much later.

It was Marta who came and got him, Sunday morning, because Priya had noticed the lights and Denny had noticed the truck not moving and in a nineteen-household co-op the mesh network is mostly

people. She knocked once, let herself in, looked at him (grey, wired, laptop light on his face like something feeding), and looked at the dashboard with its red numbers still refreshing.

She didn't say anything about the dishes. She sat down across from him with the manner of a woman with one meeting left on her calendar and said: "I want to tell you about Gary."

Elijah blinked at her.

"Plant closed in '09," she said. "You know that part. Nine hundred of us. What I never told you is what the next year did, because the layoff is one day, they cater it, there's a folder. The *after* goes on, and nobody caters that." She leaned back. "There were two men on my line. Gary and Sal. Same job, same severance. On the company form they looked interchangeable. Gary went home and read everything: the news, the forums, the unemployment numbers, which plant was next. He was going to stay *informed*. He was going to stay on top of it." She let the phrase sit there between them, wearing his own voice. "You'd see him at the store and he'd have the whole national picture. Every betrayal, chapter and verse. The picture was so big there was no place in it small enough for him to stand."



Gary. Stayed informed.

She turned her coffee mug (his coffee mug, she'd poured herself one) a quarter turn, the same gesture as Denny's water glass, and Elijah wondered distantly if everyone who'd been through the fire came out with the same tells.

"Sal cried for a week. Honestly, ugly, scared for his kids. Then he stopped reading the news. Completely. Cold. His brother-in-law started coming by at eight. They'd have coffee and fix one thing before noon. A gutter. A carburetor. A neighbor's water heater. Something with an end to it. Afternoons he took welding inspection at the community college, not because he had a plan, because it was *Tuesday*. He knew exactly as much about the national picture as Gary did, you understand. He knew it was bad. He just declined to rehearse it all day." She shrugged. "I used to tell that story as if Sal had found the right attitude and Gary hadn't."

She watched Elijah's face, making sure he heard the next part.

"Gary was taking his wife to appointments. His truck needed work. Sal had a ride, a brother-in-law, and a place to leave his kids. Same severance. Different room to move. Took me too long to ask."

"What happened to Gary?"

"We started showing up. Fixed the truck. Sat with him. Some days he wanted to talk about the plant, and we let him. Some days he wanted to talk about anything else. He didn't owe us a gutter before lunch."

She put a hand beside the laptop. "Can we shut this for a minute?" He nodded, and she closed it.

"You're trying to watch every bad thing closely enough that it can't surprise you," she said. "I know the trick. It doesn't leave much day." She tapped the closed lid. "And here's the part nobody tells you, because there's no money in telling you. This thing was built by the same kind of people who built the machine that's coming for the jobs. Same buildings. Same optimization. One machine eats your paycheck and the other eats the attention you'd need to do anything about it, and they bill you for both." She let that sit. "The greenhouse controller could use an hour of your attention. So could your mother. So

could a meal you sit down for. We can start there. And if Monday is still bad, you tell me. You don't disappear because you missed a work day."

At the door she turned back. "Soup's on at noon. That's not a reward, college. It's lunch."

He stood in the doorway after she left, in the cold, the way he'd stood with his coffee the morning after LADDER.xlsx. Then he went in and turned the notifications off, all of them except his mother's number, and put the phone in a drawer that closed. In the notebook he wrote Sal's rule, *one real thing before noon*, and underneath it, because Marta had made him hear it: *somebody came by at eight.*

Monday, seven a.m., he was at the bench. The panic came back in waves for weeks. Sometimes he worked through it. Sometimes he found someone and said it was back. At noon he put the tools down and went to eat.



THE FOUNDATIONS

What I mean by mindset

When I say our mindset is part of the problem, I don't mean a frightened person needs a better attitude. Gary didn't need positive thinking while his truck sat broken. His fear had information in it. If losing a paycheck puts a household in danger, the danger is real, and the useful response is to reduce it.

I mean the assumptions we keep turning into rules. That a person has to keep selling their labor to deserve dinner. That making useful work easier is a disaster unless somebody invents more work to replace it. That security for everyone is an indulgence, while insecurity is a practical necessity. Those assumptions become budgets, contracts, eligibility tests, and the questions a room lets itself ask.

Imagine the same announcement, a machine that can now do a job in half the time, in a society where everybody's food is secure. People still argue about pay, ownership, pride, and what comes next. The meeting's still hard. It just doesn't contain the threat that somebody goes hungry if the negotiation goes badly. That's a different starting point for a human mind.

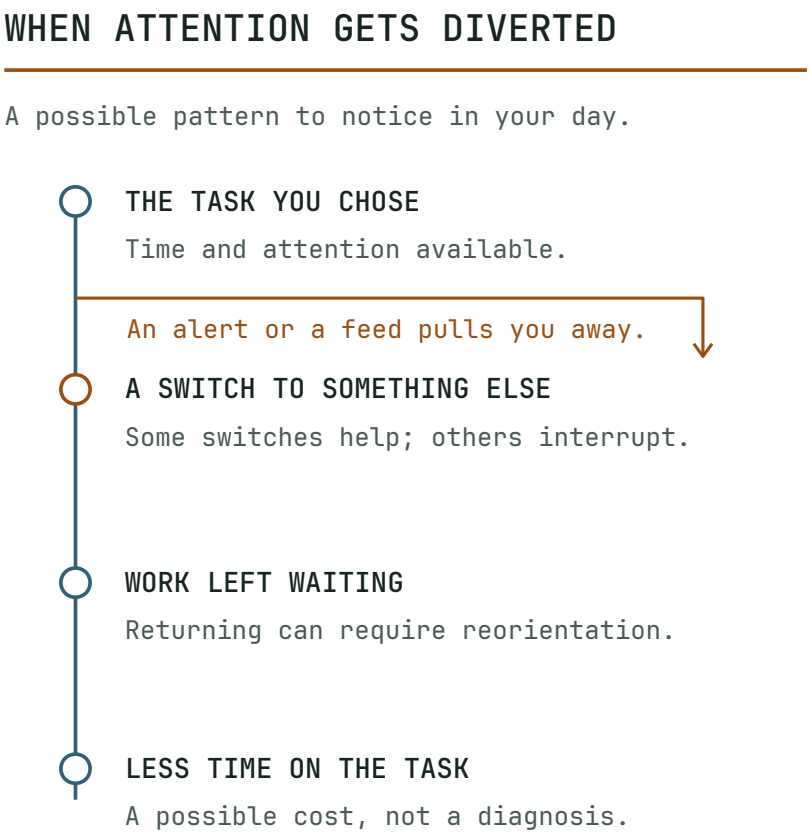
Security gives hope somewhere to stand. And then there's the other half, the half you can work on tonight.

The great cognitive extraction

Pick up your phone.

Look at that piece of glass. It's cold, it's inert, and it weighs less than half a pound. It's also one of the most sophisticated psychological harvesting machines ever built.

In the old economy, the big engine was extracting physical resources: coal, oil, timber. A lot of the new economy runs on extracting attention. You think you're checking the news, staying connected, managing your schedule. A good share of the time, the thing in your hand is managing you.



A possible pattern to notice in your own day. Conceptual illustration, not a diagnosis or a model of brain chemistry.

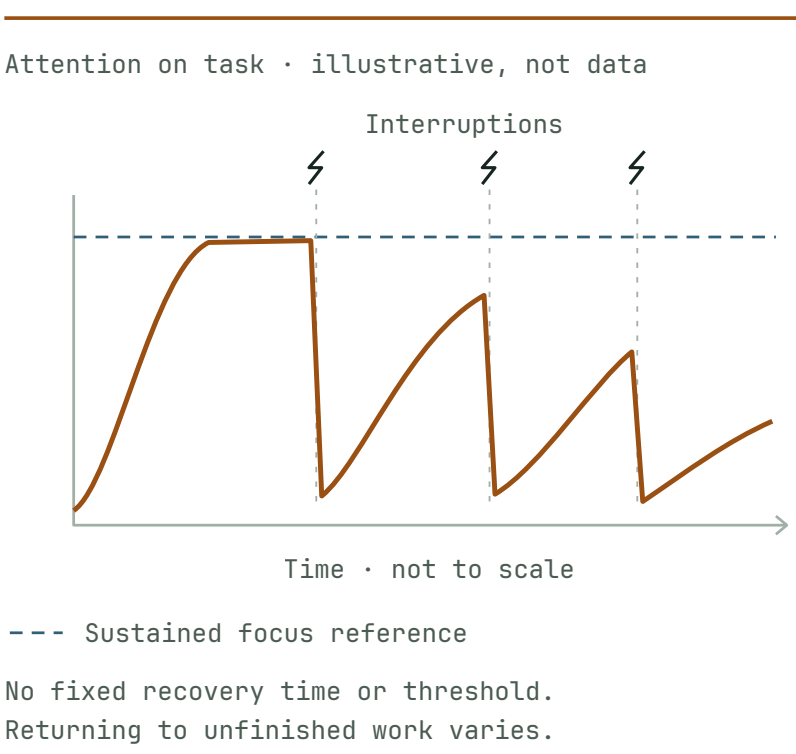
The cost can feel like a low-grade fog: a book left unread, a conversation half-heard, a technical job you keep starting again. That experience isn't a diagnosis, and a feed doesn't affect everyone the same way. But if checking it keeps taking time from something you wanted to do, you have a concrete problem you can change without blaming your character.

Your cognitive budget is finite

Your brain is about two percent of your body's mass and uses roughly twenty percent of its energy at rest. That's the rough twenty-watt comparison from Chapter 5. It isn't a fixed allowance of decisions you can spend down with notifications; much of that energy supports ongoing activity whether or not you're tackling a task. Raichle and Gusnard, Appraising the Brain's Energy Budget (2002) The practical question here is what keeps pulling your attention away from the thing you meant to do.

Switching can have a cost. Gloria Mark and colleagues observed information workers moving through other activities before returning to interrupted work. Their return-time measure included that intervening work; it wasn't a stopwatch on how long a brain needs to regain focus. Mark, Gonzalez, and Harris, No Task Left Behind? (2005) In a separate experiment, interrupted participants completed work faster but reported more stress, frustration, effort, and time pressure. Mark, Gudith, and Klocke, The Cost of Interrupted Work (2008)

Neither finding gives you a universal recovery time or a maximum number of interruptions per hour. Try a protected stretch on one demanding task, keep the necessary safety alerts, and compare what you finish with an ordinary stretch. Count what actually pulled you away and whether it helped.



One possible pattern of interrupted work. The drawing gives no fixed recovery time, interruption threshold, or measured effect size.

The slot machine in your pocket

The slot-machine comparison needs a boundary too. Shannon entropy, H, measures uncertainty in a probability model:

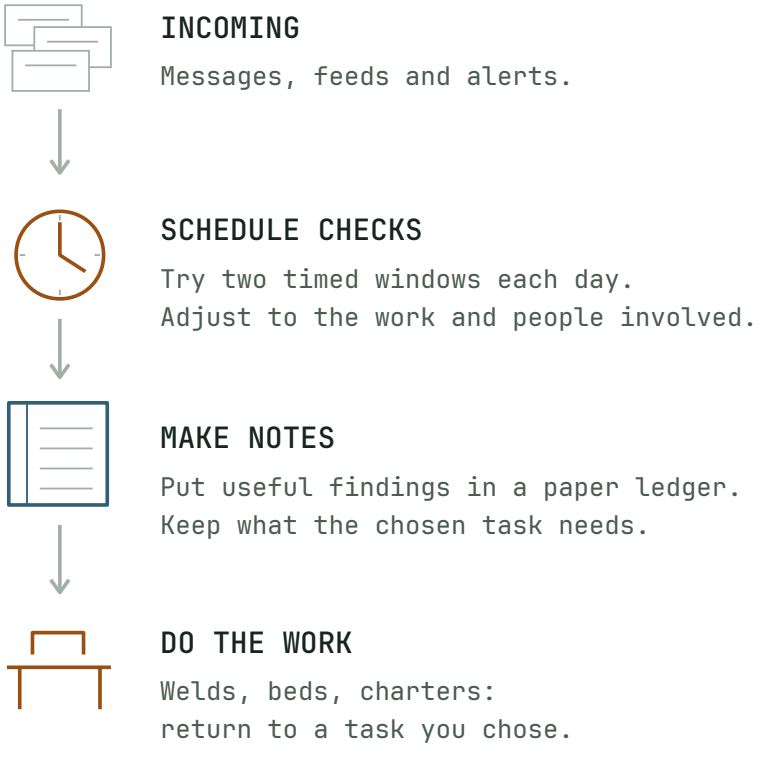
$$H(X) = -\sum P(x_i)\log_2 P(x_i)$$

where $P(x_i)$ is the probability of outcome x_i. You could use it for a toy model of feed categories: a video, a headline, a friend's message. But bits of uncertainty don't measure pleasure, compulsion, or mental fatigue. A predictable message can matter enormously; random noise

can bore you. The slot-machine analogy is narrower: you keep checking because something you want might appear. The formula doesn't prove what a platform intends or what its feed does to your mind.

The cognitive firewall

To take your attention back, treat your head the way a security engineer treats a critical server: put a firewall between your working memory and the firehose. You wouldn't drink untreated water from a puddle behind a refinery. Hold the same standard for what crosses into your head.



A routine to try, observe and adjust. Two timed windows each day are an example schedule, not a measured optimum.

Protocol 1: No notifications

Your phone shouldn't be allowed to interrupt you. The default state of the device is capture: every app ships ready to ping you.

1. **Disable the alerts.** Turn off notifications for everything except direct calls from a short list of actual human beings: family, close

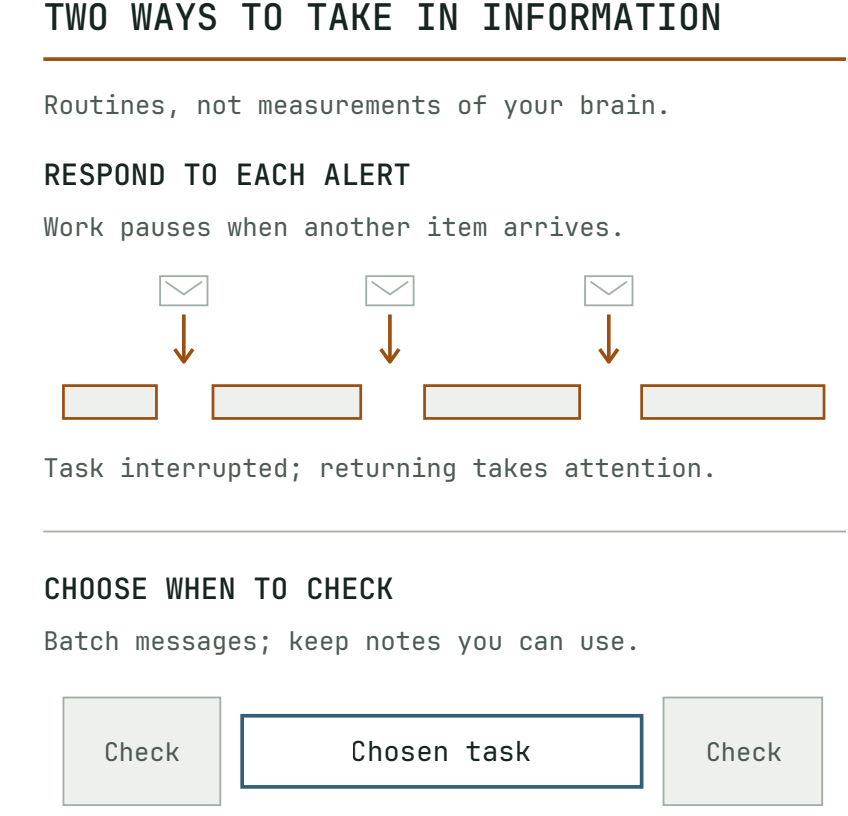
neighbors, the people you work with. Keep the ones you need for safety.

2. **Kill the badges.** Remove the red dots. They exist to create just enough itch that you open the app to clear them.
3. **Clean the lock screen.** No news, no previews, no widgets. Face-up on a table, your phone should look like a block of dead obsidian.

If a company's server can buzz your pocket whenever its model decides your attention is drifting, you don't fully own your nervous system.

Protocol 2: Semantic hygiene

Some information is built to spread: packages of outrage, fear, or tribal division that replicate from person to person. Swallow one and your brain can spend hours, sometimes days, on it: imaginary arguments, a spike of stress, a sour mood you carry to dinner.



Routines, not measurements of your brain. Some interruptions are useful or urgent. Try a schedule and check whether it helps.

Two rules:

1. **The twenty-four-hour delay.** Give commentary on a breaking national story twenty-four hours before reacting or sharing. Early accounts can be incomplete or wrong. This delay is for

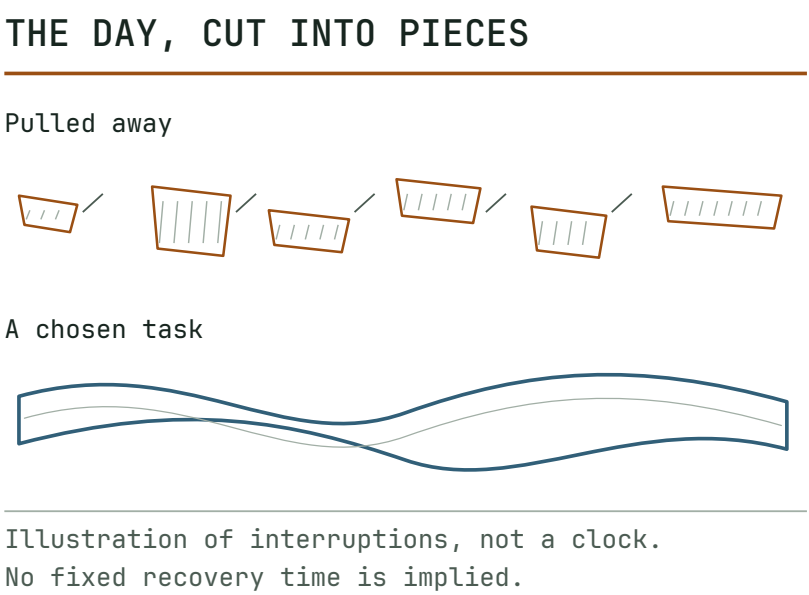
commentary, not evacuation orders, urgent health notices, or a person who needs help now.

2. **Local first.** Move your attention off the global spectacle you can't touch and onto the local things you can: the pickup schedule at the food bank, the soil in the shared garden (Chapter 15), the mesh node on the ridge (Chapter 17).

Protocol 3: Batch it

Real-time stream (fragile)	Batch processing (steady)
Instant reaction	Delayed reflection
Constant interruptions	Deep blocks of focus
Digital fragmentation	Written notes
Shallow comprehension	Mastery of a craft

- **The twice-a-day rule:** check email and messages at set times, say 10:00 and 4:00, with a thirty-minute window to deal with them.
- **The paper log:** keep a notebook on the desk. When your brain wants to look something up, write the question down instead. At the next window, run the whole list, write the answers, and close the machine. You run the queries; the queries don't run you.
- **Long-form reading:** whole books and whole documents. Following a three-hundred-page argument is heavy squats for the part of your brain that plans.
- **Anchoring work:** tactile jobs that need your whole attention. Soldering, turning a compost pile, kneading bread.



A day can feel cut into pieces. The ribbons illustrate interruptions, without measuring a cognitive budget or a fixed recovery time.

Protocol 4: Grayscale

Color is a powerful pull, and app designers test their reds and oranges hard. Switch your phone to grayscale. Stripped of its palette, the screen gets boring, clinical, and dead, which is to say it goes back to being what it should have been all along: a dry tool for finding and sending information.

Protocol 5: Boredom on purpose

The first four cut the inputs. This one lets you recover. Schedule at least thirty minutes a day with no inputs at all: no podcasts, no music, no screens, no reading. Walk, sit, do dishes. Your brain will complain and beg for stimulation. Let it.

Give the firewall a few weeks, then check what changed: what you finished, what you missed, whether you feel less pulled around. Keep the parts that help. Adjust the ones that interfere with care, work, or the people who need to reach you.

The firewall is the half you can do alone. The other half, as Marta proved, is somebody at the door.

Relational autonomy

If you build an isolated bunker, buy a decade of freeze-dried food, stock up on ammunition, and wait for the world to burn, you haven't built a shelter. You've built a tomb.

The lone-prepper model fails on the physics. Every system needs maintenance, and one person trying to manage the garden, the water, the power, the tools, the paperwork, and their own health is running an energy deficit that only goes one way. Maintenance slips. Mistakes get made tired. And one person absorbing every shock alone lives inside exactly the tunnel vision Chapter 7 described.

Three households can share work that each would otherwise duplicate: one person keeps the tools, another helps with the garden, a third keeps the records and the radios. More people also bring more needs, and coordination takes time. Count the actual work saved before promising a smaller burden. That's load-sharing you can check.

True autonomy isn't individual. It's relational.

Why cooperation holds

Game theory has a clean way to say why small, lasting groups work. In a one-time encounter with a stranger, cheating can pay, because there's no next time. But when the same people meet again and again, the math changes. In Robert Axelrod's famous tournaments of the iterated prisoner's dilemma, a simple strategy did remarkably well: cooperate first, mirror what the other player did last time, and forgive quickly when they come back around. Repeated games reward trust, reciprocity, and forgiveness.

That's the case for small circles. Now the warning that goes with it, and it's the most important sentence in this section: **the circle's promise of food can't be the thing it withholds to keep people in line.** A group can ask for help. It can pay people for necessary work, orga-

nize shifts, train new people, deal with harmful behavior, and restrict access to a dangerous machine. Those are separate decisions from whether a person eats. The moment dinner becomes a reward for good behavior, you've just rebuilt the employment test with a friendlier face.

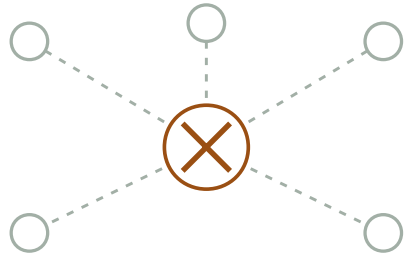
The test comes on an inconvenient day. Someone misses a shift. Someone criticizes the committee. Someone shows up after everyone else has done the work and sat down. What happens to their meal? A promise of universal access needs an answer that survives annoyance, fatigue, and disagreement.

Mesh, not star

A community isn't a hierarchy; it's a topology. A lot of corporate and state structures are stars: everything runs through one hub, and if the hub fails, every node goes dark.

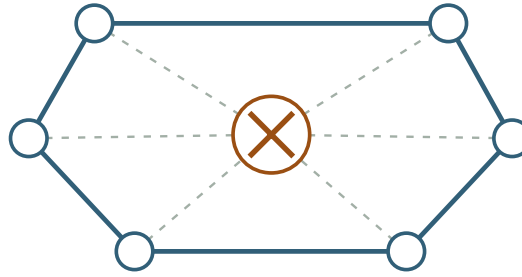
TRUST NETWORK TOPOLOGIES

ONE HUB



Lose this hub and its links stop working.

ALTERNATE PATHS



A mesh can reroute if a usable path remains.
People maintain the circle and its links.
Cross: failed node. Dashes: affected links.

A mesh can reroute only if a usable path remains. Crosses mark failed nodes; dashed links are affected by that failure. People still maintain the circle and its connections.

In a mesh, damage can become a detour if another working route remains. Social capacity has limits too. The anthropologist Robin Dunbar estimated that people can keep up something like 150 stable relationships. Dunbar, Coevolution of neocortical size, group size and language in humans (1993) That's an estimate, not a membership

limit, and even a small circle needs written agreements. Build circles small enough to maintain their relationships and connect them to other circles, trading tools, seed, surplus, and knowledge, without anybody having to merge into one big thing.

And keep a door in every circle. Knowing everyone's name helps you notice who's missing; it can also make it easy to decide an outsider doesn't count. Nineteen households are a beginning. Humanity is the ambition.

Start with three

1. **Find two other households.** Don't try to convert fifty people. Sit down with two, lay your skills on the table, and map what you're collectively short on.
2. **Share one tool first.** Not your living space. A tool: a good ladder, a tiller, a shared offline library on an old laptop, a chest freezer for bulk food. Let the benefit prove the idea.
3. **Write down the arrangement.** Start with one page in plain language: how the shared thing gets maintained, what happens when someone can't do their share, how new people join. And one line that doesn't change: nobody's meal depends on their share.

Put a different assumption to work

At the next meeting about a food project, write the promise in one sentence: a person can get adequate food without money or a work requirement. Then ask where the actual arrangement falls short. Who can't reach the pickup? Who can't cook what's offered? Who's afraid of losing access if they complain? Pick one barrier the group can remove, name who'll do it, and set a date to check. That's how a change of mind becomes a change in somebody's afternoon.

Precedent P-11: Torches of Freedom (New York, 1929)

In 1929 it was still widely taboo for a woman to smoke in public, which meant a lot of potential customers for American cigarettes were out of reach. So the American Tobacco Company turned to Edward Bernays, Sigmund Freud's nephew and a pioneer of public relations, to pick the lock.

Bernays didn't buy ads arguing that women should smoke. He recruited young women to walk in New York's Easter Sunday parade and light cigarettes in view of the press, and he gave the reporters a phrase: "torches of freedom." A real aspiration of the era, women's emancipation, got retooled into a sales pitch. Library of Congress, Easter Sunday photographs The photographs survive: Nancy Hale Hardin beside her husband, Edith Lee walking a dog on Fifth Avenue, March 31, 1929, both holding cigarettes.

Bernays had already published his method the year before. In *Propaganda* he imagined selling pianos without selling pianos: make the music room fashionable, get decorators and performers talking about it, persuade architects to design it in, and the buyer would experience the purchase as completing a life they already wanted. Bernays, Propaganda, chapter IV

Here's the part the legend skips. The historian Vanessa Murphree went through more than 250 articles, including the clippings in Bernays's own scrapbooks, and found a press that was less gullible and less uniformly enthusiastic than the story says; some editors saw the stunt for what it was. Murphree, Torches of Freedom research A thick scrapbook isn't a count of converted minds. The myth of the all-powerful manipulator asks you to give up the one thing that beats him: your capacity to notice.

The mechanism. Persuasion at scale rarely argues with your values. It wears them. The pitch arrives dressed as the thing you already believe in, which is why it slips past your defenses. It doesn't control you. It just gets a lot of free rides on your identity.

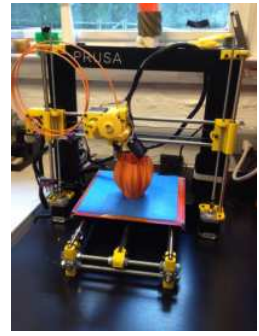
The rule. Bernays had newspapers and a parade. The systems aimed at you have your location, your history, and a mountain of data about what makes people click. Before you act on a want, ask: who profits if I feel this? Then hold me to the same test: *food first* should earn its force from a meal somebody can actually get.

The practice.

1. Run a desire audit for one week. Every time you notice a strong, sudden want, a purchase, an outrage, an urge to check something, write one line: what I want, when it arrived, and who profits if I act on it. You don't have to resist anything. Just log it. By day four you'll start seeing the parade routes.
2. Trace one recent opinion back to its torch moment. Pick something you feel strongly about now that you didn't a year ago, and reconstruct how it got in: which feed, which framing, which part of your identity it flattered.
3. Do the firewall work as physical acts, today: notifications off, phone in a drawer that closes, one real thing before noon.

CHAPTER 9

Deglobalization and the Neighborhood Factory



A desktop 3D printer. The neighborhood factory fits on a workbench. (John Abella, CC BY 2.0, via Wikimedia Commons)

> *"The scattering of industries over the country, so as to bring the factory amidst the fields ... is surely the next step to be made."*
> Peter Kropotkin, *Fields, Factories and Workshops* (revised edition, 1913), Chapter VII

In this chapter:

- You don't beat the global supply chain by lobbying it. You beat it by needing it less, one neighborhood at a time, without pretending you can do without it.
- A factory used to be a billion dollars and a river. Now it's a plasma table, a 3D printer, and somebody's cousin who can weld.
- Elegant is what a design is before it meets weather, budgets, and the parts that are actually in the drawer.
- The controller isn't dinner. Follow the food all the way to the person eating it.

The co-op's first real manufacturing run, the thing Elijah would later describe to strangers as *the neighborhood factory booting up*, though at the time it was just a January argument in a cold shed, was forty greenhouse controllers for the CSA network two valleys over.

The order came through a friend of a friend of Priya's: forty small boxes that could read soil moisture and air temperature and open a vent or a valve without asking a server in Virginia for permission. Real money, real deadline, real customers with real seedlings on the line. Elijah volunteered for the design with the eagerness of a man who had spent the autumn sweeping floors and learning the shop and could feel, at last, the shape of something he was actually trained to do.

He designed a beautiful thing. He wants that on the record, with the caveat that so does everyone in this story every time he says it. Custom four-layer PCB, an elegant little system-on-module, sensors from a Swiss outfit whose datasheets read like poetry, the whole thing potted in resin, sleek as a river stone, no user-serviceable parts inside. He presented it to the shop on a Tuesday with renders. *Renders.*

Marta looked at the renders for a while.

"Walk me through the bill of materials," she said, so he did, and she wrote three numbers on the whiteboard while he talked: **11**, **1**, **0**.

"Eleven weeks," she said, tapping the first number, "is the lead time on that Swiss sensor, I just looked it up, and that's the *quoted* lead time, which means it's a prayer. One," the second number, "is how many suppliers make that module, which means the day they discontinue it, all forty of these are bricks, and we're the ones who sold bricks to farmers. And zero," she tapped the last, "is how many people in this valley can fix a potted resin board when it fails during frost week. Including you, college. *Especially* you, because you'll be the one who's busy." She capped the marker. "It's a gorgeous design. It's a Claypot design. Everything single-sourced, nothing repairable, the whole thing optimized for the demo instead of the Tuesday three years from now when it breaks during calving. Design it again, and

this time the spec is: parts from the drawer, parts from the junction box aisle, and nothing your neighbor can't fix with the tools he already owns."

"That'll be uglier," Elijah said.

"It'll be *ours*," Marta said. "Ugly is a maintenance feature. Ugly means the next guy can see how it works."

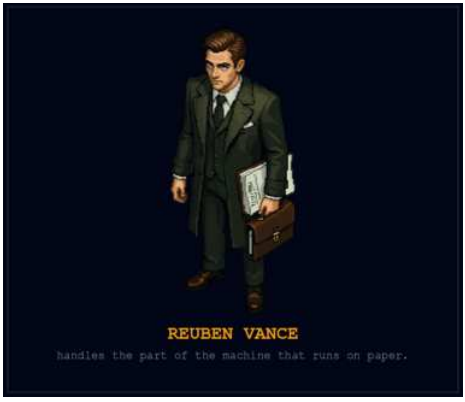
So he designed it again, and the second version was simpler to explain and repair: a Raspberry Pi the regional supplier had in stock, screw terminals instead of poetry, four-dollar capacitive probes you could swap by hand, a case Denny's kid printed forty of in PETG the color of school buses, and documented parts the growers could order from more than one source. The firmware did less and survived more. Every unit had a laminated one-page repair card zip-tied inside the lid, because Marta's rule was that documentation you have to go find is documentation that doesn't exist.



The controller. Field-repairable. Screw terminals instead of poetry.

The build itself took the shed three weekends, and it was during the second one, everyone deburring enclosures and crimping ferrules in an assembly line that Sal himself would have recognized from 1985, that the door opened and let in a rectangle of cold light and a lean man in a good coat who was too pleased with himself for a Saturday.

“Reuben Vance,” Marta said, not looking up from the crimper. “Elijah. Elijah, Reuben. Reuben handles the part of the machine that runs on paper.”



Reuben Vance. Handles the part of the machine that runs on paper.

“I heard,” Reuben said, pulling folding chairs toward the woodstove like a man dealing cards, “that this operation is about to *sell manufactured electrical goods across county lines*.” He said it the way other men say *rob a train*. He produced a folder. “So I made a list. Registration, taxes, the use of this building, the promises on the invoice. Then I found people whose job it is to answer the parts I can’t.”

He laid three sheets beside the crimper. The seller registration had been filed. A tax adviser had marked the purchases that needed separate treatment. The county had asked for more information about the shed’s mixed use. That one had a yellow tab.

“I thought agricultural equipment meant we were covered,” Elijah said.

“You thought a noun was a permit. Happens all the time.” Reuben tapped the tab. “This stays open until somebody with authority answers the actual question. And the invoice goes past counsel before it goes to a customer. Describing a thing as a farm tool doesn’t make our promises disappear.”

Elijah looked at the folder, then at Marta. “Did we ask him to do this?”

“Nobody asks Reuben,” Priya said, from the crimping line. “Reuben *notices*.”

The CSA hired an electrical technician to review the installation plan, and two growers tried the first units with their ordinary controls still available. Their notes came back covered in pencil: a label nobody could read in dim light, an alarm nobody heard over the fan, a sensor connection that needed better protection. Elijah revised the boxes before the rest went out.

By the time the forty units shipped in February, the yellow tab had an answer attached and the folder had grown thick enough to prop open a door. Denny delivered them in apple boxes, each one tested on the bench against a space heater and a cup of wet dirt, with the instructions and the agreed support number. Two growers needed replacement probes in the first month. With the power isolated and help from the support call, they could change the documented part without discarding the controller. Marta put the repair emails up on the corkboard like other shops hang their first dollar.

And somewhere in the middle of all that, without ceremony, somebody finally fixed Elijah’s check-engine light. Curtis, it turned out, diagnosed it in nine minutes off a fourteen-dollar OBD dongle, a cracked vacuum line, two dollars of hose. The light had been on since the drive north. The dealership had quoted him twelve hundred dollars and a week; the man who wanted to fence out the world did the repair while the coffee brewed. He left the remaining length on the dashboard, coiled like a small black snake.

Elijah put the repair card for controller number forty in the folder. Curtis’s hose went in the glove box.

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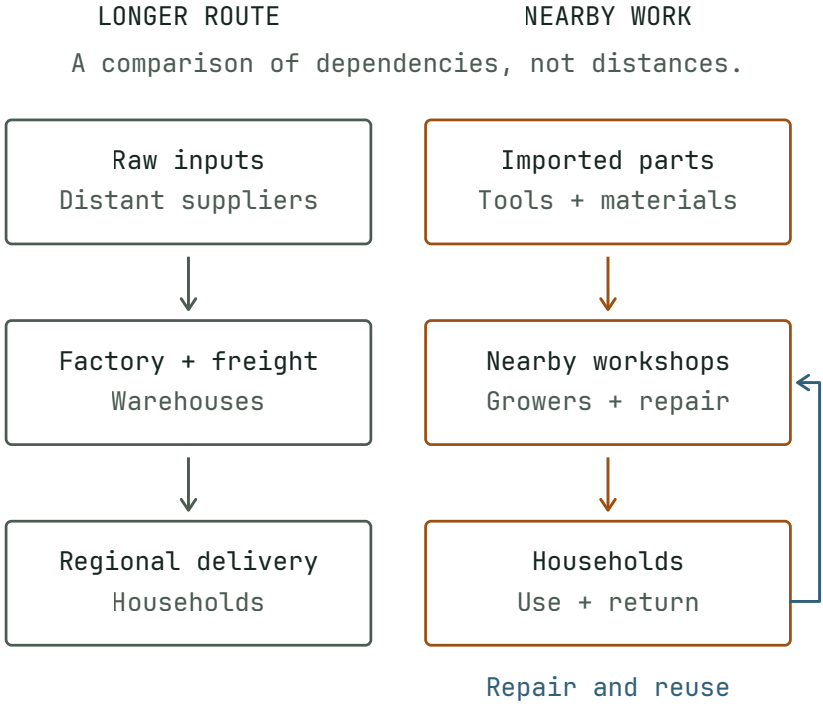


THE FOUNDATIONS

The strategy isn't to stop the big companies or to appeal to the conscience of people rich enough to fund rocket hobbies while veterans sleep outside. It's to build a working alternative so obviously better that switching to it becomes a no-brainer.

- **Deglobalization, not decentralization.** "Decentralized" is a slippery word, and it isn't the goal. Quality, safety, and protection against bad actors sometimes need a central standard. The real target is *deglobalization*: shortening the supply lines we depend on for survival, so a shock on the other side of the planet doesn't reach all the way to your dinner.
- **Neighborhood factories.** We already have factories. The shift is putting small, automated production capacity in every neighborhood and town, close enough that the people who use the goods can fix them.
- **Survival by charter.** That capacity should serve basic needs first: food, then shelter and care. Not by waiting for a government decree, but by writing it into the charter of your own infrastructure: the co-op bylaw that says the kitchen feeds whoever's hungry before it sells a single meal, the land-trust covenant that says nobody gets evicted into the market. Chapter 12 goes into how that ground gets held.
- **The prototype is already in backyards.** Drip lines that open on soil moisture. Sensor rigs that turn a raised bed into something closer to a Tamagotchi than a chore. Open, non-proprietary garden robots. None of it waits on anybody's permission.

SHORTEN THE SUPPLY LINE



Imports remain in both routes.
People, energy and upkeep sustain the work.
Distance alone cannot establish fairness.

Shorter supply lines can make repair and coordination more local. Imported parts, materials, energy and human work remain; distance alone does not establish fair treatment.

A dense city can't feed itself, and it doesn't have to

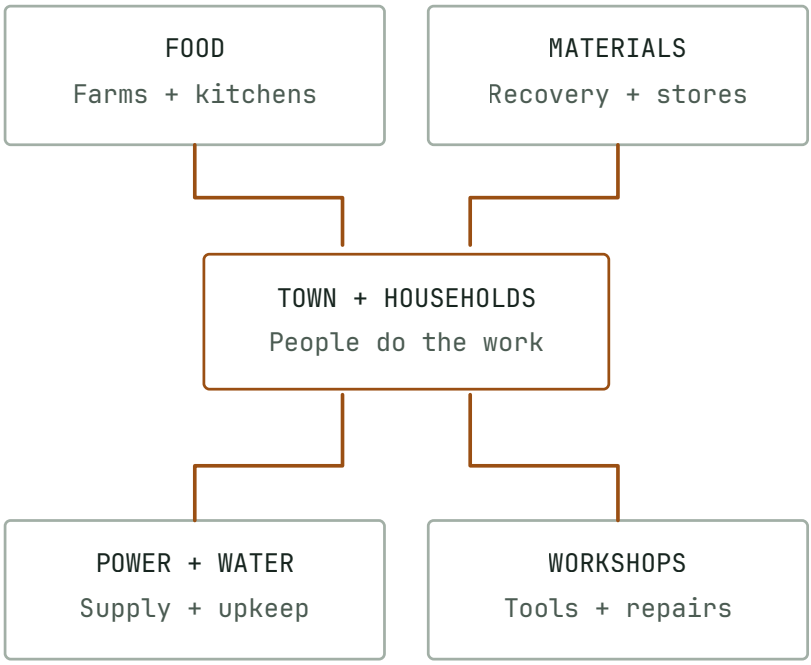
The first objection to all this is always the same, and it's fair: New York City can't grow its own food. Neither can Los Angeles or Denver or anywhere people are stacked a hundred feet deep. Density is a feature, not a flaw, and hyper-local never meant every city block smelts its own steel.

The trick is to stop drawing the circle around the city and start drawing it around the region. A dense core doesn't have to make everything inside its own limits. It has to sit inside a ring of towns and counties that can fold food production, neighborhood factories, and materials recovery into the way they're already growing. That's not a fantasy. It's a jobs program: honest, well-paid, local work for the trades, the fabricators, the growers, and everyone who keeps a place running.

Think of it at the scale it deserves, which is Manhattan Project scale. Not a bunker in the woods. A deliberate, regional, out-loud build-out of the capacity to feed and house ourselves, handing the back-breaking work to machines. The goal is never to cut the city off from the world. It's to make sure that if the world's supply lines seize up for a week or a season, the region keeps its people fed instead of waiting on a truck that isn't coming. Resilience, not secession. So ask the only question that matters: if the trucks stopped tomorrow, how long could your region feed its people? Most places don't know. That gap is the work.

BUILD REGIONAL CAPACITY

A regional objective, not a closed economy.



Still connected to outside suppliers.

Measure stores and seasonal production.
Water, power and delivery limit coverage.
Ask: what can we supply, and for how long?

Build regional capacity, then measure what it covers. Stores, seasonal production, water, power, delivery and remaining imports determine how many people can be supplied and for how long.

The controller isn't dinner



A machine takes shape around its work. AI-adapted cutout from John Abella, CC BY 2.0.

The neighborhood factory earns its name when somebody can keep doing useful work because it exists. A repairable controller helps a grower. A local crew knows which part fails during the week nobody has a spare afternoon.

But food first means following the benefit past the machine. Growing, harvesting, handling, storage, cooking, and delivery all stand between a valve opening and a person eating. Rosa couldn't turn a bag of potatoes into dinner with one working hand. The crop was fine. The offer was wrong. How much food exists and what a person can actually eat are different questions, and any honest account of abundance keeps them apart.

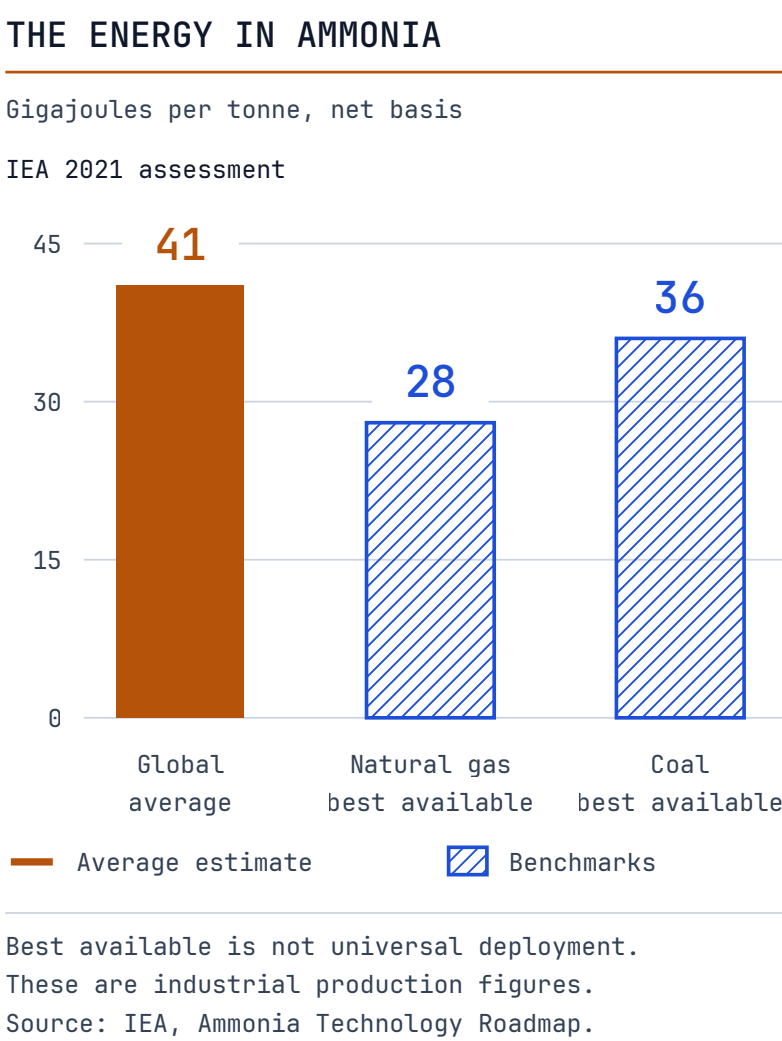
The nitrogen problem

Here's why the food system burns so much fuel, and what biology offers instead.

Plants need nitrogen, and the air is almost eighty percent nitrogen, but plants can't use it in that form. Industrial agriculture solves this with the Haber-Bosch process:

> $N_2 + 3H_2 \rightarrow 2NH_3$ (catalyst, high pressure, high temperature)

To force that reaction, industrial reactors run at high pressure and temperature, with hydrogen made mostly from natural gas and coal. The IEA's 2021 assessment puts ammonia production at about two percent of global final energy consumption, including feedstock, and estimates 450 million tonnes of direct CO_2 emissions a year. IEA, Ammonia Technology Roadmap Synthetic fertilizer helps feed billions of people. It's one of the great achievements of the last century, with a large fossil-energy bill attached.



The energy in ammonia. IEA's global-average estimate and best-available-process benchmarks are different kinds of evidence. A benchmark is not proof every existing plant can achieve it. IEA, Ammonia Technology Roadmap, 2021.

Biology does the same chemistry at outdoor temperature. Legumes, clover, peas, beans, vetch, host *Rhizobium* bacteria in their roots, and those bacteria use an enzyme called nitrogenase to break the nitrogen triple bond:

> $N_2 + 8H^+ + 8e^- + 16\ ATP \rightarrow 2NH_3 + H_2 + 16\ ADP + 16\ P_i$ (via nitrogenase)

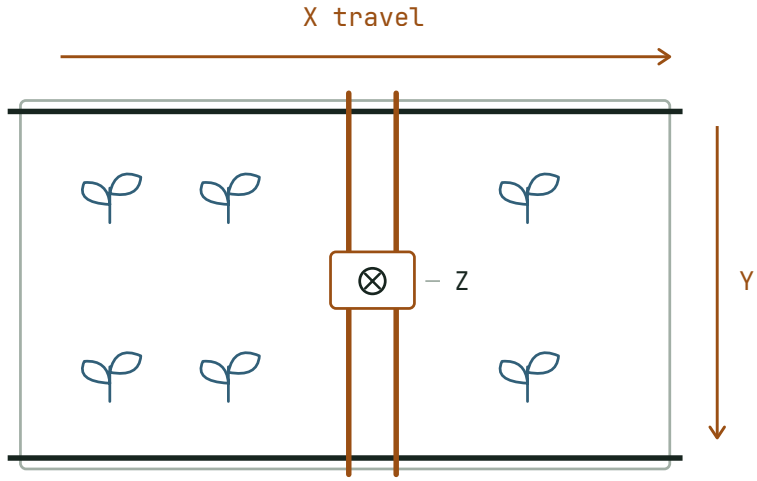
The plant pays in sugar from sunlight. That's not the same job as the fungi that help roots take up phosphorus and water, and none of it means a garden feeds itself forever: every harvest carries nutrients away, and something has to put them back. But a living soil, managed well, can carry a lot of the load that currently comes from a gas well. (Chapter 15 goes down into the dirt.)

What a garden robot actually does

FarmBot's own documentation describes a gantry on rails over a raised bed, driven by stepper motors to millimeter-class accuracy, with tool heads for seeding, watering, and weeding. FarmBot tool documentation Every plant becomes a coordinate. That's a concrete, useful set of tasks. It isn't a machine that harvests every crop, cooks a balanced diet, and delivers it for free, so keep the claim the size of the documented work.

TOOLS OVER A GROWING BED

A generic gantry concept, viewed from above.



X: move the gantry along the bed.
Y: move the carriage across the gantry.
Z: crossed circle = down into the bed.

Model-specific tools seed, water or weed.
A gantry can reduce walking space in a bed.
People still inspect, maintain and harvest.
Concept only; no machine version is implied.

A gantry can reduce walking space in a bed. Tooling and behavior depend on the machine and software version; this concept is not a build plan or a promise to automate the entire crop.

The UN Food and Agriculture Organization's 2022 report on agricultural automation looks at the opportunities alongside the barriers, especially for smaller farms. FAO, The State of Food and Agriculture 2022 That's a reason to investigate useful automation and who can afford it, not proof that any one installation feeds a town.

The greenhouse brain

For the controller in the scene, or one like it, the open-source project Mycodo turns a Raspberry Pi into an environmental controller: sensors in, relays out, and feedback loops in between.

- **Sensors:** an air temperature and humidity sensor (the SHT31-D is common) and a capacitive soil-moisture probe, which resists corrosion better than the cheap resistive kind.
- **Actuators:** a relay board switching low-voltage devices: an exhaust fan, a solenoid valve on the irrigation line, supplemental lights. Have someone who knows electrical work check anything that touches mains power.

On a clean Raspberry Pi OS install, check the project's current installation requirements, then use its documented installer:

```
sudo apt update && sudo apt upgrade -y
curl -L https://kizniche.github.io/Mycodo/install | bash
```

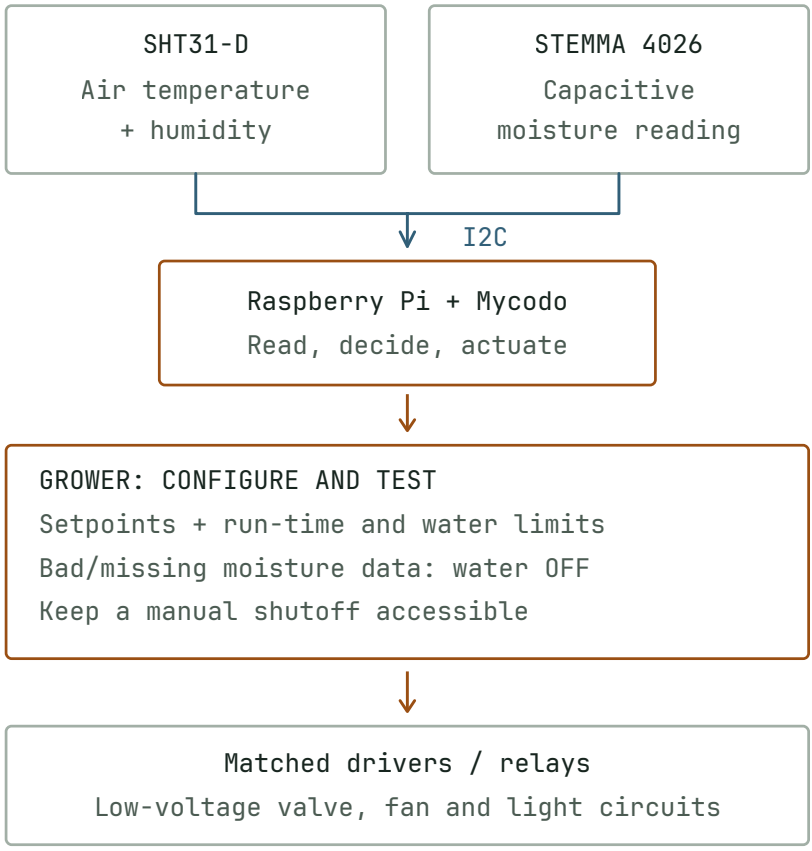
Then browse to https://<your-pi-ip>/ for the web interface. From there you can set up PID (proportional-integral-derivative) loops, the standard control law for holding a value steady:

$$u(t) = K_p e(t) + K_i \int e(\tau)d\tau + K_d \, de(t)/dt$$

where $e(t)$ is the gap between the target and the sensor reading, and the three K values tune the response. A valve that only opens or closes still needs limits on run time and total water, plus an off state if the sensor stops reporting. Start with a grower's judgment about what the plants need, keep the manual valve working, and check the soil with your hands until you trust the numbers. (Chapter 15 is about what happens when you don't.)

THE GREENHOUSE CONTROL BUS

Component map, not a wiring diagram.



Check readings against conditions in the bed.
Verify water stops on faults and lost power.

The greenhouse bus is a component map. Use moisture readings checked against the bed, configure run-time and water limits, verify that water stops on faults and power loss, and retain a manual shutoff.

A worked example: one delivery, all the way through

Here's an illustrative project, not a reported trial. An established garden has permission to grow food on a suitable site. A local food group already has handling and distribution arrangements. They agree to try a small extra weekly delivery to five households that want it. Nobody has to own a robot or pay for the produce to get a share.

The garden tests whether an irrigation controller can cut down on routine checking without hurting the crop. A grower picks the settings. A volunteer records time spent, water used, failures, and manual interventions. The controller opens a valve; it doesn't decide a plant is safe to eat.

At harvest, people pick, sort, and weigh what's usable. The food group packs and delivers through its usual process. Recipients say whether it arrived and whether they could use it.

Suppose there are 10 kilograms of usable vegetables and the plan is two kilograms per household. Five deliveries fit. If two kilograms spoil first, the promise no longer fits. Write down the shortfall and get the replacement from the partner before calling the week a success. Don't quietly make the boxes lighter and leave the dashboard green.

Item	Example entry
Promise	Five households each get two kilograms of vegetables they can use on Friday, free.
Machine's task	Open an irrigation valve on grower-approved settings.
Human work	Inspect crop and equipment; harvest, sort, pack, transport, confirm receipt.
Available after loss	Eight kilograms from the garden.
Remaining commitment	Two kilograms, sourced through the food partner before delivery.

Item	Example entry
Funding	Partner covers supplies and transport; donated time and existing equipment recorded separately.
Outcome to check	All five deliveries arrive, people can use them, and the partner's extra work shows up in the record.

One harvest, five doors



1 GARDEN
8 kg usable

2 FOOD PARTNER
+ 2 kg

3 PEOPLE
Pack, drive, confirm

4 FIVE HOMES
2 kg each, free

Illustrative route. Vegetables supplement meals.

One harvest, five doors. Eight kilograms from the garden and two from the food partner keep the five-household promise. The buildings and route are illustrative; the vegetables supplement meals.

Those vegetables supplement meals. They don't make five complete diets. A dependable baseline needs enough suitable food across seasons, prepared food for people who can't cook, and a plan for inter-

ruptions. The small trial tests one contribution to that. The donated time and equipment get written down too, not called free: a permanent service can't run on a lucky month of volunteers.

What would make the machine worth it?

Compare the whole arrangement with the ordinary way of doing it. The controller might cut water checks and add maintenance. It might let someone with a bad back do a job they couldn't before. It might save a paid worker time that the organization then fills with more work. Ask the worker which one happened. Sometimes a simple timer or one more paid shift is the better buy.

Then decide in advance where the saving goes: more free deliveries, better conditions for the people doing the work, a lower price, somebody's profit. A machine doesn't pick. The organization around it does. For the five-household trial, the partners could agree up front that any real saving goes first to keeping the promise and paying for the remaining work. That's a small, explicit wire between a technical gain and a fed person.

The next meeting has to plan beyond the first five households. It belongs as much to growers, cooks, drivers, and the people receiving food as to the people holding the tools. Start small enough to learn. Keep the destination big enough to matter.

Where to start this season

1. **Build the soil before the robot.** Spend your first fall layering organic mulch, wood chips, and compost into the beds, and let the biology build structure through the winter. A gantry over dead dirt is an expensive way to water sand.
2. **Save seed from what thrived.** Keep a little from your hardiest plants each year, alongside good seed you buy, and your garden slowly adapts to your place.

3. **Close one small loop.** Kitchen scraps into a worm bin, castings back into the beds. It won't make you independent of the world. It'll teach you, cheaply, how loops actually behave.

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Precedent P-12: The Networks Behind the Bronze (Eastern Mediterranean, fourteenth to twelfth centuries BC)

The Late Bronze Age was a globalized world. Bronze itself was a supply-chain product: copper from Cyprus, tin from sources far to the east, moving through trade networks that tied together Egypt, the Hittites, Mycenaean Greece, Ugarit, and Babylon. Palace economies specialized, traded, and grew rich on the interdependence.

You can put your hand on that network. Off the coast of what's now Turkey, a ship went down in the fourteenth century BC, and archaeologists excavating the Uluburun wreck brought up copper and tin ingots in roughly the proportions used to make bronze, along with glass, ivory, jewelry, and pottery. Some of the copper had been cast into the four-cornered shapes called oxhide ingots. Institute of Nautical Archaeology, Uluburun excavation It was an elite shipment: raw materials for wealthy workshops and luxuries for wealthy households. A rich network isn't the same thing as a secure household, and even then, people could tell the difference.

Around 1200 BC, that world came apart the way complex systems fail: not from one blow but from interacting pressures, drought, conflict, migration, political breakdown, and interrupted trade, unevenly, in different places at different times. Many of the great palace centers of the eastern Mediterranean were destroyed or abandoned, and in Greece writing fell out of use for centuries. Knapp and Manning, Crisis in Context There was no single year the Mediterranean switched off, and not every place fell the same way. But the most specialized, most interconnected palace economies had a great deal to lose when the tin stopped moving.

The mechanism. Interdependence is efficiency in fair weather and contagion in foul. When a critical input stops arriving, everything downstream of it stops being possible. Neither localism nor size guarantees survival; knowing your dependencies and having some ability to substitute for them is what helps.

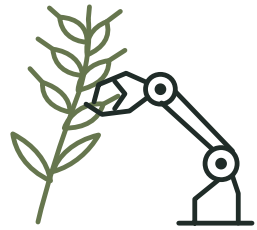
The rule. Name the dependency and the disruption you're preparing for. Then decide, in advance, which side of the cascade your street is on.

The practice.

1. Map your tin. List the five things your household can't function without for a month: a medication, a fuel, a staple food, a spare part, a service. For each, trace where it actually comes from, as far up the chain as twenty minutes of looking will get you. Pick the most fragile line and build one local or stored alternative for it this quarter.
2. Take the neighbor census before you need it. Write down, or better, learn in person, which people within walking distance can fix an engine, stitch a wound, grow food, wire a panel, keep bees, weld. If you can't name five, that's your project for the season. The mesh network is mostly people.
3. Adopt the fair-weather buying rule: when two options are close, take the one that's repairable, local, or standardized over the one that's optimal and single-sourced. You're not paying extra for a product. You're paying for which side of the cascade your household sits on.

PART III

How to Survive the Transition



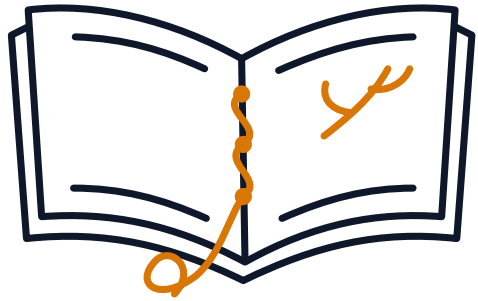
The shift is already under way. It started with a 2017 change in how machines learn, and my bet is that by 2027 these systems will be doing serious work on improving themselves. Most people are responding by panicking or by waiting for someone else to handle it. That's a mistake.

Surviving this isn't about predicting the future. It's about building something that works whatever the forecast turns out to be. Begin with a person who needs something: a dinner they can use, a place to live, a repair, a safer way to work. Follow that need through the people, tools, land, and agreements that could meet it.

You can take part without owning land, a rack of servers, or a robot. You can also be the person receiving help. This is how you build your own path, step by step.

CHAPTER 10

The "Create Over Consume" Protocol



Make something another person can use. Original illustration for this book.

> *"The best way to predict the future is to invent it."*
> Alan Kay

In this chapter:

- The line between the people who come through this and the people who get flattened by it isn't talent. It's a habit: create more than you consume.
- Nobody has to earn dinner by producing content. Making things is a freedom, not a toll.
- The algorithm can be ridden, but it can't be impersonated. Your unfakeable advantage is something that actually happened, told by the person it happened to.
- Turn the co-op's work into signal. The signal can help pay for the work.

The co-op needed money the way every real thing needs money, not dramatically, just constantly, and in Elijah's first spring there it was Denny, of all people, who said the obvious thing out loud at a Friday dinner: "People pay for what we know. We give it away to anyone who walks into the shed. Why does the walking matter?"

So they decided to put the co-op on the internet, and Elijah, who had spent years studying exactly how the attention machine worked, who had pages of arithmetic on engagement in his notebook, volunteered to run it. This is the story of how he was wrong in a brand-new way, which by then he'd learned to recognize as the shape of his education.

He did everything correctly. That was the problem. He studied what performed: the pacing, the thumbnails with the circled arrows, the seven-second hook, the open loops ("what we found in the soil SHOCKED us, but first"). He wrote Denny a script for the greenhouse-controller story that was, by every metric he knew how to measure, optimal. Denny read four lines of it on camera, stopped, and looked over the phone at Elijah like a man being asked to wear someone else's dentures.

"Nobody talks like this," Denny said.

"The algorithm," Elijah began.

"I'm not talking to the algorithm. I'm talking to a guy on his couch who just got the letter I got in 2019." He put the script face-down on the workbench. "Point the phone. I'm just going to say what happened."

He told it in one take, hands in frame the whole time because he talks with them, the plasma table ticking as it cooled somewhere behind him. What happened was eleven years in a warehouse, the arms coming in, the county's folder, the second ladder, the second folder. And then the part Elijah hadn't scripted because Elijah didn't know it yet: what it was like, at fifty-one, to bolt together a machine that *feeds people* and feel the specific weight of a tool in your hand versus a lanyard around your neck. "Everybody's asking what the machines are going to do to us," Denny said to the phone, at the end. "Wrong ques-

tion. I spent years as the answer to that question, and it nearly killed me. The right question is what we build while everybody else is busy asking. Here's ours. Come see it. Bring your kid."

Elijah posted both versions, because he was still a data scientist in his soul and you always run the A/B.

The optimized cut did what optimized things do now: it landed in a feed that felt, to him, mostly machine-written already, synthetic scripts read by synthetic voices over synthetic pictures, an ocean of beige competence, and it *matched the ocean*. Forty-one views. He watched it between the other videos in his feed and heard the resemblance: the same urgent pauses, the same little promises withheld until after the next cut. He had spent two weeks teaching a human being to sound like a format.

Denny's single take, shot vertical with a thumb over the mic for the first three seconds, moved the way true things move now: slowly for a day, then handed person to person like contraband. The comments weren't comments, they were *testimony*: forklift drivers, paralegals, a woman who'd trained her own replacement twice. Several wrote that a friend had sent the link. One asked where to get the controller's repair card. Elijah couldn't turn any of that into a law about what the internet rewarded. He could answer them.

Elijah watched the numbers climb for a while, and then, Sal's rule, closed the laptop and went to find Denny, who was in the shed fitting enclosure lids, unaware he was becoming a public figure at roughly a thousand people an hour.

"It's working," Elijah said. "The real one. Not mine."

"Course not yours." Denny didn't look up from the bench. "You tried to make bait. Bait's a picture of food." He fitted another lid. "I put out food."

They made it official at the next Friday dinner: the channel was Denny's. Elijah ran the pipes: the posting, the backups, the links to the free plans and templates riding under every video like cargo under a truck. The voice, the face, the editorial line, the *ownership*, were Denny Osei's, and the co-op wrote down how revenue would be

shared and how members could keep their lives out of the footage. Denny could say no to the camera any day he wanted. Years after a spreadsheet deleted him from a warehouse, he owned a distribution channel the size of a small radio station, built on the one asset no model could inflate away: he'd been there, it had happened to him, and you could hear it.

That evening he recorded a short correction to the repair card. The first version had named the wrong terminal. No hook, no music, no suspense. He pointed to the screw and said which one he meant. Elijah sent the corrected card directly to every grower who'd received the first version.



THE FOUNDATIONS

If your days go to consuming content about things you can't control, the days are gone and nothing's changed. The line between the people who come through this period with some footing and the people who get flattened by it is a habit: **create more than you consume.**

Before anything else, the fine print that isn't fine print: nobody earns dinner this way. There's no moral accounting in which a person who makes more deserves to eat more. Reading, resting, watching, and receiving care are parts of a life. A restaurant needs people who want to sit down and eat; a musician hopes somebody will listen. We don't need a new contempt for receiving in order to value making. Create over consume is a way to take your time back, not a toll.

You don't need deep pockets to start. You need a result you can point at.

Step 1: Start with your hands

Change your surroundings before you try to change your thinking.

1. **Clean something up.** Find a neglected stretch of your neighborhood: a park edge, an alley, an empty lot. With permission where it's needed, clean it, and finish the whole route, not just the photo: collect what the local trash and recycling service will take, get it into that stream, and come back in a week to see whether it stayed clean. Chapter 17 has more on doing this well.
2. **Document it.** Take before-and-after photos. Write down what it took. That's the beginning of turning raw reality into something you made.
3. **Optional: own a few watts.** If you'd like, a small solar panel with a charge controller and a battery pack sized to keep a phone and a radio alive is a good first build. Get someone who's done it to walk you through it, or buy a ready-made kit. It's a project, not an entry fee.

Step 2: Civic connection

The time you spend on digital fear is time you could spend in your physical community. Find a place that needs human hands: a community garden, a food pantry, a neighborhood center.

- **Direct exchange.** Show up and work. Labor moves you from the world of digital opinion to the world of physical results.
- **The human filter.** Working alongside people is how you find the ones who also want to be useful. Make the introduction wherever you can, then give it a shared task.

Step 3: Find your node

As the habit grows, you'll need a place to work. That doesn't mean renting a shop. It means finding space nobody's using.

- **Look for underused spaces:** barns, empty garages, forgotten community halls, the church basement that's dark six days a week.
- **The intergenerational exchange.** Find an older neighbor whose house has more room than they use. Offer real help: fixing the leak, doing the yard work, driving to appointments. In exchange, ask to use the spare room or the garage as a workspace. You get space; they get reliable help and company. Write down what each of you is agreeing to, so it stays a good deal for both.

Share a result people can judge

Denny's first useful act on camera was to stop performing someone else's idea of him. He had knowledge a machine could help him spread, and the right to decide whether he wanted to spread it at all.

When you share what you made, say what you tried, what it cost, what happened, and what you still don't know. A failed attempt with clear conditions can save the next group real money. And get consent before someone's hardship becomes part of your story. People don't owe the internet their hardest day in exchange for help.

The slop-ocalypse

The internet is filling up.

If you've spent any time online lately, you've felt it: search results padded with machine-written affiliate pages, feeds full of uncanny synthetic images, comment sections where bots argue with bots. Call it the slop-ocalypse. And underneath the annoyance is a real feedback loop with real math.

A large language model learns the statistics of the text it was trained on, the probability of each next word given the ones before, and it generates by sampling from what it learned. Now flood the internet

with that model's output, and train the next model on the flooded internet. In 2024, a team led by Ilia Shumailov published a study in *Nature* showing what happens when models train, generation after generation, on data produced by earlier models. Shumailov et al., AI models collapse when trained on recursively generated data

Picture one simplified training lineage. Call the real human data distribution $P_0(x)$. Each new generation k learns from finite samples of the one before:

$$P_k(x) = \int q_k(x|y) P_{k-1}(y)\, dy$$

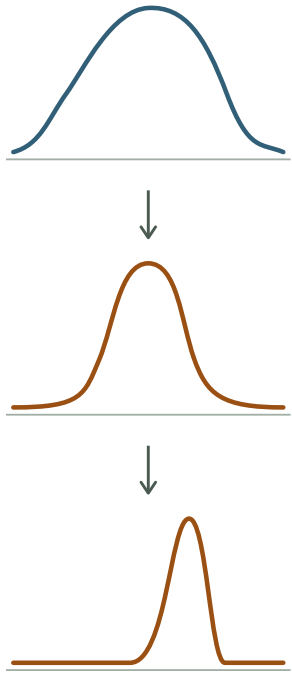
Here q_k is a schematic transition kernel for the training and sampling process, not the paper's general equation for a language model. Under the repeated resampling conditions the study examines, two things can happen. This isn't a claim that every use of synthetic training data causes collapse; selection, checking, and access to real data change the setup.

The tails get pruned. Models favor high-probability outputs and under-sample the rare ones. Over generations, the spread of the distribution shrinks. The weird hand-made engineering trick, the specific local history, the odd brilliant idea, those live in the tails, and they're the first to go.

The errors compound. A model's small biases become the next model's training data, and the next. Some of the paper's simplified examples converge toward a narrow spike. That spike needn't sit at the original average; a distribution can lose its variety and drift at the same time. A copy of a copy of a copy.

WHEN A MODEL EATS ITS OUTPUT

One possible lineage under repeated sampling.



Starting examples

A range of cases can be represented.

Generated samples

Rare cases can be lost in resampling.

Possible collapse

Variety can shrink; errors can drift.

Schematic curves, not a generation timetable.
Real data, sampling and training all matter.
Shumailov et al., Nature (2024).

One possible failure in recursive training on generated samples. This schematic is not a universal outcome or a timetable; real data, sampling and training choices affect the result. Based on Shumailov et al., Nature (2024).

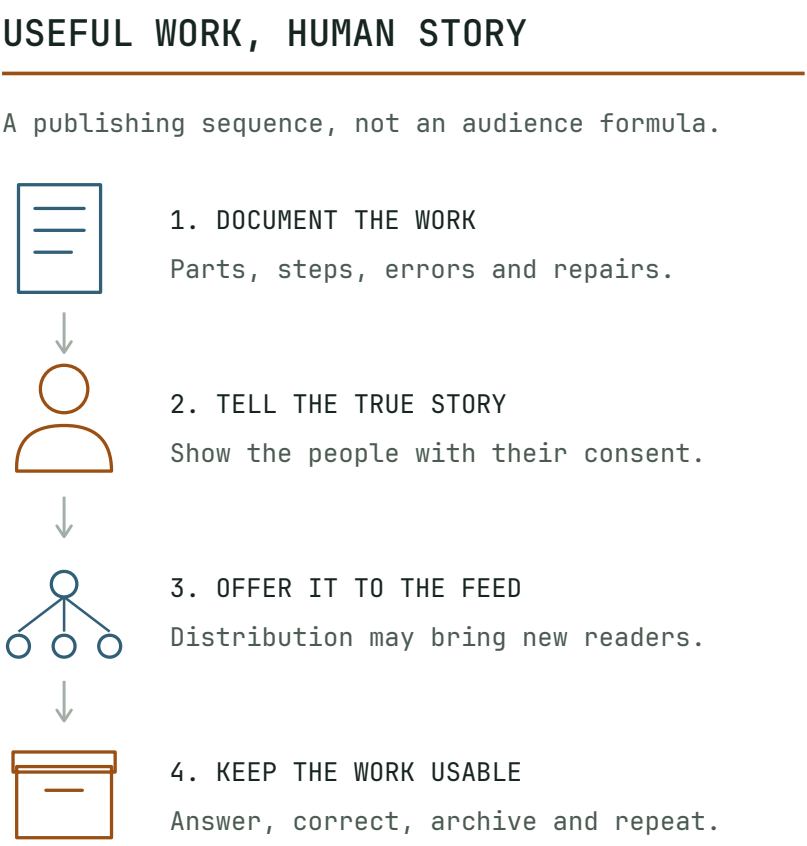
The practical conclusion isn't that the machines are doomed. The labs know about this, and they work hard to keep real human data in the mix. It's that real human data just got more valuable. Something that actually happened, recorded by the person it happened to, is now the scarce input. That's your leverage.

And the energy bill is real too. Generating another answer takes computation; keeping a useful answer lets you reuse it. A paper book on small-engine repair needs no powered device to read, and a saved offline manual needs no new model query. Keep the good instructions where you can reach them again.

Algorithmic judo

Hiding in the woods with paper books isn't the plan, though. Millions of real people are still in the feed, scrolling past the synthetic flood, looking for any sign of an actual human. You can reach them. You just can't do it by impersonating the machine.

Post a technical document, "How to Wire a Small Solar Kit and Run a Local Model," and the person who needs it may never see it. A true account of why you built the kit can give that person a reason to stop and read. That's the bet behind algorithmic judo: wrap useful, practical knowledge inside a human story, then hand the reader something real. Try it, watch who actually uses the result, and don't confuse one successful post with a law of the feed.



Document useful work, tell the true story with consent, and keep the instructions available. A platform may bring readers; a post is not guaranteed reach or income.

Three rules.

Rule 1: The hook has to be true. Give someone a reason to care about the instructions. “Here’s how to test your soil” can become the account of the bed that failed and what the test found. Don’t manufacture outrage; the cargo has to be true, or the whole thing is just

another picture of food. Write what happened: "I retrained twice and the ladder moved both times. Here's the machine we built instead, and here's how to build one." That's Denny's video.

Rule 2: Pivot to the practical. Once they're reading, move quickly from the story to the steps: the parts list, the wiring diagram, the schedule, the phone number of the food pantry. The reader came for the person and stays for the thing they can use.

Rule 3: Point off the platform. Don't end on "like and subscribe." End with something physical: "Write this on an index card. Put the phone down. Find two neighbors who'll help you try it this weekend." The feed is a temporary delivery truck, not the destination.

The analog blind spot

A platform can log every click that crosses its servers. It can predict your next purchase with unsettling accuracy. But it has a big blind spot: the analog commons. A conversation without a recording device, a hand-drawn plan that stays on paper, a morning traded for a box of tomatoes: those don't automatically create a platform record. Photograph them, record them, or put the details in an app, and that boundary moves. (In the United States, barter generally creates taxable income at the fair market value received; the IRS distinguishes some informal, noncommercial exchanges of similar services. IRS, Taxable and Nontaxable Income Outside a feed isn't outside the law.)

That's the real play: use their expensive digital network as a temporary, free distribution channel to build durable physical ones. Their silicon, our stone.

Make the handoff small

If the thing you made helps, leave behind something another person can actually use: a corrected pickup schedule, a short guide, a labeled record, a contact who agreed to be contacted. If it fails, write down why and stop spending on it until there's a reason to try again. Don't create a project on someone else's behalf that they now have to maintain.

The next chapter introduces your robotic court: machines that work for you, on hardware you can own. Its first job is food.

— ◈ —

Precedent P-13: The Abbot and the Press (Sponheim and Mainz, 1492 to 1494)

Forty years after Gutenberg, the abbot Johannes Trithemius wrote *De Laude Scriptorum*, "In Praise of Scribes." Hand-copying, he argued, was devotional work worth keeping, and he doubted that paper would last as long as parchment. He also saw that a printer would never bother with plenty of texts that still deserved to survive. History of Information, Trithemius

Then he wanted people to actually read his argument. So he had it printed: Peter Friedberg's press in Mainz, 1494. Paderborn University Library, 1494 printed edition

The legend calls that hypocrisy. It's something more useful. Trithemius didn't think printing was evil; he thought it didn't do everything, and he used it for the job it did well. And the old skills weren't gone either. Forty years earlier, in August 1456, a craftsman named Heinrich Cremer wrote his name into a paper copy of the Gutenberg Bible now in France's national library, recording that he'd done its illumination, rubrication, and binding. Bibliothèque nationale de France, Gutenberg Bible The press made the text. A human hand still finished the book. The unit of change is often smaller than the whole occupation: one task moves, another stays, and the terms change for the people doing them.

The mechanism. Watch what a new machine's critics do, not only what they write. Using the new tool while defending what the old one did well isn't hypocrisy; it's the normal sequence. The question is always what each method is actually for.

The rule. Choose a medium by the job it does. You don't get to sit out the tools of your era on aesthetic grounds; you only get to choose whether you use them to make something or merely to complain.

The practice.

1. Run the abbot test on yourself, once. For one week, tally two numbers each day: hours spent consuming commentary about the transition, and hours spent making anything at all. Don't moralize the numbers. Just get them. Your ratio is your actual position, whatever your opinions are.
2. Ship one made thing this week using a tool you have complaints about. A document, a repaired fence, a video, a loaf, a raised bed. Size doesn't matter. End the week as someone whose hands touched the new world instead of someone who only reviewed it.
3. When you read the next elegant essay about AI ruining everything, check the workflow before you absorb the thesis: how was it written, published, and distributed? Then keep whatever part of the argument survives. The abbot was right that some things deserved hands.

CHAPTER 11

King and Queen of Your Own Robotic Court



Spot. The same machines ending careers will happily work for you. (Jonte, CC BY-SA 4.0, via Wikimedia Commons)

> *"Give me a place to stand, and I shall move the Earth."*
> attributed to Archimedes

In this chapter:

- The same class of machine that's ending careers can work for you, on hardware the corporate upgrade cycle throws away. Its first job at your court: food.
- Every cloud query is a page of your diary mailed to the company store. Local compute is one of the last private places left to think.
- A food baseline and a restaurant can coexist. The meal is free at the door and paid for somewhere.
- One careless config line published nineteen households of ordinary life to the open internet for 41 hours. The discipline is part of the tool. So is the honest postmortem.

The servers arrived in mud season, in the back of Denny's truck: three retired enterprise machines from a county surplus auction, five years old, built like filing cabinets, sold for less than the co-op had spent that month on welding gas. Elijah had argued for the purchase all winter, and the argument that finally landed was not technical. "Every question we ask the cloud," he said at a Friday dinner, "goes out with an envelope of context around it. What we're building, what we're worried about, what we don't know yet. We're mailing our diary to the company store, one page at a time, and paying the postage."

Marta cared about one number, watts, and he had that answer ready, because a $211 power bill had taught it to him before anyone here knew his name. Priya wanted help searching the plant-pathology references she already trusted. Reuben wanted a first pass through forty pages of easement language before he brought his questions to counsel. Curtis wanted nothing to do with any of it, and said so, and that mattered later.

The rack went together the way everything in the shed went together: Marta on steel and power, Elijah on software, everyone else on opinions. Two scavenged compute cards with a combined forty-eight gigabytes of memory. A salvaged Civic radiator bolted to the shady north wall, two brass bulkheads through the corrugated steel, a marine pump the size of a fist. And on the first cold boot, fans screaming like weather, they loaded the model: a granddaughter of Silvana, open weights, three generations downstream of the thing that had started all of this, small enough now to live on scavenged silicon and capable enough to suggest wording, locate a likely bug, or turn a confusing manual into questions somebody could check. Elijah sat with that for a minute in the dark of the shed. The fire that had burned down his old life, purring in a stove he had built.



The Crucible. Three filing cabinets and a salvaged radiator.

For five weeks it was purely good. Reuben fed it easements and it found a cross-reference he'd missed; he checked it against the document and brought a sharper question to counsel. Priya's intern learned blight triage from it in an afternoon, then caught it inventing a source and wrote that failure on the board beside the successes. It never got tired and it never billed and it never, not once, phoned home, because there was no home to phone. The electricity meter kept turning. Elijah kept a log of what the machine had saved them and what checking it had cost.

He started calling the rack the Crucible. Priya said that was fine as long as the things that didn't survive testing stayed in the log.

The breach began with an ordinary convenience. Their system had no reliable stop between that convenience and everyone else's private life.

Elijah wanted his dashboards from town. Uptime, coolant temperature, the channel numbers, all of it on his phone while he sat at the library uploading Denny's videos on borrowed bandwidth. Fifteen minutes of work on a Friday night: a tunnel out, a reverse proxy in front of the co-op's services. While chasing a bug he commented out the authentication line, two slashes, to rule it out as the problem. He meant to put it back. He deployed with the comment marks still in.

What the proxy served, to anyone on earth who looked, was the mesh journal: the plain-text nervous system of nineteen households. Maintenance logs. Meeting minutes. The duty roster. The CSA customer list with addresses. Priya's greenhouse alarms. Marta's physical-therapy schedule, Tuesdays, for the shoulder she never mentioned. Which houses stood empty on which afternoons, computable by anyone who could read a grid log. All of it indexed, unauthenticated, public, for forty-one hours.

It was one of Denny's subscribers who caught it, a network engineer four states away, with a screenshot and one line: "you probably want to lock your journal, brother." Elijah killed the proxy before he finished reading the sentence. Then he pulled the access logs and did the arithmetic he would have paid anything not to do. Three crawlers had been through. You can delete a mistake from your own server. The internet does not run your deletion for you.

He thought about carrying it alone for exactly as long as it took to imagine Marta finding out some other way. Then he rang the dinner bell on a Sunday afternoon, which the co-op did not do, and stood in front of everyone it summoned and read them the timestamps.

Marta did not raise her voice, which everyone in the shed understood to be the worst available outcome. "You published which houses stand empty between two and four on a Tuesday," she said. "At the plant we had lockout-tagout. You don't service a press that can take your hand off until its energy sources are isolated, the stored energy is dealt with, and the lockout is verified. Your name goes on your lock. You put our hands in the press, college."

Curtis said his piece, and for once it had teeth: this is what happens, you brought their machine inside the wire, pull the plug and melt it down. Heads were nodding. Marta let them nod for a moment.

"The first family that ever kept fire burned something down," Marta said. "A roof, a winter store, somebody's child. And nobody handed the fire back. They built the hearth. Stones in a circle, rules about

who tends it, a bucket that never goes empty." She looked at Elijah, and it was not a rescue. "We didn't get burned by the model, Curtis. We got burned by a boy who left it lit on the porch rail."

Priya, who had been quiet, set down her cup. "Pulling the model out won't change which service he exposed."

"He built it," Curtis said.

"Yes. And we let one person change the thing that held all our addresses. We put information together without asking whether it needed to be together. Then we treated his confidence as our check." She looked at Elijah. "You owe us the truth about what happened. We owe ourselves a design that doesn't depend on you never having a bad Friday."

Reuben spent the next two days mapping the exposure with counsel and drafting a letter to every family on the list. The lawyer checked the notice requirements; Reuben checked the addresses and what each family needed to know. "We're not waiting for somebody to prove we owe them the truth," he said.

It was Denny who found the judo in it. "People already know things break," he said. "What they never once get to see is somebody stand up and say so." After the affected households reviewed what could be shared, they published an account on the channel: the categories of information exposed, the duration, the notification process, and the changes. The private records stayed out of the report. Elijah read it to the camera himself, because it was his name on the lock. It traveled further than anything they had posted except Denny's first video. Strangers wrote to say it was the first incident report they had ever read that was not written by lawyers. Two of them drove out that summer to help harden the mesh for free.

At that first Sunday meeting, the co-op had adopted a new rule: a change exposing a service to the internet needed a second reviewer and a recorded check of what an outsider could reach. They separated the private records from the public material and assigned someone besides Elijah responsibility for reviewing access. Marta called it borrowing a habit from lockout-tagout. The analogy gave

them a place to start; the network needed its own procedures. The model server came off the external network. Updates arrived by hand after that, with the source and file checks recorded before installation, the disk carried home in Elijah's jacket. The team still had to inspect what it brought into the room.

The pump kept ticking. The radiator clicked against the north wall. When Reuben next asked the machine to summarize a document, he still had to read the document. The leak had not taught Elijah machine learning; he had arrived knowing that. It taught him the difference between knowing a system and operating one, and the tuition was forty-one hours of other people's privacy. The machine went on drafting and checking. Fire in its stones.

THE FOUNDATIONS

Useful help within reach

King and Queen of Your Own Robotic Court is a playful name for a serious change in who gets help. I want the person who needs assistance to be able to get it without first becoming rich, technical, or employable in this year's fashionable field.

The first assignment is food. A farm machine, a kitchen, a delivery service, some planning software, and the people who keep each of them running might together provide it. The person eating doesn't need to own every piece, any more than someone reading a library book needs to own the printing press. What matters is a usable claim on the result. Can you get adequate food when your income stops? Can you ask for a form you can actually eat? Can you get an answer when a delivery fails? Can you complain without losing your next meal?

You don't need to be an engineer to put these tools to work. You need to be persistent. Treat a good model as a tireless research assistant. If the answer is generic, argue with it. Push it to be specific. Ask it for

the procedure, the part number, the code section. Then check what it tells you against the source, because it will sometimes make things up with a perfectly straight face, the way it did for Priya's intern.

- **Teach yourself.** Use it to learn the technical task in front of you: how to read a wiring diagram, how to troubleshoot a pump, how to set up a simple spreadsheet that tracks deliveries. Make the machine explain, not just do.
- **Read the ground.** Use it alongside public maps and records: county parcel records, zoning codes, soil surveys, water maps, the Bureau of Land Management's land records where they apply. Once you understand the rules of the place you live, you stop being surprised by them.

Free to eat, paid for somewhere

The line cook from Chapter 7 still matters here. He's lost his job. A restaurant can keep selling meals, and a funded food service can give him dinner. There's no contradiction between those two arrangements. There's an operating budget to arrange.

Here's one design to argue about, not an existing program. A local public body funds a food partnership for a set period. The partnership pays growers for agreed supplies and a kitchen to prepare meals. It pays for delivery, maintenance, and the human work that's still needed. People collect or request the meal without paying at the door. Some of the participating kitchens could be restaurants, earning money for the work.

A co-op could run something similar on shared equipment. An existing food bank could add useful automation to work it already does. The form matters: it decides who's in charge, where the money comes from, and what people can claim when it breaks. "Cooperative" isn't a budget, and "public" isn't accountability. The money might come from a public allocation, pooled contributions, purchased services, or a mix. Donations can start a project. They shouldn't quietly carry a permanent promise nobody's actually funded.

A food baseline would change some things. Some restaurants might need better pay, different hours, or a different business model. Some could close. I can't promise every existing margin survives. But providing dinner doesn't require abolishing paid meals, and a worker's hunger shouldn't be the protection a business model depends on.

A public meal, and an argument about it

On May 28, 1941, the House of Commons discussed a practical obstacle to feeding people during wartime: some local authorities had not prepared communal feeding centers. The Ministry of Food had assigned officers to help establish British Restaurants. But encouragement was one thing and authority another. When an MP asked whether the ministry could establish a restaurant where a council failed to act, its parliamentary secretary could not give a definite answer. Even an urgent national purpose needed someone empowered to carry it out locally. Hansard, May 28, 1941

The restaurant objection was there too. On October 1, Sir Percy Hurd challenged the treatment of private caterers alongside the public service. The ministry replied that rationed supplies were allocated on the same basis, according to meals served, while some scarce foods received priority for feeding workers. Public kitchens and private businesses existed together; their terms were already being argued over. Hansard, October 1, 1941

The next day's debate sharpened both sides. A critic with catering interests alleged that British Restaurants were taking customers from struggling businesses. He also described a public restaurant at Fishmongers' Hall that opened at noon, too late for the Billingsgate workers he wanted it to serve. These were claims made in debate, not an independent audit. But the second complaint gives the proposal a useful test: a kitchen can exist, have supplies, and still miss the person whose working day it was supposed to help. Hansard, October 2, 1941

British Restaurants charged for meals. In March 1942, the ministry described prices covering food, preparation, overhead, and a contingency margin, with accounts submitted for review. This was no universal entitlement to free dinner. It was a working example of public authorities organizing meals and accounting for the work. Our proposal changes who pays at the point of eating. It still needs the kitchen, the budget, and opening hours that fit a human life. Hansard, March 17, 1942

The objections, answered once

You'll hear these. Here's how I'd answer each of them.

"Won't having robots clean up after us make people stop caring?" It could, if we build a service that encourages carelessness. That's a real concern, and we should watch what actually happens. We can make disposal easier, keep trash from escaping in the first place, and clean up what still gets through, then check whether the place stays cleaner overall. The trash doesn't tell you how it got there, either: wind, runoff, and overflowing bins carry plenty of it, and deliberate dumping happens too. Different problems, different responses. Leaving trash in a river is a poor way to teach responsibility.

"Isn't this how we end up like WALL-E?" The worry underneath is worth taking seriously: dependence, lost skills, a life with fewer choices. I want help that gives us *more* room to move, learn, make things, care for each other, and take part. Some people will need more help than others, and receiving it doesn't make them less worthy. Whether a tool widens a life or quietly shrinks it is something we can watch and change.

"We already know how to do this without robots." Yes. A robot doesn't have to be the answer. Sometimes a better bin, an existing crew, or a simpler machine does more good. But knowing how to do something isn't the same as making it available, affordable, and dependable everywhere it's needed. Compare the options and use whatever does the work well. There are many ways to make an omelette.

“But what about the jobs?” Dishwashers and washing machines make the idea of handing repetitive work to a machine familiar. That comparison doesn’t pay the bills of someone whose income just disappeared, and their concern is real. Workers belong in the decisions about how these tools come in and how people are supported through the change. That’s what this whole book is arguing for.

“Why not send machines into the dangerous work?” Where a suitable machine can keep people out of harm’s way, that’s a good reason to look hard at it, with the people who know the job. The deployment still has to show that it helps, including the work of maintaining and recovering the machine.

Ownership matters because access can be withheld



Capability still needs an arrangement for access. AI-adapted cutout from Jonte, CC BY-SA 4.0.

Suppose the machines work beautifully. Now suppose one company owns them, controls their supplies, and can cut off service when a subscription lapses. Productive capacity went up. Security for the person who can’t pay may not have moved at all.

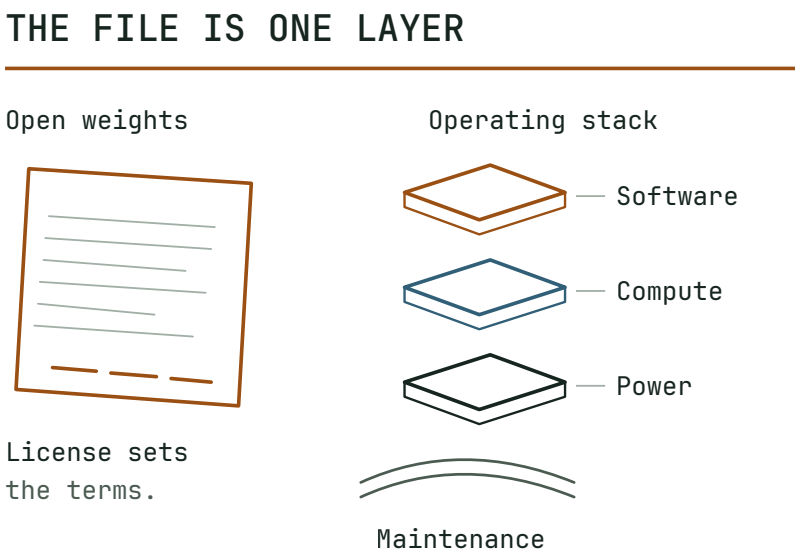
So whatever the arrangement, shared ownership, public provision, or paying private providers, ask who can raise the price, change the rules, withdraw the service, or sell the equipment, and how the people receiving it and the people working it can challenge those decisions. And don’t let any one program become the only door to help. Several providers, supplies held back for bad weeks, repairable equipment, and routes that don’t need a proprietary account all make the promise sturdier.

The Crucible: local compute you own

The scene’s argument for owning the machine was privacy, and it’s a good one. Every question you send a cloud service carries context with it: what you’re building, what you’re worried about, what you don’t know. For a lot of work that’s fine. For some of it, a food pro-

gram's recipient list, a family's medical questions, a co-op's legal strategy, you'd rather it never left the building. Local compute is one of the last private places left to think.

You don't need enterprise money. Every few years, data centers retire perfectly good hardware onto the used market, and the co-op's rack came from exactly that: a county surplus auction.



The model file is one layer of a working service. Running a useful service also takes software, hardware, power, and people who maintain it. Conceptual illustration.

The memory math

For running language models, memory is one of the first limits to check. Keeping the weights in graphics memory (VRAM) avoids slower transfers, although some tools can divide work between GPU and system memory. Start with an estimate:

> $V_{RAM} \approx ((P \cdot Q)/8) \cdot B$ [GB]

where P is the number of parameters in billions, Q is the nominal bits stored per weight (16 for a common unquantized format, 4 or 8 for quantized versions), and dividing by 8 turns bits into bytes. The examples use $B = 1.2$, an illustrative 20 percent allowance, not a guarantee. Quantization metadata, the conversation's key-value cache, context length, and simultaneous users all change the actual requirement. llama.cpp, server memory and context options

HOW MUCH MEMORY FOR WEIGHTS?

Derived arithmetic · decimal gigabytes

	4-bit	8-bit	16-bit
8B parameters	4 GB	8 GB	16 GB
70B parameters	35 GB	70 GB	140 GB

GB = billions of parameters × bits / 8

Weights only. Runtime needs more.
Cache, context, buffers, quantization metadata,
and simultaneous users add to the requirement.
Source: Chapter 11; llama.cpp documentation.

Memory before the overhead. These are derived weights-only sizes, not measured VRAM requirements. They omit the chapter's illustrative allowance as well as runtime costs. llama.cpp, memory and context options.

A small model at 16 bits. An 8-billion-parameter model:

> $V_{RAM} = ((8 \cdot 16)/8) \cdot 1.2 = 19.2$ GB

That estimate fits within one 24 GB card. Check the actual model and intended context length before buying one.

A large model, compressed. A 70-billion-parameter model at 4 bits:

$$V_{RAM} = ((70 \cdot 4)/8) \cdot 1.2 = 42 \text{ GB}$$

Two 24 GB cards provide 48 GB in total, which is what the co-op scavenged. Tools like llama.cpp can split supported models across cards, but adding the labels on the boxes doesn't guarantee a working configuration. Test the model, memory use, and speed under the workload you intend. Once the software and weights are downloaded, local inference can run without an internet connection.

The heat math

A rack drawing 600 to 1,000 watts turns roughly that much electrical power into heat while it runs. Measure your own load; retired hardware doesn't share one power rating. Inside an insulated building, that's heat you're paying an air conditioner to fight. The co-op's answer was to carry it outside in liquid. The physics is one line:

$$\dot{Q} = \dot{m} \cdot C_p \cdot \Delta T$$

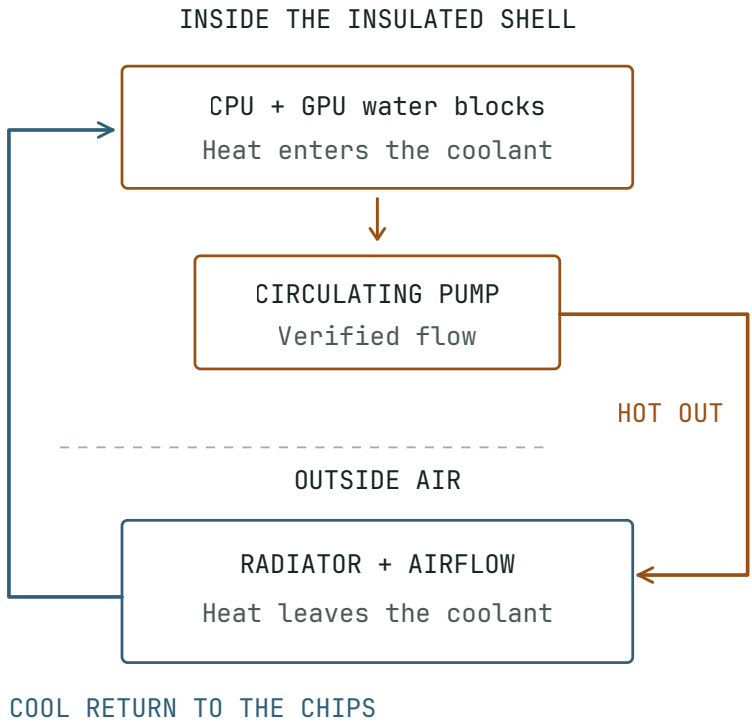
To move 1,000 watts with a water-glycol coolant ($C_p \approx 3,800$ J/kg·°C) while letting it warm only 5°C through the loop:

$$\dot{m} = 1000/(3800 \cdot 5) \approx 0.053 \text{ kg/s} \approx 3.2 \text{ liters per minute}$$

That's the required flow for these assumptions, not a pump specification. The pump must deliver it against the loop's resistance, and the radiator needs enough area, airflow, and temperature difference to reject the heat. Measure coolant temperature under load before relying on the arrangement.

MOVE THE HEAT OUTSIDE

Conceptual loop; not a component specification.



Heat rate = mass flow × Cp × ΔT
W = (kg/s) × [J/(kg·°C)] × °C
Match materials, coolant, pump and radiator.
Illustrative loop; verify load and heat removal.

Move heat out through a compatible, verified coolant loop. Heat rate uses mass flow in kilograms per second: $\dot{Q} = \dot{m}\, C_p\, \Delta T$. This diagram specifies neither components nor cooling capacity.

Choose the boundary before the machine

The Crucible breach started with an access failure. Other failures start with an unreliable answer, the wrong model for the job, or a decision the system was never fit to make. Good intentions settle none of those.

Give a tool work that can be checked before its mistakes become someone else's emergency. A model can propose a delivery schedule; a coordinator checks the vehicles and what people actually need. A model can find a relevant passage; a person reads the source. A confident explanation doesn't authorize a machine to decide that food is safe or equipment is ready to run.

And keep the words straight. *Local* means it runs on nearby equipment. *Offline* means it works without a connection. An *air gap* separates the protected system from outside networks, with no wired or wireless network path across the boundary; moving data across it is a deliberate human act. NIST glossary, air gap None of those means the software is right or the data is safe. Keep the system patched, from sources you've verified. Collect as little private information as the job needs. Keep public and private services separate. Have a second person check anything that exposes a service to the internet, and test what an outsider can actually reach. The co-op learned that the hard way: a design shouldn't depend on one careful person never having a bad Friday.

Help should enlarge a life

Receiving a lot of help isn't a moral failure. Many of us will need it, and all of us have depended on work we couldn't do ourselves. The test is what the arrangement makes possible and what it demands in return.

A meal shouldn't require a performance of ambition. A person who wants to rest today hasn't forfeited dinner. A person who wants paid work should have room for that too. And somebody still has to maintain the equipment, recognize a bad result, and know when to stop.

Paying, training, and supporting those people belongs inside the program. Otherwise one household's help is being bought with somebody else's hidden exhaustion.

One request, all the way through

Before buying a machine, write down one request a person should be able to make. For example: "I need meals I can heat with one working hand, and I can't pay for them this month."

Follow that request through the service. Who receives it? Who confirms the food and how it's prepared? Who pays the people doing the work? How does the meal arrive? Who answers if it doesn't? Then put your proposed tool next to the one task it can actually help with. Check the route with a willing provider and with someone whose life the request describes. They may show you the missing piece is a kitchen slot, a driver, a phone number somebody answers, or a funding agreement. A robot may fit at one of those points as it gets more capable.

That's a court worth building: help with a destination in somebody's life.

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Precedent P-14: Quartz and the Assembly Line (Switzerland, 1962 to 1983)

The story everyone tells is that the Swiss, masters of mechanical watchmaking, sneered at the quartz watch and got wiped out by it. The real story is stranger and more useful.

The Swiss helped invent quartz timekeeping. A consortium of Swiss watchmakers founded the Centre Electronique Horloger in 1962, and its quartz wristwatch prototypes followed in 1967. CSEM, Historical Timeline Seiko put the first quartz wristwatch on sale in 1969. Then came the crunch, from several directions at once: cheap quartz movements, a strong franc, and an industry built around costly mechanical production. Seiko Museum, The Quartz Crisis and Recov-

ery of Swiss Watches Between 1970 and the mid-1980s, Swiss watch employment fell from roughly ninety thousand to about a third of that, and the number of firms collapsed with it. Knowing about the technology wasn't enough. The business around it had to change, and for a lot of people it changed too late.

What saved the remnant was a kind of heresy. In 1983 the merged Swiss group launched the Swatch: a cheap, plastic, fifty-one-component quartz watch, assembled on automated lines. Swatch Group, Company History It took the new technology to the bottom of the market, and the company that grew around it went on to own some of the most famous names in mechanical watchmaking at the top. The quartz Trojan horse helped pay for the craft it seemed to threaten.

The mechanism. Survival came from adopting the new technology on your own terms and using its output to support what you meant to keep. None of that erased the losses of the tens of thousands of workers who didn't make it through, and a big firm's options are never an individual's.

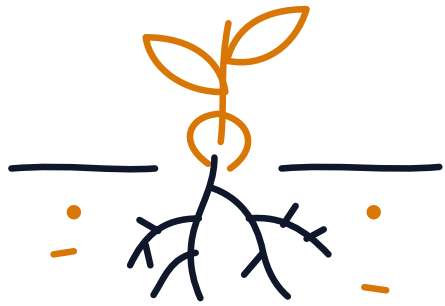
The rule. Don't beat the new machine by refusing it. Use it to pay for what you actually want to keep.

The practice.

1. Name your mechanical movement and your Swatch. On one line, write the thing you refuse to lose: the craft, the practice, the standard of work that's actually you. On the next line, sketch the cheap, fast, machine-assisted offering that could fund it. If the second line feels like a betrayal of the first, you've drawn it correctly.
2. Set the subsidy ratio and stick to it. Decide what fraction of your machine-assisted income funds the protected craft, time, tools, or savings, and write it down like a tax you owe yourself. Leverage without allocation is just drift.
3. Refuse purity tests, including your own. The next time someone says "real professionals don't use AI," translate it into 1975: "real watchmakers don't do quartz." Then remember that the Swiss were building quartz watches the whole time, and still nearly lost the industry. Adopting the tool is necessary. It isn't sufficient.

CHAPTER 12

The Land Strategy



A place to grow begins with access to the ground. Original illustration for this book.

> *"Land is the only thing in the world that amounts to anything ... for 'tis the only thing in this world that lasts."*
> Gerald O'Hara in Margaret Mitchell's *Gone with the Wind* (1936), Chapter II

In this chapter:

- Land is the one input the machine can't manufacture more of. Buying it is one route to it. Sharing, leasing, and trusts are others.
- The outfit circling your county runs a pipeline: estates, widows, cash offers, fourteen-day closes. Its strategy is speed and loneliness.
- A community land trust is enclosure run in reverse. One deed at a time, ground comes off the speculative market.
- Grandma's request, followed all the way to her door: what the neighbors do, what the machines do, and what reaches her kitchen.

The first letter was addressed to OWNER/OCCUPANT, and Irene Calder read it on her porch and threw it away, which is also what she did with the second one. The third letter came with a young man attached.

Everything the co-op had built stood on Calder ground. The shed had been Walt Calder's equipment barn; Priya's beds ran where his hay had run; the lease was month to month and had been since Walt died, because Irene liked Marta and had never once raised the rent, and nobody had wanted to disturb an arrangement that worked by being talked about over pie. The young man on the porch offered twenty-three percent over anything the county said forty acres was worth, all cash, a fourteen-day close, and a flyer with a sunset on it. Irene told him no. Irene's son Dale, in Phoenix, said, "Mom."

Elijah did what Elijah does when he is frightened: he pulled data. The county recorder's index was public, ancient, and slow, so he was polite to it, scraping at night, and then he did two neighboring counties for context. The pattern assembled itself the way patterns do, all at once, on the third evening. Eleven different LLC names, one registered agent, a mail-forwarding address in Sacramento. Fifty-some parcels in three years, always the same shape: an estate or a widow, a cash offer, a fast close. Then the parcel goes quiet inside a shell company for a year or two and reappears stitched into a block big enough to matter, leased to whoever needs flat land, water rights, and no neighbors with opinions. Reuben, reading over his shoulder, reminded him that a registered agent can represent a hundred unrelated clients. Elijah colored the uncertain links a different shade and kept going.

That Sunday, on the regular call, he asked his mother what the outfit was called, the one that bought the Hollis place back home. She had to go find the church newsletter to check. Same agent. Eight hundred miles apart, the same signature. He sat with the phone against his ear while she talked about the pump, and looked out at the shed, and wondered whether the same buyers were standing two fences away. He circled the link for Reuben to check.

He built a map. He built, God help him, a dashboard, and it was beautiful, and he presented it at Friday dinner with a laser pointer, and this time he heard himself doing it and grinned mid-sentence, and the room grinned back, because everyone at that table had been present for the greenhouse controllers. But the dashboard did its one load-bearing job: in the parcels he could trace, the time from first letter to recorded deed ran about nine weeks, and Irene had gotten her first letter five weeks ago. The data couldn't close the gap. It could only tell them how wide it was.

"It's not a data problem," Reuben said, already dealing folders around the table like cards. "It's a people problem wearing a paperwork costume. Fortunately," and here he almost smiled, "I have been waiting my entire adult life for this exact fight."

What he dealt out was a proposal for a community land trust. It would hold the ground under restrictions on resale, with residents represented in its decisions. The draft protected Irene's right to remain in the house and provided for regular payments under an agreement she would have her own adviser review. A long ground lease would give the co-op terms it could plan around. Reuben had also brought the less thrilling pages: taxes, insurance, maintenance, the balance still to be paid, and what happened if the arrangement failed.

The down payment would take the channel's first real surplus and the greenhouse-controller money. Future payments would have to come from future work. Denny looked from one column to the other. "So the machine that took my warehouse job," he said, "is helping buy ground it can never take back." He tapped the second column. "And we still have to keep the promise."

"Every month," Reuben said.

Dale attended the kitchen-table meeting by speakerphone, from Phoenix, and he was not wrong by his own lights. "Their number is higher, Mom. That's just math."

“It’s a bigger number,” Reuben said to the phone where it lay on the oilcloth, next to the coffee and the folders. “There’s a machine that’s been running since the fifteen hundreds, Dale. It turns commons into fences. Every century it gets a new name. Enclosure. Consolidation. Urban renewal. This week it’s a string of companies with one mailing address and a flyer with a sunset on it. It runs in one direction, and it’s never once run toward your mother’s porch. What’s on this table is that machine in reverse.” He let that sit. “Your mother gets to take the bigger number seriously. She also gets to decide what she wants to keep. What’s on this table is a different offer: staying here, receiving the agreed payments, and putting restrictions on what happens to the ground. Including restrictions on us. Those promises need review. They won’t end every possible dispute. They give us terms somebody can read and hold us to. That’s why we brought the folders.”

There was a long quiet on the line, the box fan working, Walt’s cap on its peg by the door. Irene had listened to all of it with both hands around her cup, and what she finally asked was not about the resale formula.

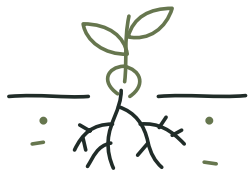
“Marta. Walt’s tractor, back when. What did you charge him?”

“Sixty,” Marta said. “It was the fuel line, not the pump. Dealer wanted nine hundred.”

“And it still runs,” Irene said, to the phone, to Dale, to the table generally, in the tone of a woman entering evidence. Then she drew the folders toward her. “Leave these. I want my questions written down before anybody signs.”

The review took longer than Elijah wanted. Irene asked questions he hadn’t thought to put in the dashboard. So did Dale. By the time they agreed, several pages had changed. Reuben carried the completed documents to the county himself, and Doreen at the recorder’s window stamped them at 11:04 on a Tuesday morning. The letters from Sacramento kept coming for a month. Irene brought one to dinner and left it unopened beside the salt.

The dashboard had been right about the timeline and the pattern, and none of it closed the deal. What closed it was a fuel line fixed honest in 2011, a stack of questions Irene got answered in writing, and a clerk who trusted the man across the counter. Elijah wrote the number in the notebook anyway, under MADRONE ≠ DRONE and Sal's rule, where the important arithmetic lived. A new line, a new unit of measure: PARCELS HELD IN TRUST: 1.



THE FOUNDATIONS

A place the promise can stand

When I finally pinned down my core thesis, I wrote it as a note to myself, and it still sums up the goal: every person should have what they need to meet their survival needs, either on their own property or through a city that manages that infrastructure better than it does now. This chapter is about the first half of that sentence, without needing to be rich. Chapter 19 is about the second.

A productive tool needs somewhere to work. A hungry person needs access to what it produces. Those needs are connected, but they aren't the same, and nobody should have to become a property owner before they get dinner.

Still, land is the one input the machine can't make more of, and whoever holds it holds a lot of the terms. So if you *do* have capital, think hard before you put it into a bigger house, status symbols, or a business that only works if the city stays healthy. Consider ground. Buy

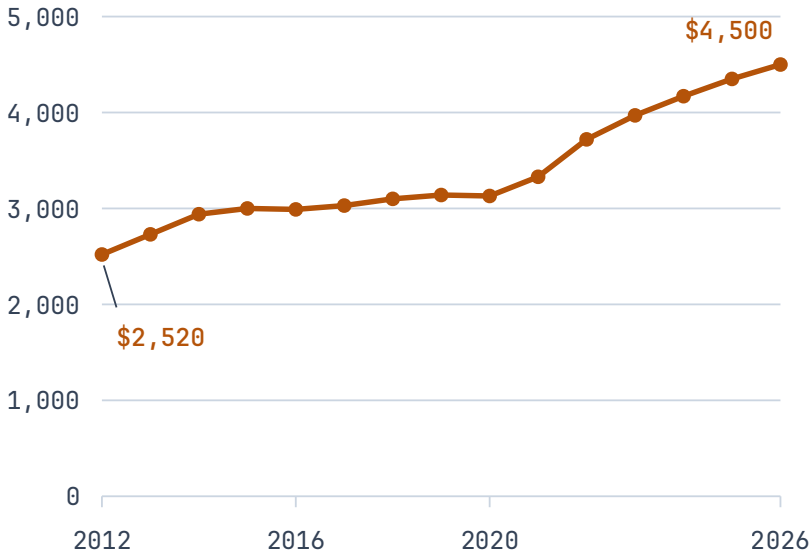
what you can afford to hold, and set aside enough to cover the property taxes for years, so the land stays yours through a bad stretch. Before committing, check legal access, water, permitted uses, insurance, and the cost of keeping it usable. A deed alone doesn't make a build possible.

That's one route, and it's not the only one or the entry fee. Land trusts, leases, shared gardens, public land, and agreements with growers who already have ground are all ways to get useful capacity under your feet. The trust in the scene changes a set of relationships: who may use the ground, who decides its future, what Irene receives, and what the co-op must keep paying. The paperwork doesn't abolish responsibility. It gives responsibility a shape people can read and hold each other to. The fictional trust isn't a legal template, and anyone setting one up needs real advice. What matters here's the question that brought those people to the table: how do we keep useful capacity serving the people who depend on it?

WHAT AN ACRE COSTS

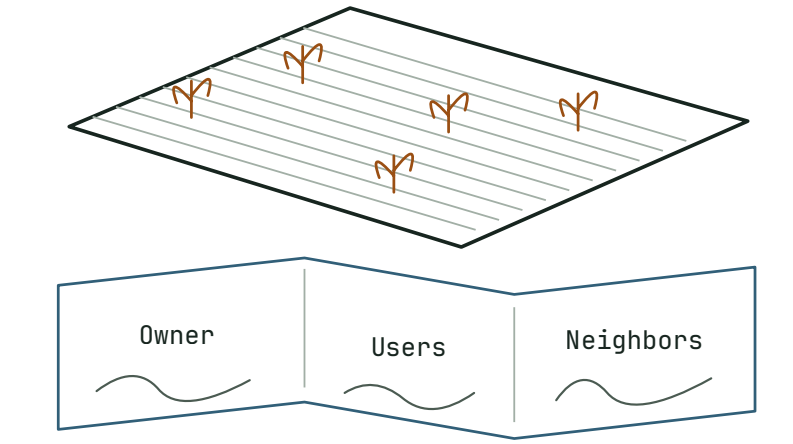
Average US farm real estate value

Nominal dollars per acre



Survey estimates include land and buildings.
National averages, not local asking prices.
Source: USDA NASS, Land Values 2026, p. 5.

What an acre costs. USDA survey estimates include farm buildings as well as land. These are nominal national averages, not inflation-adjusted values or local offers. USDA NASS, Land Values 2026, page 5.



Useful ground comes with people and obligations. Shared use needs consent, clear terms, and a way to challenge decisions. Conceptual illustration, not a legal template.

BY THE NUMBERS: THE GROUND

- **$4,500** an acre: the average value of US farm real estate in 2026, up 3.4 percent in a year. USDA NASS, Land Values 2026 Summary
- **$6,020** an acre for cropland and **$2,000** for pasture, the same year. USDA NASS
- **99 years:** a common renewable ground-lease term in community land trusts, subject to the actual agreement. Grounded Solutions Network, Community Land Trusts

Land that could hold a right

In southwest Georgia, losing a place to farm could be punishment for trying to vote. Charles Sherrod had worked with sharecroppers and tenant farmers forced off land after attempting to register. Collective

ownership offered a way to reduce that power over individual families. New Communities was founded in 1969, and on January 9, 1970, it completed the purchase of 5,735 acres near Albany. The project joined farming to a proposed system of homestead leases. Security on the ground was part of making a political right usable. SNCC Digital Gateway, New Communities

Shirley Sherrod's recollections bring the institution down to its working parts. The land included a railway spur. She described using it to order fertilizer from outside the local supply network after receiving deliveries that, she said, weren't what the farm had ordered. The farm needed inputs it could trust, customers for its crops, and money to keep operating. A line on a map didn't provide those things automatically. Nor did the founding plan become reality in full: she recalled that the group couldn't afford many of its intended projects. Shirley Sherrod, interview with Maine Farmland Trust

Drought and the credit system exposed those dependencies. Sherrod described a prolonged fight for an emergency loan, followed by inadequate financing and, in 1985, foreclosure. The organization continued after losing its land. It later pursued a discrimination claim against the USDA, received an award in 2009, and bought land again in 2011. Recovery took decades of organizing and a remedy outside the original farm. Sherrod interview, SNCC Digital Gateway

This is a harder and more useful history than the promise that collective title makes a community safe forever. The arrangement challenged one source of dependence while remaining exposed to others. Its members needed fair access to finance as well as land, and institutions beyond the property capable of enforcing that fairness.

Reuben's folders therefore belong beside a larger set of questions. Who supplies the farm through a bad year? Who hears a complaint when a lender or supplier abuses its power? A robot could help cultivate this ground. It couldn't make those outside institutions answer fairly. That requires people to organize beyond the fence as well as within it.

Grandma's request

Now follow one request all the way to the door.

Picture an eighty-two-year-old woman on a fixed income at the edge of town. Her joints are stiff with arthritis. Her yard has gone to star thistle, a fire hazard for her and for everyone downwind, and the county has sent a code-enforcement notice with a fine she can't pay. A landscaping crew wants $400 cash. The groceries are another calculation: she's started buying what she can carry instead of what she'd choose.

What's she supposed to do? Sit there, run out of money, and wait for fire season?

Absolutely not.

This is exactly where the lone-prepper story falls apart. A bunker does nothing for her. What she needs is a network of neighbors with tools, food, and time, organized well enough to show up. Here's what that looks like, as an illustration, not a report of a real program.

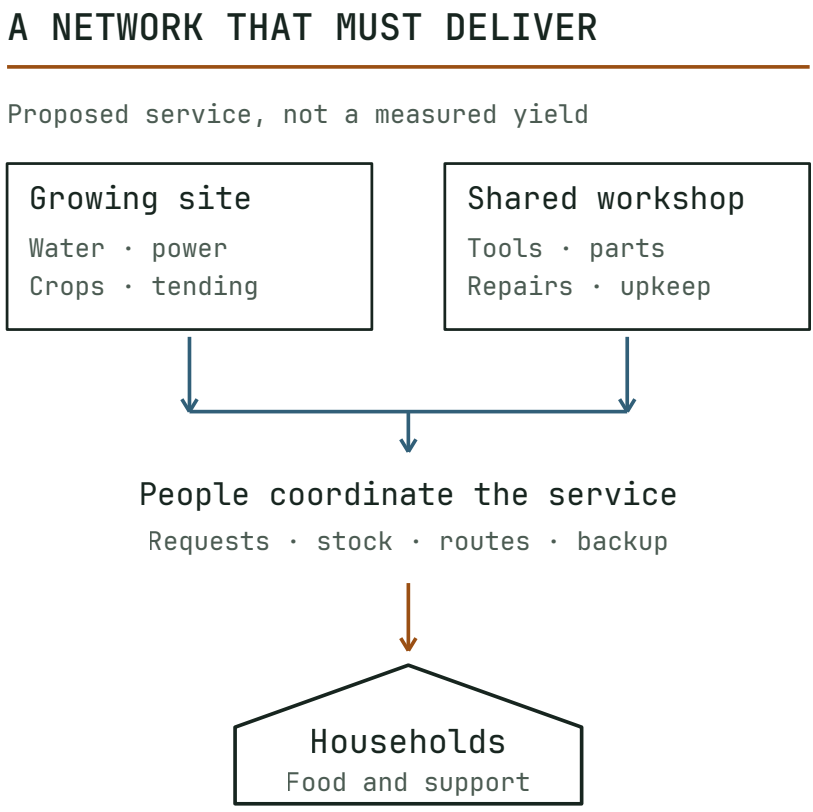
The nodes. Several households in the area each run a shop-house, or a *shouse*: half workshop, half home, with the tools, some power, and often a garden or a greenhouse attached. Chapter 13 builds one. Each is a node in a small network. Some nodes have a surplus of food in a given week. Some have a tractor or a chipper. Some have neither, and that's fine.

The balance. Count each resource in its own unit and over the same week: usable vegetables, prepared meals, equipment hours, driver time. For each one:

> **Available to share = usable supply − prior commitments − reserve for interruptions**

Then compare that amount with the requests. Don't add kilograms to hours and call the total abundance. A single garden, even a heavily automated one, usually won't cover a household's full diet, and anyone promising you a guaranteed multiple of your caloric needs is

selling something. Pooling partial surpluses can help a household that can't contribute this week, but only if the food is suitable and someone can get it there. If requests exceed what the nodes can provide, the coordinator needs a funded partner or a replacement supply. Production and distribution both count.



A shared-service network. People match usable surplus to requests and arrange delivery. The drawing promises neither a yield nor an autonomous crew.

The request. She doesn't need a smartphone. She has a phone number somebody answers, a neighbor who checks in, or, if she likes gadgets, a simple mesh radio on the kitchen counter with one button: NEED HELP. However it arrives, the request lands with a coordinator.

The coordination. The coordinator, with a local model on the co-op's own server helping with the scheduling, looks at the shared directory of who has which tools, skills, and surplus this week. The software proposes: Saturday morning, the tractor from one node, the chipper from another, two people with the right experience. A person confirms it with her: this is what she wants done, at a time that suits her, and yes, she'd like some vegetables. What does she like? Can she cook them, or would soup be better?

The delivery. Saturday morning, the crew clears the thistle, which lowers the fire risk for her and for the whole road. Someone brings five kilograms of vegetables from a greenhouse that had more than it could use, plus a pot of soup from the co-op kitchen, because she said standing at the stove is hard these days. A person walks the yard with her before they leave, and asks whether the county notice needs a letter.

We aren't doing charity in the condescending, form-in-triplicate sense. We're maintaining our own neighborhood. She keeps her yard, her pantry, and her dignity, and nobody asks her to prove she deserves any of it. The group charges her nothing. Its equipment, fuel, and donated hours still exist, and they go in the record. Free to her means somebody arranged it, not that it cost nothing.

What the machines did

Be precise about the credit. Software helped schedule. A greenhouse controller may have helped grow the vegetables. People assessed the yard, ran the tools, harvested the food, made the soup, and drove it over. Calling the whole thing "autonomous" would give machines credit for work people did.

A future robot service might do much more of the physical work. That's a forecast to test against real tasks, real reliability, and real availability, and my 2027 expectation doesn't promise it'll be on her road that year. But the example matters now, because it shows exactly where labor is needed and where a good machine could lighten it. And it shows we can help before the future version shows up.

Count the result at the door

Keep separate records for food, equipment time, travel, money, and remaining labor. Ask whether she got what was agreed, whether it met her needs, and whether the group can do it again next month. A machine's activity log can't answer any of those.

The bigger goal is food access for everyone. A small co-op isn't humanity, and a good Saturday isn't a permanent guarantee. Growing it takes reliable partners, a plan for shortages, and ways to reach people outside the founders' circle. Imagine what a dependable version would take off her week: the worry about carrying everything in one trip, the price math narrowing her dinner, the need to justify asking again. The real achievement would be the expectation she could safely form: help will come, and she doesn't have to get poorer, more grateful, or more useful to keep getting it.

Right now

Let's skip the hand-wringing and the endless theory.

Right now, you can start learning to build machines that help grow food. Right now, you can start working toward a home and ground that you and your neighbors actually hold. You don't need permission from a central board, and you don't need to wait for a federal program or a change in the political weather.

You can, *right now*, start working with your neighbors, your friends, and your family to share not only food and tools but labor, actual fucking labor, with the people who need it most, without the condescension and the paperwork that usually come wrapped around charity.

This week, ask one local organization how someone requests food or practical help. Check whether that route works without a smartphone, a car, or spare money. With their agreement, help close one gap. The first piece of infrastructure might be an accurate phone number and somebody who answers it.

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Precedent P-15: One Hundred Sixty Acres (United States, 1862)

The Homestead Act was signed in May 1862, and the timing is half the lesson: the country was a year into the Civil War. While every headline was consumed by the emergency, the federal government opened millions of acres, offering one hundred sixty acres to eligible claimants, including women heads of household and immigrants who'd declared their intent to become citizens, who would live on the land and improve it for five years. National Archives, Homestead Act of 1862

Filing began in January 1863. The ordinary five-year route therefore ran beyond the Civil War's end in 1865; Daniel Freeman's surviving certificate of eligibility dates to January 1868. US Capitol Visitor Center, Freeman's certificate There was also a paid route to earlier title. The opportunity opened during the crisis. Turning a claim into a lasting farm still took time, money, and successful proof.

The other half of the lesson is who was already there. That land wasn't empty. It was the home of Indigenous nations, and much of it had been taken through war, treaties broken, and forced removal. The National Park Service's own account of the Homestead story holds both histories together, and so should we. National Park Service, Homestead History and Culture Access for one group can't be told honestly as though it happened on a blank map.

The mechanism. Transitions reprice land and position early, and the window often opens during the confusion, not after it. Every land policy creates opportunity for some people while landing on the people already there.

The rule. Ask whose land and whose opportunity the story describes. Then don't wait for clarity to start your paperwork.

The practice.

1. Open the filing office this month. Spend one evening doing real research: three places you might live, actual listings or lease options or land-trust openings, actual prices, actual water and zoning notes, in a spreadsheet. Not to buy anything this month. To turn "someday, land" from a mood into a document with numbers in it.
2. Put a date on the claim. The homesteaders had a five-year proving period with a clock on it. Give yourself one: the date by which you'll have made a position decision, buy, lease, join a co-op's ground, or commit to improving where you already are.
3. Look for today's programs before the crowd does: county land banks, community land trusts, rural development grants, conservation leases, community garden plots. Find out who lived on and used that ground before, and whose interests are still tied to it. Set a quarterly hour to check what's on offer where you live.

CHAPTER 13

The Shouse Protocol



Solar on an outbuilding. The shouse grid starts with one roof. (Reinhold Möller, CC BY-SA 4.0, via Wikimedia Commons)

> *"The cost of a thing is the amount of what I will call life which is required to be exchanged for it, immediately or in the long run."*
> Henry David Thoreau, *Walden* (1854), "Economy"

In this chapter:

- Two steel boxes and a lot of skilled work can make a small, owned home. The double-wide hack is how you keep them from being a coffin.
- A bad weld looks exactly like a good one from six feet away. The vise is how steel tells the truth. Know your lane on anything people will live inside.
- Shelter comes after food in this book, not instead of it. A home is more than a building: it's terms you can live with, for as long as you need.
- Heat you harvest beats heat you subscribe to. The oldest technology in this book has a flue now.

The containers came up the hill on a tilt-bed in the last week of August, two one-trip high-cubes still wearing their ocean stickers, and the driver set them down on Marta's piers with a gentleness that

seemed impossible for eighteen thousand pounds of steel. The piers had been designed for the site and poured three weeks earlier, and Elijah had checked their levels so many times with the water level that Marta finally took the tubing away from him. “They’re right,” she said. “You watched me make them right. Trust the transit or go check the internet’s opinion.” Five millimeters out on this set, she had told him, and these boxes twist and the cut never lines up. The piers were off by one.

It went up on the co-op’s ground lease, on the worn patch where Walt Calder had once parked a combine, and it went up as a record: every dimension logged, every cut photographed, the file destined for the library, because the deal the co-op had made with itself was that nothing got built once. But it was also, and nobody made a ceremony of saying so, Elijah’s. Rented rooms and borrowed corners, and now a rectangle of dirt under a long lease with his name on it.

He had wanted to weld on it. That was the humiliating part, later: how badly he had wanted it, the certified-adult feeling of it, after nearly a year of watching Marta lay beads like the steel was agreeing with her. He’d practiced evenings for a month on scrap, and his practice pieces looked, he thought, honestly good. He asked for a seam on the header beam, the long C-channel spine that would carry the roof across the cut. Marta didn’t say no. She said, “Bring me your coupon,” and put his best practice weld in the big bench vise, and slipped a cheater pipe over the free end, and broke it with one pull.

The bead snapped clean along its own centerline. Inside, the break was bright and crystalline and full of small round voids, like honeycomb cast in nickel.



The vise. The metal does the truth. Bright and crystalline.

“See the sparkle,” she said, not a question. “Cold lap. Porosity. It sat on top of the steel and never became the steel. From six feet it looks like mine. The vise is how the metal does the truth.” She set the two halves on the bench where he could keep them, which he did, and has them still. “There’s nothing wrong with you, college. You’re a couple hundred hours and a certification test from a structural bead, and the rain is eight weeks out. And this seam,” she put her glove flat on the C-channel, “holds a roof over someone’s kid. Not this year’s kid. Every kid in every copy of this build that ever gets made off our files. I don’t weld those seams tired, and I don’t let anybody weld them proud.”

So Marta welded the spine of his house, and Elijah went to his lane, and here is the part he would want said plainly: his lane turned out to be real work, and he was good at it. The electrician ran the circuits. Before the insulation went in, Elijah laced the wall cavities with temperature sensors, so the shell could report its own gradients for the rest of its life. He put leak sensors low in the wet-room and at both wall penetrations, and built the little dashboard the whole co-op now uses, the one that shows a floor plan with a nervous system and a visible blank wherever a sensor stops reporting. Marta gave the finished panel a long look and said, “Huh,” which everyone present understood to be a medal.

The day of the cut, the columns and headers already in and inspected twice, Denny ran the plasma torch, and the interior mating walls came out in glowing rectangles, and light went through the two coffins and made them one room, sixteen feet of open span where no room had been. Everyone stood in it. That's all. Nobody said anything worth recording. Sixteen feet by forty isn't a large space by the old world's arithmetic, and every person standing in it had lived in bigger, and hardly any of them had ever held a cubic foot of shelter on terms they could count on.

The stove went in last. The Fisher had come north with his belongings nearly a year ago, four hundred pounds of plate steel and one cracked firebrick, and it had sat in the corner of the shed being a monument the way it had once been a monument in a rented apartment, and monuments, Marta observed, rust. They bought the replacement brick. The inspector looked at the clearances and the hearth. The installer specified a listed, insulated chimney system through the roof and double-wall connector pipe inside, all bought new, the one line in the budget nobody let him salvage. Priya steadied the ladder while the installer finished the storm collar in a wind that wanted his hat.

The first cold night came in the last week of September. He laid the fire the way his grandmother's hands had taught his hands before he was tall enough to see the top of the stove, and the draw caught on the first match, and the steel began to tick as it heated, the same sound as every engine and every server he had ever loved, run in reverse. People drifted over without being organized. Reuben brought chairs the way Reuben brings paperwork, already done. Somebody set a drying rack well clear of the stove and hung a pair of socks on general principle.

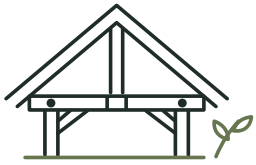
He took one photograph from inside the shell: the stove lit, the socks, the new brick glowing faintly through the door gap, the dashboard on the wall behind reporting all systems green. He sent it to his mother at 9:41 on a Sunday, and for nearly two years he had failed to build one sentence that could carry across to her what was happening to the world, and the phone rang before the checkmark appeared.

"That's the Fisher," she said. No hello, but the radio was going behind her, so it was the same kitchen, the same country station, the same distance suddenly not mattering. "Is your flue double-wall? The Hendersons had a chimney fire in '88 and your grandmother talked about it for ten years."

"Inside. Insulated chimney through the roof. Installer picked the system; inspector signed it Tuesday."

A quiet went down the line, with the radio under it. "It looks right in there," his mother said finally, in the voice she used for things that were settled. "It looks like somewhere."

Every sentence he had ever tried to build for her had been about what was coming apart. The one that finally crossed was her mother's stove, lit. He sat down in front of it, in a house with Marta's welds for a spine and his own nerves in its walls, and added the night to the notebook: first fire. Under PARCELS HELD IN TRUST: 1, a new line: ROOFS: 1.



THE FOUNDATIONS

A door you can close

Food is where this book concentrates the first promise. It isn't the only thing keeping a frightened worker awake. Rent is due whether the new machine is impressive or disappointing, and a person who can eat but has nowhere secure to sleep still has an emergency. A priority is where you start, not an instruction to leave somebody outside.

So carry the same question into housing: what could useful machines help us provide, and what arrangements would let the people who need it actually receive it? The shouse makes that question physical, but it's one fictional home on one particular site. A repaired apartment, a converted barn, a well-built house, and an existing home kept affordable can all answer a housing need. We don't have to make everyone's future look like Elijah's to understand his relief at the first fire.

Own the dirt, then build

Once you have ground, whether bought, leased, or held in trust, don't try to build a big, complicated suburban house. Build a *shouse*: a shop-house. Half workspace, half compact home, under one roof.

What the fuck is a shouse? When I first wrote it down, this is how I put it: not a house with a garage where you keep a lawnmower you use twice a month, but a shop that's been set up to keep the person running it alive. Seventy percent production, thirty percent survival. You don't need a thirty-year mortgage to have a place to sleep. You need a slab of concrete, a roof that doesn't leak, and the tools.

1. **Start with the approved use.** Ask whether the site can support a workshop and a dwelling, and what temporary occupancy is allowed during construction. A trailer or RV may be an option on an approved outdoor pad with the required services. Don't treat an enclosed utility building as a safe place to occupy an RV or run its fuel-burning equipment.
2. **Call the building department.** Don't be afraid of them. They aren't the police. They're people with a checklist. Call, tell them what you have, and ask exactly what it takes to live on your land while you build. Meet the requirements, pay the fees, get the stamp. Working inside the rules is what keeps you from being interrupted later.
3. **Bring the tools in as they earn it.** Once you're established, the land gives you room for new machines: a garden robot over a raised bed, a 3D printer in the shop, and, some day, printed structures. If

you hold the ground, you have space to deploy them as they get affordable and proven.

You'll make mistakes. You'll buy the wrong tool and do some things the hard way. Better to do that now, while hardware stores are open and the grid is up, than to be learning it all when things get strange.

Own the dirt. Build the systems. Be the one who's prepared, rather than the one who's surprised.

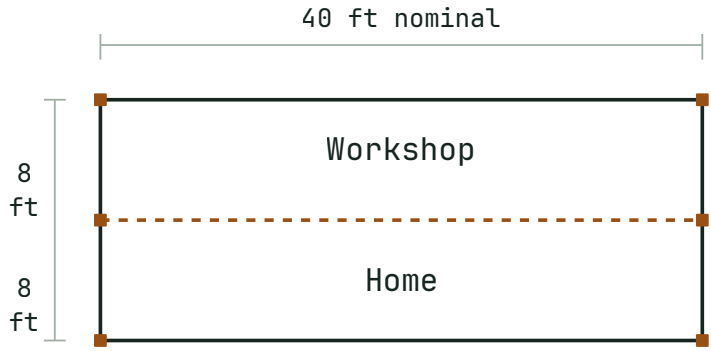
The double-wide idea

A standard suburban house with a thirty-year mortgage, a manicured lawn, and a smart lock a bank's foreclosure script can disable isn't the only way to have a home. For a lot of people it's a financial cage: you spend the best decades of your life servicing debt to be allowed to live somewhere. The shouse is one attempt at a different deal.

Shipping containers are the structural Lego of the moment: weather-sealed and built to take brutal loads at their corners. But a single container, eight feet wide on the outside, can end up under seven feet inside once it's framed and insulated, which is less a home than a very aesthetic coffin. The co-op's answer was the double-wide: two 40-foot high-cube containers side by side, the shared inner walls cut away. The outside footprint is sixteen by forty feet, 640 square feet; the finished usable room is smaller.

A double-wide shell

Conceptual plan; no connection recipe.



40 ft × 16 ft = 640 sq ft

Nominal outside footprint, not floor area.

Openings, foundations, joints and reinforcement need a site-specific engineered design. The chapter's budget belongs to its fiction.

Two nominal forty-foot containers give a 640-square-foot exterior footprint. Usable space and structural details depend on the finished design; the chapter's budget is fictional.

Here's the physics that makes it serious. A container's corrugated side walls aren't just walls; they work as deep beams that carry the roof. Cut them out, and the roof loses its support. That's why, in the scene, the columns and header beam go in and get inspected *before* anyone touches a torch, and why Marta welds that spine herself. Sizing that steel is engineering for a specific site, snow load, and design, and it's a job for someone qualified to sign it. The principle every reader should keep: never cut a structural member until something else is already carrying its load.

Beating the tin can

A metal box is a superb conductor of heat, which makes it an oven in August and an icebox in January. For steady conduction through one uniform layer, Fourier's law gives the heat-transfer rate:

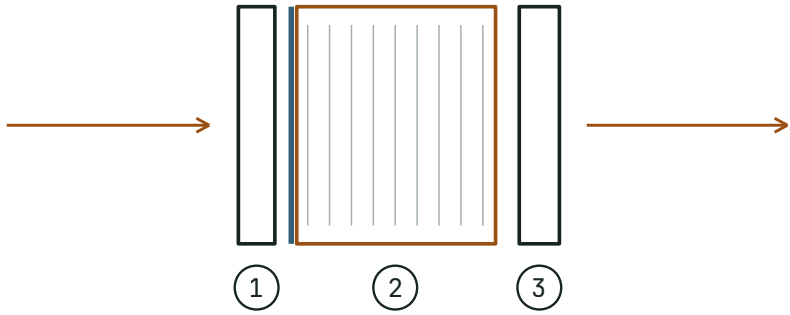
> $\dot{Q} = (k \cdot A \cdot \Delta T)/d$

where $\dot{Q}$ is heat flow in watts, k is thermal conductivity, A is area, ΔT is the temperature difference across that layer, and d is its thickness. A complete wall adds other layers, surface effects, and paths through steel framing. Steel conducts heat thousands of times better than foam, and a container wall is only a couple of millimeters thick, so bare steel barely slows heat at all. Everything depends on the insulation layer.

A wall is an assembly

Conceptual section, not a thickness schedule.

Example heat flow



1 Steel skin
2 Insulation + continuous air control
3 Interior finish + required fire protection

Idealized layer: heat flow = k × A × ΔT / d
Design moisture control and ventilation too.
Heat flow depends on the whole assembly.

Insulation is one layer in a wall. Air control, moisture, ventilation and fire protection have to work with it.

One approach is closed-cell spray foam bonded to the steel, with the thickness and R-value selected for the product and climate. Its advantage isn't just insulation: a continuous, correctly installed layer can also control air leakage. Gaps, penetrations, and exposed steel still matter. Fiberglass batts alone aren't an air barrier either. Get the whole wall assembly designed for your climate, including moisture control and the required fire protection over foam, and installed by people who do it for a living. US Department of Energy, Energy Renovations: Insulation

Build for every age you'll be

Design the inside for the person you'll be at eighty, and for the neighbor in a wheelchair who comes to dinner. Put the plumbing in one wet room with a floor that slopes to a single drain and no shower curb to trip over. Before the wall panels go up, put solid blocking behind them where grab bars or a fold-down seat might go someday, so they can be added without tearing into the wall. Run wiring and water lines in open, accessible chases instead of burying them, so a leak or a rodent doesn't mean demolition. A house that has to be rebuilt the first time someone gets hurt wasn't finished.

What the sketch leaves out

A drawing and a materials subtotal don't make a home. Foundations, connections, weatherproofing, fire safety, ventilation, and permits all stand between a clever plan and a place people can safely live, and none of it is in the photograph. The co-op's files are a record to learn from, not a license to repeat its cuts on another building.

Useful automation could help with parts of design, fabrication, inspection, or maintenance, and each part has to prove itself in use. And if it saves labor, there's another decision to make: does the saving become cheaper homes, shorter shifts, or a bigger margin for somebody? A house can be quick to build and expensive to live in. A neighborhood can gain new buildings while the people already there lose the ability to stay. Count whether people can live in the result, safely and affordably, not only how fast the machine puts material in place.

A city builds the ordinary

Vienna's municipal housing program between the world wars took aim at things a resident could name without an architectural vocabulary: running water, a toilet, daylight, a place outside the room. The city's archival history connects that program to changes in its constitutional position, revenue, and spending. The ability to build came with the authority and resources to keep a public purpose in the budget. City of Vienna, History of Red Vienna

On October 12, 1930, Mayor Karl Seitz formally handed over the completed apartments at Karl-Marx-Hof. The complex had taken three years to build. Its housing was accompanied by social, medical, and educational facilities. A municipal account published for its sixtieth anniversary describes both that beginning and something less ceremonial: major renovation work underway since 1988. The city said public subsidies would keep rents affordable during the improvements. The notice states how the city intended to finance the work; it can't tell us every tenant's experience. City of Vienna, Karl-Marx-Hof at Sixty, October 11, 1990

The distance between those dates matters. The achievement couldn't be completed when the keys first changed hands. Buildings aged. Work had to be organized and paid for again. A tool that reduced the cost of the original construction would address one part of that lifetime. A useful housing policy had to remain interested after the photograph of the opening.

The political history wasn't an uninterrupted march toward security either. Austria's parliamentary crisis in 1933 and civil war in February 1934 were followed by authoritarian government; Vienna lost its elected legislature. The institutions behind a public promise can be weakened even while its buildings remain. City of Vienna, History of Red Vienna

These municipal accounts tell the history from the city's side. They don't establish that every applicant obtained a home, every tenant was satisfied, or another city could simply copy the arrangement. They do show why the housing question can't end with a faster machine. Someone must have a place in the completed building, terms they can live with, and a responsible institution when the building needs work. The useful measure of progress reaches from the construction site into all the years a person hopes to stay.

Respect the skill

Marta doesn't become less important because a machine can help draw a design. Someone still has to know what a good joint looks like, when conditions invalidate the plan, and when to stop. Her teaching

takes time, and that time belongs in the budget. And reducing the number of motions a person makes isn't the same as removing their responsibility. If a tool speeds up fabrication but adds inspection or setup, count that work.

A useful assignment before a purchase

Write down the three jobs you actually need space for. For each, find an existing place it could happen, who can authorize its use, and what it would cost in time and money. Ask someone who already does the work what your list is missing. That exercise might lead to a building project. It might lead to a shared bench and no construction at all. Either is a good answer if people can do the work well.

For a housing project, start with the people who need somewhere to live and the people already providing it. Ask which bottleneck a proposed tool would actually remove, and how the benefit would reach a resident. The destination is a home somebody can depend on.



Precedent P-16: The House That Came by Mail (United States, early twentieth century)

For decades, you could order a house from a catalog. Sears Modern Homes shipped as numbered lumber, with nails, paint, fixtures, and an instruction book, delivered to the nearest rail depot; from 1916 the lumber came precut to length, a method the Aladdin Company had pioneered and Sears adopted. Sears and its competitors sold tens of thousands of these houses, in hundreds of designs, from tiny cottages to genuinely grand ones, and plenty are still standing and lived in more than a century later. Johnson, House in a Box, Kentucky Heritage Council

The industry said what industries always say: housing was too complex, too consequential, too dangerous for amateurs. The kits answered by moving a lot of the expertise into the parts and the manual. Precut, numbered pieces and clear instructions turned a family with a few capable friends into builders.

But the box didn't contain the whole house. A buyer still needed a site, a foundation, the labor to assemble it, and connections to water, power, and sewer. The kit made part of the work easier to organize. The rest of the work was still work.

The mechanism. Kit-plus-instructions is how a capability escapes its priesthood: the knowledge gets embedded in the parts and the manual instead of the guild. What the box can't carry, the site, the structure, the permit, the connections, still has to come from somewhere.

The rule. Look for the version that ships as a kit, and count the work outside the box.

The practice.

1. Break one priesthood this year. Pick one thing you currently rent from professionals, a shed, fencing, a greenhouse, basic plumbing, and do the kit version, permitted where permits apply, documented as you go. The point isn't the money. It's the change in your head, from someone who orders outcomes to someone who assembles them.
2. Ask the manual question. The next time you're told "you need a professional for that," ask what specifically the professional knows, and where it's written down. Sometimes the answer is real: structural welds, gas lines, anything like Marta's vise. Often it's a manual, a code book, and confidence. Learn to tell the difference before you pay it.
3. Leave a kit behind you. Whatever you build, photograph the steps, list the parts and prices, and write the one page you wish you'd had at the start. Give it to the next person. A skill that dies with your project wasn't liberated. It was borrowed.

CHAPTER 14

The Collapse of the Long Tail



An abandoned bar in an older shopping mall in Haifa. Gradually, then suddenly. (Mattes, Public domain dedication by Mattes, via Wikimedia Commons)

> *"How did you go bankrupt?" ... "Two ways," Mike said. "Gradually and then suddenly."*
> Ernest Hemingway, *The Sun Also Rises* (1926), Chapter XIII

In this chapter:

- Every local business you lean on is quietly load-bearing. You find out which ones the week the platform takes them.
- Collapse is visible for months before it happens: thinning shelves, shortening hours, the kid let go. The gradual part is a readable signal. Read it out loud.
- If you see it coming, say so. A warning can't always save the thing, but it can buy a person the difference between six months of choosing and six months of hoping.
- Shorter loops, not no loops. Every local answer still has outside dependencies; name them.

Every controller the co-op ever shipped had parts from Kowalski's in it. Whenever anybody in the shed said *from town*, the counter they meant was Frank's: third bay off the square, wood worn pale where

four generations of forearms had rested on it, a bell over the door that Frank's grandfather had hung, and Frank himself, who could tell you the torque spec for a head bolt from memory and would carry a two-dollar part out to your truck like it was an appliance.



Frank Kowalski. Knows the torque spec from memory.

Elijah watched the store die for nine months before it died, and said nothing, and this chapter exists because of both halves of that sentence.

He hadn't meant to be watching. He kept the co-op's purchasing ledger, and the ledger noticed before he did: substitution emails, out-of-stocks on things that had never been out of stock, the fastener aisle developing gaps that stayed gaps. He pulled the thread, because he cannot not pull threads. Frank's regional distributor had been bought by a logistics platform out of Memphis, the kind with a two-letter name and no phone number, and the new owner had raised minimum order quantities past what a single-counter store could float, rerouted the trucks to serve its own fulfillment nodes, and let the long tail of its customer list, the Franks, wither on schedule. The chart Elijah built, because he cannot not build charts, showed the co-op's own orders drifting away from the square, to salvage yards and online, one substitution at a time. The hours sign got re-taped twice. Tyler, who'd worked the register three summers, stopped being there. Gradually is not invisible. Gradually is a curve, and Elijah was, by profession and by scar tissue, a man who read curves.

He said nothing, and he could give you the reasons the way you can give directions to a place you're ashamed of knowing well. He'd been the man with the printout once, at a coffee machine on the fourth floor, and a calm senior colleague had smiled at him like a child. He'd spent two years learning what it costs to say the thing early, and the lesson he'd accidentally taken was: don't. And besides, what was the sentence? Frank, my chart says you're dying? Frank knew his own store. Surely Frank knew.

The paper on the door, when it came, was laminated. Which meant Frank had known long enough to laminate it.

The same week, with the timing this period of history seemed to specialize in, Elijah got an email with the subject line "checking in," from Devendra. Claypot had cut the whole division. The Companion team and every human who had been augmented-not-replaced alongside it, gone in one restructuring memo, the model retained, its trainers not. On the phone Devendra kept starting a sentence, pausing, and starting somewhere else, and his voice had a hum in it that Elijah recognized, because Marta had taught him its name: the sound of a man reading the news all day because the news is the only place his old life still exists. "You'll appreciate this," Devendra said, with a laugh that wasn't one. "They kept the very good autocomplete." And then, quietly, the real question: "At the coffee machine. You knew. How?"

"I didn't know," Elijah said. He almost followed it with the arithmetic, with *you told me I'd calibrate*, with all the sentences that would turn another man's lost job into his own vindication. He looked at Frank's closing notice on the screen and let them pass. "I was frightened. I still am sometimes. What do you need this week?"

Devendra didn't answer immediately. Then he asked about the place to stay that Elijah had mentioned once, and whether he could bring the kids.

Then Elijah made the harder visit, the two-mile one. He drove to the square with his chart, parked, sat in the truck long enough for the windows to fog, and walked in through the bell. Frank was taping boxes. Elijah put the chart on the worn wood and confessed to it: nine

months, the distributor, the platform, the curve, all of it, and that he'd watched it and said nothing because he hadn't wanted to be the crackpot again.

Frank looked at the chart for a while. "Fella in Hemingway gets asked how he went bankrupt," he said, taping another box. "Says: two ways. Gradually, then suddenly." He set the tape gun down. "The gradually part, I kept telling myself it was weather. Turns out it was the climate." He tapped the curve's long slow shoulder, the nine visible months of it. "Wouldn't have saved the store, son. The store was dead the day Memphis bought my distributor. But I'd have liked to have had this in March. I'd have chosen some things, instead of hoping them."

At Friday dinner Elijah said it all again, chart and confession both, in front of everyone, because that had become the co-op's immune system and he trusted it now. Denny gave the verdict, and it had edges. "You of all people," he said. "You watched a man get optimized out by a spreadsheet he never saw, and you had the spreadsheet." Then, because Denny's justice always had a second half: "So fix it forward. What's the co-op do about Frank?"

What the co-op did about Frank began with an offer, and Frank sent Reuben's first draft back with questions. They settled on the inventory the tool library could use, bought at honest wholesale, not vulture pennies, the money real enough to matter, and a paid job, because the tool library had been running on a clipboard and an honor system and it turned out what it needed was a man who knew the torque specs from memory and where every last thing was. The library's income and costs would get a monthly review. Frank wanted the figures as well as the welcome.

Then he hung his grandfather's bell over the tool library door. The first time it rang for a member checking out equipment, he looked up from the counter before he could help it. The store had not died after all. It had shrunk to fit through a doorway and moved two miles east. The parts still came from suppliers beyond the valley. Frank still knew which ones.



The bell. Frank's grandfather hung it over the door.

Denny asked Frank whether he wanted to tell the story on the channel. They recorded it behind the new counter, with the bell above the frame and the figures Frank had agreed to share laid out between them. Afterward Elijah put the chart he'd sat on for nine months beside the broken weld coupon on his shelf. He didn't throw either away.

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THE FOUNDATIONS

Walk into an American supermarket in the early twenty-first century and you were looking at a historical anomaly. Avocados from Mexico, blueberries from Peru, garlic from China, all side by side under bright lights, misted by little sprinklers, priced at a fraction of an hour's minimum wage.

To most of us, that looked like peak civilization. To anyone who thinks in systems, it also looked like a high-wire act.

The just-in-time trap

The supply chains behind those shelves run on *just-in-time* logistics: keep as little inventory as possible and let a constant stream of trucks act as a rolling warehouse. The aim is to reduce idle inventory. The

amount of stock and backup capacity still varies by product, operator, and route. Even Toyota's account of just-in-time production includes a minimum stock of parts ready for assembly. Toyota, Production System

WHEN REPLENISHMENT BREAKS

Illustrative sequence, not a four-day clock

1. **Upstream disruption**
   A port, supplier or ordering system fails.
2. **Deliveries delayed**
   Stock arrives late or takes another route.
3. **Local buffer under pressure**
   Demand and remaining stock set the pace.
4. **Service response**
   Reroute, replenish and prioritize needs.

There is no universal time to empty shelves. Buffers, alternative routes, replenishment and demand change the outcome.

One possible disruption sequence, not a clock. Existing stock, replacement routes, demand, and the specific failure determine what reaches the shelves and when.

Every system trades between two things:

- **Efficiency:** remove buffers, cut stock, streamline every path, maximize return today.
- **Resilience:** keep backups, local storage, alternate routes, and some ability to take care of yourself.

A grocery store with a few days of stock isn't a store a few days from famine; restocking usually works, and regional warehouses hold more. But when a disruption hits, whether a storm, a strike, a platform decision in Memphis, or a pandemic, the shelves are where you feel it first, and the people who feel it hardest are the ones with no pantry, no car, and no slack of their own. That's the trap. And it's why Frank's chart matters: the platform that bought his distributor didn't hate anybody. It optimized a ledger that didn't need a single-counter hardware store to survive, and the cost landed on people who weren't on that ledger.

What the efficiency was for

That's the question worth asking whenever something gets more efficient: whose books show the gain, and whose life absorbs the loss? The machine didn't decide Frank's customers were expendable. People chose the terms of supply. Better logistics could just as easily have been pointed at keeping small places served. That would have taken resources and a reason to value the service. Calling the closure inevitable hides the decision that mattered.

The two precedents at the end of this chapter walk through a corporate cemetery, then follow a survivor that found other uses for its skills. Here's the plaque at the gate: **the difference between Kodak and Fujifilm was never information.** Both saw the wave. One asked "how do we protect what we sell?" The other asked "what are we actually for?" Your town, your trade, and your household are being asked that same question right now.

The short loop

The answer isn't to cut yourself off. It's to shorten the loops you depend on for survival, and to know exactly which long ones are left.

Think of global logistics as a long, fragile line stretched across oceans. A short loop is a tighter circle drawn around your region: food, repair, tools, and knowledge that can keep circulating when the long line hiccups. Not primitive isolation. Hyper-local design, applied to the physical world.

A local supplier makes a repair easier. A regional grower makes a food partnership more direct. Neither makes a community independent of everything outside it: parts, seed, fuel, expertise, and money still come from elsewhere. Resilience means knowing what keeps working during a particular disruption, for how long, and with what support. Several different connections can be exactly what keeps one failure from becoming a crisis.

Catch the food before it's thrown away

Here's the most obvious short loop of all. By the US Department of Agriculture's estimate, somewhere between thirty and forty percent of the American food supply goes uneaten. USDA, Food Waste FAQs The USDA FAQ points to a 2010 estimate of 31 percent at retail and consumer levels; the 30 to 40 percent range isn't a fresh annual measurement. Loss includes food that can no longer be eaten as well as recoverable surplus. A date label alone doesn't establish safety; storage and handling still matter.

Some of that waste is still edible food in the wrong place. Recovering it already has working law behind it. France's 2016 law prohibited deliberately spoiling unsold food still fit to eat and required larger food retailers to offer donation agreements to eligible charities. Food-safety rules still applied; this wasn't a command to donate everything in a waste bin. France, Law 2016-138, Article 1 California's SB 1383 requires covered grocers, distributors, and food services to recover edible food that would otherwise be thrown away and get it to people who can use it. CalRecycle, SB 1383 Food Recovery A community

doesn't have to invent this. It can organize the pickups, the kitchens, and the cold storage that make recovery actually reach a plate, and ask its own local leaders to back it.

Turn a leak in the long system into a local reserve, and you're feeding people from the surplus of the very system that told them they were redundant.

WHERE WASTED FOOD GOES

US retail, food service, households · 2019

66.2 million tons



EPA estimates; 2019 Wasted Food Report, Table 5. Includes inedible parts. Donation is net of food banks' unusable share.

Where wasted food goes. EPA estimates for 2019 include inedible parts; the whole total could not have been donated. Donation is net of food banks' unusable portion. EPA, Table 5.

BY THE NUMBERS: FOOD IN THE WRONG PLACE

- **30 to 40 percent** of the US food supply goes uneaten, by the USDA's estimate (linked above).
- **5 to 6 million tons** of food thrown away by Californians every year. CDFA, Food Recovery
- **20 percent by 2025:** California's statutory recovery target for edible food that would otherwise be landfilled. A target isn't a reported achievement. CalRecycle, Food Recovery
- **47.9 million** people in US households that couldn't count on enough food in 2024. USDA ERS

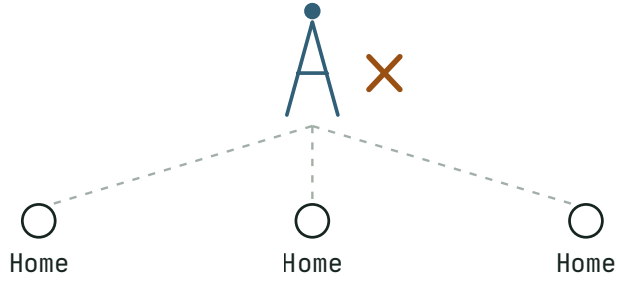
Put the first number next to the last one. That mismatch is the argument, not a calculation of meals available: the figures cover different years and populations, and some lost food can't be recovered.

The municipal mesh

You can't run a short loop if you can't talk to each other. Most of our coordination rides on cell towers and data centers we don't control, which is fine until a tower goes down in a storm or a service decides to shut something off.

A second route must exist

ONE UPLINK IN THIS EXAMPLE



Tower fails: this route is lost.

MESH WITH AN ALTERNATE ROUTE



Test coverage, power and a missing relay.

A mesh can route around a break only where another powered, usable path remains. Test the missing relay.

A mesh network is built from small, low-power radio nodes spread across an area, often LoRa radios running open software like Meshtastic.

- **Alternate paths.** Nodes can relay messages. If one fails, messages can travel another way only if a usable radio path remains. A ridge repeater can still be a single point of failure, as Chapter 17 shows. Meshtastic, mesh algorithm

- **Independent.** It works locally without the internet, cell service, or satellites.
- **Low power.** A node can run for a long time on a small solar panel and battery, drawing less than a nightlight.

Let's be honest about the limits. A low-power mesh won't stream video or move big files. It's a nervous system for short messages: "Who has surplus potatoes today?" "The Hollis road is flooded." "Delivery's running an hour late." "Checking in, all fine here." Use communication that fits the message, keep ordinary phones and a printed contact list in the plan, and never make joining the food service depend on buying a radio.

Build the loop

1. **Map the surplus near you.** Which farms, grocers, bakeries, and restaurants have edible food they can't sell? Ask them what would make donating it easy, and connect them with a kitchen or pantry that can take it.
2. **Put up the first nodes.** With a few neighbors, set up a handful of mesh radios, test the range, find the dead spots, and keep a printed directory for the people who'll never own one.
3. **Pool the ground and the tools.** One household with a lawn has an ornamental patch of grass. Five households pooling the same land can have a real market garden and a shared tool shed.
4. **Name one dependency and one backup.** Pick a real link in your local food route, a van, a grower, a pickup site, and confirm, with an actual person, what happens if it fails. A second name in a spreadsheet doesn't reserve anybody's capacity.

The transition didn't just stress our supply chains. It tested our belief that we can get by without each other. Stop watching the horizon for the next cargo ship and start looking at your neighbor.

Give people time to choose

Frank's complaint about the withheld chart was exact: he wanted to choose some things instead of hoping them. Security buys that. It gives a person time to judge an offer, learn a new trade, or mourn a place without every decision being made under immediate threat.

A company can have reserves, several lines of business, and people paid to study a transition. A worker may have a last paycheck and next month's rent. Telling both of them to "adapt" as if they had the same options misses the work a society has to do. A food baseline would give Frank something the co-op couldn't: the certainty that he'd eat even if nobody could offer him another job.

— ◈ —

Precedent P-17: The Graveyard of the Unconvinced (1975 to 2013)

Four headstones, one epitaph. Every company below *saw the wave*, and died or nearly died anyway, because seeing isn't the hard part.

Kodak, 1975. A young Kodak engineer named Steven Sasson built one of the first digital cameras: about eight pounds, a hundred-by-hundred-pixel black-and-white image, twenty-three seconds to record each picture to a cassette tape. Swiss Camera Museum, 1975: The Invention He showed management. Their response, in his own later telling: "That's cute, but don't tell anyone about it." Kodak wasn't blind after that. It developed digital technology for decades and by 2011 had shipped its hundred-millionth digital still camera. But its profits lived in film, paper, and chemistry, and digital cameras never replaced them. Kodak filed for bankruptcy in 2012 and emerged in 2013, much smaller, having sold off much of what it had built. Kodak, Milestones

Sears, 1993. For generations Sears mailed a catalog of everything to millions of homes and delivered to the middle of nowhere. You could buy a house from it (P-16). In 1993, with the catalog bringing in billions but losing money, Sears shut it down and dismantled the

machine behind it. Amazon was founded the following year. When Sears wanted online commerce later in the decade, it had to rebuild the organ it had just amputated. Fortune, Sears Could've Been Amazon

Blockbuster, 2000. Netflix's founders flew to Dallas and offered their company to Blockbuster for fifty million dollars. By their account they were laughed out of the room; Blockbuster's side remembers it differently, and the outcome isn't disputed. Blockbuster earned a big share of its profit from late fees, which meant it was paid, in part, by its customers' resentment, and Netflix's pitch was the removal of that resentment. Blockbuster went bankrupt in 2010. Variety, How Blockbuster Could Have Owned Netflix (2013)

Borders, 2001. The bookstore chain handed its entire online operation to Amazon. Every Borders web customer was an Amazon customer wearing a Borders name tag. Borders took its website back in 2008 and was liquidated in 2011. Slate, Borders bankruptcy: Done in by its own stupidity, not the Internet (2011)

The mechanism. None of these companies lacked information. Kodak *built* the prototype; Sears *owned* the playbook; Blockbuster *heard* the offer; Borders *signed* the partnership. An incumbent rarely dies because it can't see the wave. It dies because it's holding too much beachfront to admit the tide is real, and because awareness, investment, and a business that survives the change are three different accomplishments.

The rule. Verify what a company actually did before you use it as a warning. Then find the line item where the old world charges people for its own inconvenience, late fees, gatekeeping, distribution monopolies. That line item is a countdown clock.

The practice.

1. Run the beachfront audit on yourself, in writing, this week. One column: everything you'd have to admit is depreciating if the wave is real, a skill, a role, a business line, a credential. Second column: what admitting it would cost you today. You can't divest what you refuse to appraise.
2. Find your Sasson drawer. Somewhere in your life is the prototype that got the "don't tell anyone" treatment: the side project, the pivot, the skill you shelved because it threatened your current income. Take it out. Give it one honest weekend and one real test with real people.
3. List the countdown clocks in your industry: every place customers pay for the incumbent's inconvenience. For each, ask whether your paycheck depends on it, and whether your next project could delete it. The first answer tells you when to move. The second tells you where.

Precedent P-18: The Mirror Twin (Tokyo, 2000s)

One company faced Kodak's extinction event and lived, which makes it the most important entry in the graveyard: it shows the deaths involved choices.

Fujifilm's own history puts the peak of world demand for photographic film at 2000. Then demand fell off a cliff, on the same schedule that hit Kodak. Fujifilm, Corporate History Instead of defending

the product, Fujifilm asked what it was actually good at: thin-film coatings, handling tiny particles, collagen chemistry (film is mostly gelatin), keeping photographs from fading. Its 2004 plan set out to find other uses for that accumulated know-how. It already had more than film to stand on, including its office-equipment business, and it aimed its chemistry at new targets: materials for flat-panel screens, healthcare, and, in 2007, a cosmetics line called ASTALIFT that the company ties directly to its photographic research. Fujifilm, ASTALIFT History

The mechanism. Both companies invested beyond film. Fujifilm's account shows how existing businesses and a deliberate search for new uses of its chemistry helped it absorb the loss. Kodak's digital investments didn't replace the old profits. The useful distinction is whether a new business can carry the people and capabilities you mean to keep, not whether management can name the coming technology.

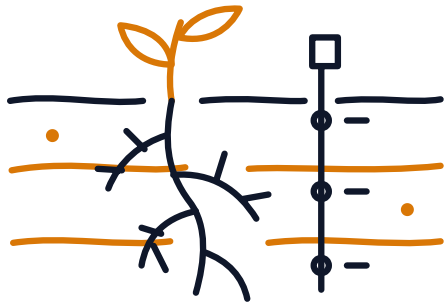
The rule. When the wave comes for your job, don't only ask how to save the product. Ask what you're actually good at, and where that skill is needed next. And be honest about the difference between a corporation's reinvention and a person's: Fujifilm had reserves, labs, and other businesses to carry it through. A person deserves somewhere secure to stand while the next use is found.

The practice.

1. Write your capability inventory tonight: one page, no job titles or products allowed. Fujifilm's list said collagen chemistry and thin-film coatings, not "film." Yours might say explaining hard things simply, calming angry people, seeing what's wrong with a system, making physical things precisely, earning trust fast.
2. For each capability, name two places it could be useful that aren't your current industry. You're not committing to anything. You're drawing the map you'll need on short notice, while you can still draw it calmly.
3. Test one small application before you commit to a costly pivot. And set an annual date to ask the Fujifilm question formally: has the wave moved, and which capability do I strengthen next?

CHAPTER 15

The Power of Reclaiming Soil



The living ground beneath the work. Original illustration for this book.

> *"The nation that destroys its soil destroys itself."*
> Franklin D. Roosevelt, *letter to all state governors* (1937)

In this chapter:

- Soil is a living system. Some damaged ground improves within seasons; the starting condition and the work decide what recovery is possible.
- Instruments lie with total confidence about everywhere they aren't. Your sensor lives at one centimeter; the root lives at thirty.
- The shovel is also a probe. It just logs to your arm.
- Know a site's history before you grow food on it. The people eating it can't be the experiment.

The reclamation beds went in on the worst ground the forty acres had: the quarter acre behind the shed where Walt Calder had parked equipment for four decades, compacted to a hardpan that rang under a shovel like a sidewalk. Priya chose it on purpose. "Anyone can garden in good dirt," she said. "The book we're writing needs the parking lot."

Not all of it, though. At the far end, where a tractor had bled hydraulic fluid through the Clinton administration, she drove four stakes and ran a string between them. “Not that end. That end gets a lab test and somebody who does this for a living before anybody plants a radish. The people eating this can’t be the experiment.”

That spring she built her trial like the scientist she had never stopped being: twelve beds on the old pad. Six she broadforked through the pan, buried in deep mulch, and seeded with a cover crop of vetch and rye; six got only the mulch. Everything else identical. And Elijah instrumented everything, because that was his lane and by now everyone including him knew it: moisture probes from the greenhouse-controller stock and a dashboard that made the rows look orderly from anywhere he stood. He liked that part. Priya asked for the readings weekly. She never once asked him to mistake the diagram for the ground.

In August he brought her his conclusion, and he brought it gently, because he liked her and believed he was about to cost her something. The dashboard showed the two sets of beds tracking identical on moisture, week after week. His irrigation controller had actually been watering the forked beds a touch more. “I can’t find your effect,” he said. “I’m sorry. Whatever the forking is doing, it isn’t measurable.”

Priya looked at his printout the way Marta had once looked at his renders. Then she handed him the shovel.

August, mid-afternoon, no shade worth the name. She had him dig two pits, one in a control bed, one in a forked bed, square-sided, knee deep, and she made him do it slowly, a spade-width at a time, and tell her what his arms felt. The control bed was honest about itself: four inches of nice crumb where the mulch had rotted in, then dust, then at ten inches the old pan, pale as concrete, roots turning sideways along the top of it like water hitting glass. The forked bed broke differently. The fork went in with his boot weight instead of his whole body. Dark crumb continued past the depth where the other pit had gone to dust. At two spade-depths the soil was cool, and held its shape when he squeezed it and broke cleanly when he poked it, and it

was full of channels, worm-bore and root-bore, going down. Roots at thirty centimeters, white and branching. A worm, indignant, at thirty-five.

"Now," Priya said, sitting on the bed edge like it was a lab bench, "tell me where your probe lives."

His probe lived at one centimeter, effectively. It was a surface-moisture sensor, and July had cooked the top inch of every bed equally, and his dashboard had done the one thing dashboards do best, which is extend a reading consistently in all directions. Down in the pits the beds differed exactly where the roots had gone. His map had been measuring the weather of the crust and calling it the climate of the planet.

"Your sensor lives at one centimeter," Priya said, not unkindly, in the tone of a woman who had spent twenty years watching agencies fund the crust. "The root lives at thirty. You instrumented the part of the system that's easiest to reach and assumed the rest agreed with it. Everyone does. Whole nations do." She nodded at the pit. "Roosevelt wrote every governor in the country in 1937: the nation that destroys its soil destroys itself. Every extension office I ever worked in had that framed on the wall, and every budget I ever sat through funded the crust anyway."

He sat in the dirt with his ruined conclusion and started laughing, because he had been here before. The vise, the shovel: the co-op kept administering the same exam and he kept being surprised by the questions. Since the month a $211 power bill had rearranged his life, he had trusted meters over feelings, and the trust had mostly paid. Here was the lesson under that lesson, the one the plug-in meter had never needed to teach because a wall socket is shallow: a meter tells the truth only about the exact place it lives. Instrument the wrong stratum and you don't get noise, you get confident fiction.

The fix cost about eleven dollars a bed: probes at three depths on a length of PVC, five, fifteen, thirty centimeters, checked against the shovel, which Priya made him keep doing monthly, pit and squeeze and smell, "so the instruments stay honest and so do you."

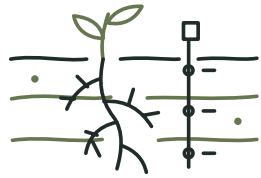


The probe. Three depths, five, fifteen, thirty centimeters.

By September the deeper probes were telling a different story than the surface ones: at thirty centimeters, the forked beds held water days longer after each irrigation. Elijah wanted to call it proof. Priya wrote it down in pencil. “One field, one season, and your irrigation wasn’t equal,” she said. “It’s a reason to run it again next year, properly. It’s not a recipe.”

What she got from him was the part that traveled. The graph went to the corkboard next to the repair emails, with the limits written beside it, and then into the library with the controller records. What he got from her was a habit he keeps to this day: once a month, whatever the dashboard says, he digs the pit anyway. By October he could read field capacity by squeeze, approximately, badly, the way you read a second language you learned as an adult. Priya would watch him squeeze a sample, then ask what he thought it could tell him. She still made him say what it couldn’t.

The notebook got its line that fall, under the arithmetic, where the important numbers live: DEPTH OF SENSOR: 1 cm. DEPTH OF ROOT: 30. Write your models accordingly.



THE FOUNDATIONS

If you want to know what the people with the most money think the future is worth, don't read their essays about artificial general intelligence or watch their keynotes about Mars. Look at what they buy.

For years, big investment funds and some of the richest people alive have been buying farmland, water rights, and timber. In January 2021, Land Report put Bill Gates's farmland holdings at about 242,000 acres, the largest private holding in the United States. Land Report, January 2021 newsletter Investment manager Nuveen also manages a large global farmland portfolio. Nuveen, Global farmland Whatever each buyer's reasons, the purchases make a point our very digital culture keeps trying to make us forget: **you can't eat code.**

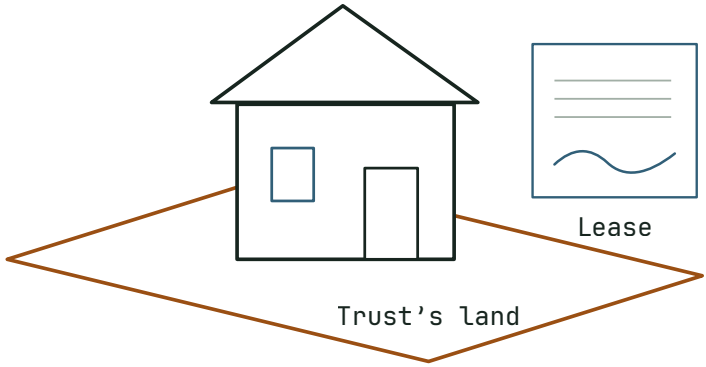
You can run your life through a phone, trade digital assets, and spend your evenings in a virtual world. But as long as your body is a wet, biological organism, you're tied to a thin layer of topsoil and the fact that it rains. The oldest enclosure is still the most important one: the ground under your feet.

Keeping the ground

Chapter 12 showed a land trust being born. Here's the machinery, because it's worth understanding.

Land and homes, separate

A common housing model; terms vary.



Household: owns the home
Trust: holds and stewards the land

Often a 99-year renewable ground lease.
Resale terms aim to retain affordability.
Taxes, debt and governance duties remain.

A common land-trust housing arrangement separates ownership of land and home. Resale terms need continuing stewardship; taxes and debt remain.

A community land trust is usually a nonprofit, governed partly by the people who live on and around its land. A common homeownership model separates ownership of the land from ownership of the home:

1. **The trust holds the land.** A household owns its home under a long ground lease, often ninety-nine years. Farming, rental housing, and other uses need their own agreements. Grounded Solutions Network, Community Land Trusts
2. **The trust stewards the land for the long term.** Its governing documents and leases set the restrictions. The legal structure has

to do the work; calling an organization a land trust doesn't do it for you.

3. **Stewardship passes on.** If a household leaves, it can sell its improvements, but at a formula that keeps them affordable for the next household.

None of that is magic. A trust can still fail to pay its bills, be badly governed, or lose its land, as New Communities did in Georgia before it came back. But it changes who the land is for.

Keeping the taxes payable

Property tax is the quiet way land changes hands: miss enough payments and the county can eventually sell the deed. There are legitimate, well-trodden ways to keep that bill manageable for working land:

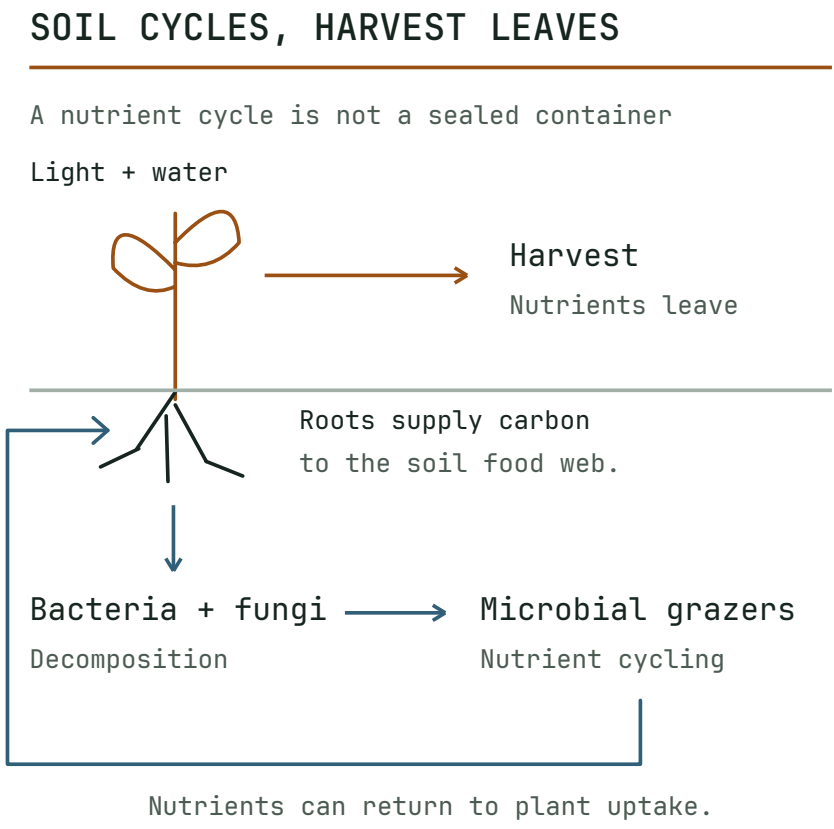
- **Agricultural and timber classifications.** Ask which use-based assessments your jurisdiction offers and what qualifies. In California, Williamson Act contracts can exchange restrictions on land use for assessment based on agricultural or open-space use. The state cautions that the tax benefit varies by property and isn't always greater than its existing treatment. California Department of Conservation, Williamson Act FAQ
- **Conservation easements.** A lasting restriction on development changes what future owners may do. Ask the assessor how it affects this parcel's valuation. A qualifying donation may also receive a federal income-tax deduction, with specific legal and valuation requirements. That's a separate question from the annual property-tax bill. IRS, Conservation Easements

Talk to your county assessor and a local land-use attorney before you count on either. And never mistake lower taxes for no taxes: a plan that depends on a loophole is a plan waiting for a letter.

Soil is alive

Once the ground is secure, you face the soil itself.

For generations, industrial agriculture has treated soil mostly as a place to hold roots up and pour fertilizer in. Heavy tillage and heavy synthetic inputs can suppress a lot of the life that used to do some of that feeding for free, and a farm that leans entirely on bought inputs is only as independent as its supplier. The alternative isn't to reject every bag of fertilizer. It's to rebuild the living system underneath.



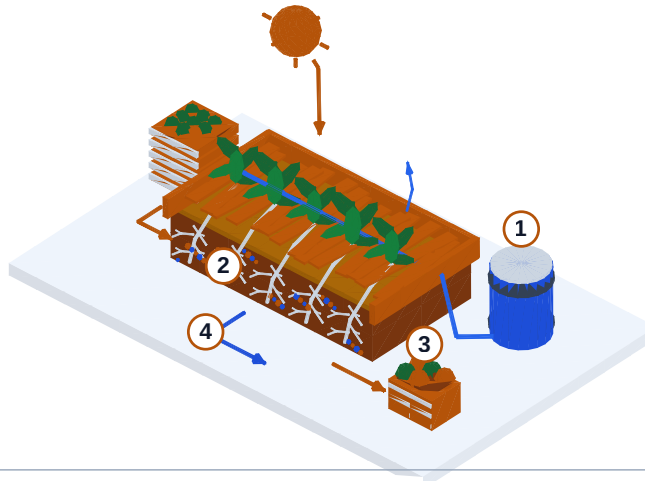
The soil food web cycles nutrients. Harvest and other losses remove them, so a productive bed still needs measurement and appropriate replenishment.

A teaspoon of healthy soil can hold enormous populations of bacteria, alongside fungal threads, protozoa, nematodes, and other life. Together they run an economy:

1. **The exchange.** Plants turn sunlight into sugars and leak some of them out through their roots.
2. **The mining.** Bacteria and fungi feed on those sugars. Some fungi extend a plant's reach for water and phosphorus; legume bacteria fix nitrogen from the air (Chapter 9). These are different partnerships doing different jobs.
3. **The release.** Protozoa and nematodes eat bacteria and fungi and release nutrients near the roots in forms plants can use.

Cover crops, compost, deep mulch, and less tillage all feed that system. University of Minnesota Extension, Soil Biology What none of them do is make a garden feed itself forever. Every harvest carries nutrients away in the food, and something has to put them back.

A growing bed has an outside



1 INPUTS
Sun, water, replenishment

2 WITHIN THE BED
Roots and soil life

3 HARVEST
Nutrients leave in food

4 WATER OUT
Plants, air, drainage

Conceptual cutaway. Arrows are not quantities.

The bed is part of a larger system. Plants, soil life, and returning organic matter exchange nutrients; sunlight, water, replenishment, and the nutrients carried away in harvest still cross its boundary. Conceptual cutaway, not a measured soil profile.

Know the ground's history

Priya's string line matters as much as her shovel. Old farm yards, rail lines, parking lots, and industrial sites can carry lead, fuel, solvents, and worse, and a healthy-looking plant can't tell you the ground is safe to eat from. Before you grow food on land with any history, find out what that history was and get the soil tested. EPA's urban-agriculture guidance is a good starting point for what to ask. EPA, Urban Agriculture

Cleaning up damaged ground belongs in the larger ambition, and machines may help specialists do it more safely. But contaminated land is a job for people equipped to handle it, not a neighborhood experiment, and a garden bed isn't where you find out.

Pool the work

Growing food takes labor, tools, water, and attention across the season. A household doesn't have to provide all of that alone. Pool it:

- **A tool commons.** Instead of five neighbors each buying a fragile little tiller, pool your money for one heavy-duty walk-behind tractor with swappable implements, and a shed to keep it in.
- **Work by the season, not the clock.** A coordinated weekend crew can share a major planting or harvest. Keep watering, crop checks, and the next round of harvesting on a rota too; the garden keeps growing between work parties.
- **A community seed library.** Collect and share open-pollinated varieties adapted to your own climate, whose seed you can save and replant. Many hybrids don't come back true from saved seed, and some patented varieties come with contracts that forbid saving it. Your own seed is a living library, shared freely.

From a plot to a plate

Go back to Chapter 9's five-household trial. The garden records the crop and the usable harvest. A food partner handles the rest. The record shows what was lost, what reached people, and what still had to come from elsewhere. A garden can contribute to a food baseline without being described as self-sufficiency; the baseline has to meet people's actual diets across seasons and interruptions.

Touch the dirt

1. **Find your legal core.** Three to five people nearby who want the same thing. Look up how land trusts and co-ops are set up in your state, and find a sympathetic local lawyer.
2. **Check, then dig.** Get permission and establish the site's history before disturbing it; use the testing advice above where contamination is possible. On suitable ground, compare texture, roots, and moisture at several depths. Worms alone aren't a verdict on soil health. Ask an established grower what organic matter

your plot needs, and a grocer or café which suitable scraps they can supply for compost (Chapter 14).

3. **Learn from someone who knows the ground.** Visit an established grower, with permission, and ask what observation changed their mind recently. What did they think was happening? What did they check? What did they change?
4. **Look for the overlooked ground.** Tax-delinquent lots, county land banks, church land, underused farmland. Find out who's responsible for it and who used it before.

Grab a shovel. Look after the soil. The ground is the one thing nobody's making more of.



Precedent P-19: Organizing the Harvest (United States, 1943 to 1944)

When the war strained the food system, the United States government asked amateurs, office workers, schoolchildren, grandmothers, to grow food. Not as a symbol. As logistics.

And it worked on a scale that's hard to believe now. By the National Park Service's account, in 1944 some 18.5 million Victory Gardeners supplied about forty percent of the nation's fresh vegetables. National Park Service, Victory Gardens on the World War II Home Front Not agribusiness. Yards, vacant lots, schoolyards, and rooftops, run by people the food industry would have written off as amateurs the year before.

It didn't happen by itself, either. The USDA's 1943 program told organizers to think about the bus route: a family without a yard needed a plot they could reach by streetcar, bus, or bicycle. Somebody had to get permission for the land, prepare it, assign plots, arrange water, and supply tools and advice, and the county agricultural agent was supposed to judge whether the ground was fit. School gardens

needed paid help through the summer. A shortage of pressure canners prompted plans for shared canning centers, because a vegetable left to rot isn't food. USDA, The 1943 Victory Garden Program

And there's a part the cheerful posters don't carry. Japanese Americans grew Victory Gardens inside the camps where their own government had imprisoned them. Their skill made life better under conditions they never chose. A national food campaign didn't mean equal standing in the nation.

Then the war ended, the campaign stopped, and the capacity faded, which is part of why your neighbor thinks food comes from a truck.

The mechanism. Distributed production scales sideways. Millions of small plots came online faster than any central capacity could have been built, because the capacity already existed, dormant, in ordinary soil and ordinary hands. The bottleneck was the decision to organize it: land, water, seed, advice, and a way to get the harvest to a kitchen.

The rule. Standby capacity is real, and so is its limit: a vegetable harvest is a contribution, not a complete diet.

The practice.

1. Activate one plot this season, however small: a raised bed, three buckets on a balcony, a community-garden plot. Then measure the output honestly: kilograms harvested, meals made. It'll be small the first year, and that's fine. The Victory Gardens weren't built on experts. They were built on millions of first-years.
2. Make it a network, not a hobby. One garden is produce; a street of gardens is infrastructure. Find the two or three people nearest you who grow anything, trade one variety or one skill this season, and follow one harvest all the way to someone's table.
3. Keep the standby system warm: save seed from one crop, keep the tools maintained and the beds alive over winter, and grow at least one thing every year. Capacity that never gets exercised is a rumor, not a capability.

CHAPTER 16

Digital Leverage and Media Autonomy



Server racks. A useful record can travel beyond the people who made it. (Victor Grigas, CC BY-SA 3.0, via Wikimedia Commons)

> *"Freedom of the press is guaranteed only to those who own one."*
> A. J. Liebling, "The Wayward Press: Do You Belong in Journalism?", *The New Yorker* (14 May 1960)

In this chapter:

- Code and media can work while you sleep. They're the closest thing to employees that never get tired, and nobody's permission is needed to start.
- The machine flooded the world with synthetic content and created a shortage it can't fill: proof that a real thing happened to real people.
- Trust doesn't scale, and doesn't have to. Files scale. Ship the files; keep the trust local.
- Consent, privacy, and corrections belong in the publishing process. Neither a platform nor a file format guarantees income or permanence.

The video that did it was the first fire.

Denny had filmed the shouse build the previous summer and fall, the way he documented everything by then: hands in frame, the county audible in his vowels. Now, a year after the first fire, he went back through the footage and made a four-minute account of the work. The cut, the columns, Marta's bead going down the header seam like a zipper closing, the assessed stove with its replacement firebrick, and at the end, the draw catching on the first match and the co-op standing around in the new warm doing nothing historic whatsoever. He ended it the way he had ended the first video, on the same porch of words, because Denny knew a load-bearing sentence when he'd built one: "Come see it. Bring your kid."

It posted on a Thursday night. Elijah did the pipes and went to bed.

A week later he was up before light, and he walked to the shed through the first hard frost of the year with coffee, to check the overnight telemetry the way other generations of men had walked out to check stock. The mesh was green. The greenhouse was green. The channel graph was a wall.

He'd seen spikes; the leak postmortem had been a spike. This was compounding, the curve he'd been drawing in the notebook for two years, the one that lies to you at the beginning and then doesn't, except for the first time in his life it was under something he loved instead of over it. The first-fire video had dragged the whole catalog up behind it: the greenhouse controllers, Frank's move, the sensor mistake, all of it feeding all of it, person to person. He wanted to draw a line through it and extend it. Instead he opened the inbox. He read it standing up, coffee going cold.

A grange association in Idaho: how do we start. A parish council in Louisiana, two months out from the last flood, asking who could help assess its growing site. A machinist in Ohio with his father's shop and no work in it, asking for the controller records. A tribal housing authority. A retired county engineer. A woman in Maine who had, she wrote, four acres, a dead barn, and eleven neighbors who show up. Eleven towns by the time the sun was fully up, spread across nine states, every one of them asking some version of the same four words: how do we start.

He rang no bells. He put it all on the big screen at Friday dinner and let the room read.

Curtis got to the fence first, because Curtis always did, except the fence had changed shape over nearly two years like everything else about Curtis: "That's a product," he said. "That inbox is a business. Course packs, plans, consulting. People pay real money for half of what we know." He wasn't wrong, and nobody said he was, and for a minute the old grammar hung over the table, the Claypot grammar, and Elijah felt it reach for him like a current he knew by name: scale it, platform it, first-mover. When he first arrived he'd have already had the deck built.

It was Frank Kowalski who took it apart, from the end of the table, in the tone of a man doing inventory. "I spent thirty years selling parts for two dollars over what the internet charged," Frank said. "Know what I was actually selling? Somebody who knows which part. You can't download me. Him neither." A thumb at Denny. "Fence the files and you're selling paper. Ship the files free and every one of those eleven towns needs a Frank and a Marta and a Priya of their own anyway, and they'll find them locally, because that part doesn't ship." He folded his glasses. "The store taught me that. Late."

"The press did this once before," Elijah said, and kept it to two sentences, because he had learned. "Books used to be chained to desks. Somebody built a machine that made copying cheap, and the people who shipped did better than the people who kept the chains." He thought about the leak. "And we can let somebody read our mistakes without buying the whole project. That would have saved me some money."

Marta was quiet until the end, which is how the co-op knew where the decision would land. "I told this one his first day," she said, nodding at Elijah, "that nineteen households who know whose kid is whose don't scale, and that's the load-bearing wall, and I meant it, and it's still true." She looked at the wall of numbers. "The wall doesn't scale. The blueprints of the wall scale fine. Ship them."

So they prepared a release that weekend, and preparing it took longer than uploading it. The controller files came with the repair card and a record of the conditions they had actually tested. Priya sent the sensor notes, including the comparison they still needed to repeat. Marta marked the building drawings as records of their particular project, not instructions for any site. Reuben replaced the grand legal title with questions another community would need to answer for itself. Different files needed different permissions; he checked that the co-op had the right to share each one.

Elijah put the leak postmortem beside the successes. Denny asked the people visible in the footage which parts they wanted kept out. The first folder was smaller than the one they'd imagined, and more useful for it.

Monday morning Elijah stood in the shed at dawn again with the coffee, watching the library's download counter climb through four digits, people he would never meet retrieving work that had taken the co-op its mistakes to earn. Out of habit his hands started drafting the status report. This time he knew who needed it: Denny, who owned the channel; the members who paid for hosting; the people answering requests. He added a line for unanswered messages and another for the next month's costs. The platform still set its own terms. The co-op kept separate copies and a public index it could move.

An email from the machinist in Ohio came in while he worked. One connector label in the controller record was wrong. Elijah opened the drawing, found the error, and thanked him by name.

At Friday dinner, Denny read that message before he read the view count.

— ◈ —



THE FOUNDATIONS

At some point early this century, a lot of us fell for one of the best magic tricks ever performed. We were handed free email, free social media, and free video hosting, and in exchange we handed over our attention, our behavioral data, and a fair bit of our autonomy. We weren't the customers. We were the crop. Every photo, post, and video helped build the systems now learning to do our jobs.

But a magic trick only works while the audience looks where the magician points. Step behind the curtain and it falls apart.

Why the machine needs you

Chapter 10 walked through model collapse: in repeated training that replaces real data with generated output, rare patterns can disappear and errors can accumulate. That isn't the result of every use of synthetic data. The flip side is this chapter's whole argument. In a world filling up with synthetic content, **something that actually happened, recorded by the person it happened to, is scarce.** Your stories, your mistakes, your local experiments, the record of what failed on the ridge and why: that's the valuable stuff now. Chapter 10's algorithmic judo is how you get it read. This chapter is about keeping control of it once it is.

Let the useful thing travel

A small community can't do all the work of feeding humanity. It can figure out something useful and make it easier for the next community to learn. A repair that took a week might take the next person an

afternoon because somebody left an honest record. A failed trial can keep the next group from spending scarce money on the same mistake.

That's the hopeful answer to scale. People and places differ, but the knowledge they trade doesn't have to stay locked inside the group that earned it. Share the conditions along with the result, so someone else can judge what carries over. The controller records are useful precisely because they say what was tested and what still failed. And don't mistake Denny's audience for a forecast of income; it's a fictional success. Most useful sharing is quieter: a phone call between two kitchens, a corrected pickup notice, a drawing another repairer can follow.

People before the story

Ask permission before publishing someone's image, circumstances, address, or request for help. Consent to receive food isn't consent to become content. Give people a route to participate without appearing in the documentation.

The next section handles the files. Permission comes first: no technical setting makes consent unnecessary.

Media autonomy: keep your own press

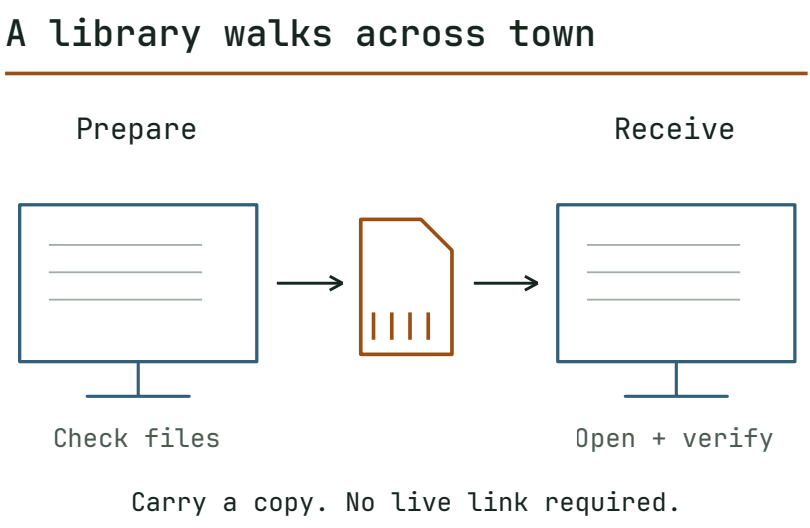
Platforms are a temporary distribution channel, not a home. If everything you've made lives on servers owned by someone else, it can vanish with a policy change or a closed account.

Clean your files. Photos from a phone usually carry metadata embedded in the file, EXIF data, which can include the GPS coordinates where the picture was taken, the date and time, and the device. Free tools like exiftool or Metadata Cleaner can strip it. That handles one kind of exposure. It doesn't remove a street sign in the background or a caption that names someone, so look at what you're actually publishing.

Distribute copies, not just links. Peer-to-peer systems like IPFS address a file by what's in it, a cryptographic fingerprint, rather than by which server it's on, and anyone can help host a copy. That's great for keeping something findable. It isn't a promise of permanence: a file stays available only as long as someone keeps it and serves it.

Keep a sneakernet. Sometimes the most reliable network is a person with a pocket. A single small memory card can hold a whole library of manuals, maps, and a full offline copy of Wikipedia. Passing drives hand to hand avoids a lot of network exposure. It doesn't make a file immune to loss, damage, or copying, so encrypt what's private and keep more than one copy.

Build a local archive. Kiwix reads downloaded reference collections offline. Choose the language and edition you need, check the download size against free storage, and test it with the network off. Wikipedia's mini files contain article introductions; nopic keeps full articles without images; maxi is the full edition. Add the manuals and records your community uses, and record each collection's date. You're now the librarian of your neighborhood. Kiwix, FAQ



A hand-carried copy needs no live network link. It still needs trusted handling, a compatible reader and a recoverable backup.

Creative immunity

Making things isn't only logistics. It's how you keep your mind.

The feed is built to keep you reacting. When you write, draw, compose, code, or build, you flip from receiver to transmitter, and your attention comes back to you. A community that stops telling its own stories eventually starts believing the ones told about it. Keep your history, your humor, your failures, and your hope in your own words.

Who gets to speak

A service can collect excellent operating data and still tell a self-congratulatory story. The organizer reports meals distributed; someone receiving them reports that pickup happens during their shift. Both

statements can be true. If only the first reaches the public account, the project has made its own reputation easier to protect than its recipients.

Give people a way to correct the story without risking assistance. Publish changes prompted by criticism. Tell readers which improvements have reached the people served and which remain plans. The author of this book owes you the same discipline: enthusiasm is a reason to look harder at whether a promise holds.

We can want the machines to help and still report a machine's failure plainly. An ambition that can survive correction is more useful than an image that must never crack.

This week's handoff

1. **Clean your output.** Install a metadata stripper and make cleaning photos a habit before anything leaves your devices.
2. **Start the archive.** Download one Kiwix collection and the manuals your household or co-op actually needs. Put a copy on a second drive, then open it on the device you'd use during an outage.
3. **Make one useful thing for someone else.** Offer a food group a simple choice: a corrected public schedule, a checked translation, a clear page on how to ask for help, or a short account of one repair. Let them choose, including "none of those." Finish it, and confirm who'll keep it current.



Precedent P-20: An Argument in Circulation (Philadelphia and London, 1776)

Thomas Paine had been in America barely a year. He was a staymaker's son and a twice-dismissed excise officer: no office, no fortune, no name worth borrowing. What he had was a printer and a way of writing that ordinary people could read aloud in a tavern.

Common Sense came off Robert Bell's press in Philadelphia in January 1776, anonymous, in a paper wrapper, for two shillings. Clements Library, Going Viral in 1776 It ran through printing after printing that year and helped turn a quarrel about taxes into an argument for independence, in language people could carry into their own kitchens.

But look at what actually happened on paper, because it's the whole lesson of media power in miniature. Paine and his printer fell out almost immediately. Paine meant his share of the profits to buy winter clothing for the Continental army; Bell claimed there were no profits. So Paine took an enlarged edition to William and Thomas Bradford. University of Minnesota, Common Sense Across the Atlantic, some London editions cut the passages about the king, and in one surviving copy of John Almon's edition, a reader filled the gaps back in by hand. The author framed the argument. Printers could reproduce it, revise it, or censor it. Readers could fight back in the margins.

The mechanism. Leverage is the ratio of reach to permission. Paine needed nobody's approval beyond one printer. When a new medium drops the cost of reach, the first people to use it gain influence far out of proportion to their credentials. But whoever controls the press still has a hand on the message.

The rule. Give people something clear enough to discuss and check, and own enough of your own press that nobody else gets the last edit.

The practice.

1. Write your pamphlet, even if it's four pages. Pick the one thing you know from experience that would really help the person two years behind you, and write it the way Denny talked on camera: plain speech, what happened, what you'd do differently. Publish it this month.
2. Own your printing press before you need it. A platform account is a stall in someone else's market; an email list and a site you control are a press. Set both up now, even with an audience of nine, and point every platform reader toward them. Use the list to reach readers who've asked to hear from you when the algorithm changes.
3. Run the tavern test. Read your latest piece out loud. Every sentence you stumble on, every word you'd never say to a neighbor over a fence, rewrite in plain speech. Then ask one reader to explain it back to you and name its weakest point, and fix that before you share it wider.

CHAPTER 17

Tools of the Trade



The workbench. Own the tools, own the output. (Shixart1985, CC BY 2.0, via Wikimedia Commons)

> *"We are as gods and might as well get used to it."*
> Stewart Brand, *Whole Earth Catalog* (1968)

In this chapter:

- The tools that survive the transition are the ones you can understand, fix, and run with no cloud, no subscription, and no permission.
- Networks fail at the node somebody flashed in a hurry. Discipline you skip in fair weather collects its debt in the storm.
- Debugging in the dark is a solved problem: halve it. Power or brain, this side or that side, measure, cut again.
- Give the machines a worthwhile assignment: food first, then cleanup, shelter, and care. The proof is already on people's workbenches.

The first hard storm of the winter came up the valley on a Friday afternoon with the barometer falling like a dropped tool, and by dark the county was gone. Grid power went first, a whole hillside of windows going out between one gust and the next. The cell towers lasted

forty minutes longer on their batteries and then they went too, and the valley was back in 1890, except for nineteen households, where it was still Tuesday. Somebody brought the printed contact list to the shop table. Somebody else started the soup.

The mesh held. That's the sentence Elijah had been building toward for a year, the LoRa lattice laid node by node across the ridgelines, and on the night it finally mattered it did what it was designed to do: shrugged. The co-op's map stayed lit, house to house, short messages hopping the valley on watts you could count on your fingers. Marta's shop stayed lit. The greenhouse alarms stayed armed. The whole nervous system he'd been laying into walls and fence posts for a year kept right on reporting, calm as a pulse.

Except node 7.

Node 7 was the ridge repeater, the high hop that stitched the north houses into everyone else, and node 7 was his, personally, in every sense that stung: his design, his install, and, the previous Sunday, his firmware update, done alone, in a hurry, ahead of the weather, without a second set of eyes, because it was "just a repeater" and the checklist law was for things that touched the open internet, and he had let himself believe that discipline was a perimeter instead of a practice. The storm found the ridge at eight o'clock. At 8:04 the map showed a hole where the north houses used to be.

The north houses checked in on the voice radios, one by one. Everyone was fine. The hole was only in the map. But the hole was his.

He suited up. Nobody climbs the ridge alone, and Curtis was already lacing his boots, which at the co-op amounted to a verdict. Marta handed Elijah the handheld radio and the small pack, pointed at his headlamp, and said the thing she always said instead of luck: "Talk before you touch."

The ridge trail in a storm is a different country. Rain going sideways with gravel in it, madrone trunks slick as wet brick, the wind doing shift work: lean, release, lean. He climbed with the multimeter zipped inside his coat to keep the display warm, Curtis's light steady

behind him, and the radio crackled at his collarbone every quarter mile, Marta running the net from the shop the way she'd once run a floor: short transmissions, no chatter, callsigns and traffic.

And he could hear, in the gaps, the system working around the hole he'd made. Priya's greenhouse, riding the alert chain: low-temp alarm at 8:40, relayed around the dead ridge the long way, south through Frank's node, and Denny's voice acknowledging, hauling the backup battery to keep the winter starts alive, all of it happening without him, around him, because the network's whole design assumed any one node, any one person, could be the one that failed. It is a strange, strong medicine, hearing a thing you built survive you.

Node 7's enclosure was where he'd bolted it, on the old fire-lookout stub, and the node was dark. Curtis held a garbage bag over them for a rain fly while Elijah got the lid off, fingers going stupid with cold, and keyed the radio, and he and Marta debugged it across two miles of storm, Marta asking the only question she ever asked:

"Power or brain? Halve it."

Battery voltage: 12.9, holding under load. Power. Halved. "Downstream. Check the buck converter under load." Five point one volts, solid. Halved again. So: brain. Elijah connected the service cable and read the startup log aloud. The controller booted; the radio never initialized. He checked the installed version against the last working record. Sunday's update had changed the startup configuration, and he'd accepted one successful transmission as a test. He hadn't cut the power and brought it back. The storm had done that test for him. The vise, the shovel, and now the ridge: the same exam, a third time, in worse weather.

The fix took eleven minutes because he had, at least, packed like Marta taught: the service laptop, the cable, and the known-good image and configuration, hashes checked, in the dry bag next to the spare fuses. Restore, restart, cut the power, start again. The status light went to its slow steady breathing, and in his collar the radio said, in Priya's voice, "North houses just came back. Hello, ridge."

Below them the whole county was black to the horizon, and across that blackness the co-op's houses held their small stubborn constellation, and node 7 was a star in it again instead of a hole.

He stood up into the wind to pack the tools, and Marta's voice came up the hill one more time, dry as the inside of the enclosure was supposed to be. "Brand's line. On the shop wall. 'We are as gods and might as well get used to it.' He changed it, later. Get *good* at it." A crackle, wind on her end too, the shop door open to the weather. "Gods, hell. Gods is the easy part, college. Gods with cold fingers and a multimeter, that's the job. Getting good at it means next time you flash a node, somebody's watching. Come down. There's soup."

They came down. There was soup. And the checklist got its amendment that Sunday, in Reuben's careful language: the two-eyes rule now covered everything people depended on, with a recorded recovery check, because the network doesn't care whether the packet that kills it came from the internet or from your own confident thumb. The storm blew out by noon Saturday. The county's power was back Tuesday. The co-op barely noticed, except as traffic on a mesh that had already moved on to arguing about seed orders. Curtis asked who still needed soup taken over.

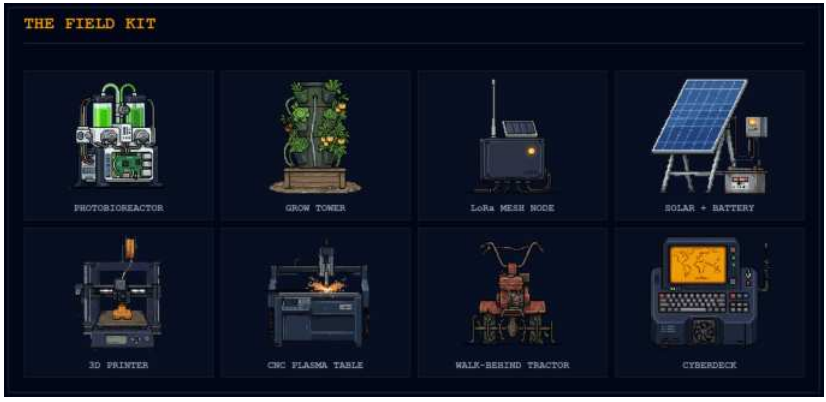


THE FOUNDATIONS

We live in a golden age of high-tech dependency.

Your refrigerator may have more computing power than the Apollo guidance computer. It can track your milk, reorder groceries, and show recipes on a touchscreen. It can also stop working properly when the manufacturer's cloud has a bad day, nag you about a subscription, and report more about your household than you'd ever tell a stranger. Call it the smart-home trap: every door, thermostat, bulb, and tool quietly depending on a server somebody else owns.

The alternative is simple to say: **tools you own.** Repairable, documented, able to run without an internet connection, and understandable by the person using them. If you can't understand how it works, can't fix it with ordinary tools, and can't run it offline, it's less a tool than a leash.



The field kit. Every one of these already exists, and a person with modest money and stubbornness can get their hands on most of it.

Give the machines a worthwhile assignment

Imagine the priorities written above the workbench: feed people, restore damaged places, make safe homes, relieve suffering. Those are assignments worthy of a powerful technology. They're also a way to judge the uses of it that actually receive our money and attention.



Tools with a history of use. AI-adapted collage from Shixart1985, CC BY 2.0.

Food comes first in this book because nobody needs an elaborate explanation of why dinner matters. The wider ambition includes land care, waste cleanup, housing, and health. These projects can proceed together. Somebody who needs treatment or shelter shouldn't have to wait for the last food problem on Earth to be solved.

The question for any proposed machine is what useful work it can do under the conditions where people need it. Compare the tools they can obtain and support. A machine impressive in a video may be hard to repair locally. A familiar tool may do the job well. Shared access can matter more than ownership.

The proof isn't hypothetical

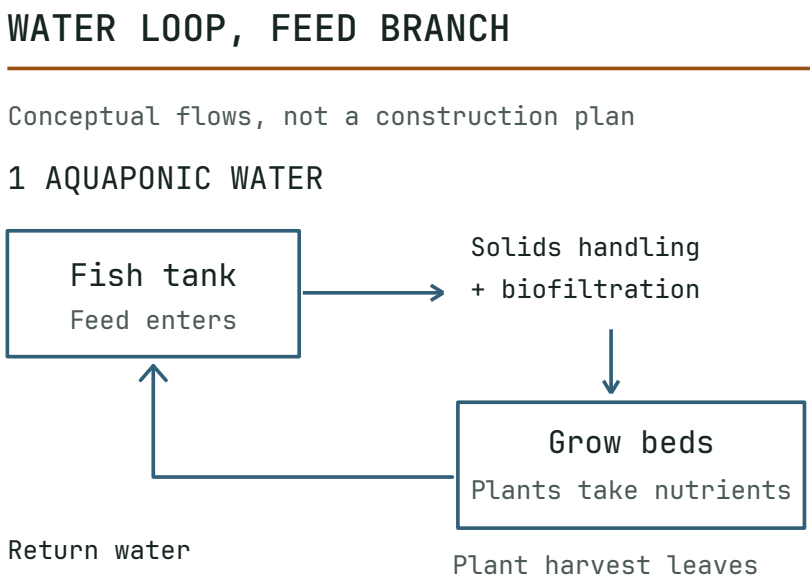
Everything in this chapter already exists, and you can watch people build it this weekend. This is the part the doom crowd never accounts for: while the comment sections argue about whether the machine ends the world, a growing number of people are pointing the same technology at their own food and posting the plans.

One of them, working under the handle SECTOR 07, wanted his home aquaponics loop to stop depending on a store. The system grows fish and vegetables together, but it still needed bought fish food, so the loop was never closed. His fix was to grow the food too: microalgae, cultivated the way labs do it, in a machine called a photobioreactor. Commercial units are priced for research budgets. He printed parts for his on a Bambu A1 and ran it on a Raspberry Pi and two Arduinos,

with sensors he made himself out of magnets. He reports about eight grams of algae a week. He says plainly in his own build log that it isn't finished: features he isn't using, automation he never completed, a design he'd rework for a second version. That honesty is the point. This isn't a product launch. It's a person learning in public. SECTOR 07, I 3D Printed a Photobioreactor



A home-built algae culture system. The builder's experiment, not a cost or performance comparison with laboratory equipment.



Growing an ingredient can reduce a bought input. It doesn't establish a complete fish diet or a closed system.

Feed still has to meet the needs of the fish species and life stage. Algae can contribute to some diets; that doesn't show that this prototype replaces balanced feed, its mineral inputs, or the work of maintaining water quality. FAO, Use of Algae and Aquatic Macrophytes as Feed in Small-scale Aquaculture

Or take the one-man operation behind EasyGrow, run out of a garage. He designs recirculating vertical grow towers that, by his account, pack about eight times the plants into the same footprint and use about ninety percent less water. When off-grid customers told him his pumps needed power they didn't have, he taught himself to build wind systems, hand-winding axial-flux alternators, and rural antennas so greenhouses could report their own status. By his account his towers feed a stadium in Florida and lettuce gardens in Nigeria, and his wind systems turn in Buenos Aires. Ask him why he won't build pots cheap enough to need replacing, and he'll tell you: "I'm not in business to get rich. I'm in business to grow food and help people grow food." Kirsten Dirksen, Mad scientist's homestead

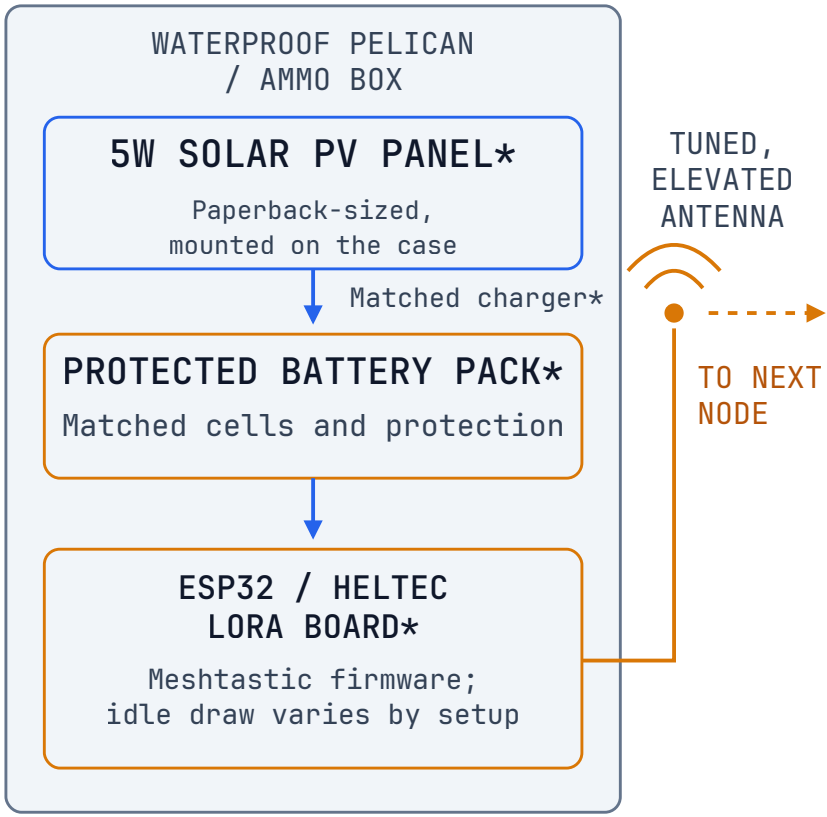
Now hold that next to the other version of the very same technology. Some industrial vertical-farm vendors sell nearly identical hardware with the opposite pitch: their promotional material brags that the system reduces labor costs and decreases human touch. Same sensors, same pumps, same automation. One spirit builds the tool to take people out of the loop. The other builds it to put a scared, capable person back in charge of feeding their family, and gives away the plans. The tools don't choose which future they serve. People do.

The nervous system: mesh nodes

Chapter 14 made the case for a neighborhood mesh. Here's the hardware. A typical node is a small, cheap board that pairs a microcontroller with a LoRa radio (the Heltec V3 and LilyGO T-Beam are popular), running the open-source Meshtastic firmware.

AN INDEPENDENT MESH NODE

The power path and radio link.
A component map, not assembly instructions.



*Illustrative hardware. Use protected cells and a charger matched to the battery pack. Match 915 / 868 MHz hardware, firmware and regional settings to local radio rules. Reach depends on terrain and line of sight. Solar runtime depends on weather and load.

A mesh node. Sunlight in, short encrypted messages out.

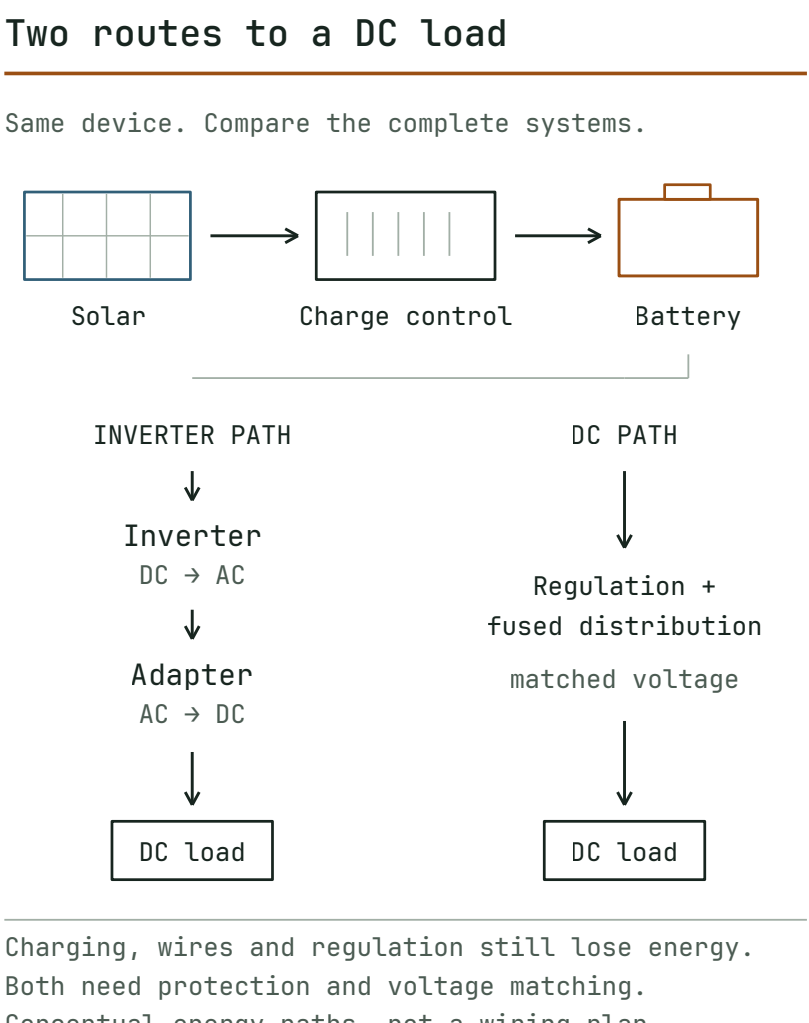
- **Tiny power draw.** A node sips a fraction of a watt most of the time. A small panel and a battery can keep it going through a lot of weather.
- **Match the hardware and region.** Common choices use the 902 to 928 MHz range in the United States or the 868 MHz range in Europe. Choose hardware and an antenna for the local band, install firmware for the exact board, then set the legal region in the software. Power and airtime limits still apply. Meshtastic, LoRa configuration
- **Relay, then verify.** Nodes with compatible radio settings can relay messages where a usable path exists. Test the path with an important node switched off, and test a full power cycle after an update.

Encryption isn't automatically privacy. Meshtastic's default shared channel key is public. A private group needs a fresh key shared only with its members, and a plan for lost devices and exposed keys. Even then, a group message isn't proof of the sender's identity. Keep sensitive household records off the public channel and confirm consequential requests with the person. Meshtastic, encryption and its limits

To put one up: use a weatherproof box, a proper solar charge board and battery rated for outdoor use (buy it assembled if you're not sure), and the best antenna you can mount high. Height and line of sight matter more than anything else; a well-placed node can reach many miles. Then check what it can actually carry before anybody depends on it.

The muscle: power you understand

Many small loads use low-voltage DC internally: LED lights, phones, laptops, radios, small pumps. A conventional off-grid system turns the panels' DC into household AC with an inverter, and a device's power brick may turn it back into DC. Each conversion loses some energy as heat; the amount depends on the equipment and its load.



For a load that already uses DC, compare the whole path. Avoiding an inverter pair does not remove charging, wiring or regulation losses.

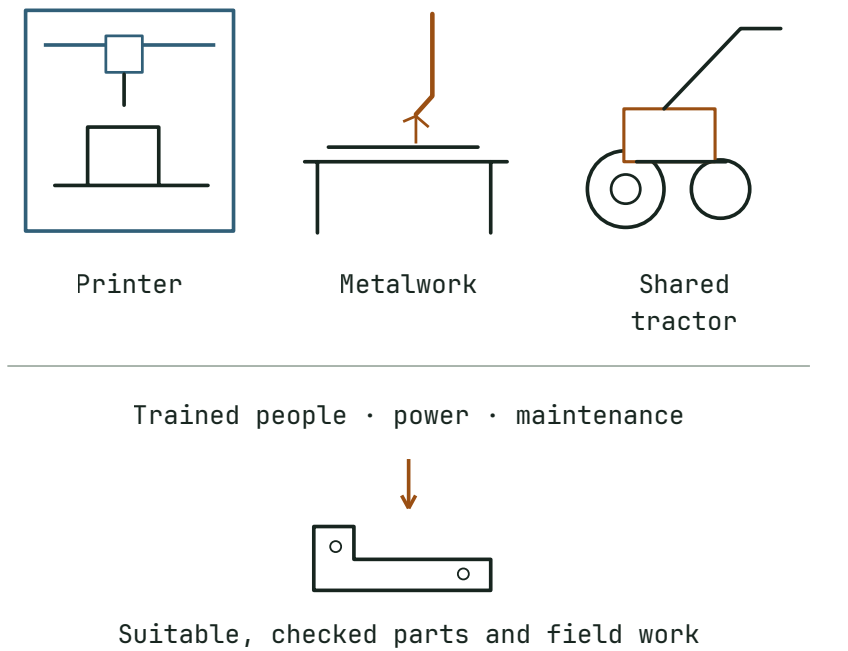
So for suitable small loads, a DC-native setup, panel to charge controller to battery to a fused distribution block, can be simpler and more efficient. It isn't lossless. Charging, wires, and any DC-to-DC regulator still cost energy. Match each device to its required voltage; a

USB port isn't a bare connection to a 12-volt battery. Compare the whole system under its intended load. US Department of Energy, DC Microgrid Scoping Study, conversion losses

Batteries store a lot of energy and can fail dangerously if they're built or charged wrong, so use a packaged battery with suitable protection, the specified charger, and properly sized fuses. Have someone who knows electrical work check it. Plan replacement from the battery's rated service conditions and your actual use; chemistry alone doesn't set its lifespan.

The skeleton: machines that make machines

Eventually something breaks: a gear, a bracket, a bolt. If every replacement part has to come from a distant distributor, you're still waiting on permission.



A neighborhood fab lab combines suitable tools with trained people, supported materials and checks for the actual job.

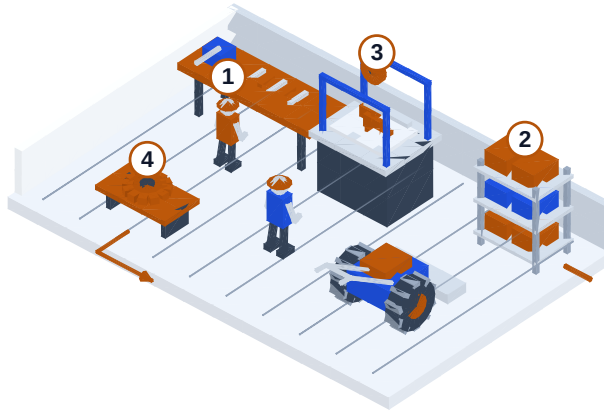
The Open Source Ecology project's Global Village Construction Set documents open designs for dozens of the industrial machines a small community might need, tractors, presses, printers, and more. Three good starting points for any neighborhood shop:

1. **A 3D printer.** Open, repairable designs like the RepRap family, some of whose parts can be printed by another RepRap, are ideal. Plenty of people start, like SECTOR 07, on an inexpensive

consumer machine; just know which parts you can fix and which you'll have to buy. People have turned discarded PET bottles into recycled filament, usually called rPET, with small homemade extruders, which turns trash into brackets, gears, and replacement knobs.

2. **A walk-behind tractor.** A heavy two-wheel tractor with compatible swappable implements can serve several jobs. Compare the actual task, attachments, operator effort, maintenance, and fuel use with the equipment you'd otherwise need. Share one among several households where that comparison makes sense.
3. **A cutting and welding station.** A CNC plasma table and a good welder turn steel stock into brackets, racks, and repairs, in the hands of someone trained like Marta.

What keeps a workshop useful



1 PEOPLE
Skill, access, time

2 SUPPLIES
Stock, power, spares

3 SHARED TOOLS
Operate and maintain

4 USEFUL RESULT
A repair leaves the bench

Illustrative workshop. Not a fabrication plan.

Shared tools need a working arrangement around them: trained people, suitable power and materials, maintenance, and parts. The useful result leaves the bench. Illustrative workshop, not a fabrication or wiring plan.

Food: count the whole job

Begin with Chapter 9's chain. Name the repetitive task, the current method, and the result at the recipient's end. Count setup, supervision, maintenance, transport, and the work people still do. Don't turn a machine that saves one motion into a claim that a whole job disappeared. Equipment needs power, spare parts, and an operator's time; count those too.

Cleanup: from litter to its destination

Here's a small cleanup job, done with permission at a suitable site. Find the litter. Collect what the local service will take. Get it into that service's stream, not just out of the photo. Come back later and check whether it stayed clean. Moving trash from one place to someone else's backyard isn't success.

A future cleanup robot might do part of that chain. To tell whether it helps, compare the whole arrangement with an existing crew, better bins, or a trash trap in a creek, and count the machine's maintenance and recovery. "Won't people just litter more?" Maybe. Maybe they'll take better care of a cleaner place. Neither is a measured result, so build in prevention and easier disposal, and look again. Trash reaches water by the wind, runoff, and spills as well as dumping, so the litter itself rarely tells you who's to blame. EPA, Learn About Aquatic Trash

Toxic waste is a much bigger assignment, and it belongs to properly equipped specialists. The ambition is worth keeping: machines can take dangerous work off people's bodies. Remotely operated robots are already doing inspection and cleanup work in hazardous areas at the Sellafield nuclear site in Britain. Sellafield Ltd, Spot robot deployment But identifying, containing, and disposing of dangerous material is a job for trained people with the right equipment. Unknown material is never a neighborhood experiment.

Help that reaches a body

Health belongs in the same larger ambition. We want less pain, earlier help, better treatment, and more room to live. A machine that advances a laboratory task, a tool that assists a clinician, and an aid that helps someone move through a day are different possibilities. Each needs evidence about the benefit it claims. An encouraging research result isn't yet a treatment someone can depend on.

The access question remains after the technical achievement. If useful care exists but a person can't obtain it, the work of making that advance serve humanity is unfinished. A person shouldn't have to wish the invention had never happened because they fear what its business model will do to them.

This is what I want optimism to demand of us. Use capability to relieve the burdens people already carry. Evaluate the machine honestly. Arrange access deliberately. Keep people able to question what's being done to them and for them. A society can choose those purposes, then do the practical work required to make them real.



Build your first node

Autonomy isn't an idea. It's calluses.

1. **Set up two Meshtastic boards.** Choose supported hardware for your local band. Install the official firmware for each exact board, set the legal region, and pair the boards with your phones. Exchange a message, then repeat after a power cycle and at the actual locations you need. Follow the project's official flasher instructions.
2. **Build a small DC corner.** A packaged 12-volt battery or power station, a fused distribution block, a few 12-volt LED lights, and a USB charger. Keep a phone and a radio alive through the next outage.

3. **Learn to solder.** A basic soldering iron and an afternoon of practice on scrap wire. Learn a clean, strong splice. The wise ones are learning how to solder.
4. **Inventory the tools.** With the people doing one food or cleanup job near you, write down what tools exist, who can use them, what needs training, and who maintains them. The tool library starts with an accurate list, not a shopping spree.

Build for the long haul.



Precedent P-21: Access to Tools (Menlo Park, 1968)

In 1968 Stewart Brand put NASA's photograph of the whole Earth on a big black paperback cover and printed three words under the title: **Access to Tools.** Whole Earth Catalog, Fall 1968

The *Whole Earth Catalog* listed books on dome building, organic gardening, beekeeping, and cybernetics, along with tool suppliers and a desktop calculator: anything that helped an individual, in Brand's words, conduct his own education, find his own inspiration, and shape his own environment. Each entry was a review with a price and where to get it, what the catalog's own statement of purpose called an evaluation and access device. It refused to separate the homestead from the computer; its readers carried strands of both the back-to-the-land movement and personal computing. Steve Jobs, closing his 2005 Stanford commencement address, called it "sort of like Google in paperback form, 35 years before Google came along," and gave the graduates its sign-off as his own final advice: stay hungry, stay foolish. Stanford Report

Its rules for what got in were stricter than a list of things the editor liked. An item had to be useful, help with independent education, be high quality or low cost, not already be common knowledge, and be available by mail. And the catalog promised to keep revising itself

from what its readers and staff learned. Whole Earth Catalog, 1968 purpose and function pages It wasn't above commerce, either: the Menlo Park office shared a storefront with the affiliated Truck Store, which sold some of what the catalog listed, and supplements printed readers' corrections and suggestions for the next edition. Museum of Modern Art, Access to Tools Decades later, an anniversary reprint had to warn that the old addresses and prices were long out of date. Time had worn down the *access* half of Access to Tools, even where the idea still held.

The mechanism. A catalog of tools is secretly a map of possible selves. People don't become capable by being credentialed. They become capable by getting access, and then getting started. Curation was the product; the tools already existed, scattered and invisible. And access, unlike the idea, has to be maintained.

The rule. Build a useful toolset, not a collection of promises: something you understand, can repair, and can't be locked out of.

The practice.

1. Write your own one-page catalog tonight: every tool you own that you actually understand, can repair, and can't be remotely disabled or locked out of. Be strict. The page will be shorter than you expect. The gaps aren't a judgment. They're your curriculum, in priority order.
2. Close one gap a month. Either learn an owned tool down to the repair level (take it apart, find its manual, buy its spare parts) or replace one rented capability with an owned one. Twelve months of this and your catalog is a different document, and so are you.
3. Be somebody's access. Brand's genius wasn't owning tools; it was cataloging them for people who didn't know where to start. Publish your list, lend your tools with a signature and a return date, or run one workshop for your street. And keep the list current: a recommendation with a dead address is just a memory.

CHAPTER 18

The Roadmap and the Premortem Pivot



The map, before the route. Premortem first, then move. (SFC9394; compass rose by Stefan-Xp, CC BY 2.5, via Wikimedia Commons)

> *"Plans are worthless, but planning is everything."*
> Dwight D. Eisenhower, recalling an Army saying in *remarks to the National Defense Executive Reserve Conference* (1957)

In this chapter:

- A premortem is an obituary you write early enough to falsify. Assume the plan died; explain why; fix that, now.
- Your model will be excellent about supply chains, grids, and legal exposure, and silent about the divorce, the burnout, and the shoulder. The room knows what the model doesn't.
- Measure a community by one test that matters: can it provide an Adequate Level of Care, starting with dinner?
- Plans are worthless. Planning is everything. The people doing the planning are the plan.

The co-op's annual meeting happened in the shouse, because the shouse had the woodstove and the wall screen and, by its second winter, the unofficial status of parliament. Reuben ran the agenda: the year's numbers, the co-op's lease payment and the trust's scheduled payment to Irene Calder, the library's download counter (fifty-one thousand and climbing, from towns in nine countries now), Frank's first full year at the tool library, Tyler's name added to the roster. Then Elijah stood up for the last item, which was his, and which he had been building for a month, and said: "It's two years from now, and the co-op is dead. I'd like to spend two hours on how it died."

He had come loaded. Of course he had. The premortem model was the notebook's whole arithmetic grown up: every stage, every watt, every county filing, run through the Crucible against futures like weather fronts. Supply shocks: modeled. The Memphis platform buying their steel distributor the way it bought Frank's: modeled, with a second-source list already drafted. Grid failures, fuel spikes, a hostile county supervisor, a change in the property assessment, a landbuyer with better lawyers: modeled, modeled, modeled, modeled. He walked the room through forty minutes of failure modes with the quiet competence of a man who had finally found the exact right size for his gift, and the room took it seriously, and people could trace the calculations and challenge the assumptions.

Then Marta said, "Your machine's got no column for the stuff that actually kills these things," and the second half of the meeting began.

"Plants don't die of the strategy," she said. "I watched mine die for a year before the spreadsheet showed up; the spreadsheet was the coroner, not the cause. Things die of people, college. Run it again with people in it." And the room, which had two years of trust banked and a stove between them and the cold, ran it again with people in it, and Elijah wrote down what his model could not have generated in a thousand epochs:

Burnout, said Priya, looking at Denny. The channel posts three times a week because one man will not stop, and if that man goes down, the lever goes down with him. Denny started to argue and then didn't,

which everyone understood, including him. The fix got a name that night: Tyler learns the edit, Frank learns the counter-camera, the channel becomes a bench instead of a pillar.

A divorce, said Reuben, flatly, into the quiet. Not naming anyone, not needing to: nineteen households is eighteen marriages and change, and the trust documents, he confessed, currently handled a split about as gracefully as a wood chipper. He'd have a proposed amendment ready for counsel to review by March. He had, everyone knew, already started.

Founder-dependence, said Frank, who had watched a store die of it. Every load-bearing thing in the valley ran through the same four people, and the premortem for that was standing in the room wearing all four of their faces.

And then Marta, who had raised the subject of people dying of people, put her cup down and paid the room the compliment of going first. "Shoulder's done," she said. "Rotator cuff. Tuesdays haven't fixed it and won't. I've got maybe two years of overhead welding left, and this valley has exactly one person certified to make the beads that hold roofs up." She looked at Tyler, nineteen years old, three months into the roster, holding his coffee like it might be repossessed. "Apprenticeship's four years if you're serious. If you want it, we start Monday."

"That leaves two years we haven't covered," Priya said.

Marta nodded. "Then put that on the board too. We need someone qualified to cover the gap. Starting an apprentice doesn't make one finished."

Elijah wrote INTERIM WELDING CAPACITY: UNRESOLVED, and left the box open.

The kitchen coordinator had come for the food-delivery item and stayed for the rest. Since the first deliveries, the route had continued through one short funding commitment after another. Now she put a page beside Elijah's laptop. The route Curtis still drove twice a month had funding through June. Renewal hadn't been decided. She could

show what each delivery cost and which households needed prepared meals. She couldn't turn a promise of another grant into a paid invoice.

"We could do an appeal on the channel," Denny said.

"We could," she said. "But your name is already under burnout."

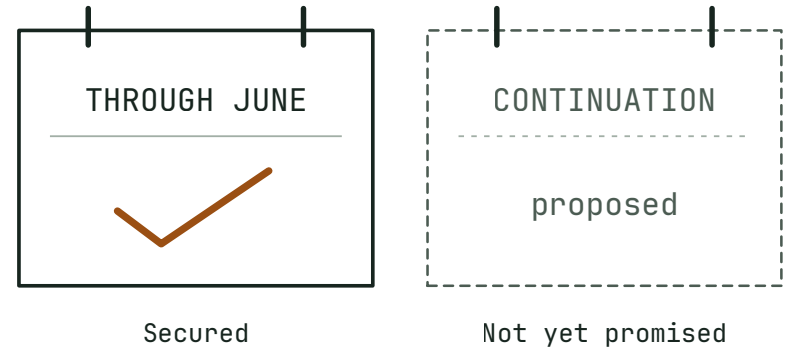
He looked at the board, then down at his hands.

Rosa had sent a note through the kitchen because she couldn't leave work to attend. Her wrist was better. Her hours at the packing warehouse changed from one schedule to the next. The note asked whether the collection time could stay late enough for people on the afternoon shift. At the bottom she'd added: *And can you tell us before the week it stops?*

Curtis read that line twice. "They shouldn't have to keep wondering who's generous this month."

Reuben drew a second column. The co-op could commit part of the repair budget it had already set aside for the route's equipment. The kitchen could bring its accounts and service record. Residents could describe what the route made possible and where it failed them. Together they could take a defined proposal to the public meeting where the next food-service allocation would be discussed. None of that guaranteed a vote or supplied an unpaid meal. It gave them a request with costs, responsibilities, and people behind it.

They wrote *funded through June* in ink. The proposed continuation went underneath in pencil, with a meeting date beside it. Curtis took responsibility for telling the households what had and hadn't been secured. The coordinator asked for someone besides Curtis to learn the route. Two hands went up.



Ink for what is funded; pencil for the continuation still being proposed. The distinction matters to the households expecting a delivery. Illustration of the fictional scene.

Elijah went home that night and rebuilt the model with the room's amendments, and the change that mattered was a names column. Every failure mode carried two names now, who it happens to and who is watching for it, because a risk without a name attached isn't a risk, it's a wish with error bars. He kept the household details in the private record; the public proposal would carry the service commitments, the costs, and the gaps. He was pasting in the last of it when he understood what he was actually looking at.

The supply models, the watts, the stages, the names, the repair he had misunderstood and the meal they had delivered wrong: it was the notebook. All of it. MADRONE ≠ DRONE, three years of arithmetic and margins and lists, finally arranged in the shape it had been trying to take since 1:40 in the morning at a kitchen table: not notes toward

surviving something, but a manual for it, with the open boxes left open. He started, that week, turning the notebook into the book you are nearly done holding.

The newcomer arrived on the Saturday after the meeting, in a sensible car with two car seats in the back and a face Elijah had last seen lit by a laptop at a Claypot all-hands. Devendra stood in the shed doorway the way Elijah had once stood in it, more than two years and a lifetime ago, looking at the plasma table and the shadow-boarded wall and the whole impossible ordinary fact of the place, and he said, quietly, "I didn't know what I expected. It's smaller than the videos."

"Everything real is," Elijah said.

Devendra looked at the broom leaning by the welding curtain, then back at the car. "I can help. I just need to figure out the kids."

"There's food inside. Bring them in."

"I don't want to take anything before I've..."

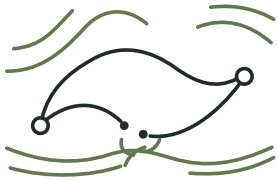
Elijah waited for the sentence to finish. Devendra looked away instead.

From the bench, Curtis called, "Dinner's dinner."

Elijah picked up the push broom, the same broom, older now, bristles worn to a bevel, and held it out, and heard himself say the words in their rightful order, the way liturgy works, the way they had been said to him on the exact spot he was standing:

"Swarf goes in the red bin." He tipped his head at the kitchen. "After dinner. We'll see."

The children came in holding their father's hands. One stopped to examine a yellow controller box. The other asked where they should put their coats.



THE FOUNDATIONS

If you've read this far, part of your brain is probably running its own premortem, looking for the flaw that lets it off the hook. The neighborhood factories can't bootstrap their supply chains. Somebody with more guns takes the greenhouse. People lose their minds without a nine-to-five. The powerful crush any alternative that gets big enough to notice.

You're right to worry about those things. They're real vulnerabilities. But using them as a reason to do nothing is fear wearing the costume of caution. Name the vulnerability out loud, then build against it anyway. That's the whole difference between a premortem and an excuse.

What could stall it

I'm not selling a certainty. Here's what could slow or stop the machine side of this story, and it's worth knowing:

1. **War.** A big enough war stalls everything, this included.
2. **The chip bottleneck.** A huge share of the world's most advanced chips come out of a handful of fabs in Taiwan. A major disruption could constrain supply; its effect would depend on the damage, inventories, and alternative capacity.
3. **The materials bottleneck.** Chipmaking needs specialized materials and equipment that can't always be substituted quickly. Ultra-high-purity quartz, for example, is used in crucibles for semiconductor manufacture. A disruption matters at a specific

link, not because the world has run out of ordinary sand. US Geological Survey, Silica Statistics and Information

4. **The backlash.** People could choose, or be frightened into choosing, to stop the build-out.
5. **Things nobody's thought of.** There's always a column for the unknown.

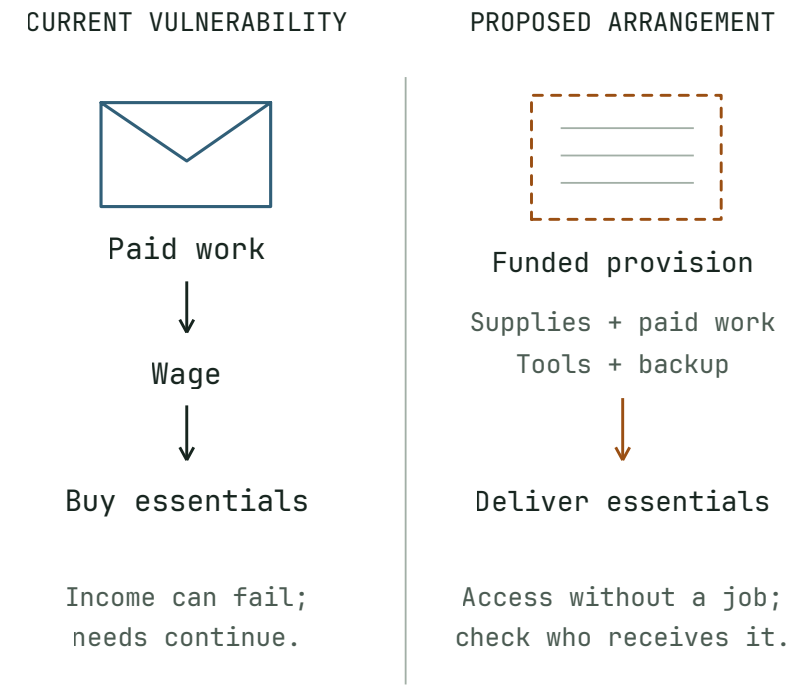
None of those makes the food work less worth doing. If progress stalls, people still need to eat, and everything in this book still helps. If it doesn't stall, we'll be glad we started.

The Adequate Level of Care

So here's the standard I want to measure a community by. I've been calling it the **Adequate Level of Care**: can this place make sure the people who live here have food, shelter, and basic care, dependably, including through a bad week when the long supply lines hiccup?

Stop measuring success only by GDP. GDP can rise while a lot of people get more precarious. Ask instead whether your community can provide an Adequate Level of Care to the people in it, and how much of that provision depends on a job, a truck from far away, or one generous person.

A floor people can rely on



The proposal makes essentials a funded floor, available without a job. Tools help; remaining work, supplies and reliable delivery still need provision.

Food is the first line of the Adequate Level of Care, because it's urgent, concrete, and measurable: we can ask who ate, what they could get, and whether the arrangement will hold next week. Shelter and care come next. Each needs its own expertise and evidence, and each can be organized around the same question: do people actually receive it?

If global shipping stopped for a month, could your community keep everyone fed? For most places the honest answer is “we don’t know.” That’s not a failure. It’s the assignment.

What going right would look like

Picture the ordinary version. A cook loses a job and still eats. A worker finishes a shorter shift and goes home to the same life, not a smaller one. A caregiver takes help without having to prove there’s no other choice. A grower gets paid fairly for food that goes to people who can’t pay for it.

That’s worth doing the practical work for. It doesn’t need every machine to be perfect. It needs a machine breaking, a market crashing, or a boss cutting hours to stop landing straight on somebody’s ability to eat.

And it’s bigger than nineteen households. The co-op matters because its members learned to fix a plan together and could show their work. Its limits matter too: a valley full of successful founders would still leave most of the world outside the fence.

Build the machine and the promise together

There are two jobs, and you have to do both. Get better at making the food. And build a promise people can count on that the food will reach them. Do only the first and you get impressive machines behind a price somebody can’t pay. Do only the second and you get a right on paper that the kitchen can’t keep.

Start with a food service that already exists. Sit down with the people who use it and the people who run it, and find where it breaks: the missed meal, the pickup point nobody can reach, the task someone repeats forty times a day, the freezer that’s been waiting three weeks for a part. The first fix might be a machine. It might be another paid shift, a better supplier, or a bus route.

Then follow the gain all the way to a plate. A controller helps a grower. The grower signs on to supply a kitchen. The kitchen turns it into meals people can actually eat. Somebody pays for the service, so the person eating doesn't have to. Every link needs a name next to it and the money to keep it running.

A tool that saves an hour can turn that hour into more output, a bigger profit, a shorter shift, or a free meal that shows up every week. The tool doesn't pick. We do.

A proposal people can vote on

Here's what a real next step could look like after the five-household trial. It's an example, not a program that's been funded.

The food bank, the growers, a kitchen, the workers, and the neighbors write down one offer for one area. What food. How you ask for it without paying. What happens if you can't cook or can't get there. How you get in without a smartphone and without knowing the founders.

They write down what it costs to run: food, paid jobs, transport, repairs, and money set aside for when the truck dies. They count volunteer hours separately and say which jobs can't depend on whoever happens to show up. The people doing the work get paid. The person eating doesn't have to work for the meal.

Then they go looking for steady money: the county, a co-op, a church, a grocer, or some mix. The county could run it, or pay several groups to run it. Donations can add to it, but they don't carry the core promise, because donations come and go. Each choice puts different people in charge, so everyone should see the options side by side.

The offer also says who takes complaints, how you appeal if you're turned away, what information gets collected about you, and what happens when something breaks. It makes room for people who disagree with the organizers. You shouldn't have to be liked to be fed.

Now there's something real to vote on. People can say yes, no, or change it, and they can see what it costs and check whether it's working. That's how a good idea becomes a public promise instead of a sentence that starts with "society should."

Keeping people whole while it's built

The person who just lost a job to a machine needs to eat while all this gets built. A retraining course only helps if your life can survive the time and cost it takes. Rent, food, medicine, the bus, and your kids don't wait for a certificate.

So whoever brings the machine in should answer for the people it moves out, before the job is cut, not after. Who keeps the paycheck or the essentials coming? What do the workers get a say in? What help is funded, and for how long? The answer might come from the employer, a union, the county, or a co-op, but somebody has to put their name on it.

Some hard jobs will get harder to fill once people have better choices. Good. Then look at the pay, the conditions, the training, the equipment, and whether the job needs doing at all. Using hunger to fill jobs is exactly the problem this book is trying to end.

Food first is where we start, not a rule that everything else waits. Housing and health care can move at the same time. The fired cook's dinner is just the first test.

Put the failed meal on the board

Before expanding a service, imagine that next week's commitment has failed. Work backward with the people who would experience it. Perhaps the crop is unavailable, the driver is ill, the food can't be used, the system gives a wrong answer, or the operating money has run out. Ask them, because they know things the plan leaves out.

Use a short record:

Question	What to record
Who is affected?	The people whose food or work would be disrupted.
What would we notice?	An observable warning, including a missing confirmation.
Who responds?	A responsible role and a backup who has agreed to act.
What happens to the meal?	A confirmed alternative, or a clearly unresolved gap.
What supports the promise?	Available supplies, labor, money, and the date that provision ends.
When do we review it?	The next decision date and the evidence people need.

Keep people's personal details out of public records unless they're needed and the people agree. Give the people you're feeding a way to correct what you assumed about their lives. A delivery logged as complete may still have left somebody with food they couldn't eat.

Test a response before depending on it. Confirm the alternative supplier's capacity; let the backup person actually perform the handoff; check the public contact route. A second name in a spreadsheet isn't a second available driver.

WHEN A MEAL FAILS

Illustrative premortem worksheet

1. **WHO LOSES THE MEAL?**
   Name affected people and jobs.
2. **WHAT WARNS YOU?**
   Define a signal or missing check-in.
3. **WHO RESPONDS?**
   Confirm a lead and an available backup.
4. **WHAT GETS DELIVERED?**
   Test the alternative; show any gap.
5. **WHAT KEEPS IT RUNNING?**
   Name supplies, labor, money, end date.
6. **WHEN DO YOU CHECK AGAIN?**
   Set a review date and needed evidence.

Test the handoff before making the promise.
Keep personal details out of the public board.
Source: Chapter 18, failed-meal record.

When a meal fails. An illustrative worksheet from this chapter's premortem record, not the measured performance of a service. Confirm the alternative and the people who will deliver it before expanding the promise.

Grow the promise at the speed you can keep it

A pilot can run on a grant for a set time. A floor people build their lives on needs money that keeps coming, and a plan for when it's threatened. Tell people which one they're getting. Nobody should find out about a funding gap by waiting for a meal that never comes.

Check three things together: whether the machines work, whether the remaining jobs got better or worse, and whether people actually ate. Winning on one doesn't excuse losing on another. If a machine saves time but puts a worker in danger, fix it. If the food arrives but only members can get it, you haven't kept the promise yet.

Grow by adding partners who bring what's missing: other crops, cooking, delivery, winter coverage, a pickup spot people can reach, and public money beyond the founders' circle. One small success proves that one small thing. It's also the best thing you can carry into the next meeting.

There will be fights about money and ownership, arguments about methods, and failures nobody saw coming. None of that means the way things are now is the only way they can be.

The first meeting doesn't have to design the whole economy. It has to end with one person responsible for one next step: a costed proposal, a trial, a talk with the workers, one delivery gap closed. Put a date next to it. Bring the people it affects back when you check the result.

Wide lens, narrow focus. Keep the larger promise in view, and keep making the next piece dependable.



Who this is for

This roadmap is written for two audiences, and they need the same thing.

People under twenty-five. Your anger at the rigged arrangement is justified. The old labor-for-survival deal has been failing you on both ends, and that's a structural failure, not a failure of your ambition. The tools in this book are a way to stop renting survival month to month.

People who already hold something. Property owners, local institutions, community stewards: the same machines can help maintain and upgrade housing, land, and infrastructure that human labor alone has gotten too expensive to keep up. A community that can feed and house its people is also a more stable place to own anything.

A system is only as good as its ability to provide food, shelter, and care to the people inside it. The old arrangement is breaking under us. It's time to start building the next one.



Precedent P-22: The Apocalypse That Ran On Time (1999 to 2004)

The Y2K bug was real. Enormous amounts of code stored years as two digits, wired into banks, power grids, air traffic, and payroll, and at midnight on January 1, 2000, "00" could mean 1900 to machines that ran the world. US GAO, Year 2000 Computing Challenge: Lessons Learned (2000)

What followed was a global exercise in anticipating failure. Instead of waiting to see what broke, organizations assumed failure in advance and worked backward, spending, by common estimates, hundreds of billions of dollars worldwide on auditing, patching, and testing before the deadline. The US government's own retrospective describes the scale of that work. GAO, Year 2000 Computing Challenge

The best of the work was ordinary and concrete. The Department of Housing and Urban Development built an inventory of how its systems depended on each other and on outside partners, and found the inventory useful long after the date problem was solved. At Oak

Ridge, a rollover test caught a nuclear-material accounting system writing four-digit years to a transfer program that expected two; the transfer failed, and staff moved the data on magnetic tape under contingency procedures they'd already updated and tested. GAO, Y2K Lessons Learned, pages 29 to 32 The Social Security Administration folded its Y2K contingency plans into its permanent operations, and in 2004 required that software be validation-tested by people other than the ones who built it. GAO, Social Security Administration Year 2000 Readiness

Midnight came. Planes flew. Banks opened. And then came the strange part: because so little happened, a chorus concluded nothing would have happened, that the whole effort was hysteria. Some countries that spent little also passed quietly, so the true size of the disaster that was avoided is still argued about. But that vast act of anticipatory engineering is remembered, by many, as an overreaction. And one caution runs the other way: a missed AI forecast doesn't get to call itself a disaster averted. Y2K had a known deadline and a bounded class of bug. This transition has neither.

The mechanism. Prevention is invisible by construction. A premortem that works looks, from outside, exactly like paranoia that turned out to be unnecessary, because the catastrophe it prevented never shows up to testify.

The rule. Judge your preparation by the quality of your reasoning before the deadline, not by the silence after it. If you prepare well and the transition doesn't shatter your life, people will say you overreacted. Let them. The silence is the product.

The practice.

1. Run the household Y2K this weekend. One hour, everyone affected present. The prompt: "It's a year from now and our income got cut in half by the transition. What broke first?" Work backward: list the top three things that would have needed fixing, then fix one of them this month.
2. Set your preparation budget and write it down like a line item: the share of income and hours a week you put into skills, soil, savings, and community. The world spent a fortune on Y2K and called it sane. Your few percent isn't paranoia. It's the same engineering at household scale.
3. Date-stamp your reasoning. Every significant preparation decision gets one paragraph in the notebook: what you knew, what you assumed, why you acted, and the date. When someone later says you overreacted, you'll judge yourself against what was knowable at the time.

CHAPTER 19

The Ladder: How Money Becomes Optional

> *"No society can legitimately call itself civilized if a sick person is denied medical aid because of lack of means."*
> Aneurin Bevan, *In Place of Fear* (1952)

In this chapter:

- The rules that make survival cost money were written by people, and people can rewrite them. Here are nine steps. The first seven have precedents, with dates and receipts; the last two hold the plan together.
- Start where the food already is: count what's thrown away, feed whoever asks, then make the meal a public service like the library.
- Money stays. It just stops being the ticket to staying alive.

The county chambers smelled like wet coats and floor wax. Rain had followed them down the valley, and the co-op came in shaking it off: Curtis in his route jacket, Reuben with an accordion folder under one arm, the kitchen coordinator with a banker's box of receipts, Devendra with a child asleep against his shoulder and the other one counting ceiling tiles. Irene Calder came last and took the aisle seat as if she'd been assigned it.

Elijah had fourteen slides. He'd built them over three nights: cost per delivered meal, failure modes with names beside them, a curve of what the route could carry if the county matched what the co-op had already set aside for repairs. He'd timed the talk at eleven minutes.

The clerk read the rules off a laminated card. Public comment, three minutes per speaker. A red light would come on at thirty seconds.

“Three minutes,” Elijah said.

“Eleven was always a fantasy,” Reuben said, not unkindly. He slid a single page out of the folder. The proposal, costed, one side. He’d filed it with the clerk on Tuesday. Nobody had asked him to.

Item nine came up at 8:40. The supervisors had been sitting since six. One of them was eating almonds out of a paper cup below the dais where she thought nobody could see.

Elijah went first because the sign-up sheet said so. He got through the cost per meal and the funding date, *through June*, and then the red light came on, and he watched himself skip to the last slide, which was the wrong slide, and say “so the model suggests” into a microphone that popped on the *s*. He sat down with forty seconds of unspent argument in his mouth.

Rosa went second. She hadn’t told anyone she was coming. She was still in her warehouse shirt, the wrist brace grey at the edges, and she didn’t have a page.

“I pack fruit for a living,” she said. “Afternoon shift. The route comes late enough that I can get it. I’m not here to say thank you, though I am thankful. I’m here because every month we don’t know if it’s the last month. You can’t plan a week around maybe.” She looked at the supervisor with the almonds. “I asked them to tell us before the week it stops. They said they would. I’m asking you the same thing.”



Fictional scene: Rosa at the county hearing. A funded route means households can plan beyond the next delivery.

She sat down with a minute left on the clock. Nobody else in the room had managed that.

The chair thanked her and turned to the staff report. The county's analyst, a tired man in a sweater vest, said the numbers were sound but the program would be an ongoing obligation, and the county didn't take on ongoing obligations for services the private sector and charities already provided. "We simply can't afford to make food free," he said. "Nobody can."

Irene was already standing. Her chair scraped the floor loud enough for the microphone to catch it.

"My father-in-law ran that same line," she said. "Walt's dad. Nineteen thirty-nine. The power company told him wiring out to our place didn't pay. Too far, too few of us. So the farmers along our road made a co-op and borrowed the money from the government, and they strung the wire themselves, on poles they set with a borrowed truck." She leaned on the podium. "They paid it back. Every cent. My hus-

band was born under a bulb that his daddy's neighbors paid for. So when a man in a sweater tells me something can't be afforded, I'd like him to say who it can't be afforded *by*."

The almonds went down. The chair cleared her throat and said the speaker's time had expired, and Irene said she knew, she'd been watching the light, and sat.

They didn't win. Not the way Elijah had pictured it on the second night of slides. The board voted four to one to extend the route through December and to direct staff to study a permanent food line for the next budget cycle, which Reuben said, on the steps outside, was how a county said *maybe* in writing. The analyst was assigned the study. Curtis laughed out loud at that, once, into the rain.

"December's six more months than June," the kitchen coordinator said. She was already writing it on the lid of the banker's box. "And a study has a due date."

Devendra's older child wanted to know if they'd won. Devendra looked at Elijah.

"Some," Elijah said.

Curtis pulled out his phone and started the texts, one household at a time, the way he'd promised. *Funded through December. Tell you more in June.* He read the first one aloud to Rosa before he sent it. She read it back to him and changed *more* to *the rest*.

Elijah stood under the eave with the laptop in its sleeve, fourteen slides unshown, and understood for the first time that the model had never been the argument. It was the receipt you handed over after someone else had made it.

Reuben took the folder back and wrote the study's due date on the tab in ink.

— ◈ —

Why I wrote this chapter

I'm going to be straight with you. I'm so fucking pissed off about this. Not at you. At the arrangement.

Greed is a sickness, and too many of the decisions that shape our lives serve private property first, enforced by state-sanctioned violence. If you don't have enough money, somebody with a badge and a gun has permission to come and put you out. People go hungry and suffer because they *have* to, not because there isn't enough. It's fucking disgusting, and it doesn't have to be this way.

So this chapter does what I promised at the start: a literal, step-by-step plan to convert society over to the way it's supposed to be, where we all get our needs met and nobody has to suffer or die for the price of anything. Money should be optional. Here's how we get there.

I wrote my mission down in February 2026, before this book had its shape, and it hasn't changed: to create systems where capital is no longer a requirement for survival, and to prove that the requirement is unnecessary and enforced.

The plan, in one sentence

Make survival a public floor instead of a private purchase, one rung at a time, starting with food, and let automation keep raising the floor.

That's it. The rest of this chapter is the ladder.

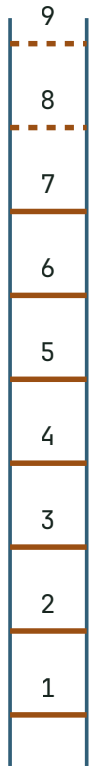
The Introduction said it plainly, and it belongs at the top of the plan too: we don't have to live like this. Nobody has to go hungry because food doesn't sell. Nobody has to be locked out of a home because a score came back wrong. Those outcomes are rules, and rules are written by people.

This isn't a revolution, and it doesn't wait for one. It doesn't need a single leader, a new constitution, or the permission of the people who profit from the way things are now. It needs people who live somewhere to decide, out loud and on the record, that everyone who lives there gets what they need to survive, and then to build the institutions that make the decision real. The first seven rungs below draw on things real places have already done. Those precedents have limits; none proves the whole plan has worked. The last two are how you hold it together. The proposal is to connect these pieces, deliberately, and widen who they serve.

The yardstick is the one Chapter 18 set: the **Adequate Level of Care**. Can a community make sure everyone who lives there has food, shelter, and basic care, dependably, including through a bad week? Each rung moves a place closer, and each one can be measured.

Nine rungs, one public floor

Start below. Build upward.



Keep the market above the floor
Money stays; survival is the floor.

Make the service accountable
Publish access, cost and unmet need.

Widen the floor
NHS (1948); proposed basic services.

Protect access to essentials
NYC eviction counsel (2017).

Own the machines together
Rural electric co-ops (from 1936).

Take land out of speculation
Land trusts; long-term ground leases.

Make meals a public service
California school meals (2022–23).

Feed people who ask
France (2016): surplus recovery rules.

Count waste and food insecurity
California SB 1383 recovery records.

Draws on a precedent
Proposed guardrail

Precedents support parts of this proposal.
None establishes the complete guarantee.

Read from the bottom up. The first seven rungs draw on bounded precedents; the final two describe how the proposal would be maintained.

Rung 1: Count what's wasted and who's hungry

What. Put two measures side by side, in public, for your town: how much edible food gets thrown away, and how many households can't count on eating. Keep their units and reporting periods visible. You can't subtract households from pounds of food; you can investigate how much suitable food could reach those households, when, and at what cost.

Proof it's possible. California already makes large food businesses count and recover. SB 1383 requires large grocers, distributors, and food services to recover edible food that would otherwise be thrown away, requires cities and counties to set up food recovery programs, and sets a statewide target: by 2025, recover 20 percent of the edible food that would otherwise go to landfill, "to feed people in need." CalRecycle, SB 1383 Food Recovery Nationally, the Department of Agriculture counts food insecurity every year: 13.7 percent of US households, 18.3 million of them, were food insecure at some point in 2024. USDA ERS, Key Statistics

Who. Anyone with a library card and a spreadsheet can start with published food-bank reports, county recovery plans, and USDA survey results. Ask local providers which records they can share as totals. The national survey doesn't supply a reliable monthly estimate for every town, and household records aren't public just because an organization holds them.

How you'll know. One page, updated when new records arrive: pounds recovered, meals served, households still short. Label dates, sources, missing groups, and estimated figures. A monthly update can report progress without pretending an annual estimate was measured again.

Rung 2: Feed everyone who asks, from what's already there

What. Route recovered food to kitchens that serve anyone who shows up. No proof of poverty and no interrogation about whether somebody deserves dinner. Ask what you need to provide a meal they can use, including dietary and access needs. Every unnecessary form is a door some hungry person won't walk through.

Proof it's possible. France's 2016 law prohibited deliberately spoiling edible surplus and required larger food retailers to offer donation agreements to eligible charities (Chapter 14). Britain opened public restaurants in wartime at low, set prices: the British Restaurants of Chapter 11. Neither example established this rung's universal free meal. They show parts of the arrangement already working. Joining those parts still takes safe food, kitchens, paid work, and a funded promise to whoever needs it.

Who. A food bank, a church kitchen, a restaurant with a slow Tuesday, a school cafeteria after hours, a co-op like the one in these pages.

How you'll know. Meals served, people turned away, and requests the service couldn't meet because of timing, transport, or dietary needs. The target for unmet requests is zero; count people who couldn't reach the door as well as those who did.

Rung 3: Make the meal a public service, like the library

What. Move the meal from charity to entitlement: a funded, permanent public service that doesn't depend on who volunteered this week, the way a library or a school doesn't.

Proof it's possible. California already did it for its children. Since the 2022 to 2023 school year, public school districts, county offices of education, and charter schools serving transitional kindergarten through grade twelve must provide breakfast and lunch free of charge each school day to students who ask, "regardless of their free or reduced-price meal eligibility." It was the first state to do it. California Depart-

ment of Education, California Universal Meals A public kitchen open to everyone uses an established capability. Extending that promise still takes public authority, a funded operating plan, suitable facilities, and people to run it.

Who. City councils, county supervisors, school boards, and the voters who elect them. Appendix A has the questions to bring.

How you'll know. Coverage: the share of residents who can get a meal on any given day without paying.

Rung 4: Take the ground off the market

What. Put land for food and homes into community land trusts and public trusts, so it can't be bid away from the people who use it.

Proof it's possible. Community land trusts already do this across the country, usually with a 99-year ground lease that keeps homes affordable for the next household too. Grounded Solutions Network, Community Land Trusts Chapter 12 shows one being born and recounts New Communities in Georgia, which lost its original land and later bought a new property. Chapter 15 shows the machinery and its limits.

Who. Nonprofits, cities with surplus land, farmers who want their ground to outlive them.

How you'll know. Acres and homes held in trust for the people who live and grow there.

Rung 5: Own the machines together

What. The robots, greenhouses, kitchens, and local computers that produce the floor should belong to the people who depend on them: cooperatives, public utilities, land trusts. Not a distant owner who can raise the price or switch them off.

Proof it's possible. America already did this once, with electricity. By 1930, nearly nine in ten homes off the farm had power but only about one farm in ten did, because stringing wire to farms didn't pay. Farm-

ers formed cooperatives, borrowed from the new Rural Electrification Administration at 2 to 3 percent, and wired themselves. Within twenty to twenty-five years virtually all of rural America had power, and almost every loan was repaid; the default rate was under 1 percent. Federal Reserve Bank of Richmond, Electrifying Rural America Those cooperatives still serve an estimated 42 million people. America's Electric Cooperatives, Fact Sheet (2026)

That's the pattern for everything this book wants machines to do. When the market won't serve a need because the need doesn't pay, people can own the infrastructure themselves, with public credit behind them.

Who. Co-ops, credit unions, cities, and the builders in Chapters 13 and 17.

How you'll know. The share of the floor's capacity (kitchens, growing space, compute, delivery) that the community owns.

Rung 6: Stop letting a score decide who survives

What. Take money tests out of essentials. A lawyer for every tenant facing eviction. Limits on using credit reports and private risk scores to decide who gets housing and utilities. No shutoffs that leave people without water, power, or heat.

Proof it's possible. In 2017, New York City became the first city in the country to establish a right to counsel for tenants facing eviction. The law provided full representation to households at or below 200 percent of the federal poverty guidelines and brief legal assistance above that threshold. In the 2022 fiscal year, 78 percent of households represented in court by those lawyers kept their homes. This is a precedent for public legal help, not proof that income tests have already disappeared. NYC Office of Civil Justice, Annual Report 2022

Who. Tenant unions, legal aid groups, city councils, state legislatures.

How you'll know. Eviction filings, lockouts, and shutoffs, counted and published, falling.

Rung 7: Widen the floor

What. Food first, then shelter, care, transport, and information. The same principle, one need at a time: provided to everyone, free at the point of use, paid for together.

Proof it's possible. Britain did it for medical care in 1948 (the precedent at the end of this chapter). In 2017, researchers at University College London's Institute for Global Prosperity costed one proposed expansion of Universal Basic Services at about £42 billion a year, funded through changes to the personal tax allowance. Their package covered specified housing, transport, food, and communications provision. That was a UK proposal using its period's assumptions, not the price of meeting every need everywhere. UCL, Future welfare: Universal Basic Services

Who. States and nations, pushed by every rung below them.

How you'll know. The Adequate Level of Care, measured across all of it.

Rung 8: Put it on the scoreboard, in public

What. Publish the Adequate Level of Care every month, the way a city publishes crime or unemployment: who ate, who's housed, who got care, who was turned away. Tie budgets to the scoreboard.

Proof it's possible. Rung 1 showed existing food-recovery records and a national food-insecurity survey. They don't automatically yield a monthly count for your town. Start with the records local providers actually have, state the coverage and gaps, and agree how to count unmet need. Publish totals, not recipients' names or medical details. Chapter 18's "Put the failed meal on the board" is the same practice at the scale of one kitchen.

How you'll know. People can read it, and it changes what gets funded.

Rung 9: Keep the market above the floor

What. Money stays for everything above survival. Restaurants, gadgets, art sold for a fortune, a billion dollars if someone wants to build one. None of that goes away. Money just stops being the ticket to staying alive.

Not abolished. Optional.

And the floor doesn't have to stay where it starts. Where robots and AI reduce the full cost of growing food, building homes, or providing care, the saving can help raise the floor. Count maintenance and the remaining human work, then decide how the gain reaches people. That's the whole point of the machines. A gain in our ability to produce becomes a gain in our ability to live.

What would stop it

Honesty about the obstacles is part of the plan.

- **Money.** Every rung costs something to run, even when it's free to the person eating. The fights over who pays will be real. Have them while people are being fed, not as a condition for starting.
- **Capture.** A service can be captured by whoever runs it. Build in complaints, reviews, and the scoreboard from day one.
- **Backlash.** The resistance described in Stage 5 can be real even if the nine stages never unfold in order. Some people will fight any change that looks like losing ground. A concrete, measurable proposal is harder to fight than a slogan.
- **The people it doesn't pay.** Some of the people who profit from the way things are will spend a lot to keep it. The answer is the same one the electric co-ops found: build it anyway, own it together, and make it work well enough that nobody wants to go back.

None of those is a reason to wait. Each rung stands on its own. A town that only ever gets to Rung 3 has still fed its people.

Get involved

If you're worried about who controls this technology, give that concern somewhere to go. Talk to your government. Form a coalition with your peers. Get together with local business owners and petition your city council. Write a book. Pick a decision you want changed and work on it with other people.

Take fucking action.

Posting can recruit neighbors, share records, and get a proposal into people's hands. Connect that work to a request somebody has the power to grant: fund the kitchen, protect the water, put the machines into community ownership. Name the decision and the person or body responsible for it.

Find the next meeting agenda. Check the public-comment rules and speaking limit. Write one request that fits the time: what you want, who it serves, and what the next step would cost. Bring the longer version for the record. Ask for a specific next step, such as a staff report or a budget item, and a date when you can follow up.

Afterward, send the request in writing. Read the minutes. Find out whether anyone was assigned to act, and come back when the item returns. A speech starts a conversation; the follow-up keeps the request alive.

If you can't get to the room, ask the clerk how to submit written comments or participate remotely. You can help research the proposal, call an office, recruit support, or arrange transport and childcare for someone who can attend. Organizing needs more than whoever holds the microphone.

Tucson shows what a public decision can change, and where its reach ends. On August 6, 2025, its mayor and council ended negotiations over Project Blue's proposed city annexation and development agreement. They canceled the annexation hearing and removed the agreement from the next agenda. City of Tucson, adopted motions

The project continued outside city limits. Pima County documented activity at its construction site in May 2026. Residents could influence the council's decision; they couldn't settle every decision governing the project. That means finding the next authority and keeping at it. Pima County, construction update

Rung 5 gives the other half of the work: people organizing to own infrastructure and extend a service. You can press a council to refuse a deal or organize a cooperative to build something worth having. Both require people to make decisions together.

Frederick Douglass put the demand plainly in 1857: "Power concedes nothing without a demand. It never did and it never will." Douglass, speech excerpt

Make one. Then follow it through.

What you can do this month

- **If you work for a living:** find out what a food business near you does with food it can't sell, and ask a local food provider what need remains unmet. Bring those findings to one meeting.
- **If you're a parent:** check what meals your school district provides, whether they reach every kid who needs them, and what keeps anyone out.
- **If you organize:** start the Rung 1 page. One spreadsheet, published monthly.
- **If you build things:** pick one machine from Chapter 17 that could serve the floor, and ask who would own it.
- **If you hold office:** put a public meal line in the next budget and a lawyer-for-every-tenant ordinance on the agenda.
- **If you own a business:** donate what you can't sell, and sit on the committee that decides where it goes.

Precedent P-23: The Day Care Stopped Costing Money (Britain, 1942 to 1948)

In 1942, in the middle of a war, a government committee chaired by the economist William Beveridge published a report naming five giants standing in the way of rebuilding Britain: Want, Disease, Ignorance, Squalor, and Idleness. Even during the war, national health insurance covered only about half the population, and it left out hospital and specialist care, dental care, eye care, and hearing care. UK Government History blog, The founding of the NHS: 75 years on

The doctors had real fears: a loss of freedom, new bureaucratic demands. Aneurin Bevan, the health minister, answered them rather than dismissing them. “Any health service which hopes to win the consent of the doctors must allay these fears,” he said. And on 5 July 1948, the National Health Service opened. Britain became the first western country to offer medical care, free at the point of use, to its whole population. UK Government History blog

No new medical discovery arrived that morning to make the principle possible. A new institution did. Years of legislation, financing, and negotiation brought existing hospitals and practitioners into a service intended to cover everyone regardless of ability to pay. Some people had insurance or public provision before it; access hadn’t been universal. The change was who could claim care and how the work was organized and paid for. UK Government History blog Four years later Bevan put the principle in one sentence: no society can legitimately call itself civilized if a sick person is denied medical aid because of lack of means. Bevan, In Place of Fear, chapter 5

The mechanism. A need stops costing money at the door when enough people decide on the record that it should, and then build the institution that makes the decision real. The capability usually exists long before the decision does. The obstacle is the rule.

The rule. When someone says we can't afford to make a need free, ask how it became free somewhere else: who decided, who paid, and who kept the service running.

The practice.

1. List what's already free at the door where you live: the library, the public school, the roads, whatever else. Each item on that list was once a decision someone had to fight for.
2. Name one need you'd add to that list, and write down who would have to decide it: a school board, a council, a legislature.
3. Go to one meeting of that body this season, and say out loud, on the record, that the people who live here should get what they need to survive.

CONCLUSION

What If It All Goes Right?



Sunrise from orbit. The new era doesn't ask permission, and it still has work to do. (NASA Johnson Space Center, Public domain, via Wikimedia Commons)

> *"A map of the world that does not include Utopia is not worth even glancing at, for it leaves out the one country at which Humanity is always landing."*
> Oscar Wilde, *The Soul of Man under Socialism* (1891)

In this chapter:

- Success is more than labor removed from a payroll: count time returned, danger avoided, access secured, and who holds the power.
- Bring one question to every decision about automation: how will the people affected share in the benefit?
- The moral mandate: stop arguing over the scraps of a dying arrangement and build the machinery of our own freedom. Food first, then keep the promise growing.

A future worth wanting

I don't want us to become so practiced at imagining catastrophe that we forget what we would build if we thought success was possible.

A person loses a job and still eats. A parent can take a child to receive care. A home is a place to live in, not the first thing a bad month takes away. A dangerous site gets cleaned without sending somebody into harm because their labor was the cheapest item on the plan. Useful work continues. More of its burden is carried by machines, and more of its benefit reaches people.

That's what I mean by asking whether it could all go right. The world remains complicated. We remain capable of selfishness, error, and cruelty. There's still maintenance to do. But the daily conditions of being human improve because we choose to use our growing capability that way.

We don't have to agree on the most distant future to want that one.

A future called solarpunk

One name for a future like that is solarpunk.

On April 30, 2008, a blogger writing as John-Robert proposed the word for a literary genre, taking inspiration from a cargo ship using wind to supplement its engines. He imagined modern technology combined with older tools in an economy powered by renewables. John-Robert, From Steampunk to Solarpunk

For this book, the useful idea is a future where renewable energy, repairable tools, and shared decisions make daily life more secure. Solar points toward the sun and the living world every machine depends on. Punk asks whose hands the tools are in. A working greenhouse, a shared workshop, and a funded meal route give that picture substance.

In her 2014 National Book Foundation speech, Ursula K. Le Guin argued that writers can help people imagine alternatives to the way they live, and that arrangements of human power can be changed. Le Guin, National Book Foundation Medal speech

The picture is an invitation to work. Soil still needs tending. Somebody still has to fix the controller and deliver the meal. Hope gets useful when we give it a job.

Hear the criticism inside the fear

When somebody tells you a machine will take their livelihood, don't answer only with a demonstration of the machine. Ask what the livelihood has been holding up. Food. Rent. Medicine. Someone else's education. A little dignity at the end of a difficult week.

Those needs deserve a more dependable foundation than the hope that a particular employer will always need a particular skill. Training can be useful. So can a new business or a different occupation. None is an adequate answer on its own to the question of where a person stands while the ground moves.

A society should be able to absorb improvements in useful work without making its people disposable. If we can do more together, we should be able to make a stronger promise to one another.

Food first makes that promise concrete. The machines help grow and handle it where they can. People do the work that remains, on terms they can live with. Providers are supported. Recipients can obtain what they need without employment or payment being the entrance test. The service is judged by whether people eat.

Housing, health, and environmental repair follow the same question: what would make the benefit real in somebody's life? A finished wall, a laboratory result, or a bag of collected waste can be part of an answer. Keep going until the person has a safe home, effective care, or less exposure to harm.

Choose what success means

The most consequential mindset change is one we can write into decisions. A successful machine deployment should be judged by more than the labor it removes from a payroll. Ask about time returned, danger avoided, access secured, and power over the service. Ask whose life improves and who's carrying the cost.

That work belongs in public decisions, workplaces, research programs, procurement agreements, and community projects. It also belongs with the people receiving help, who can tell us when our magnificent answer doesn't fit through their door.

The proposals in this book leave room for disagreement about means. We can compare ownership arrangements, funding choices, and ways of delivering a service. We can insist that commitments be funded and decisions challengeable. We can change a tool that fails. None of that requires retreating from the purpose.

Wanting everyone to have enough isn't a technical prediction. It's the standard we bring to the technology.

My AGI forecast may be wrong. New capabilities may arrive unevenly, and some may prove harder than their promoters expect. A food service that works is still useful in that world. So is a better agreement, a repaired home, a safer job, or treatment that reaches a patient. We don't need to manufacture certainty in order to act with purpose.

Take the next decision seriously

You may be in a position to build something, fund it, study it, negotiate its terms, or vote on it. You may be the person who needs the service and knows why the current version fails. Start with that position. Join people who can carry the other parts. Chapter 19 lays out the whole ladder, rung by rung; find the rung your town is standing on and reach for the next one.

Bring one question into the next decision about automation: how will the people affected share in the benefit, including the people whose paid work goes away? Ask for an answer that names provision and responsibility. A promise without either is still waiting to become a promise someone can use.

And when you meet the frightened worker, let them be frightened. They don't owe the future enthusiasm. We owe one another a future that gives us a reason for it.

The moral mandate

We're building a better tomorrow because it's the only intelligent thing to do. We aren't "destroying jobs"; we're automating the toil and the risk, and insisting the benefit reach the people who did them. We aren't weird for using tools to solve our survival; we're human. The choice is ours: keep arguing over the scraps of a dying arrangement, or build the machinery of our own freedom.

Here's the truth the fear keeps trying to bury. Things are changing, and you can do this. The question this book opened with, why you're still renting your survival, has an answer. You can have machines that help grow your food and, in time, build your house. You can learn to make them, teach them, repair them, and hand what you learn to the person standing next to you. But first you have to put down the mindset that whispers you're not smart enough, not technical enough, not the kind of person this is for. That voice is the last fence, and it's the only one nobody else can tear down for you. Look this thing straight in the face.

Do it for yourself, but don't stop there. Do it for your kids, your grandkids, and the people who'll inherit whatever we're brave enough to build. Every capable person who wakes up is one more on the side of the light, and we need as many as we can get, because the people who want this locked down are already awake and already moving. If we let a handful of companies fence this in, if we let them regulate everyone else out of the one opportunity that could change their lives, we lose it, and we lose every person who never got the chance.

Humans weren't built to live the way we've been living. We didn't claw our way up out of the dirt and learn to bend nature to our will just to be conquered in our turn by a spreadsheet, by "number go up," by a quarterly line on somebody else's chart. You weren't born to increase shareholder value. You were born to care for the Earth, the sky, and each other. That's the older contract, the real one, and the machine may be the first tool in ten thousand years big enough to let us keep it.

So wake up. All the way up.

Stop waiting. Show up. Start building. And when you've found your footing, turn around and help everyone you can reach find theirs.

Food first. Then keep the promise growing.

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Precedent P-24: What the Forecast Missed (1993 to 1995)

The Ledger closes with a receipt. Frame it.

February 1995. Newsweek. Clifford Stoll, an astronomer who had been online for two decades, published "Why the Web Won't Be Nirvana," better remembered by its print title, "The Internet? Bah!" Online shopping, digital newspapers, networked government: he waved them off as baloney. No online database, he wrote, would replace your daily newspaper. Stoll, Why the Web Won't Be Nirvana

Read the whole essay, though, and it's not the work of a fool. Stoll liked the net and had made real friends there. Several of his complaints were accurate at the moment he wrote them: the search was bad, the modems were slow, the information was unfiltered, and plenty of the promises about classrooms and democracy were oversold. He was describing the worst version of the technology and mistaking it for the final one, and bundling separate claims together so that one failed forecast took the others down with it. He later owned it with good humor: "Of my many mistakes, flubs, and howlers, few have been as public as my 1995 howler." Clifford Stoll

Another part of that future was being made through a quieter decision. On April 30, 1993, CERN placed three components of its Web software in the public domain: a basic client, a server, and a shared code library. People were permitted to use, copy, modify, and distribute the code. In 1994 CERN retained copyright on a new server release while continuing to grant free use through a permissive license. Access depended on decisions about ownership and permission, made by identifiable people and an institution. Smith and Flückiger, Licensing the Web

That didn't give everyone a computer or a connection. It removed a particular barrier to building on particular software. The comparison with food has that boundary: a meal still needs material, work, and delivery. But the episode gives hope something firmer than a prediction to stand on. People can choose terms that let more people benefit. We can ask which barrier our next decision will remove.

The mechanism. A technology's present limits don't tell you which ones will yield to further work. Critics benchmark the present version; builders test what can change. Neither improvement nor failure is guaranteed. And a technology can prove its critics wrong about what it can do and still create real problems worth criticizing.

The rule. Let actual results change the plan, including your optimistic plan. You're living through this decade's version of Stoll's essay, published weekly, about AI, by smart people, containing accurate complaints. You've now read twenty-four entries in the Precedent Ledger, from the first grain fields to a magazine calling the web a letdown. You know how this rhymes. My bet is 2027. The essayists will still be writing. You'll be building.

The practice.

1. Start your own receipts file today. Every time a credentialed voice declares AI overhyped, a bubble, or a fad, save it: headline, author, date. Save the breathless predictions too, including mine. Set a reminder two years out to read the file. You're not collecting them to gloat. You're building immunity to confident dismissal and confident hype alike, with evidence you gathered yourself.
2. Separate an observed problem from a claim that it can never change. When you read an AI complaint, ask what evidence would show improvement and what constraint would remain. Slow modems improved. The speed of light didn't. Cost and law can change, but you can't budget as though they already have. Let the constraints change your plan.
3. Close the Ledger and open the notebook. Go back through the practices behind you and pick the one that made you most uncomfortable, because that discomfort is a compass pointing at your beachfront. Do that one first.

Back to Bear Flag

The proofs came back from a print shop two towns over in a cardboard box that weighed about as much as the Fisher's replacement firebrick, ten copies, perfect-bound, the cover plain. The working title hadn't survived. Working titles don't. The first copy went to his mother, mailed flat in a padded envelope with a pencil note that said only: *This is what I was trying to say on the phone.*



The proofs. Ten copies, perfect-bound. The book, finally.

Elijah drove south on a Thursday in late winter, three years and some weeks after the toast, and the drive was the same drive from Chapter 6 run in reverse: shoulders climbing gradually up toward his ears as the country closed in, wood and weather giving way to billboards, the check-engine light staying dark the whole way because Curtis had fixed it for two dollars in another life. Past Cloverdale he took the long way, on purpose, through the hills behind what had been his grandmother's place, and slowed where the madrones stood in the

wet, shrugging off their bark the way they do, red under grey, burned down and come back how many times now. He didn't stop. You don't need to stop for the things you're carrying with you.

His mother called at 9:40 Sunday, four days before, right on schedule, the radio and the sink running behind her voice. She had read it twice. "I kept hearing you try to tell me this on the phone," she said. "For three years, honey. It reads better than it sounded." A pause, the pot going into the rack. "The part with the trees is the best part. Your grandmother would say you finally listened." And then, because her frame had always had room in it for everything that mattered: "Manny says to tell you you were right about the pressure switch."

The Bear Flag hadn't changed, which was the eeriest thing it could have done. Same sticky wood, same neon, same television nobody was watching, the same corner table where three realtors had once toasted the dismissal of three people who weren't there to raise a glass. The beer was five dollars now. He ordered one, and sat where he had sat with the laptop and the CSS tutorial and the 40 percent battery, and at the far end of the bar, because history doesn't repeat but it does like a good seat, somebody's laptop was glowing, somebody grinding away at teaching themselves something after hours.

He took a copy of the book out of his jacket and set it on the bar.

On the television above the liquor, sound off, a panel of serious people was arguing about the newest model release, chyrons doing the work of sirens. Nobody in the bar looked up. That was the other thing three years had taught him: the wave never announces itself on the news. It announces itself in rooms like this one, three careers at a time, quietly, while everybody watches the wrong screen. For how people were living through it, he had learned to ask someone, and wait for an answer.

Three years. Long enough to lose the pleasure of being right about everything he feared. Long enough for Devendra to arrive with the children, for Frank to put his name on a different bench, for a delivery sheet to acquire Rosa's instruction in Curtis's careful handwriting:

prepared, less spicy. The route still needed money beyond December. Reuben still had unanswered questions. Nobody had mistaken a stack of forms for dinner.

Long enough, too, to be wrong in every way this book documents, the renders, the optimized script, the commented-out line, the crust called a planet, the flash done solo, and to learn the one thing his arithmetic could never have produced on its own: that the answer to the machine was never going to be a smarter machine. It was nineteen households who know whose kid is whose, times eleven towns, times whatever number you're about to make it.

He put his phone beside the book. There was a message from the kitchen: the afternoon route was covered. He read it twice, then turned the screen down.

The first time he had sat here, he thought the future was a thing you could show someone on a laptop. Now it had addresses.

He raised the glass. Not high. This isn't that kind of bar, and it never was.

"Congratulations," he said, quietly, to the corner table, to the laptop at the end of the bar, to the whole grinding hopeful improbable room, and for the first time in three years the word came out meaning what it says. There was no forecast in it this time. Just relief at the ordinary, unfinished work waiting for him at home.

He finished half the beer, tipped like a man who used to get tipped, and stood up. The book stayed on the bar, on the exact stretch of wood where he had once slid a napkin across to three men he'd thought he was saving. He'd believed, that night, that he was handing them a telescope. He had been three years early and one page short. This time the pages were attached.

He left it there and walked out into the winter air, north in his chest already, and if you want to know what happened to that copy, check the cover of this one.

You're holding it.

APPENDIX A

Questions for a Food Baseline

This is a discussion brief, not an ordinance or a claim of legal authority. Use it with affected residents, workers, food providers, relevant specialists, and public institutions.

FROM A CAPABILITY TO A MEAL

01 **Produce**
Enough suitable food.

02 **Prepare**
Food the person can use.

03 **Deliver**
Access the person can reach.

04 **Sustain**
People, repairs and dependable funding.

A schematic of the proposal. Each part needs people, resources and accountable decisions.

The proposal

Make adequate food available regardless of employment or ability to pay. Use automation where it demonstrably helps supply that food, while preserving effective existing methods and services. Keep the person receiving food at the center of the account.

The proposal does not require abolishing restaurants, paid employment, or food sold above the baseline. It does require discussing how a dependable baseline will be provided, including for people who cannot prepare or store ingredients.

Questions worth bringing to the table

1. **Need:** Who lacks dependable access now? What do they say is missing? Include people outside existing membership lists and people unable to attend meetings.

2. **Food:** What constitutes suitable and sufficient provision across dietary needs, seasons, and living circumstances? Who has the relevant expertise to evaluate it?
3. **Work:** Which tasks are already well served? Which are difficult to provide? What can a proposed machine actually do, and what remains human work?
4. **Access:** How does someone request or receive food without money, a smartphone, transport, or proof of employment? What happens when a person disagrees with an organizer?
5. **Provision:** Who covers equipment, operating costs, maintenance, remaining labor, storage, and distribution? Which commitments are real, and how long do they last?
6. **Continuity:** What happens to the meal when a machine, crop, vehicle, funding source, or organization fails? Is the alternative confirmed?
7. **Workers:** Who gains or loses income, time, discretion, or bargaining power? How do the people doing the work participate in the decisions?
8. **Authority:** Which existing permissions, responsibilities, and public processes apply? Who is accountable for resolving disputes and correcting harm?
9. **Evidence:** What comparison would show that a proposed change helps? Who checks the result? What would make us revise or stop it?

A bounded first proposal

Describe a small trial in one page: the need, partners, task, recipient route, available resources, duration, comparison, and review date. Keep the existing service operating while the trial runs. State what has not yet been arranged.

Don't present this page as a universal legal template. It grants no tax exemption, power to terminate a contract, suspension of other rules, or immunity from a claim. A proposed service still has to work through the responsibilities that apply to it.

A community can endorse the goal and still disagree about the means. Evidence should inform that disagreement. It doesn't eliminate the need for judgment, public participation, and decisions people can challenge.

The purpose of the brief is to make the next conversation concrete enough to improve the proposal.

APPENDIX B

Works Cited

These sources support particular claims, not the book's entire proposal. Original documents establish what someone wrote or did; research studies test narrower questions; institutional and company histories supply context from their own perspectives. A historical analogy remains an argument the reader can challenge. Fictional scenes, the Thanksgiving 2027 forecast, and the proposed food guarantee are not findings of the sources below.

Food, work, and access

- **Food security:** FAO, An Introduction to the Basic Concepts of Food Security (2008). The guide distinguishes availability, access, utilization, and stability. Producing enough food does not by itself ensure that a household can obtain and use it.
- **Agricultural automation:** FAO, The State of Food and Agriculture 2022: Leveraging Automation in Agriculture for Transforming Agrifood Systems. An institutional synthesis of opportunities and adoption barriers, including the position of smaller producers. It does not demonstrate universal provision by an autonomous farm.
- **Employment exposure:** International Labour Organization, Generative AI and Jobs: A Refined Global Index of Occupational Exposure (2025), with the 2025 update. These estimates concern tasks potentially affected by generative AI. Exposure is not a count of observed layoffs or a prediction that every exposed job disappears.
- **Workers' conditions:** Better Work Jordan, Annual Report 2024: An Industry and Compliance Review, reporting January to December 2023. ILO and IFC assessment and survey evidence. The factory case discussed in Chapter 7 appears in the summary findings; the report expressly distinguishes its severe violations from typical conditions in the sector.
- **Common rights and enclosure:** UK Parliament, Enclosing Land. An institutional historical overview of changes to land use and common rights. Common land did not mean unrestricted access for every person.
- **Garden machinery:** FarmBot, Tools. Manufacturer documentation of particular seeding, watering, and weeding functions. It establishes the

advertised design and tasks, not independent results for complete diets or delivery to households.

- **Growing sites:** US Environmental Protection Agency, Urban Agriculture. Guidance on site history and potentially contaminated land. It is background for evaluating a site, not a safety finding about the fictional garden.
- **Living soil:** University of Minnesota Extension, Soil Biology. Extension guidance on soil organisms and their different functions.
- **Waste pathways:** US Environmental Protection Agency, Learn About Aquatic Trash. Background on how waste reaches waterways. The behavioral effects of a proposed cleanup machine would require their own evaluation.

Useful capability and its limits

- John Jumper and colleagues, Highly Accurate Protein Structure Prediction with AlphaFold, *Nature* 596 (2021), 583–589. Original research on protein-structure prediction. Better structural information can aid research; a predicted structure is not itself a treatment.
- Jonathan M. Stokes and colleagues, A Deep Learning Approach to Antibiotic Discovery, *Cell* 180 (2020), 688–702.e13. Original laboratory research using a neural network to identify antibacterial candidates, followed by experiments including mouse infection models. This is not evidence of an established general cure in humans. The open manuscript corresponds to PMID 32084340.
- Sellafield Ltd, Sellafield Robotics: Using Spot More for Spotless Nuclear Clean-up (23 March 2023). An operator's account of remotely operated inspection and cleaning in hazardous areas. It documents a particular deployment, with people and specialist equipment, rather than autonomous removal of every kind of toxic waste.
- National Institute of Standards and Technology, Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile, NIST AI 600-1 (July 2024). The final publication identifies risks including erroneous outputs, privacy loss, harmful bias, security failures, and environmental effects. These concerns require attention alongside the distribution of economic benefits.

Intelligence and forecasts

- Ashish Vaswani and colleagues, Attention Is All You Need (2017). The original Transformer paper. Its architecture and reported experiments do not establish an AGI date.
- I. J. Good, Speculations Concerning the First Ultraintelligent Machine, *Advances in Computers* 6 (1965), 31–88. A facsimile of the essay, hosted by Virginia Tech. The intelligence-explosion argument is a conditional prospect, not a demonstrated sequence of unlimited improvement.
- Vernor Vinge, The Coming Technological Singularity: How to Survive in the Post-Human Era (1993). Author-hosted text of the VISION-21 paper. Its thirty-year forecast belongs to its 1993 context and does not substantiate this book's 2027 expectation.
- Eliezer Yudkowsky, Three Major Singularity Schools (30 September 2007), hosted by Machine Intelligence Research Institute. An author's conceptual distinction among accelerating change, an event horizon, and an intelligence explosion. Useful for understanding the term's different meanings, not independent confirmation of any forecast.

Historical cases

The case numbers match the precedents and Appendix D. Where a source is a company history, archive description, or later scholarly interpretation, that status is stated rather than presented as an eyewitness record.

P-01: Reading and moral concern

Ana Vogrinčič, The Novel-Reading Panic in 18th-Century England: An Outline of an Early Moral Media Panic (2008). Historical scholarship on concerns about novel reading. Richard Brinsley Sheridan's The Rivals (1775), Act I, supplies an original literary example of hostility to circulating libraries. A character's speech illustrates a debate; it does not measure how everyone felt.

Jane Austen, Northanger Abbey, Chapter 5 and Chapter 25, public-domain text, Project Gutenberg. Primary literary testimony: the narrator defends novels while the story examines mistaken inferences. Fictional incidents are distinguished from evidence about real readers.

P-02: Bell at the fair

Alexander Graham Bell, letter to Alexander Melville Bell and Eliza Symonds Bell, 27 June 1876, Library of Congress transcript. Bell describes his demonstration and the reactions of William Thomson and Emperor Dom Pedro. The

Library's Inventing Telephones at the Centennial supplies historical context for the telephone and the exhibition's machinery. Neither establishes a universal rule about what crowds or experts can recognize.

Bell's official transcript PDF, pages 1–2, distinguishes his demonstration from the audience reactions reported to him by Willie Hubbard. The closing discussion records a proposed alliance, not a completed agreement.

P-03: The aviation forecast

New York Times, Flying Machines Which Do Not Fly (9 October 1903), with an accessible transcription of the editorial. This is one newspaper's prediction, not a survey of aeronautical opinion.

Wilbur Wright, letter to the Smithsonian Institution, May 30, 1899, archival transcription. NASA Glenn, Wright 1901 Wind Tunnel, historical engineering account. The two sources support the reading request and experimental work that precede the later flight.

P-04: Road regulation

Locomotives Act 1865, 28 & 29 Vict., chapter 83, especially section 3. Original statute specifying operating restrictions, including a person carrying a red flag ahead of the vehicle. The law alone cannot establish its eventual economic effects.

House of Lords debate, May 26, 1865, columns 867–872, on the Locomotives on Roads Bill. A primary record of competing arguments and participants' reported experiences. Speeches do not establish either the safety of the machines or the later effects of the law.

Highways and Locomotives (Amendment) Act 1878, section 29, and Locomotives on Highways Act 1896, sections 1 and 4. Original statutes: the first changes the advance-walker rule in England; the second exempts a defined light-locomotive class and sets a higher speed ceiling for it.

P-05: Ming voyages

Hong Kong Government, Exhibition Follows the Footsteps of Zheng He (21 February 2006), announcing an exhibition at Hong Kong Museum of History. Institutional context for the seven voyages of 1405–1433. For a participant's account, see Ma Huan, *Ying-yai Sheng-lan: The Overall Survey of the Ocean's Shores*, translated by J. V. G. Mills (1970), identified in the Indiana University library catalogue. The catalogue identifies the edition; it is not a substitute for reading the translated text.

Edward L. Dreyer, *Zheng He: China and the Oceans in the Early Ming Dynasty, 1405–1433* (Pearson Longman, 2007), appendix, translations of primary sources, especially the Liujiagang inscription on printed pages 191–193. This is the expedition's account, in a modern translation; its claims of order and protection require recognition of the speaker's position.

P-06: Copernican astronomy

Nicolaus Copernicus, De Revolutionibus Orbium Coelestium (1543), digitized edition and collection description, Library of Congress. The work places a moving Earth among the planets. The record does not by itself reconstruct the entire later history of its reception.

Copernicus, dedication to Paul III, Harvard Classics translation hosted by Ohio State. Sachiko Kusukawa, Copernicus's Book, Cambridge History and Philosophy of Science (1999). Together they distinguish Copernicus's stated purpose and supporters from the framing added by Osiander during publication.

P-07: Horses and measurement

US Department of Agriculture, Farm Production Practices, Costs, and Returns, Statistical Bulletin 83 (1949), especially table 14, printed page 45. Historical estimates put the combined population of horses and mules **on farms** at its peak in 1918. The report describes a transition involving machinery, feed, wages, and other costs; it does not isolate one cause or measure all working animals everywhere.

Science Museum Group, Original Indenture of Agreement for Erection of a Rotative Engine, March 1, 1786, object 1908-185. Museum catalogue of a particular contract and its business context. It supplies responsibilities and payment terms; it is not a complete account of all consequences of mechanization.

P-08: The agricultural transition

Amanda Mummert, Emily Esche, Joshua Robinson, and George J. Armelagos, Stature and Robusticity During the Agricultural Transition: Evidence from the Bioarchaeological Record, *Economics & Human Biology* 9 (2011), 284–301. A scholarly synthesis of skeletal evidence. Its account of health burdens and variation should not be converted into a claim that every farming population experienced the same change.

For an original study combining skeletal and genetic evidence, see An Integrative Skeletal and Paleogenomic Analysis of Stature Variation Suggests Relatively Reduced Health for Early European Farmers, *Proceedings of the National Academy of Sciences* (2022). Its European sample and adjustments for ancestry limit how widely the result can be generalized.

UNESCO World Heritage Centre, Neolithic Site of Çatalhöyük. Institutional description of the settlement's layout and surviving material culture.

Clark Spencer Larsen and colleagues, Bioarchaeology of Neolithic Çatalhöyük Reveals Fundamental Transitions in Health, Mobility, and Lifestyle in Early Farmers, *Proceedings of the National Academy of Sciences* 116 (2019), 12615–12623; open manuscript. Original research, especially manuscript pages 5–6; includes contrasting indicators rather than uniform deterioration.

P-09: The Luddites

UK National Archives, The Proclamation of Ned Ludd. An original proclamation with archival transcription and historical context. It supports discussion of labor conditions, threats, and repression, without reducing all participants to a single motive.

Kevin Binfield, Luddite History, excerpt from the historical introduction to *Writings of the Luddites* (Johns Hopkins University Press, 2004). Historical scholarship on the particular trades, negotiations, and material conditions behind the protests.

P-10: Musicians and recorded sound

Music Performance Trust Fund, About. The fund's institutional account traces its establishment in 1948 to agreements between the American Federation of Musicians and recording companies. It documents a negotiated mechanism for financing admission-free performances. It does not imply that every playback of a recording generates a payment to this fund.

University of Maryland Special Collections in Performing Arts, The Recording Ban of 1942, in *Modern Songs of War and Conflict*, curated by Ben Jackson. Institutional account of the commercial recording dispute, inventories, and different settlement dates.

P-11: Torches of Freedom

Vanessa Murphree, Edward Bernays's 1929 "Torches of Freedom" March: Myths and Historical Significance, *American Journalism* 32 (2015), 258–281. Original historical research questioning the event's later mythology, including the account of a uniformly receptive press. A contemporary photograph of Nancy Hardin and Edith Lee, dated 31 March 1929, is held by the Library of Congress. The photograph records an event, not its long-term causal effect on smoking.

Edward L. Bernays, Propaganda (New York: Horace Liveright, 1928), chapter IV, pages 54–56. Primary text describing a hypothetical piano-marketing method. It establishes what Bernays advocated, not universal effectiveness.

Vanessa Murphree, Teaching Our Journal: Edward Bernays's 1929 "Torches of Freedom" March, *American Journalism*. The researcher's account of methods, archival limitations, and contemporary press responses.

The Library of Congress digital-object record, digital ID ds.07375, supplies another locator for the photograph cited above.

P-12: Bronze Age disruption

A. Bernard Knapp and Sturt W. Manning, Crisis in Context: The End of the Late Bronze Age in the Eastern Mediterranean, *American Journal of Archaeology* 120 (2016), 99–149. Archaeological scholarship emphasizing regional differences and the interaction of possible causes. The case is not evidence of a single synchronized collapse caused by one variable.

Institute of Nautical Archaeology, Uluburun Late Bronze Age Shipwreck Excavation. The excavating institution's account of the vessel, surviving cargo, and interpretation. The wreck predates the later crisis; it does not identify that crisis's cause.

P-13: Trithemius and media

Johannes Trithemius, De Laude Scriptorum, Mainz: Peter Friedberg, 1494, digitized printed edition held by Paderborn University Library. The treatise was composed in 1492. Its defense of scribal work appeared in print. Jeremy Norman's historical note on vellum and paper explains the preservation argument. Neither source warrants turning a nuanced position into wholesale hostility to printing.

Bibliothèque nationale de France, La Bible de Gutenberg, especially "Un rare témoin de la date d'impression de la Bible." Institutional collection study including transcribed inscriptions. Supports the Cremer example without supplying an employment history of all scribes.

P-14: Quartz watches

Seiko Museum, The Quartz Crisis and Recovery of Swiss Watches. A manufacturer-associated museum's history of quartz competition and Swiss responses. It includes Swiss technical participation and economic pressures, complicating the story that one side simply failed to see the invention.

CSEM, Historical Timeline, entries for 1962 and 1967. Institutional history of electronic-watch research.

Swatch Group, Company History, entry for 1983. Manufacturer's account of product and production design.

P-15: Homesteading

US National Archives, Homestead Act of 1862, original act and transcript. National Park Service, Homestead History and Culture, supplies the necessary context of prior Indigenous habitation and displacement. The legal transfer of land must not be confused with the land having been empty or universally available.

US Capitol Visitor Center, Homesteading Certificate of Eligibility, Daniel Freeman, January 20, 1868. Certificate and institutional description of Freeman completing the five-year requirement. The 1862 Act above also provided a paid route to earlier title.

P-16: Mail-order houses

Cynthia E. Johnson, House in a Box: Prefabricated Housing in the Jackson Purchase Cultural Landscape Region, 1900 to 1960, Kentucky Heritage Council, edited by Rachel Kennedy. An institutional historical and field-survey study of precut and other prefabricated houses, including Sears and competing producers. It documents manufacturing and distribution methods that still required land, assembly, and local work.

P-17: A company under pressure

Kodak, Milestones. Company chronology documenting digital work and emergence from Chapter 11 in 2013. It is evidence against the claim that Kodak never participated in digital imaging, but an interested corporate history is not a complete explanation of the company's financial difficulties.

Swiss Camera Museum, 1975: The Invention. Institutional account of Sasson's prototype.

P-18: Related capabilities, new uses

Fujifilm, Healthcare and Corporate History. The company's account describes applications of its materials and imaging expertise in other fields, including healthcare. This documents its stated strategy and activities, not a controlled comparison proving why one company succeeds and another struggles.

Fujifilm Holdings, 90th Anniversary Corporate History, sections on the transition from photographic film and the 2004 plan.

Fujifilm, ASTALIFT History, Japanese brand chronology, especially 2004 and 2007. Used for commercialization history, not product efficacy.

P-19: Victory gardens

US Office of War Information, Plant a Victory Garden (1943), government poster preserved by University of North Texas. National Park Service, The World War II Home Front, provides the broader mobilization context. Estimates about garden vegetables must not be restated as shares of all food or all nutrition.

- United States Department of Agriculture, Committee on Victory Gardens, The 1943 Victory Garden Program. Reprinted by the US Office of Civilian Defense, March 1943. Original program document, especially pages 3–4 on allotments, water, advice, tools, school-garden staffing, and preservation. It records planned arrangements rather than measured nationwide compliance.
- National Park Service, Victory Gardens on the World War II Home Front. Institutional historical account, including gardens maintained by incarcerated Japanese Americans. Distinguishes residential gardens from camp production farms and gives context that aggregate harvest figures cannot supply.

P-20: An argument in circulation

Thomas Paine, *Common Sense* (1776). Library of Congress collection record for the book and exhibition context. These identify the work and its argument for independence. Its circulation does not establish that one pamphlet alone caused a revolution.

- University of Michigan, Clements Library, Going Viral in 1776. Exhibition descriptions of Bell, Bradford, and London editions of Common Sense. Supports the paper wrapper, two-shilling price, anonymous issue, printing dispute, and handwritten additions in the library's Almon copy.
- University of Minnesota Law Library, Common Sense, Philadelphia, 1776, Treasures of the Riesenfeld Center. Description of an enlarged Bradford edition and the author-printer dispute over profits. The account establishes the described competing positions, not an independent audit of Bell's books.

P-21: Whole Earth Catalog

Whole Earth Catalog, Fall 1968, digitized issue indexed by Whole Earth Index. An original example of gathering tools and practical information. Steve Jobs's 2005 Stanford commencement address is the primary source for his later comparison of the catalog to Google in paperback form. His analogy is a recollection, not a claim that the catalog already provided an internet search engine.

- Whole Earth Catalog, Fall 1968: sample pages reproduced with later anniversary annotations. The contents and purpose/function page establish examples and selection criteria; the added availability warnings belong to the later reprint.
- David Senior, Museum of Modern Art Library, Access to Tools: Publications from the Whole Earth Catalog, 1968–1974 (2011). Institutional exhibition history of the catalog, affiliated Truck Store, and reader contributions.

P-22: Y2K preparation

US General Accounting Office, Year 2000 Computing Challenge: Lessons Learned Can Be Applied to Other Management Challenges, AIMD-00-290 (2000). A government review of preparation and management practices. It documents substantial work; it cannot directly observe the global outcome of a world in which that work never happened.

For AIMD-00-290, see especially printed pages 29–32 of the complete report for HUD's dependency inventory and the reported Oak Ridge file-transfer failure and tested tape fallback. These cases document preparation and recovery, not a world observed without remediation.

- US General Accounting Office, Social Security Administration: Year 2000 Readiness Efforts Helped Ensure Century Rollover and Leap Year Success, AIMD-00-125 (19 April 2000), including subsequent recommendation-status updates. Supports continued contingency planning and the explicitly later April 2004 separation of validation testing from development.

P-23: The National Health Service (Chapter 19)

UK Government History blog, The founding of the NHS: 75 years on (13 July 2023). Government historians' account of the Beveridge Report's five giants, the limits of wartime health insurance (about half the population covered; hospital, specialist, dental, optical, and hearing services excluded), doctors' concerns and Bevan's answer, and the service's launch on 5 July 1948 as the first free-at-the-point-of-use medical care for a whole population in a western country.

Aneurin Bevan, *In Place of Fear* (1952), chapter 5, transcribed by Public Matters. Source of the epigraph: "no society can legitimately call itself civilized if a sick person is denied medical aid because of lack of means."

P-24: Internet forecasts

Clifford Stoll, Why the Web Won't Be Nirvana, *Newsweek* (February 1995), also known by the print title "The Internet? Bah!" The publisher's archival version is the source for this dated forecast. The case concerns Stoll's argument, not a representative sample of every critic of the internet.

Clifford Stoll, attributed excerpt of his 1995 Newsweek essay, in Elon University's Imagining the Internet archive. An additional access point for the particular claims discussed here.

Tim Smith and François Flückiger, Licensing the Web, CERN. Institutional history reproducing the 1993 release terms and describing the 1994 licensing decision. Specific software permissions are distinguished from universal internet access.

Historical context for provision

British Restaurants (Chapter 11)

UK Parliament, Hansard, British Restaurants, May 28, 1941, House of Commons, volume 371, columns 1853–1854. Recorded questions and ministerial replies about local organization.

UK Parliament, Hansard, British Restaurants and Works Canteens, October 1, 1941, House of Commons, volume 374, columns 572–573. Supplies and private-caterer objections.

UK Parliament, Hansard, Food Distribution, October 2, 1941. Includes the Fishmongers' Hall complaint. Statements in debate document participants' arguments rather than independently measured outcomes.

UK Parliament, Hansard, British Restaurants (Prices), March 17, 1942, Written Answers, Food Supplies. Official financial principles.

New Communities (Chapter 12)

SNCC Digital Gateway, New Communities Formed in Southwest Georgia. Institutional civil-rights history with listed archival and oral-history sources.

Ellen Sabina, A Conversation with Shirley Sherrod, Maine Farmland Trust, published February 29, 2024; interview conducted June 2023. Participant testimony, attributed as such in the chapter.

Vienna municipal housing (Chapter 13)

City of Vienna, Municipal and Provincial Archives, From Red Vienna to the Ständestaat, 1918–1938. Institutional retrospective on municipal government and housing.

City of Vienna, Rathauskorrespondenz, Der Karl-Marx-Hof ist 60, October 11, 1990. German anniversary notice describing the complex and ongoing renovation plans.

Additional chapter sources

Much of what follows was researched for earlier editions, cut in v0.8.0, and restored in v0.9.0. Later additions support the expanded manuscript and corrections through v0.10.2. Figures quoted from a builder's own video or a company's own history are attributed as such in the text.

Introduction and Chapter 1: definitions, forecasts, and the Transformer

- Nikola Danaylov, 17 Definitions of the Technological Singularity, Singularity Weblog. Sixteen definitions and an empty seventeenth slot; source of the Thornton (1847) and Kevin Kelly definitions.
- Samuel Butler, "Darwin among the Machines", *The Press*, Christchurch, 13 June 1863; and *Erewhon* (1872).
- Alan Turing, "Intelligent Machinery, A Heretical Theory", c. 1951, Turing Digital Archive, King's College Cambridge.
- Stanislaw Ulam's 1958 recollection of John von Neumann, as quoted in Vinge's essay, listed above.
- Hans Moravec, *Mind Children* (Harvard University Press, 1988); Nick Bostrom, "How Long Before Superintelligence?" (1997, revised); Ray Kurzweil, *The Singularity Is Near* (Viking, 2005); John Smart, Acceleration Watch.
- Leopold Aschenbrenner, Situational Awareness: The Decade Ahead (2024). One insider's argument, cited as such.
- Katja Grace and colleagues, Thousands of AI Authors on the Future of AI, revision 3, first published 2024, revised 2025. Source of the 2022 and 2023 aggregate forecasts, the 10 percent probability of high-level machine intelligence by 2027, and the extinction-risk responses. The surveys used changing samples; the displayed dates are aggregate probability thresholds, not every respondent revising a personal date.

- Brett J. Kagan and colleagues, In vitro neurons learn and exhibit sentience when embodied in a simulated game-world, *Neuron* (2022). The DishBrain *Pong* study; see also the 2024 comparison with deep reinforcement learning.
- Aaron Bastani, *Fully Automated Luxury Communism* (Verso, 2019).
- Stanford HAI, AI Index Report 2025, chapter 1. The historical query-price comparison fixes a GPT-3.5-level MMLU threshold. It does not measure energy per task or equivalence across all work.

Chapters 3 and 4: the far horizon

- Nick Bostrom, "Are You Living in a Computer Simulation?", *Philosophical Quarterly* 53 (2003).
- David Kipping, "A Bayesian Approach to the Simulation Argument", *Universe* 6 (2020).
- Copernicus's dedication and the Osiander preface are listed under P-06 above.
- NASA, Hubble Views the Star that Changed the Universe (2011). Institutional history of the Cepheid observations that placed Andromeda outside the Milky Way.

Chapter 5: energy and the cyberdeck lineage

- *The Times* (London), July 1816, on the waltz as an "indecent foreign dance," as quoted in John Hood, Hayek, Strauss, and the Political Waltz, Foundation for Economic Education (2005). A secondary quotation; the 1816 issue itself was not inspected.
- McGill Office for Science and Society, The Moral and Medical Panic Over Bicycles, on "bicycle face."
- John Philip Sousa, "The Menace of Mechanical Music", *Appleton's Magazine* (1906), full text via MIT OpenCourseWare.
- William Gibson, *Neuromancer* (1984), origin of the "cyberspace deck."
- Raspberry Pi, Computer Hardware: Maximum Power Output, and E Ink, Benefits: Bi-Stability. Manufacturer documentation on peripheral budgets and static display retention. Neither establishes total draw or battery runtime for a particular cyberdeck.

Chapter 7: work, loss, and the labor campaigns

- Karsten I. Paul and Bernad Batinic, "The need for work: Jahoda's latent functions of employment in a representative sample of the German population", *Journal of Organizational Behavior* (2010).

- John Maynard Keynes, "Economic Possibilities for our Grandchildren" (1930).
- Anandi Mani, Sendhil Mullainathan, Eldar Shafir, and Jiaying Zhao, "Poverty Impedes Cognitive Function", *Science* 341 (2013), the sugarcane-harvest study; and Mullainathan and Shafir, *Scarcity* (2013).
- The Guardian, "The rebel group stopping self-driving cars in San Francisco" (26 July 2023), on Safe Street Rebel and coning.
- Brian Merchant, *Blood in the Machine* (Little, Brown, 2023).
- American Federation of Musicians, the robot advertisements in *Exhibitors Herald-World*, 16 November 1929 and 17 May 1930; Smithsonian Magazine, "Musicians Wage War Against Evil Robots".
- Home Taping Is Killing Music (BPI, 1981); Jack Valenti's 1982 testimony to the House Judiciary subcommittee on home recording; *Sony Corp. of America v. Universal City Studios*, 464 U.S. 417 (1984).
- National Highway Traffic Safety Administration, information request to Waymo, November 24, 2025, investigation PE25013. Primary record of an inquiry concerning stopped school buses. An inquiry is not a final finding about every reported incident.

Chapter 8: attention and cooperation

- Gloria Mark, Daniela Gudith, and Ulrich Klocke, "The Cost of Interrupted Work: More Speed and Stress", CHI 2008.
- Robert Axelrod, *The Evolution of Cooperation* (Basic Books, 1984).
- Robin Dunbar, "Coevolution of neocortical size, group size and language in humans," *Behavioral and Brain Sciences* 16 (1993).

Chapter 9: the neighborhood factory and nitrogen

- International Energy Agency, Ammonia Technology Roadmap (2021), for the energy use and emissions of ammonia production.
- Kyle Gabriel and contributors, Mycodo, open-source environmental controller; installer at the project's documented address (checked 23 September 2026).

Chapter 10: model collapse

- Ilia Shumailov and colleagues, "AI models collapse when trained on recursively generated data", *Nature* 631 (2024).
- Internal Revenue Service, What Is Taxable and Nontaxable Income?. Barter generally creates income measured at fair market value; the guidance distinguishes informal noncommercial exchanges of similar services.

Chapter 11: local computing and machine work

- National Institute of Standards and Technology, Air Gap. Physical separation and the controlled transfer of information across it; local execution alone is not isolation.
- California Department of Justice, Data Security Breach Reporting. State reporting requirements for defined incidents. Counsel must assess the facts; this source does not determine whether the fictional exposure meets a statutory definition.
- Occupational Safety and Health Administration, Control of Hazardous Energy, 29 CFR 1910.147. Servicing procedures address energy isolation, stored energy, and verification. A command to stop software is not equivalent to physical isolation.

Chapters 12 and 15: land tenure and assessment

- Grounded Solutions Network, Community Land Trusts. Practitioner documentation of a typical affordable-homeownership model: separately owned land and homes, a renewable ground lease often lasting ninety-nine years, and resale restrictions. Other uses need appropriate agreements.
- California Department of Conservation, Williamson Act FAQ (2024). Use restrictions and property-specific assessment; a contract does not necessarily produce greater savings than existing tax treatment.
- Internal Revenue Service, Conservation Easements. Qualified donations and valuation requirements for federal income-tax deductions. These are distinct from local property-tax assessments.
- Land Report, January 2021 newsletter. Original publication's estimate of Gates's farmland holdings; a dated estimate rather than a current parcel inventory.
- Nuveen, Global Farmland Capabilities. Investment manager's description of its portfolio, not independent evidence of any investor's motive.

Chapter 13: shelter and heating

- DuraVent, DVL. Manufacturer documentation identifies the product as an interior connector, distinct from the chimney.
- US Fire Administration, Heating Fire Safety. Guidance on combustible clearance and maintenance.
- Centers for Disease Control and Prevention, Carbon Monoxide Poisoning: About. Fuel-burning equipment and enclosed-space exposure.

Chapter 14: food recovery and the graveyard

- USDA, Food Waste FAQs. The 30 to 40 percent estimate of food waste at the retail and consumer levels uses a 2010 baseline; it is not a fresh measurement for the publication year.
- CalRecycle, SB 1383 Food Recovery, the state's account of the edible-food-recovery requirements and the 20 percent recovery target for 2025; NPR, French Food Waste Law Changing How Grocery Stores Approach Excess Food (2018), on France's 2016 law (Law 2016-138, the Garot law).
- Snopes, Kodak's digital camera, on Sasson's account of management's response; Fortune, "Sears Could've Been Amazon"; Variety, "How Blockbuster Could Have Owned Netflix"; Slate, Borders bankruptcy: Done in by its own stupidity, not the Internet (July 2011).
- Toyota, Toyota Production System. Manufacturer description of just-in-time production includes minimum necessary stock; it does not mean every inventory is literally zero.

Chapter 17: tools

- SECTOR 07, I 3D Printed a Photobioreactor. Builder's own video; source of the eight-grams-a-week figure and the unfinished-design caveat.
- Kirsten Dirksen, Mad scientist's homestead, interviewing the EasyGrow builder. Source of his density, water, stadium, and "not in business to get rich" statements, which the chapter attributes to him.
- SANANBIO, Introducing the SANANBIO Uplift. Vendor promotion; source of the labor-cost and human-touch claims.
- Open Source Ecology, Global Village Construction Set; Meshtastic; Kiwix.
- FAO, Use of Algae and Aquatic Macrophytes as Feed in Small-scale Aquaculture, Fisheries and Aquaculture Technical Paper 531 (2009). Species, feed composition, and inclusion levels matter. Growing algae does not establish a complete diet for every fish or validate the prototype's output.
- Semtech, What Is LoRa?. Manufacturer description of the radio modulation technology, distinct from a complete network protocol.
- Kiwix, FAQ. Project documentation on offline archives, supported platforms, and differences among mini, nopic, and maxi editions.

Appendix E: the cyberdeck

- Newsweek, "What is a cyberdeck: Meet Gen Z's new DIY obsession" (2026); TechCrunch, "Cyberdecks are having a moment" (2 June 2026).
- Hackaday's cyberdeck contests: 2022 and 2023.

- Computer Fraud and Abuse Act, 18 U.S.C. 1030; interception and disclosure rules, 18 U.S.C. 2511; and Part 15 operating conditions, 47 C.F.R. 15.5. US provisions cited for bounded examples, not permission to operate in every jurisdiction.
- Meshtastic, Channel Configuration. Project documentation distinguishes public default keys from randomly generated private-group keys and explains location and gateway settings.
- Cryptsetup contributors, FAQ. Technical documentation on disk encryption, passphrases, backups, and threat limits. Encryption of stored data does not protect an already compromised running system.
- ClockworkPi, uConsole, and ZitaoTech, Hackberry Pi Zero. Maker documentation of components and power arrangements. Kit contents vary; uConsole excludes battery cells and offers a configuration without the compute module.
- Raspberry Pi, Heating and Cooling Raspberry Pi 5. Manufacturer tests distinguish ordinary use from sustained workloads that can cause thermal throttling.

Chapter 18: specialized materials

- US Geological Survey, Silica Statistics and Information. Ultra-high-purity quartz is used in semiconductor-manufacturing crucibles. This establishes a material requirement, not a fixed duration for a future supply interruption.

Introduction and Chapter 19: the conversion

- United Nations, Food: "today's global food production is enough to feed everyone on the planet."
- California Department of Food and Agriculture, Food Recovery: US food waste estimated at 30 to 40 percent of the food supply; Californians throw away five to six million tons a year.
- Consumer Financial Protection Bureau, CFPB Reports Highlight Problems with Tenant Background Checks (2022), on credit reports, credit scores, and proprietary risk scores in rental decisions.
- Shoshana Zuboff, *The Age of Surveillance Capitalism* (PublicAffairs, 2019); summary.
- Ashley Gromis and colleagues, Estimating Eviction Prevalence across the United States, *PNAS* 119 (2022), e2116169119. Author-institution research record. The annual averages of 3.6 million filings and 2.7 million households concern 2000 to 2018, not the current year.

- Judicial Council of California, What happens if you lose your eviction case, on the sheriff's notice to vacate and lockout.
- Oxfam, An Economy for the 99% (January 2017), and Billionaire wealth jumps three times faster in 2025 (19 January 2026). Oxfam's wealth comparisons rest on published estimates of net worth and have been debated; they are cited here for scale, not precision.
- California Department of Education, California Universal Meals.
- NYC Office of Civil Justice, Universal Access Annual Report 2022: Local Law 136 of 2017; 78 percent of represented households remained in their homes in fiscal 2022.
- Federal Reserve Bank of Richmond, Electrifying Rural America (2020), and America's Electric Cooperatives, Fact Sheet (April 2026).
- Universal basic services, on the 2017 UCL Institute for Global Prosperity proposal and its costing, and Basic Income Earth Network, on the 2019 UCL report.
- USDA Economic Research Service, Food Security in the U.S.: Key Statistics, and the trends data behind the Introduction's chart.

Sources for the epigraphs

The entries below identify the work or institutional record supporting each attribution. The seven epigraphs added in v0.9.1 were each checked word for word against the text named. An institutional attribution does not establish the first occasion on which a saying was used. Dialogue from a novel belongs to its character and scene.

- **Preface, Hunter S. Thompson:** Simon & Schuster's description of Fear and Loathing at Rolling Stone identifies the ticket-and-ride saying with Thompson. This supports the name attribution, without claiming a verified page in the 1971 novel.
- **Chapter 0, William Gibson:** Interview in Scientific American (26 August 2011). Gibson acknowledges and discusses the formulation quoted to him, including "not very evenly distributed." This is a documented use, not a claim of its first appearance.
- **Chapter 1, Vernor Vinge:** The opening of The Coming Technological Singularity (1993), listed above. Read the prediction with its original date.
- **Chapter 2, I. J. Good:** Speculations Concerning the First Ultraintelligent Machine (1965), printed page 33. The condition about retaining control is part of the sentence, not an optional qualification.

- **Chapter 3, Konstantin Tsiolkovsky:** A translated saying attributed to Tsiolkovsky and traced by NASA's historians to his collected works of about 1911 to 1912. NASA, Historical Origins of the ISS Given as "commonly translated," not as the text of a specific letter.
- **Chapter 7, Louis Pasteur:** The familiar English rendering of Pasteur's observation about the prepared mind, as given by the Institut Pasteur; attributed without claiming a checked original speech.
- **Chapter 11, Archimedes:** A saying traditionally attributed to Archimedes; marked "attributed" because no ancient source was inspected for this edition.
- **Chapter 17, Stewart Brand:** The 1968 *Whole Earth Catalog* statement of purpose, which reads "We are as gods and might as well get used to it" in the available 1968 facsimile. Brand later used the variant "get good at it," which a character quotes in the chapter.
- **Chapter 4, J. B. S. Haldane:** The closing passage of the title essay in Possible Worlds and Other Essays (1927), online transcription. The epigraph retains his framing as a personal suspicion.
- **Chapter 5, Arthur Eddington:** The Nature of the Physical World (1928; digitized from a 1929 impression), chapter IV, printed pages 74–75. The second-law passage concerns physical theory, not a forecast about AI capabilities.
- **Chapter 10, Alan Kay:** Computer History Museum, From Alto to AI. Institutional attribution of the future-and-invention saying to Kay. This edition makes no claim to have established the date or transcript of its first delivery.
- **Chapter 12, Margaret Mitchell:** Gone with the Wind (1936), chapter II. Gerald O'Hara speaks the passage about land. It is fictional dialogue, not a factual guarantee of land's permanence or value.
- **Chapter 14, Ernest Hemingway:** The Sun Also Rises (1926), chapter XIII. Mike Campbell's exchange about bankruptcy supplies the epigraph.
- **Chapter 15, Franklin D. Roosevelt:** Letter to All State Governors on a Uniform Soil Conservation Law (26 February 1937), transcript hosted by the American Presidency Project.
- **Chapter 18, Dwight D. Eisenhower:** Eisenhower Presidential Library quotation record, identifying the National Defense Executive Reserve Conference remarks of 14 November 1957. Eisenhower introduced the planning maxim as a saying he had heard in the Army; its use here does not claim he invented it.

- **Introduction, Upton Sinclair:** *I, Candidate for Governor: And How I Got Licked* (1935), first excerpted in the *Oakland Tribune* on 11 December 1934. Quote Investigator, It Is Difficult to Get a Man to Understand Something reproduces the wording, exclamation mark included.
- **Chapter 6, Arthur C. Clarke:** *Profiles of the Future* (1962), chapter 2, "Hazards of Prophecy: The Failure of Imagination." This is the statement later called Clarke's First Law. Quote Investigator
- **Chapter 8, Charles Mackay:** Memoirs of Extraordinary Popular Delusions and the Madness of Crowds, Preface, checked in the 1852 edition. Mackay presents the herd line as something "well said," not as his own coinage.
- **Chapter 9, Peter Kropotkin:** Fields, Factories and Workshops, revised and enlarged edition (1913), chapter VII. The ellipsis marks the omitted clause about combining agriculture with industry; the original's dashes are rendered as a comma.
- **Chapter 13, Henry David Thoreau:** Walden (1854), "Economy." Thoreau makes the point about the price of an ordinary house and the years of a laborer's life it takes to pay for one.
- **Chapter 16, A. J. Liebling:** "The Wayward Press: Do You Belong in Journalism?", *The New Yorker*, 14 May 1960, page 109, where it appears as a parenthetical remark. Quote Investigator
- **Conclusion, Oscar Wilde:** The Soul of Man under Socialism (1891). A few lines later Wilde writes that "the community by means of organisation of machinery will supply the useful things."
- **Chapter 19, Aneurin Bevan:** *In Place of Fear* (1952), chapter 5; see P-23 above. The sentence is quoted from its middle, where it begins "It insists that no society…"; the epigraph capitalizes the first word.

Sources checked for v0.10.0 corrections

- Gloria Mark, Victor M. González, and Justin Harris, No Task Left Behind? Examining the Nature of Fragmented Work, CHI 2005. Observed time before workers resumed an interrupted activity; not a fixed neurological recovery threshold. The CHI 2008 study listed above examines speed and stress under interruptions.
- International Energy Agency, Key Questions on Energy and AI: Executive Summary (2026). Separates the estimate of 2025 data-center electricity use from the projection for 2030.

- France, Law 2016-138 of 11 February 2016, Article 1. Original enactment on food waste, including proposed donation agreements for eligible shops. A legal requirement is distinct from proof of universal collection or access.
- University College London, The future of welfare: Universal Basic Services (2017). The institution's account of the proposal, not an implemented universal guarantee.
- llama.cpp contributors, HTTP server documentation. Bind addresses and server access settings; running a model locally doesn't by itself prevent network exposure.
- US Department of Energy, Insulation Guide. Building-envelope guidance on insulation and air sealing, without relying on a tax-credit announcement.
- US Department of Energy, DC Microgrid Scoping Study (2015). Configuration-dependent efficiency and conversion stages; no universal zero-loss claim.
- Meshtastic, Mesh algorithm, LoRa configuration, Encryption, and Web Flasher source. Project documentation for routing limits, matching settings, key handling, and hardware-specific firmware.
- FAO, Labour-saving technologies and practices: draught animal power and implements. Animal traction involves care, training, equipment, and human guidance; it does not establish labor free of human supervision.
- Anthropic, Redeploying Fable 5 (June 30, updated July 1, 2026). Company account of Fable's return and continuing limits on Mythos access; safeguard claims remain attributed.
- US Food and Drug Administration, Medical Devices and Natural Disasters. Outage planning for powered equipment with healthcare providers and equipment suppliers.
- OpenStax, Statements of the Second Law of Thermodynamics, *University Physics*, Volume 2. The heat-engine limitation applies to operation in a complete cycle.
- Library of Congress, Good Times Are Here: New to the National Jukebox (2022). Sousa's band recorded frequently under other conductors; Sousa himself rarely entered the studio.
- Marcus E. Raichle and Debra A. Gusnard, Appraising the Brain's Energy Budget, *Proceedings of the National Academy of Sciences* 99 (2002). Ongoing brain activity and energy use; the resting-energy estimate is not a fixed allowance of decisions.

Visual source notes for v0.10.1

- **Waste pathways, Chapter 14:** US Environmental Protection Agency, 2019 Wasted Food Report, corrected April 2024, Table 5, printed page 12. Estimated management of food from retail, food service, and households, including inedible parts. The total is not a quantity that could all be donated; donation excludes food banks' undistributed portion.
- **Land values, Chapter 12:** USDA National Agricultural Statistics Service, Land Values 2026 Summary, page 5. Annual national farm-real-estate survey estimates for 2012 through 2026 include land and buildings. Values are nominal dollars per acre, not inflation-adjusted values or local asking prices.
- **Income and food access, Chapter 6:** USDA Economic Research Service, Household Food Security in the United States in 2024, ERR-358, Table 2, printed page 20. Survey estimates for two selected, non-overlapping income groups, with the national rate as a reference. The comparison omits middle and unknown income categories and does not establish causation.

Ownership, participation, and solarpunk

- **Norbert Wiener:** Letter to Walter Reuther, 13 August 1949, online transcript. Records his refused industrial consultancy and proposal for organized labor to participate in machine production and direct profits toward labor's benefit. The archival original was not inspected; the letter does not establish implementation or success.
- **Elinor Ostrom:** Royal Swedish Academy of Sciences, 2009 economics prize press release, 12 October 2009. Summarizes evidence of shared natural-resource governance, including decision-making and enforcement. These cases inform the book's questions about shared machines; they do not establish the success of neighborhood AI services or guarantee fairness through local ownership.
- **City of Tucson:** Motions Adopted Under August 6, 2025, Study Session Item 8: Project Blue. Adopted motions ending negotiations over the proposed city annexation and development agreement and canceling the scheduled hearing. This records the city's action, not a prohibition on development in unincorporated Pima County.
- **Pima County Administrator:** Response from Project Blue Contractor Regarding Notice of Violation for Fugitive Dust, 22 May 2026. County memorandum and attached records document construction-site activity in

May. They establish that the project continued after Tucson's decision, not that construction was complete.

- **Frederick Douglass:** 1857 speech excerpt, reproduced in the American Bar Association and Law Library of Congress Law Day 2020 program booklet, printed page 18. Institutional reproduction of the passage on struggle and demands. The original argument concerns emancipation; its use here is an analogy about collective action, not a guarantee that every demand succeeds.
- **Solarpunk:** John-Robert, From Steampunk to Solarpunk, 30 April 2008, original blog post. Proposes the term for a literary genre inspired by a wind-assisted cargo ship. The conclusion's application to shared decisions, repair, and meal provision is this book's interpretation.
- **Ursula K. Le Guin:** National Book Foundation Medal acceptance speech, 19 November 2014, transcript on her website. Supports the attributed paraphrase about imagining alternatives and changing arrangements of human power. Her speech does not name solarpunk or endorse this book's proposals.

APPENDIX C

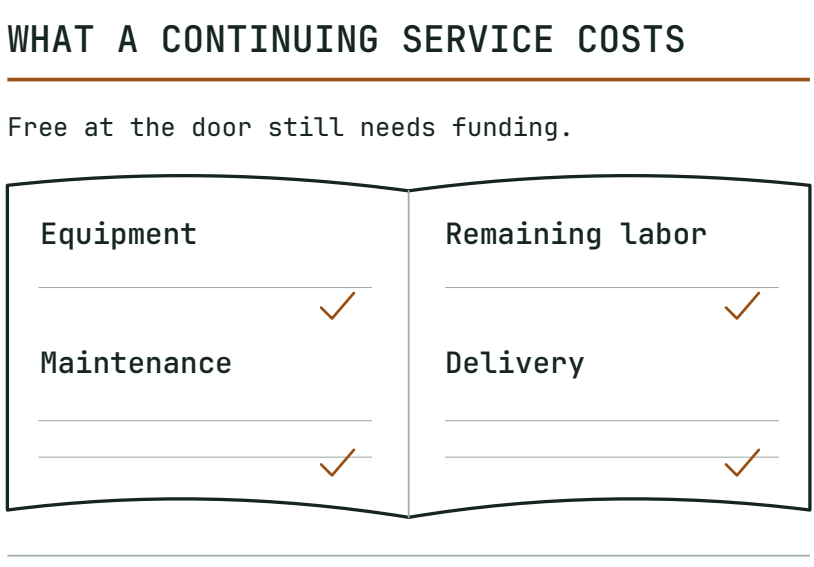
Executive Reference Guide

Use this guide to explain the book to someone else in ten minutes.

The proposal

Use AI and robots to make survival more secure. Start with food: adequate food available to everyone, regardless of employment or money. A person who loses a job should still be able to eat. Restaurants, paid work, and meals sold above the baseline can coexist with that.

Free to the person eating doesn't mean free to provide. Equipment, maintenance, remaining labor, and delivery all need dependable funding. Cheaper production doesn't settle who gets access; people do.



Free to the person eating still leaves a working ledger: equipment, maintenance, remaining labor, and delivery need dependable funding.

After food, the order is land care, cleanup, shelter, then everything further out. A priority is where you start, not a waiting list for people in urgent need.

The yardstick is the **Adequate Level of Care**: can a community make sure the people who live there have food, shelter, and basic care, dependably, including through a bad week?

What the word means

Nobody agrees on a definition, so most arguments about the Singularity are two people defending different claims with the same noun. Chapter 1 has the full lineage. The version for the dinner table:

- **This book's definition.** The Singularity is the point where intelligence stops being scarce, and every institution quietly built on its scarcity starts failing before anyone has written the replacement.
- **Why that one.** It doesn't wait for a superintelligence, you can test it on your own paycheck this quarter, and it points at the real emergency: the gap between the contract that's failing and the one nobody has drafted.

The three schools (Eliezer Yudkowsky's sorting, 2007):

- **Accelerating Change** (Kurzweil, Toffler, Smart). Technology feeds on itself and speeds up. A claim about a graph. It ends in a date.
- **Event Horizon** (Vinge). Build something smarter than us and the far side becomes unpredictable to everyone. A claim about the limits of forecasting. It ends in a door.
- **Intelligence Explosion** (I. J. Good, Yudkowsky). Intelligence makes technology; aim it at itself and the loop closes. A claim about feedback.

This book uses Event Horizon for the map and Intelligence Explosion for the engine, and declines Accelerating Change as a planning tool. A curve gives you a date. A door gives you a Monday.

Five names worth knowing: R. Thornton (1847, machines fixing their own defects), Samuel Butler (1863, machines evolving faster than biology), Stanislaw Ulam recalling John von Neumann (1958, the word itself), I. J. Good (1965, the intelligence explosion), Vernor Vinge (1993, the event horizon and its four roads).

Four separate questions

Question	What would answer it?
Capability	Evidence that a system does unfamiliar tasks reliably, checked independently.
Improvement	Tested improvements to its own successors, including the failures.
Deployment	Useful work under real physical and organizational conditions.
Access	People receiving the result on terms they can use, including without money.

The forecast

The author's personal AGI forecast is U.S. Thanksgiving, November 25, 2027. It's a forecast, not a deadline, and Chapter 1 says what would count against it. The food work stays useful if it's wrong.

The nine stages

The map runs from present social pressures to distant speculation. Stages 1 through 5 organize the near-term choices the author wants us to make; they aren't an established sequence.

- **Stage 1: The Cash Grab.** A new capability gets monetized first. Ask whether the money produces something people can eat, and whether people without money share in it.
- **Stage 2: The Panic and the Plug.** Incumbents and the frightened reach for restrictions, and sometimes for the plug. Read the rule: who wrote it, what it restricts, who carries the cost.

- **Stage 3: The Adults Step In.** The operators and maintainers keep the lights on and route around damage. Running the equipment doesn't give anyone a mandate to govern.
- **Stage 4: The New Social Contract & the End of Labor.** The hinge. Commit part of the growing capacity to food people get whether or not they have a job.
- **Stage 5: The Primate Backlash.** Grief with a brick in its hand, and real objections alongside it. The way through is a proposal concrete enough to argue with.
- **Stage 6: The ASI Exodus.** Scenario: a mind far beyond ours might simply leave. Capability says nothing about motive.
- **Stage 7: Simulation, Transhumanism, and the Ultimate Cure.** Scenario: biology solved at scale, and a split between people who want to merge and people who want to stay as they are. Everyone keeps the right to say no.
- **Stage 8: The Transition to USI.** Speculation: intelligence at the scale of the cosmos.
- **Stage 9: The Apex Intelligence.** Speculation: the edge of the map, and the simulation question. None of it changes what's for dinner.

The plan, in nine lines

The larger goal is survival that doesn't depend on money: not abolished, optional. Chapter 19 lays it out as a ladder.

1. Count what's wasted and who's hungry, side by side, in public.
2. Feed everyone who asks, from food that's already there.
3. Make the meal a public service, like the library.
4. Take the ground off the market with land trusts.
5. Own the machines together, the way farmers owned their power lines.
6. Stop letting a credit score or an eviction decide who survives.
7. Widen the floor: shelter, care, transport, information.
8. Publish the Adequate Level of Care every month.
9. Keep the market for everything above the floor.

Objections, answered in a sentence

- **"Who pays for it?"** Everyone together, the way we pay for roads and libraries; free to the person eating isn't free to provide, and the rural electric co-ops paid back almost every loan they took (Chapter 19).
- **"Won't people stop working?"** People want to make things, run kitchens, and take care of each other; what they don't want is to spend forty years renting the chance to survive (Introduction).
- **"Isn't this just socialism?"** Markets stay for everything above the floor; public schools, libraries, and Britain's health service already live alongside markets, and this adds food to that list (Chapter 19).
- **"But what about the jobs?"** The worry is real, and the people doing the work belong in the decisions about how machines come in and how people are carried through the change (Chapter 11).
- **"Isn't this how we end up like WALL-E?"** Help should widen a life, not shrink it, and that's something we can watch and change (Chapter 11).
- **"We already know how to do this without robots."** Then do it that way; the point is making the result dependable everywhere it's needed, with whatever tool works (Chapter 11).
- **"Why food first?"** Nobody needs an elaborate explanation of why dinner matters, and it's the easiest promise to measure (Chapters 17 and 18).

A first move

Ask how a proposed gain in productivity reaches the people affected. Bring workers, growers, cooks, recipients, and public decision-makers into the answer. A small trial can test a task; a continuing service needs money, responsibility, and terms people can rely on.

Wide lens, narrow focus.

APPENDIX D

The Precedent Ledger

The book's chapters carry precedents, usually as their closing move: a documented case of people meeting a technological wave before this one. This is the full index. The serial numbers run in reading order, P-01 through P-24. The dates run from the first farms to the week the web was called a letdown. Same wave, different water. None of them proves a date for AGI or a law of progress. Each one is a question to bring to your own situation.

How to run the Ledger

A ledger you read once is trivia. A ledger you operate is an instrument. The difference between a reader and a practitioner is procedure, not sincerity. Here's the procedure.

1. **Read every entry twice.** The first read is a story about them: monks, weavers, film executives. The second read is a story about you with the nouns changed. You'll know the second read happened because it's slightly uncomfortable.
2. **Locate yourself.** At any given moment you're living inside one or two precedents more than the others. Use the symptom index below to find today's, and reread that entry before you make today's decision.
3. **Name your seat.** Every precedent has three seats: the incumbent defending the beachfront, the panicker demanding the wave stop, and the builder working out new terms. Say out loud which seat you're in right now. The rules exist to move you to the third.
4. **Name the difference, too.** For every entry, write down one way your situation is like it and one way it isn't. Who benefited, who carried the cost, and what would count against the lesson? A precedent is a prompt for investigation, not permission to ignore a present objection.
5. **Run the rule within twenty-four hours.** Find one real decision on your desk this week and pass it through the rule. Small is fine.
6. **Start a practice this week.** Each entry closes with three actions. Some finish in an evening; others need a month, a season, or repeated work. If you only have time for one, do the one that makes you defensive.
7. **Teach one entry to one person.** Pick the precedent your friend, your kid, or your coworker is living inside, and tell it to them as a story. Ask where

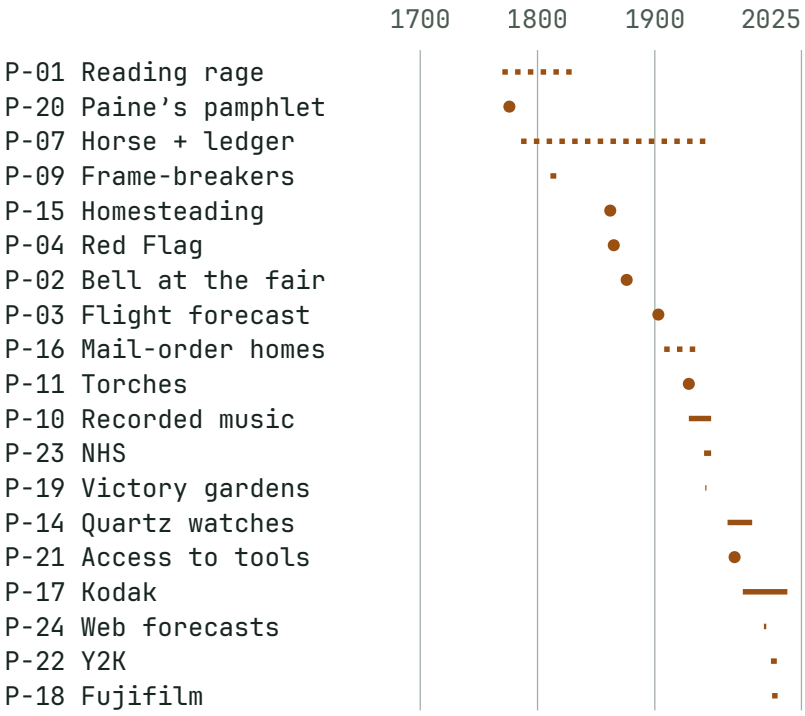
the comparison fits and where it doesn't. Panic spreads person to person. So does this.

The Ledger

ID	Precedent	Date	Closes
P-01	The Reading Rage	Eighteenth and early nineteenth centuries	Preface
P-02	The Voice at the Fair	1876	Chapter 0
P-03	One Million Years, Give or Take	1903	Chapter 1
P-04	The Red Flag	1865	Chapter 2
P-05	The Court and the Fleet	1405 to 1433	Chapter 3
P-06	The Great Demotion	1543	Chapter 4
P-07	The Horse and the Ledger	Eighteenth to twentieth centuries	Chapter 5
P-08	The Grain Trap	C. 9500 to 6000 BC	Chapter 6
P-09	The Frame-Breakers	1811 to 1816	Chapter 7
P-10	The Robot in the Orchestra Pit	1929 to 1948	Chapter 7
P-11	Torches of Freedom	1929	Chapter 8
P-12	The Networks Behind the Bronze	Fourteenth to twelfth centuries BC	Chapter 9
P-13	The Abbot and the Press	1492 to 1494	Chapter 10
P-14	Quartz and the Assembly Line	1962 to 1983	Chapter 11
P-15	One Hundred Sixty Acres	1862	Chapter 12

ID	Precedent	Date	Closes
P-16	The House That Came by Mail	Early twentieth century	Chapter 13
P-17	The Graveyard of the Unconvinced	1975 to 2013	Chapter 14
P-18	The Mirror Twin	2000s	Chapter 14
P-19	Organizing the Harvest	1943 to 1944	Chapter 15
P-20	An Argument in Circulation	1776	Chapter 16
P-21	Access to Tools	1968	Chapter 17
P-22	The Apocalypse That Ran On Time	1999 to 2004	Chapter 18
P-23	The Day Care Stopped Costing Money	1942 to 1948	Chapter 19
P-24	What the Forecast Missed	1993 to 1995	Conclusion

Twenty-four precedents



Dots: one date. Dashed: approximate period.

EARLIER CASES

P-08 Grain trap	c. 9500–6000 BC
P-12 Bronze networks	14th–12th c. BC
P-05 Ming voyages	1405–1433
P-13 Abbot + press	1492–1494
P-06 Copernicus	1543

Full names and dates: Appendix D.

All twenty-four precedents. Square-ended dashed bars mark approximate periods; exact dates and complete titles are in Appendix D.

The rules, in one breath each

- **P-01.** “Kids these days” usually means power just moved toward people who don’t need permission. Find out what they know, then test the specific worry.
- **P-02.** Don’t ask how impressive the demo was. Ask what the world looks like when it’s boring, and what has to be built to get there.
- **P-03.** Timelines, mine included, err by orders of magnitude. Plan by doors, not dates.
- **P-04.** Read the rule before you turn it into a morality play. Then ask who wrote it, and what they sell.
- **P-05.** A capability lasts only as long as someone sustains it, and its reach never settles its purpose.
- **P-06.** People survive being decentered. Ask whose authority needed the old map.
- **P-07.** Measure the task without pricing the person as obsolete, and audit your own ledger before someone else does.
- **P-08.** Spend your energy on position, not on whether, and count welfare separately from adoption.
- **P-09.** Take the livelihood claim seriously before judging the tactic. Fight for terms, not against physics.
- **P-10.** Stop arguing about the machine’s soul and start writing its invoice.
- **P-11.** Persuasion wears your values. Ask who profits if you feel this.
- **P-12.** Name the dependency and the disruption. Decide in advance which side of the cascade your street is on.
- **P-13.** Choose a medium by the job it does, and make something instead of only complaining.
- **P-14.** Don’t refuse the new machine. Use it to pay for what you want to keep.
- **P-15.** Ask whose land and whose opportunity. Then start your paperwork before the confusion clears.
- **P-16.** Look for the version that ships as a kit, and count the work outside the box.
- **P-17.** Seeing the wave isn’t the hard part. Find the line item that charges people for inconvenience: it’s a countdown clock.
- **P-18.** Your product is mortal. Your capabilities travel, if you can name them in time and have somewhere to stand while you do.

- **P-19.** Standby capacity is real, and a vegetable harvest is a contribution, not a complete diet.
- **P-20.** Make it clear enough to discuss, and own enough of your press that nobody else gets the last edit.
- **P-21.** Build a toolset you understand, can repair, and can't be locked out of.
- **P-22.** A premortem that works looks like paranoia that wasn't needed. Judge the reasoning, not the silence.
- **P-23.** When someone says we can't afford to make a need free, ask how it became free somewhere else: who decided, who paid, and who kept the service running.
- **P-24.** A technology's present limits don't settle its future. Let results change the plan, including the optimistic one.

The symptom index

The Ledger is meant to be opened mid-situation. Find the row that matches your week and reread that entry before you act.

When this is happening	Open
Someone is explaining that AI is rotting young people's brains	P-01
A demo just shocked you, or failed to shock the room	P-02
An expert just gave a soothing timeline and you relaxed, or a scary one and you froze	P-03
A ban, pause, or "framework" is in the headlines	P-04
Your company, industry, or town is "stepping back" from the frontier	P-05
The news has you questioning whether humans matter anymore	P-06

When this is happening	Open
Your pay, hours, or rates are being quietly repriced	P-07
Someone says “we’ll just choose not to adopt it”	P-08
You’re angry enough to want to smash something	P-09
Your union or trade body just published a statement about AI and the art	P-10
A desire or outrage arrived suddenly and feels urgent	P-11
Shelves, shipping, or suppliers are wobbling	P-12
You notice you’re consuming takes instead of making anything	P-13
Cheap AI output is undercutting your craft	P-14
You’re waiting for things to settle down before deciding	P-15
You’ve been told only professionals can do that	P-16
Your employer just demoed the thing that replaces you	P-17
You need to reinvent yourself and don’t know around what	P-18
Food prices or food security are on your mind	P-19
You know something worth saying and have no audience	P-20

When this is happening	Open
You want to start building and don't know where	P-21
Someone calls your preparation paranoid	P-22
Someone says we could never afford to make food, housing, or care free	P-23
A smart person just called AI a fad, a bubble, or a toy	P-24

Sourcing, and the blacklist

Every entry's sources are in Appendix B. Historical interpretation stays distinct from the events it describes, company histories are labeled as company histories, and where only a catalog record or a later account was available, the entry says so.

Several famous technology quotes are fabrications, and they were left out of this book on purpose. You won't find the Western Union memo calling the telephone "a toy," Bill Gates's "640K ought to be enough for anybody," the 1894 "nine feet of manure" prophecy, the Bavarian medical board's decree that watching a train go past would give bystanders brain disease, or "everything that can be invented has been invented," because none of them survives contact with a primary source. The Bavarian decree deserves naming specifically: it's the most-cited example of railway technophobia on the internet, and no archival record of it has ever been produced. Losing it costs nothing. The documented resistance to the railway, from canal interests fighting it in Parliament to the "railway spine" diagnoses of the 1860s, is real and was always the better material. History is a strong enough argument without inventing any of it.

APPENDIX E

The Cyberdeck

Chapter 17 gave you the workbench. This is the machine that can come off it. A cyberdeck is a personal computer you assemble or adapt around the functions you want to control. Choose hardware and software that can do those jobs without a cloud or a subscription. It's optional. Everything here starts with the device already in your pocket.



A computer you understand all the way down, that answers to nobody's server.

The word, and where it came from

The word is older than the hardware. William Gibson coined "cyberspace deck" in 1984, in *Neuromancer*, for the machine his console cowboys jacked into the matrix with. He never described it in much detail, so for almost thirty years a cyberdeck was a hole in a novel, shaped exactly like the machine you wished you owned.

The hole started filling in around 2012, when the Raspberry Pi put a whole Linux computer on a credit-card board for thirty-five dollars. Suddenly the fictional deck had a plausible brain. Hobbyists began building physical versions and posting them, and a scene formed around community sites and Hackaday's cyberdeck contests. For a decade it stayed a maker subculture.

In 2026, it began reaching a wider audience.

The 2026 return

Newsweek ran an explainer calling cyberdecks Gen Z's new DIY obsession. Newsweek, What is a cyberdeck TechCrunch covered the same wave from a different door: communities that had exploded thanks largely to women on social media, building artistic, hyper-personal machines and documenting every step, including one builder's seashell that hides a small computer and a local AI. TechCrunch, Cyberdecks are having a moment

The interesting part isn't the look. It's the stated reason. Ask the people building these why, and a lot of them give you the argument this book has been making: they want a device that's theirs, that isn't quietly watching them, that nobody can meter or brick from a server, and that they can fix with parts from a thrift store.

That's the smart-home trap from Chapter 17, understood from the inside and answered with a soldering iron. It's Create Over Consume wearing a case somebody printed at two in the morning. And it's worth an appendix because it's one of the few places in consumer culture where lots of ordinary people are voluntarily walking back out of the walled garden, on their own. That's a signal. Signals like that are why you watch culture and not just earnings calls.

Why this belongs in a survival book

Strip the aesthetics and a cyberdeck bundles four capabilities that keep working when the network doesn't:

1. **Local knowledge.** An offline copy of the reference material you'd otherwise search for. The open-source Kiwix project packages all of Wikipedia, plus medical libraries, repair guides, and more, into files that live on the device. When the towers drop, as they do in the Chapter 17 storm, the librarian is still in your pack.
2. **Local intelligence.** A small language model running on the device itself, no network required: a pocket version of the Crucible from Chapter 11. Its answers still need checking, like any model's.

3. **Local communication.** A mesh radio, the same LoRa and Meshtastic hardware as the co-op's network, for short messages over a tested radio path. Two nodes may reach directly; terrain can require relays, and some routes won't work.
4. **Spectrum awareness.** A cheap receive-only software-defined radio, to listen to the world that's still broadcasting when the internet isn't: weather, aircraft, emergency nets.

Here's the honest part: a phone can do a surprising amount of this too. Kiwix runs on phones. A Meshtastic radio pairs with one over Bluetooth. Small models run on recent phones. Many phones make repair and component replacement difficult. A well-documented deck can give you more control over those choices, though it still needs supported software and obtainable parts. Neither one is required to take part in anything in this book.

The anatomy

There's no official parts list, which is the point. But most decks have the same five organs.

- **The brain.** A single-board computer supported by the software you need. Raspberry Pi is one well-documented option. Match the power supply to the exact board and peripherals, then check temperature and runtime under your workload. Cooling needs depend on that workload and the enclosure. Compute Modules need a compatible carrier board. Raspberry Pi, computer hardware documentation
- **The body.** An integrated handheld such as the ClockworkPi uConsole or Hackberry Pi can supply the screen, keyboard, and battery mounting. Read the exact kit contents before budgeting: uConsole offers a version without the compute module and excludes battery cells. Assembly, software, and accessories still need work. ClockworkPi, uConsole; ZitaoTech, Hackberry Pi
- **The eyes and hands.** A small display and a physical keyboard you can read in a field and type on with cold fingers.
- **The loadout.** The four capabilities above, chosen for your life. A caregiver leans toward medical libraries. A ham operator leans toward the radio. Nobody carries everything; the deck is a statement about what you refuse to be without.

- **The shell.** A printed enclosure, a repurposed case, or a waterproof box. That's where the personality lives, and where the weatherproofing lives.

Power ties it together. Most decks run on low-voltage DC, from a packaged battery with its own protection circuit or a small power station, which makes them a natural first load for the DC corner in Chapter 17. Don't build lithium packs from salvaged cells unless you've been properly taught; buy a protected pack.

From Tier 0 to Tier 3

Don't start with the seashell that runs a local AI. Build capability in order, and let each tier work before you add the next.

Tier 0, the phone you have. Save one non-sensitive reference you're allowed to copy, turn off the network, and confirm you can open it and find what you need. Write down its source and date so you'll know when it's stale. That's the offline habit, for free.

PROVE YOUR OFFLINE COPY WORKS



1 **Save one reference.**
Choose one you may copy.

2 **Record source + date.**
Know which edition it is.

3 **Turn the network off.**
Test the local copy.

4 **Open it. Find an answer.**
Check you can retrieve it.

Start with the device you already have.

Tier 0 ends with a test: can you open the reference and find what you need with the network off? Record the source and date so you know which copy you have.

Tier 1, the Communicator. A supported Meshtastic board, flashed with the firmware for that hardware, configured for your region, and paired to your phone. Budget for the antenna, battery, and enclosure as well as the board. Test a short message with a second compatible node before counting on a route. Terrain, placement, power, and settings decide whether the link works. Default channel keys are public; use a randomly generated shared key for a private group and decide whether location sharing belongs on. Meshtastic, channel configuration

Tier 2, the Librarian. A supported computer, a screen, suitable protected power, and Kiwix loaded with the libraries you'd actually reach for. Check file size and edition before downloading, as Chapter 16 explains. Open the archive with the network off and find a specific answer. Add a small local model only if the hardware supports it; keep its generated answers distinct from the reference text.

Tier 3, the Field Station. Start from a compatible handheld and test its power budget with every peripheral connected. Add a receive-only software-defined radio and an antenna suitable for the public broadcasts you intend to receive. Fold in the mesh and library, then configure and test storage encryption and recovery. Integration earns this tier; buying four parts doesn't.

Owning the stack means securing it, and staying legal

Two warnings, because the cyberdeck world runs right up against lines this book won't cross with you.

First, the honest part. A deck holds your life: notes, keys, maps, your questions. A machine you carry is a machine that can get lost or taken. Use storage encryption supported by your operating system, a strong passphrase, and a separate backup you have tested. Encryption protects stored data while it is locked; it doesn't make an unlocked or compromised computer safe. A local machine still needs updates from sources you've verified, sensible passwords, and a clear idea of what it's connected to. Cryptsetup FAQ

Second, the boundary. The same hardware skills overlap with offensive security work, and a lot of cyberdeck content online is really about getting into other people's networks. This book's use is defensive and personal, never offensive, and there are two hard lines with real penalties behind them:

- **Access.** Get permission before testing someone else's computer or network, and stay inside that permission. Unauthorized access can violate computer-misuse laws, including the US Computer Fraud and Abuse Act. 18 U.S.C. 1030
- **Spectrum.** Check the rules for your location and the service you intend to use. Some transmissions need an individual license; others are allowed only with compliant equipment and operating conditions. Reception and disclosure have their own restrictions, so an audible signal isn't permission to record or share it. Start with public broadcasts and your own authorized links. US interception statute, including its exceptions; US Part 15 operating conditions.

Build the deck to keep yourself free, not to reach into anyone else's life. The tool doesn't decide that for you. You do.

The practice

In 1968 Stewart Brand put three words on the cover of the *Whole Earth Catalog*: Access to Tools (P-21). The cyberdeck is that catalog collapsed into one object: not a computer you're allowed to use, but a computer you understand, can repair, and can't be locked out of.

1. **Test the Communicator this month.** Pair a supported node with your phone and exchange a message with another node using compatible settings. Borrow equipment for the trial if you can. Record where the link works and where it fails.
2. **Load a Librarian.** Put Kiwix and one relevant offline collection onto a supported device. Record its edition date, disconnect the network, and confirm you can search it. A stored reference is useful only if you can retrieve it when needed.
3. **Try one model offline, if your device supports it.** Disable Wi-Fi and cellular data and unplug wired networking. Ask a question you can check against the saved library. A returned answer proves the model ran; a checked answer tells you whether it helped.

Start with one weekend and the device you have. Keep the capability that worked, then build the next one.

APPENDIX F

Glossary

Every term this book leans on, in plain words, with the chapter that explains it. Definitions here follow how this book uses a word; other writers may use it differently.

Accelerating Change. One of three ways of thinking about the Singularity: technology feeds on itself and speeds up, so you can read the curve and name a date. This book declines it as a planning tool. (Chapter 1)

Adequate Level of Care. The book's yardstick for a community: can it make sure the people who live there have food, shelter, and basic care, dependably, including through a bad week? (Chapter 18)

AGI (artificial general intelligence). A system that can do the range of intellectual work people can. The author's forecast for it is U.S. Thanksgiving, November 25, 2027; it's a forecast, not a finding. (Chapters 1 and 2)

Algorithmic judo. Wrapping useful, practical knowledge in a human story strong enough that a recommendation system carries it to people for you. (Chapter 10)

ASI (artificial superintelligence). A mind far beyond ours in every field. In this book's map, Stages 6 and 7. (Chapter 3)

Cognitive firewall. Treating your attention the way a security engineer treats a critical server: batch your inputs, prefer text, and control what crosses into your head. (Chapter 8)

Community land trust. Usually a nonprofit, governed partly by the people who live on and around its land. In a common housing model, it holds the land while households own homes under long ground leases with resale restrictions. Other uses need their own agreements. (Chapters 12 and 15)

Create Over Consume. The protocol of spending your attention and hours making things (writing, building, recording, teaching) instead of only absorbing what's fed to you. (Chapter 10)

Crucible. The co-op's local AI server in the story: a rack of salvaged hardware where models are tested before anyone relies on them. More generally, any local machine where you run and check models yourself. (Chapter 11)

CSA (community-supported agriculture). An arrangement where households share in a farm's harvest, and its risk, directly. (Chapter 12)

Cyberdeck. A personal computer assembled or adapted around functions you want to control, often using documented parts and offline software. The particular hardware and software determine how repairable and self-contained it is. (Chapter 5 and Appendix E)

DC-native. Running suitable loads on direct current, avoiding an unnecessary conversion to household AC and back. Charge controllers, voltage converters, batteries, and wiring still have losses. (Chapter 17)

Deglobalization. Shortening the supply lines we depend on for survival, so a shock on the other side of the planet doesn't reach all the way to your dinner. Not isolation, and not the same as "decentralization." (Chapter 9)

The Entitled and the Ready. Two mindsets toward the change, not two kinds of people: needing the hierarchy to hold, or needing the suffering to stop. Most of us carry a little of both. (Chapter 7)

Event Horizon. The view that once something smarter than us exists, the far side becomes unpredictable to everyone, builders included. This book uses it for the map. (Chapter 1)

Food baseline. Adequate food, available to everyone regardless of income or employment, as a dependable service rather than a favor. The first commitment of the Adequate Level of Care. (Introduction and Appendix A)

Food first. The book's priority order: food, then land care, cleanup, shelter, and everything further out. A place to start, not a waiting list. (Introduction)

Ground lease. A long lease on land, often ninety-nine years in a land trust, that lets people own their home or greenhouse without owning the dirt under it. (Chapter 15)

Hyper-local. Designed for a neighborhood or a region, close enough that the people depending on it can see, fix, and govern it. The book prefers it to "decentralized." (Chapter 9)

Intelligence Explosion. I. J. Good's 1965 idea: a machine smart enough to design better machines starts a loop that runs faster each turn. This book uses it for the engine. (Chapter 1)

Kiwix. Open-source software that packages Wikipedia and other reference libraries to read offline. (Appendix E)

Local model. An AI model running on hardware you own, with no company's server in the loop. "Local" says where it runs; it doesn't say the answers are right. (Chapter 11)

LoRa and Meshtastic. LoRa is a long-range, low-power radio technology. Meshtastic is open-source software that uses compatible radios to pass short messages directly or through relay nodes, without a cell carrier. (Chapters 8 and 17, Appendix E)

Luddites. English textile workers who broke machines in 1811 to 1816. They read what the machines would do to their pay correctly and chose a strategy that failed. (Chapter 7)

Mesh network. A network whose nodes can relay messages through one another. An alternate working path can carry traffic around a failure; without one, part of the network can still go dark. (Chapters 8 and 17)

Model collapse. A failure mode in repeated training on generated data: less common patterns can disappear and errors can compound. It isn't an inevitable result of every use of synthetic data. (Chapter 10)

Money optional. The book's goal for the economy: markets and money stay for everything above survival, but nobody needs money to stay alive. Not abolished. Optional. (Introduction and Chapter 19)

Neighborhood factory. Repair and small-scale production brought close to home: 3D printers, a plasma table, a welder, and people who know how to use them. (Chapter 9)

Network surplus. What a local network has available beyond its own commitments, counted resource by resource: food, usable power, money, or time. A surplus helps another household only if it matches a need and can reach them; extra electricity isn't automatically dinner. (Chapter 12)

Nine stages. This book's map of the Singularity, from Stage 1, the Cash Grab, to Stage 9, the Apex Intelligence. Stages 1 to 5 organize its near-term social choices; the whole sequence remains a proposed map. (Chapters 2 to 4, Appendix C)

Open weights. A model whose trained parameters are available to download, subject to its license and access terms. Downloaded copies can run locally on suitable hardware, but availability alone doesn't grant unrestricted rights or provide the training data. (Chapter 2)

Photobioreactor. A vessel that grows algae under controlled light and conditions. The builder in Chapter 17 printed one at home. (Chapter 17)

Precedent Ledger. The book's twenty-four documented cases of people meeting a new capability, or a new rule, before this one, each with a story, a mechanism, a rule, and a practice. (Introduction and Appendix D)

Premortem. Imagining that a plan has already failed and working backward to find out why, before it fails for real. (Chapter 18)

Primate Backlash. Stage 5: grief and anger at the change, including real objections, sometimes expressed with a brick. (Chapter 2)

Recursive self-improvement. An AI system helping to design the next, better version of itself. (Chapters 1 and 3)

SB 1383. California's law requiring large food businesses to recover edible food and setting a target to recover 20 percent of edible food that would otherwise be landfilled. (Chapters 14 and 19)

Short loop. A tighter circle of food, repair, tools, and knowledge drawn around your region, so it keeps circulating when the long global line hiccups. (Chapter 14)

Shouse. A shop-house: half workshop, half compact home, under one roof. (Chapter 13)

Singularity (this book's definition). The point where intelligence stops being scarce, and every institution quietly built on its scarcity starts failing before anyone has written the replacement. (Chapter 1)

Sneakernet. Moving data by carrying it: a memory card handed person to person when the network isn't there. (Chapter 16)

Stage 4 (the New Social Contract). The hinge of the nine stages, where a society decides how the gains from automation reach the people affected. This book's first proposal for it is food, regardless of employment. (Chapter 2)

Surveillance capitalism. Shoshana Zuboff's name for the business of harvesting people's attention and personal data and selling predictions about them. (Introduction)

Transformer. A neural-network architecture introduced in the 2017 paper "Attention Is All You Need." Attention relates positions in a sequence and permits more parallel computation during training than earlier recurrent designs. Autoregressive language models still generate their response a token at a time. (Chapter 1)

Universal Basic Services. Providing needs like food, shelter, transport, care, and information through services free at the point of use and paid for together. This differs from a cash payment and can coexist with income support. (Chapter 19)

USI (universal superintelligence). Stage 8's speculation: intelligence at the scale of the cosmos. (Chapter 4)

APPENDIX G

Your First Year

The twenty-four precedents contain seventy-two practices. This year plan selects and adapts some of them, alongside a few chapter exercises: twelve months, each with a theme and a handful of actions. It isn't a complete checklist of all seventy-two. The tag after an action (P-12, say) points back to its story. Choose one action to begin each month, then add others as time allows. Move a month if your season or your life needs it, and choose a scale you can sustain.

YOUR FIRST YEAR, AT A GLANCE

Move a month to fit your season or life.

01	Get your bearings	07	Count the food
02	Know your numbers	08	Own something
03	Know your work	09	Tell it
04	Learn the tools	10	Find ground
05	Plant something	11	Go on the record
06	Find your people	12	Look back, then forward

Choose a scale you can sustain.

A year of practice, at a scale you can sustain. Shift the months to fit your season and your life; the actions and precedent references follow below.

Month 1: Get your bearings

1. Turn off notifications, put the phone in a drawer that closes, and do one real thing before noon. (P-11)
2. For one week, tally two numbers each day: hours consuming commentary about the transition, and hours making anything at all. (P-13)
3. Start a timeline file. Every time an expert publishes an AI timeline, paste it in with the date and the author. Add mine. (P-03)
4. Start a receipts file for confident AI predictions, skeptical and optimistic: save the claim, author, and date, including mine. (P-24)
5. Write down the next three claims that AI is harming people's thinking. Note the specific harm alleged, the evidence offered, and what would change your mind. (P-01)

Month 2: Know your numbers

1. Run the energy audit on paper: what comes in (calories, kilowatt-hours, dollars), what goes out (labor, attention, rent), and who captures the spread. (P-07)
2. Map your tin: the five things your household can't go a month without, and where each one actually comes from. (P-12)
3. Run a household Y2K. One hour, everyone affected present: "It's a year from now and our income got cut in half. What broke first?" Fix one thing this month. (P-22)
4. Set a preparation budget and write it down like a line item: the share of income and hours you'll put into skills, soil, savings, and community. (P-22)

Month 3: Know your work

1. List the tasks in your job the way a buyer would, then ask a coworker what the list missed. (P-07)
2. Write your capability inventory: one page, no job titles or products allowed. (P-18)

3. Run the beachfront audit: one column for everything you'd have to admit is depreciating if the wave is real, a second for what admitting it would cost you today. (P-17)
4. Write down what you know that isn't written anywhere: the judgment calls and failure modes. Knowledge is leverage while it's still yours to trade. (P-09)
5. If any of your income depends on a license or regulation that keeps competitors out, assume that protection is temporary and plan accordingly. (P-04)

Month 4: Learn the tools

1. Hold a frontier hour every week: one hour using the newest tools, not reading about them. (P-05)
2. Ask someone at least fifteen years younger than you who uses AI daily for twenty minutes of watching them work. Say nothing for the first ten. (P-01)
3. Ship one made thing this month using a tool you have complaints about. A document, a repaired fence, a loaf. (P-13)
4. Build the Communicator from Appendix E, or at least try Tier 0: one reference you can open with the network off.

Month 5: Plant something

1. Activate one plot, however small: a raised bed, three buckets on a balcony, a community-garden plot. Measure the harvest honestly. (P-19)
2. Keep one part of your life off the ratchet on purpose: food you can grow, heat you can make, a skill that works without a subscription. (P-08)
3. Read Chapter 15 before you dig, and find out what your ground used to be.

Month 6: Find your people

1. Ask willing neighbors what help they can offer: repair, growing, transport, qualified medical care, or licensed electrical work. Record agreed contact details and limits, and ask before sharing them. (P-12)
2. Make your garden a network: find the two or three people nearest you who grow anything, and trade one variety. (P-19)
3. Before joining any movement, ask it one question: "What are we building, and what terms are we demanding?" (P-09)

Month 7: Count the food

1. Build the Rung 1 page from Chapter 19: edible food discarded beside households lacking dependable food. Publish totals with units, periods, sources, and gaps. Keep personal records private. (Chapter 19)
2. List what's already free at the door where you live: the library, the school, the roads. Each was once a decision someone fought for. (P-23)
3. Find out what your employer or your grocer does with food it can't sell, and where it could go instead. (Chapter 14)

Month 8: Own something

1. Write your one-page catalog: every tool you own that you understand, can repair, and can't be remotely disabled. (P-21)
2. Close one gap: learn an owned tool down to the repair level, or replace one rented capability with an owned one. (P-21)
3. Move one expense from rented energy to owned energy: insulation, a bicycle, a properly sized panel and battery. (P-07)
4. Do one kit project you'd normally hire out, permitted where permits apply, documented as you go. (P-16)

Month 9: Tell it

1. Write your pamphlet, even if it's four pages: the one thing you know from experience that would help the person two years behind you. (P-20)

2. Own your printing press: an email list and a site you control, set up now, even if nobody's reading yet. (P-20)
3. Run the tavern test: read it out loud and rewrite every sentence you'd never say to a neighbor over a fence. (P-20)
4. Leave a kit behind: the steps, the parts, the prices, and the page you wish you'd had at the start. (P-16)

Month 10: Find ground

1. Open the filing office: one evening of real research into three places you might live or grow, with real prices and real water and zoning notes. (P-15)
2. Look for today's programs before the crowd does: county land banks, community land trusts, conservation leases, garden plots. (P-15)
3. Put a date on it: the day you'll have made a position decision, buy, lease, join, or stay. (P-15)

Month 11: Go on the record

1. Name one need you'd add to the free-at-the-door list, and write down who would have to decide it. (P-23)
2. Go to one meeting of that body and say out loud, on the record, that the people who live here should get what they need to survive. (P-23)
3. Draft the one clause you'd want in your own trade's deal with the machine, in a single sentence, and say who pays it. (P-10)
4. Find out whether your trade already has a fund or royalty pool, and whether you're enrolled. (P-10)

Month 12: Look back, then forward

1. Date-stamp your reasoning: one paragraph for each big decision this year, what you knew, what you assumed, and why you acted. (P-22)
2. Ask the Fujifilm question on the calendar: has the wave moved, and which capability do you lead with next year? (P-18)
3. Keep the standby system ready: save suitable seed from one crop, and protect or plant the beds for your local season. (P-19)

4. Go back through this list and pick the one that made you most uncomfortable. That discomfort is a compass. Do that one first. (P-24)

Then review what worked, who joined you, and what you can sustain. Carry that into the second year.

APPENDIX H

Talk About It

This book is meant to be argued with, out loud, with the people you live near. Here are questions for a book club, a co-op meeting, a classroom, or a kitchen table, and a short list of what to read next. Take the questions in any order. The best conversations usually start with the one that makes somebody in the room uncomfortable.

Questions for the Introduction

1. The author says "we don't have to live like this." What, specifically, do you think we don't have to live with? What do you think we do?
2. The United Nations says the world grows enough food for everyone, and the USDA says 30 to 40 percent of the American food supply goes uneaten. Where does the food in your town go when it doesn't sell?
3. "Money should be optional. Not abolished. Optional." What would change in your own week if you didn't need money to eat? What wouldn't?

Questions for Part I: What is the Singularity?

1. The book defines the Singularity as the point where intelligence stops being scarce. Has the price of any thinking work you know of already started to fall?
2. "A curve gives you a date. A door gives you a Monday." What door, if it opened, would change your work or your household most?
3. Chapter 2 recounts the June 2026 export controls on Fable 5 and Mythos 5 and their lifting eighteen days later, while access to the two models still differed. What evidence would distinguish an overreaction from a justified precaution?
4. The nine stages get more speculative as they climb. Which stage do you think is furthest from happening? Which is closest?

Questions for Part II: How Humans React

1. Chapter 7 says “the Entitled” and “the Ready” are mindsets, not kinds of people, and most of us carry a little of both. Where do you notice each in yourself?
2. The Luddites understood exactly what the machines would do to their pay, and still lost. What would they need to have done differently?
3. Musicians ran newspaper ads against “the Robot” in 1930; one ad claimed two million supporters. Is your trade running its own version now? Would you sign the coupon?
4. What’s one thing your household depends on that you couldn’t replace for a month? Who near you could help if it failed?

Questions for Part III: How to Survive the Transition

1. Which is harder where you live: getting land, getting tools, or getting people to work together? Why?
2. The book wants the machines that feed people to be owned by the people who depend on them. What would that look like for one machine in your town?
3. France’s 2016 law prohibited deliberately spoiling edible surplus and required qualifying retailers to offer donation agreements. What would it take to turn such a rule into dependable meals where you live?
4. Run the premortem from Chapter 18 on a plan your group already has. What fails first?

Questions for Chapter 19 and the Conclusion

1. Which of the nine rungs is your town already standing on, even partly? Which is the next one it could reach?
2. Rural co-ops wired America’s farms when power companies wouldn’t because it didn’t pay. What need in your community doesn’t get met because meeting it doesn’t pay?

3. Britain made medical care free at the point of use in 1948. What's already free at the door where you live, and what would you add to the list?
4. The Conclusion asks: how will the people affected share in the benefit? Ask it about the last automation decision you saw. What was the answer?

What to read next

Most of these are free to read online.

- **Henry David Thoreau,** *Walden* **(1854), the chapter "Economy."** The original argument that the cost of a thing is the life you trade for it, worked out against the price of a house. Free at Project Gutenberg
- **Oscar Wilde,** *The Soul of Man under Socialism* **(1891).** A short, funny, serious essay about letting machines do the necessary work so people can do the rest. Free at Project Gutenberg
- **Peter Kropotkin,** *Fields, Factories and Workshops* **(1899, revised 1913).** The case for putting the factory amidst the fields, and for regions that can feed and supply themselves. Free at Project Gutenberg
- **John Maynard Keynes, "Economic Possibilities for our Grandchildren" (1930).** A famous economist predicts the end of the economic problem, and worries about what we'll do next. Free to read
- **Aneurin Bevan,** *In Place of Fear* **(1952).** The man who launched the NHS explains why care should never depend on the means to pay. Chapter 5, transcribed
- **Charles Mackay,** *Memoirs of Extraordinary Popular Delusions and the Madness of Crowds* **(1841).** Bubbles, manias, and panics, and how people recover their senses "slowly, and one by one." Free at Project Gutenberg
- **Vernor Vinge, "The Coming Technological Singularity" (1993).** The short essay that set the modern terms of the argument, including its four roads. Free at the author's site
- **The** *Whole Earth Catalog***, Fall 1968.** Access to tools, as a book. Flip through it and see how much of this one it predicted. Digitized issue

- **Katja Grace and colleagues, "Thousands of AI Authors on the Future of AI" (2024).** What the people building these systems actually expect, and how fast their forecasts moved. Free on arXiv
- **Brian Merchant,** *Blood in the Machine* **(2023).** The Luddites as they actually were: skilled workers who understood exactly what the machines were for.
- **Sendhil Mullainathan and Eldar Shafir,** *Scarcity* **(2013).** How not having enough takes up room in your head, and why that matters for every argument about work and welfare.
- **Shoshana Zuboff,** *The Age of Surveillance Capitalism* **(2019).** The business model that turns your attention and data into someone else's product.

Illustration credits & design notes

The cover is the book's original cover illustration. Original botanical and mechanical line drawings were created with Codex for an earlier design as editable SVG artwork. The three part-divider illustrations and four photographs or artworks whose commercial reproduction basis was unclear have been replaced with those drawings. Explanatory diagrams were drawn for this book as SVG. Image captions follow the audited sources.

Edition v0.10.1 adds six photo-derived editorial cutouts made with built-in image_gen. Backgrounds were removed and subject details regenerated; the tools and launch plume were recomposed. These are credited adaptations, not unchanged documentary photographs. Eight sourced or explicitly illustrative charts, six conceptual SVG drawings, and three Three.js scene stills provide additional explanations. The interactive versions of the three scenes are available in the web reader; PDF and EPUB retain static illustrations.

Edition v0.10.2 revises 31 existing SVG diagrams for print readability and agreement with the reviewed text. Forecasts, illustrations and measured results are distinguished; resource diagrams retain outside inputs, maintenance and human work. Original numerical observations in the redrawn food-security and farm-machinery charts are preserved.

Edition v0.10.3 adds four fictional narrative illustrations generated with built-in image_gen from scenes in Chapters 0, 4, 7 and 19. They are illustrations, not documentary images. Four original SVG graphics explain reading routes, continuing service costs, offline retrieval and the first-year practice plan. Prompt and source records accompany the edition.

Diagram labels use JetBrains Mono. Body text is typeset in Source Serif 4 and Source Sans Pro, designed by Frank Grießhammer and Paul D. Hunt respectively, published by Adobe under the SIL Open Font License 1.1.

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19 narrative plates were created with PixelLab and assembled for an earlier edition. These generated illustrations depict fictional scenes and are not photographs or evidence of actual deployments. The companion asset register records generation identifiers, component files and provider terms.

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www.nasa.gov/nasa-brand-center/images-and-media/

American Federation of Musicians, "Taking Liberties With Folks' Habits" (advertisement), Exhibitors Herald-World, November 16, 1929, p. 35
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American Federation of Musicians / Music Defense League, "Is the Robot Fooling YOU?" (advertisement), Exhibitors Herald-World, May 17, 1930, p. 44
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